# Closed Hamiltonian BF Theory as the Derived Poisson Center of a Dirac Brane Probe

A formal local mixed holomorphic-topological model on $\mathbb{R}^2 \times \mathbb{C}^2$

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**Abstract**

We study the formal local mixed holomorphic-topological brane at the symplectic disk with coordinates $z_1, z_2$ and form $dz_1 \wedge dz_2$. On a stack of $N$ probes these coordinates are matrices $\phi_1, \phi_2 \in \mathfrak{gl}_N$, and the open ghost-zero system is the Dirac first-order theory

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big)$$

with first-class constraint $[\phi_1, \phi_2] = 0$ and BV/Koszul variable $\psi = A^*$ satisfying $Q\psi = [\phi_1, \phi_2]$. Closed Hamiltonian modes act through the reduced CE/PV Hamiltonian central-operation model. A full $P_0$-factorization-centre theorem for the unreduced open theory is not asserted. The cotangent modes are not polynomial descendants: in the local theorem they are Grothendieck–Matlis residue currents in

$$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1) \subset H^2_{\mathfrak{m}}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2 = \bigoplus_{a+b>0} \mathbb{C}\, \rho_{a,b}.$$

The stable polynomial one-$\psi$ classes $\Psi_{a,b}$ are PBW special-fibre open labels; they are not another basis of $\mathcal{P}$ as an $\mathfrak{h}$-module.

The finite algebraic part proves the Dirac brane reduction, the scalar-reduced degree-zero Weyl/Moyal comparison, and the scalar $U(1)$ anomaly. The degreewise stable part proves the PBW trace sector, the cochain-level CE/PV derived-centre map for $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$, the Matlis principal-part cotangent module, the compactly supported brane-line current model, and the principal-part defect model after de Rham-current contraction. The weighted Tate construction is proved only in the weighted Tate coefficient model and makes no regulator-independence claim. The first and third Costello graph normalizations are reduced open-line Weyl/Moyal boundary coefficient computations to stated order. The all-order brane-defect graph realization is conditional on a mixed brane-defect QME theorem, the radial-parts input where finite $N$ is used, and the required centrality homotopies. The unreduced cotangent factorization-centre morphism, a one-antifield descendant Moyal lift, a Rees/Fourier bridge between $\Psi_{a,b}$ and $\rho_{a,b}$, compact Calabi–Yau consequences, BKM/Igusa consequences, and sister-volume transfers are not asserted here.

## Claim-strength ledger

The status vocabulary in this ledger is: *proved in finite algebraic model*, *proved degreewise stable*, *proved in weighted Tate coefficient model*, *computed to stated order*, *conditional on named QME/radial-parts/unweighted-regulator theorem*, *reduced target constructed; unreduced lift open*, *conjectural*, *open*, and *not asserted*.

### *Main theorem-status exclusions*

| Claim a reader might infer | Status here | Replacement statement |
|---|---|---|
| Full $Z^{P_\circ}_{\mathrm{fact}}$ theorem for the open brane algebra | *not asserted* | The proved object is the reduced CE/PV Hamiltonian central-operation model. The unreduced factorization-centre morphism and centrality homotopies are open. |
| Full unreduced BV identification | *not asserted* | Only the finite ghost-zero Dirac/Koszul model and reduced principal-part extensions are used. |
| Conilpotent Koszul duality for the full formal Hamiltonian algebra | *not asserted* | Bar-cobar arguments apply only under the stated Tate-conilpotent, nilpotent, or admissible-envelope hypotheses. The full formal Hamiltonian algebra is treated by direct CE/PV computation and by conditional replacements. |
| Bare $R$-submodule Matlis annihilator wording | *not asserted* | The object is the $\mathbb{C}$-linear residue annihilator $\mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)$ inside top local cohomology, with the Hamiltonian coadjoint action. |
| Unconditional finite-$N$ radial-parts theorem | *conditional on named QME/radial-parts/unweighted-regulator theorem* | The formal Moyal coefficient theorem is algebraic. The finite-$N$ comparison uses the named Harish-Chandra–Wallach–Levasseur–Stafford radial-parts input. |
| Normalized-Moyal vanishing beyond the Poisson bracket | *not asserted* | The raw star commutator has only odd powers. The normalized bracket $\hbar^{-1}[f,g]_*$ generally has even powers $\hbar^{2s}P^{2s+1}$, including a possible $\hbar^2P^3/24$ term. |
| Unconditional Costello realization from reduced Wick diagrams | *not asserted* | The paper computes reduced open-line Weyl/Moyal boundary coefficients to first and third order. A Costello graph realization requires the analytic brane-defect graph/QME package. |
| Exactly three QME mechanisms | *not asserted* | The manuscript lists known candidate mechanisms. Exhaustiveness would require a computation of the relevant $H^1$ obstruction complex. |

### *Claim ledger*

| Claim | Status | Scope and exclusions |
|---|---|---|
| Dirac brane probe | *proved in finite algebraic model* | Finite-$N$ constrained mechanics on the matrix probe. The BV/Koszul variable satisfies $Q\psi = [\phi_1, \phi_2]$. |

| Claim | Status | Scope and exclusions |
|---|---|---|
| Stable Hamiltonian trace sector and $\Psi_{a,b}$ labels | *proved degreewise stable* | Stable connected range only. The classes $\Psi_{a,b}$ are primitive one-$\psi$ PBW special-fibre classes; they are not principal-part cotangent modes and do not realize $\mathfrak{h}^\vee_{\mathrm{cont}}$. |
| CE/PV reduced Hamiltonian central-operation model | *proved degreewise stable* | Cochain-level identity for $\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$, with $c^I \mapsto \theta^I$ and $u_I \mapsto O_I$. The conilpotent bar-cobar statement is used only inside its stated Tate-conilpotent hypothesis. This is not the full unreduced factorization-centre theorem. |
| Principal-part cotangent module | *proved degreewise stable* | Residue and Matlis/local-cohomology identification $\mathfrak{h}^\vee_{\mathrm{cont}} \cong \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1) \subset H^2_{\mathfrak{m}}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2$. Polynomial descendants do not realize the coadjoint module. The Fourier–Rees bridge is only an associated-graded label comparison. |
| Brane-line factorization currents | *proved degreewise stable* | Compactly supported de Rham-current convention $\mathfrak{h}_I = \Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h}$. The length-one source of the boundary Hamiltonian observable is the shifted-cotangent coordinate $u_{a(t)dt\otimes f}$. |
| Reduced principal-part boundary sector | *proved degreewise stable* | Reduced defect-current model after de Rham-current contraction only. This is not an unreduced open observable algebra. |
| Primitive one-$\psi$ projection | *proved degreewise stable* | Indecomposable $P_0$ shadow of the reduced comparison. It does not construct the unreduced cotangent factorization-centre morphism or its centrality homotopies. |
| Degree-zero Moyal/Weyl quantization | *proved in finite algebraic model* | Scalar-reduced degree-zero Hamiltonian sector, with the Capelli/Weyl normalization and formal Moyal commutator, modulo the distinguished scalar anomaly line $[\bar{c}]$. No one-antifield descendant lift and no graph realization are included. |
| Weighted Tate regulator | *proved in weighted Tate coefficient model* | Internal coefficient-kernel and finite-scale graphwise regulator. Independence of the admissible finite-window weight is proved; equivalence with the strict unweighted product/direct-sum graph theory is not asserted. The mixed brane-defect QME is separate. |

| Claim | Status | Scope and exclusions |
|---|---|---|
| Strict unweighted product/direct-sum regulator endpoint | *not asserted* | The positive endpoint theorem is impossible under the Costello coefficient-kernel and finite-window compatibility hypotheses: Theorem 3.74 proves that no tensor in the strict product/direct-sum pair projects to all finite-window Casimirs. Tail-sensitive ordinary cohomology of the weighted product is also not identified with the strict direct-sum dual. |
| Scalar $U(1)$ anomaly | *proved in finite algebraic model* | Closed polynomial scalar cocycle, finite-$N$ open centre, and Capelli/Weyl scalar shift. It records the obstruction to unreduced lifting; it is not a compact Calabi–Yau, BKM, Igusa, or sister-volume transfer. |
| First and third Costello graph normalizations | *computed to stated order* | Only the first and third graph normalizations are claimed. They match the formal Moyal coefficients in the stated normalization. |
| All-order Costello graph realization | *conditional on named QME/radial-parts/unweighted-regulator theorem* | Requires a mixed brane-defect QME theorem, the all-$N$ radial-parts source input, any unweighted-regulator comparison if the strict product/direct-sum theory is used, and a centrality-homotopy theorem in the same normalization. |
| Quantum descendant lift | *reduced target constructed; unreduced lift open* | Construction 3.103 constructs the reduced Moyal principal-part target. The unreduced cotangent factorization-centre morphism is not constructed here, and Theorem 3.104 rules out strict $\hbar$-adic deformations of the classical $\Psi_{a,b} \mapsto \rho_{a,b}$ projection with PBW linear leading action. |
| Stable bulk–boundary Koszul duality | *conjectural* | Strategic large-$N$ horizon only. It is not used to prove the local Hamiltonian BF/Moyal core. |
| Cross-volume consequences | *not asserted* | No compact Calabi–Yau, global BCOV, Vol III, chiral-homology, Borcherds, BKM, Igusa, $K3 \times E$, or sister-volume theorem follows from this local Hamiltonian BF/Moyal calculation without a separate matched-conventions proof. Lemma E.4 and Lemma 5.5 state the non-transfer claim; they are firewalls, not export theorems. The local theorem may be cited only as a $d = 2$ formal-disk normalization after the target volume supplies its own global datum and proof. |

## Local dictionary

*Common local rings and quotients.*

| Symbol | Meaning |
|---|---|
| $R = \mathbb{C}[[z_1, z_2]]$ | Complete local ring of the formal symplectic disk. |
| $\mathfrak{m} = (z_1, z_2)$ | Maximal ideal of functions vanishing at the brane point. |
| $\mathfrak{h} = R/\mathbb{C} \cdot 1$ | Reduced Hamiltonian Lie algebra of the formal disk. Constants are removed because they generate the zero Hamiltonian vector field. |
| $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$ | Polynomial Hamiltonian sector used in explicit trace and Moyal calculations. |
| $\bar{A}_{\mathrm{desc}}$ | Abstract polynomial label space for PBW one-$\psi$ descendants. It is not the Matlis principal-part module. |
| $[\bar{c}]$ | Distinguished scalar $U(1)$ anomaly class in $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$, realized by the constant term of the Poisson bracket and by the Capelli shift. |

*Separation rule.* The next three blocks are disjoint. PBW special-fibre labels $H_{a,b}$, $O_{a,b}$, $\Psi_{a,b}$, CE/PV labels $c^I$, $u_I$, $\theta^I$, $O_I$, and Matlis labels $\rho_{a,b}$ live in different modules unless a specific theorem supplies a comparison. The only Fourier–Rees comparison asserted here is Theorem A.8: it matches associated-graded labels after Fourier normalization through the Fourier delta Rees lattice $\mathcal{D}_\hbar/(z_1, z_2)$, not through an $\mathfrak{h}$-module identification and not through the full Cech principal-part lattice before $\hbar$ is inverted.

*PBW special-fibre open trace labels.*

| Symbol | Meaning |
|---|---|
| $H_{a,b} = z_1^a z_2^b$ | Polynomial Hamiltonian label in $\bar{A}$, with $a + b > 0$. In the PBW special fibre it labels the scalar-reduced trace observable $O_{a,b}$. |
| $O_{a,b}$ | Open ghost-zero trace class $\overline{\mathrm{Tr}(\phi_1^a \phi_2^b)}$. |
| $\Psi_{a,b}$ | Primitive one-$\psi$ stable open class, represented by the symmetrised trace with one $\psi$, $a$ copies of $\phi_1$, and $b$ copies of $\phi_2$. It lies in the PBW special-fibre open shadow and carries the corresponding polynomial action. |
| $\mathsf{F}_{\mathrm{Rees}}$ | Fourier–Rees label bridge of Theorem A.8. It sends the polynomial label $z_1^a z_2^b$ to the Fourier-delta label $\partial_1^a \partial_2^b e_0$, whose Cech realization is $(-1)^{a+b} \hbar^{a+b} a! b! \rho_{a,b}$. After inverting $\hbar$ one may divide by this Rees factor and read the principal-part label $\rho_{a,b}$. It changes polarization and is not an $\mathfrak{h}$-module identification between $\Psi_{a,b}$ and $\rho_{a,b}$. |

The shared indices on $H_{a,b}$, $\Psi_{a,b}$, and $\rho_{a,b}$ are labels, not an $\mathfrak{h}$-module identification. The span of the $\Psi_{a,b}$ is not $\mathfrak{h}^\vee_{\mathrm{cont}}$, not $\mathcal{P}$, and not a principal-part current module.

*Closed CE/PV labels.*

| Symbol | Meaning |
|---|---|
| $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ | Shifted cotangent Lie algebra of Hamiltonian canonical transformations. |
| $H_I$ | Hamiltonian generator in $\mathfrak{h}$. |
| $c^I$ | Chevalley–Eilenberg coordinate dual to a Hamiltonian generator $H_I \in \mathfrak{h}$. Under $\Phi$, $c^I \mapsto \theta^I$. |
| $u_I$ | Chevalley–Eilenberg coordinate dual to the shifted cotangent generator. Under $\Phi$, $u_I \mapsto O_I$. Boundary Hamiltonian observables come from $u_I$, not from $c^I$. |
| $\theta^I$ | Polyvector coordinate corresponding to $c^I$ under the CE/PV derived-centre isomorphism. |
| $O_I$ | Hamiltonian observable coordinate corresponding to $u_I$ under the CE/PV derived-centre isomorphism. |
| $\Phi$ | Cochain-level CE/PV map determined on generators by $c^I \mapsto \theta^I$ and $u_I \mapsto O_I$. |

*Matlis principal-part labels.*

| Symbol | Meaning |
|---|---|
| $\mathcal{P}$ | Reduced principal-part module $\mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1) \subset H^2_{\mathfrak{m}}(R)\, dz_1 dz_2$, identified with $\mathfrak{h}^\vee_{\mathrm{cont}}$. |
| $\rho_{a,b}$ | Principal-part residue current $z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$, $a + b > 0$, with the Matlis coadjoint action. |
| $z_1^p z_2^q \cdot \rho_{a,b}$ | Coadjoint action $-(pb - qa + p - q)\rho_{a-p+1,b-q+1}$. This is the Matlis principal-part action, not the PBW action on $\Psi_{a,b}$. |
| $\mathcal{P}_I$ | $\mathcal{P}_I = \mathcal{D}'_c(I) \widehat{\otimes} \mathcal{P}$, the compactly supported distributional principal-part currents on an interval $I$. |

Thus $\Psi_{a,b}$ and $\rho_{a,b}$ live in different modules:

$$\Psi_{a,b} \in \text{PBW special-fibre open homology}, \qquad \rho_{a,b} \in \mathcal{P} \cong \mathfrak{h}^\vee_{\mathrm{cont}}.$$

No ordinary $\mathfrak{h}$-module isomorphism between these two spans is used or asserted.

*Interval-current and boundary labels.*

| Symbol | Meaning |
|---|---|
| $\mathfrak{h}_I$ | $\mathfrak{h}_I = \Omega^\bullet_c(I) \widehat{\otimes} \mathfrak{h}$, the compactly supported de Rham Hamiltonian currents on an interval $I \subset \mathbb{R}$. Degree-zero test functions appear only after taking the explicit measurement notation. |
| $\mathfrak{g}_I$ | $\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1]$, the shifted cotangent current algebra on $I$, with principal-part currents in the shifted cotangent direction. |
| $B_f(a)$ | Smeared boundary Hamiltonian observable $\int_I a(t) \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt$. Its CE source is $u_{a(t)dt \otimes f}$. |
| $A^{\mathrm{Ham}}_\partial(I)$ | Reduced Hamiltonian brane-line $P_0$ factorization algebra $\widehat{\mathbf{S}}(\mathfrak{h}_I)$. |
| $A^{\mathrm{pp}}_\partial(I)$ | Reduced principal-part defect-current model after de Rham-current contraction $\widehat{\mathbf{S}}(\mathfrak{h}_I) \widehat{\otimes} \widehat{\mathbf{S}}(\mathcal{P}_I[1])$. |

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# 1 Three principles

After the symplectic primitive, scalar quotient, and residue normalization are fixed, the local model is organized by three source principles. Each is physical in origin, but each is used here only after translation into an algebraic hypothesis or target. Together they specify the hypotheses and algebraic targets for the open brane theory, the closed Hamiltonian BF theory, the closed-open comparison, and the cotangent coefficient module proved below.

## *1.1 Dirac: a brane stack is a constrained probe*

Fix the formal holomorphic symplectic disk

$$\widehat{\mathbb{C}^2}_0 = \operatorname{Spf} \mathbb{C}[[z_1, z_2]], \qquad \omega = dz_1 \wedge dz_2.$$

A stack of $N$ branes supported on the topological line probes this disk by replacing the normal coordinates by matrices

$$z_1 \rightsquigarrow \phi_1, \qquad z_2 \rightsquigarrow \phi_2, \qquad \phi_i \in \mathfrak{gl}_N.$$

The pulled-back symplectic form is

$$\Omega_N = \operatorname{Tr}(d\phi_1 \wedge d\phi_2),$$

with primitive $\mathrm{Tr}(\phi_1\, d\phi_2)$. Change of basis in the probe stack is gauge. Its moment map is

$$\mu(\epsilon) = -\,\mathrm{Tr}(\epsilon[\phi_1, \phi_2]), \qquad \epsilon \in \mathfrak{gl}_N.$$

Dirac reduction therefore imposes the first-class constraint

$$[\phi_1, \phi_2] = 0.$$

After the primitive $\mathrm{Tr}(\phi_1\, d\phi_2)$ and moment map convention have been fixed, the corresponding first-order world-line action is

$$S_\partial = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

In BV form the constraint is the Koszul differential

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0,$$

where $\psi = A^*$. Thus the open algebra is not postulated as a trace algebra. It is the algebra of gauge-invariant functions on the derived commuting-pair locus. Lemma 3.6 and Proposition 3.7 give the formal proof.

### 1.2 *Witten convention: closed fields target central operations*

Witten's open string field theory supplies the cyclic open-field-theory source convention, not a theorem imported into this local model. The closed-open expectation is that a closed insertion approaching a brane does not merely evaluate to a boundary function; it should give an operation on compatible boundary observables. In a topological direction the algebraic target for such an operation is centrality up to derived Poisson homotopies. The target tested in this paper is therefore

$$Z^{P_0}_{\mathrm{red,Ham}}(\mathrm{Obs}_{\mathrm{open}}),$$

not $\mathrm{Obs}_{\mathrm{open}}$ itself. The centrality statement proved below is the reduced CE/PV calculation in this formal Hamiltonian sector; no global open string field theory or physical centrality theorem is imported from Witten.

For the formal symplectic disk the infinitesimal closed symmetries are Hamiltonian vector fields. Constants generate the zero vector field, so the Hamiltonian Lie algebra is

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1, \qquad \{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g.$$

In the local model the closed BV algebra of canonical transformations is the shifted cotangent theory

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1].$$

The local theorem to be proved is the cochain identity

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \cong \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big) \simeq Z^{P_0}_{\mathrm{red,Ham}}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big),$$

with generator dictionary

$$c^I \mapsto \theta^I, \qquad u_I \mapsto O_I.$$

The boundary Hamiltonian observable is the image of the shifted cotangent coordinate $u_I$. It is not the image of the Hamiltonian ghost coordinate $c^I$. This distinction is the type check behind the factorization-current theorem.

The same type check blocks the unreduced lift. A theorem at the unreduced BV level would have to produce actual boundary representatives for the shifted cotangent generators and homotopies making them central against arbitrary open observables before primitive trace projection. The reduced theorem below does not do this: it identifies the CE/PV target and the reduced $P_0$ current target, while the unreduced centrality and QME compatibility remain separate obligations.

There is a sharper obstruction than a wrong label map. In the polynomial unreduced BV/Koszul category the linear Hamiltonians act by constant translations of $\phi_2$ and $\phi_1$, hence locally nilpotently on every finite polynomial observable and on its $Q$-cohomology. The principal-part cotangent module has the opposite behavior:

$$z_1^r \cdot \rho_{a,b} = (-1)^r (b+1)\cdots(b+r)\rho_{a,b+r} \neq 0, \qquad r \geq 1,$$

and similarly for $z_2$. Thus a same-action unreduced lift cannot be hidden inside polynomial one-$\psi$ representatives. A corrected boundary target must contain genuinely distributional principal-part fields and must construct new centrality homotopies there.

The QME obstruction has an independent scalar component. With the Capelli/Weyl convention used below,

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N \ \mathrm{Tr}(A) \quad \mod \ \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N).$$

After inserting the central boundary ghost $\gamma_{\mathbf{1}}$, $|\gamma_{\mathbf{1}}| = 1$, the scalar contact contributes

$$\mathrm{Ob}_{\mathrm{sc}} = \hbar N\, \omega(f,g) \int_{\mathbb{R}} a(t)b(t)\gamma_{\mathbf{1}}(t)\, dt.$$

Its associated graded class is $\hbar N[\bar{c}]$. The reduced Hamiltonian source has no constant Hamiltonian line on which the primitive $\eta(f) = -[f]_0$ could live. Known candidate mechanisms for an ordinary $\mathfrak{gl}_N$ mixed brane-defect QME proof include a closed-exchange term with associated graded $-\hbar N[\bar{c}]$, a supertrace or central-character normalization, or an unreduced scalar primitive. The present manuscript records the class; it does not prove that this list is exhaustive or that such a cancellation has been constructed.

### *1.3 Grothendieck: cotangent modes are residue currents*

The continuous dual of functions on a formal disk is not another copy of polynomial functions. It is the local-cohomology module of residue currents. For

$$R = \mathbb{C}[[z_1, z_2]], \qquad \mathfrak{m} = (z_1, z_2),$$

the Matlis dualizing module is

$$H^2_{\mathfrak{m}}(R)\, dz_1 dz_2 = \bigoplus_{a,b\geq 0} \mathbb{C}\, z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2.$$

After removing the scalar Hamiltonian, the cotangent module is

$$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1) = \bigoplus_{a+b>0} \mathbb{C}\, \rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2.$$

The residue pairing

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \mathrm{Res}_0\, z_1^{m-a-1} z_2^{n-b-1}\, dz_1 dz_2 = \delta_{a,m}\delta_{b,n}$$

identifies $\mathfrak{h}^\vee_{\mathrm{cont}}$ with $\mathcal{P}$. The coadjoint action is the lowering formula

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1}.$$

This is the cotangent sector used by the closed theory. The one-$\psi$ open classes $\Psi_{a,b}$ carry the same labels but a different $\mathfrak{h}$-module structure. They form a PBW special-fibre shadow; they do not realize the principal-part module. The no-go theorem 3.154 records the formal obstruction.

### 1.4 *Deduction of the local package*

The three principles have disjoint jobs.

(i) Dirac determines the open BV complex and the stable trace observables.

(ii) Witten's source convention motivates the closed-open target as a derived $P_0$-centre; the local CE/PV theorem supplies the proof in the reduced Hamiltonian sector.

(iii) Grothendieck determines the continuous cotangent coefficients and separates principal parts from polynomial descendants.

The rest of the paper proves the resulting package in independent lanes: the Dirac probe theorem; the stable trace and one-$\psi$ calculation; the CE/PV derived-centre theorem; the Matlis principal-part theorem; the brane-line factorization-current theorem; the reduced principal-part defect-current theorem; the degree-zero Moyal/Weyl quantum theorem; and the scalar $U(1)$ anomaly theorem. Open analytic obligations are named separately: the unreduced cotangent factorization-centre morphism, the one-antifield quantum lift, and the Costello brane-defect QME realization.

## 2 Introduction

### 2.1 *Local measurements, constraints, and brackets*

The object of study is a local constrained mechanical system. Write the real six-dimensional mixed space as

$$\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}} = \mathbb{R}_{\mathrm{brane}} \times \mathbb{R}_{\mathrm{top,normal}} \times \mathbb{C}^2_{\mathrm{hol}}.$$

Fix the defect world-line $\mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$. The transverse data is a single real topological normal direction and a formal holomorphic symplectic disk

$$(z_1, z_2), \qquad \omega = dz_1 \wedge dz_2,$$

the formal neighbourhood of $p \in \mathbb{C}^2$. The defect is a *point in* $\mathbb{C}^2$, *dragged along* $\mathbb{R}$; nothing else has been postulated.

The two normal coordinates $z_1, z_2$ describe the position of the defect inside the symplectic disk. A stack of $N$ identical probes at the defect represents each coordinate by an $N \times N$ matrix:

$$z_1 \rightsquigarrow \phi_1, \qquad z_2 \rightsquigarrow \phi_2, \qquad \phi_i \in \mathfrak{gl}_N.$$

The symplectic form pulls back to the matrix-valued two-form $\Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2)$, and the symplectic potential along the world-line is

$$\theta_N = \mathrm{Tr}(\phi_1\, d\phi_2).$$

Acting on a probe stack by an internal change of basis is a gauge operation. The gauge generator is the $\mathfrak{gl}_N$-valued moment map

$$\mu_\epsilon = -\mathrm{Tr}\big(\epsilon[\phi_1, \phi_2]\big), \qquad \epsilon \in \mathfrak{gl}_N,$$

and demanding gauge-invariance of the world-line dynamics imposes the *first-class constraint*

$$[\phi_1, \phi_2] = 0.$$

The Lagrange multiplier for this constraint is a connection $A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N$. The boundary action is the standard first-order Dirac action attached to the fixed potential $\theta_N$ and enforcing the first-class constraint:

$$S_\partial = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big). \tag{1}$$

This is the Dirac standard. The "open string field theory on $L$" is the constrained mechanics of an $N$-probe stack on the formal symplectic normal disk, with the matrix moment-map equation as its first-class constraint. Lemma 3.6 records the precise constrained-mechanics statement and identifies $A$ as the multiplier.

The BV reduction is algebraic: the constraint becomes a Koszul differential generated by a single odd matrix antifield $\psi = A^* \in \mathfrak{gl}_N[1]$ with

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0.$$

The reduced local polynomial algebra at a brane point is

$$\mathcal{R}_N = \mathbb{C}[{\phi_1}^i{}_j, {\phi_2}^i{}_j] \otimes \Lambda({\psi}^i{}_j),$$

and $H^0(\mathcal{R}_N^{GL_N}, Q)$ is, by definition, the algebra of $GL_N$-invariant local observables on the Dirac-reduced phase space of commuting matrix pairs. This algebra is the open trace algebra of the defect. The full BV master action is the standard BV lift of (1) to ghost-number-zero fields; this lift is recorded in Lemma 3.5 and Proposition 3.7.

### 2.2 *The Hamiltonian Lie algebra of the symplectic disk*

The corresponding closed system is dictated by the same canonical mechanics. A closed infinitesimal symmetry of the formal symplectic disk is a Hamiltonian vector field $X_f = \{f, -\}$. Constants act trivially: $X_c = 0$ for $c \in \mathbb{C}$. The closed Hamiltonian Lie algebra is therefore the formal power series modulo constants,

$$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1, \qquad \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1,$$

with bracket

$$\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g \pmod{\mathbb{C}}.$$

Polynomial calculations use $\bar{A}$; closed Chevalley–Eilenberg/Koszul calculations use the completed $\widehat{\mathfrak{h}}$. Continuous duality on $\widehat{\mathfrak{h}}$ produces $\mathfrak{h}^\vee = \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$, with the Taylor-monomial basis $\delta_{k,l}$ for $k + l > 0$. We use $\bar{A}_{\mathrm{desc}}$ for an abstract copy of $\bar{A}$ whose nonconstant basis vectors are represented on the brane by the first descendant cycles $\Psi_{k,l}$ (Construction 3.123).

*Remark* 2.1 *(Scalar Hamiltonian and center-of-mass term).* The quotient by constants is load-bearing. In the reduced Hamiltonian algebra of the formal disk,

$$\{z_1, z_2\} = 1 \equiv 0 \quad \text{in } \bar{A}.$$

On the unreduced brane phase space, however,

$$\{\mathrm{Tr}(\phi_1), \mathrm{Tr}(\phi_2)\} = N.$$

This is the center-of-mass symplectic pair of the stack. Every bracket statement below involving linear Hamiltonians is therefore a statement after projection to the Hamiltonian quotient by scalar trace classes. The same term reappears quantum mechanically as the Capelli shift in Remark 3.166.

BCOV [1] is the source-convention template for the closed B-model/Kodaira-Spencer field: Dolbeault polyvectors, the divergence operator, and a cotangent BV formalism on Calabi–Yau threefolds. That theory is not imported as a theorem for the present $d = 2$ surface model. Here the local Hamiltonian sector is defined and later checked directly: after fixing the shifted BV cotangent convention, the closed target is the cotangent theory of canonical transformations of the formal holomorphic-symplectic disk, namely Hamiltonian BF theory for the Tate Lie algebra

$$\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \mathfrak{h}^\vee[1], \tag{2}$$

where the shift [1] records the BV degree of the cotangent field and the Lie bracket on $\mathfrak{g}$ is the semi-direct sum of the Hamiltonian bracket on $\widehat{\mathfrak{h}}$ with the coadjoint action on $\mathfrak{h}^\vee[1]$.

The closed local observables are the continuous Chevalley–Eilenberg cochains

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{grav}} = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]\big).$$

For Tate-conilpotent, nilpotent, or admissible-envelope replacements of this Lie algebra, the Koszul dual of these CE cochains is the completed graded enveloping algebra

$$\big(\mathcal{O}^{\mathrm{cl}}_{\mathrm{grav}}\big)^! \simeq \widehat{U(\mathfrak{g})}.$$

This bar-cobar assertion is not used as an unconditional theorem for the full formal Hamiltonian algebra with its low-degree linear and conformal-symplectic modes. The PBW Rees algebra $U_\hbar(\mathfrak{g})$ has completed special fibre

$$U_\hbar(\mathfrak{g})/(\hbar) \cong \widehat{\mathbf{S}}\big(\widehat{\mathfrak{h}} \oplus \mathfrak{h}^\vee[1]\big). \tag{3}$$

Equations (2)–(3) say what the closed sector is: *the shifted cotangent theory of canonical transformations of the disk*. Proposition 3.13 below derives $\widehat{\mathfrak{h}}$ from divergence-closed polyvectors on the formal disk.

### 2.3 *Closed-open measurement and the derived Poisson centre*

A closed Hamiltonian $f(z_1, z_2)$ is *measured* by the brane by substitution of the matrix-valued normal coordinates:

$$J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}.$$

At finite $N$ one chooses a word ordering. In $H^\circ(\mathcal{R}_N^{GL_N}, Q)$ adjacent swaps are exact, $Q\,\mathrm{Tr}(\psi vu) = \mathrm{Tr}([\phi_1, \phi_2]vu)$, so the ordering ambiguity disappears modulo the constraint. In the stable connected limit $N \to \infty$, $J(f)$ is a well-defined Hamiltonian observable on the brane. The matrix Poisson bracket gives

$$\big\{J(f), J(g)\big\}_\partial = J\big(\{f, g\}_{\mathfrak{h}}\big), \tag{4}$$

after removal of the scalar Hamiltonian. Equation (4) is the ordinary canonical bracket of the probe system, the statement that the probe brane is a representation of the Hamiltonian Lie algebra of the disk. On monomials,

$$\big\{\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}\big\} = \overline{(kn - lm)\,\mathrm{Tr}(\phi_1^{k+m-1} \phi_2^{l+n-1})}. \tag{5}$$

The map $f \mapsto J(f)$ is the *moment-map shadow* of the closed-open operation one expects from the open/closed string-field formalism. A theorem-level centrality statement is not imported from Witten or from BCOV. In this paper the algebraic target is the derived reduced Hamiltonian Poisson centre of the open factorisation algebra rather than a single boundary function:

$$Z^{P_0}_{\mathrm{red,Ham}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}}).$$

Theorem 3.27 below makes this precise: the closed Chevalley–Eilenberg cochains on $\mathfrak{g}$ identify with the reduced derived $P_0$-Poisson centre of the stable open Hamiltonian algebra, and the boundary observable $J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ is the image, under that identification, of the coordinate dual to the shifted cotangent generator (*not* of the coordinate dual to $f$ itself); the closed Hamiltonian ghost becomes the Hamiltonian vector-field operation on the brane.

### 2.4 *Scalar reduction and the U(1) center anomaly*

Quotienting by constants on the closed side and by the unit trace $\mathrm{Tr}(1) = N$ on the open side discards a single scalar-axis Lie 2-cocycle class. That class is the U(1) center anomaly: the distinguished nontrivial class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ obstructing the lift of the scalar-reduced Hamiltonian Poisson bracket on $\bar{A}$ to the unreduced polynomial Hamiltonian algebra $\mathfrak{h}_{\mathrm{poly}} = \mathbb{C}[z_1, z_2]$, and the corresponding obstruction to lifting the scalar-reduced trace bracket to the unreduced brane bracket with its Chan-Paton central line $\mathbb{C} \cdot I$. Theorem 2.3 makes this precise on the closed side; Theorem 2.4 on the open side; Theorem 2.6 identifies the same class with the Moyal/Capelli $\hbar N$ shift. Every subsequent theorem touching $\mathfrak{h}$ or $\mathcal{A}_\partial$ carries the flag "before/after killing the scalar" explicitly.

*Notation.* Write $\mathfrak{h}_{\mathrm{poly}} = \mathbb{C}[z_1, z_2]$ for the unreduced polynomial Hamiltonian Lie algebra, including the constant Hamiltonians; the bracket on $\mathfrak{h}_{\mathrm{poly}}$ is $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$ and takes values again in $\mathfrak{h}_{\mathrm{poly}}$. Define

$$\omega : \mathfrak{h}_{\mathrm{poly}} \otimes \mathfrak{h}_{\mathrm{poly}} \longrightarrow \mathbb{C}, \qquad \omega(f, g) = [\, \{f, g\} \,]_0,$$

the constant Taylor coefficient of $\{f, g\}$. On monomials,

$$\omega(z_1^k z_2^l, z_1^m z_2^n) = \begin{cases} kn - lm, & (k+m, l+n) = (1, 1), \\ 0, & \text{otherwise}, \end{cases}$$

hence $\omega$ is concentrated in the unique bidegree where the bracket multidegree $(k + m - 1, l + n - 1)$ is $(0, 0)$; this is exactly the bidegree the manuscript discards as "scalar trace".

LEMMA 2.2. $\omega$ is a Lie 2-cocycle on $\mathfrak{h}_{\mathrm{poly}}$. On $\mathfrak{h}_{\mathrm{poly}}$ itself it is a coboundary; on the scalar-reduced quotient $\bar{A} = \mathfrak{h}_{\mathrm{poly}}/\mathbb{C} \cdot 1$ the same formula represents a nontrivial class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$.

*Proof.* The Jacobi identity for $\{-, -\}$ on $\mathfrak{h}_{\mathrm{poly}}$ asserts

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0,$$

and projecting each summand to its constant Taylor coefficient gives the cocycle identity

$$\omega(f, \{g, h\}) + \omega(g, \{h, f\}) + \omega(h, \{f, g\}) = 0.$$

Antisymmetry $\omega(g, f) = -\omega(f, g)$ is inherited from antisymmetry of the bracket. Hence $\omega \in Z^2_{\mathrm{Lie}}(\mathfrak{h}_{\mathrm{poly}}, \mathbb{C})$.

On $\mathfrak{h}_{\mathrm{poly}}$ the cocycle is a coboundary. Define $\eta : \mathfrak{h}_{\mathrm{poly}} \to \mathbb{C}$ by $\eta(f) = -[f]_0$, the negative of the constant Taylor coefficient. Then for all $f, g \in \mathfrak{h}_{\mathrm{poly}}$, $\delta\eta(f, g) = -\eta(\{f, g\}) = [\{f, g\}]_0 = \omega(f, g)$, so $\omega = \delta\eta$.

After projection to $\bar{A} = \mathfrak{h}_{\mathrm{poly}}/\mathbb{C} \cdot 1$ the primitive $\eta$ does not descend, since $\eta(1) = -1 \neq 0$. The cocycle $\omega$ does descend, because it already vanishes on the line $\mathbb{C} \cdot 1$. The class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ is nontrivial: any putative primitive $\bar{\eta} : \bar{A} \to \mathbb{C}$ would lift, by setting $\tilde{\eta}(1) = 0$, to a primitive of $\omega$ on $\mathfrak{h}_{\mathrm{poly}}$. The lifted $\tilde{\eta}$ would satisfy $\tilde{\eta}(\{z_1, z_2\}) = \tilde{\eta}(1) = 0$, but $\omega(z_1, z_2) = 1 \neq 0$, a contradiction. □

Theorem 2.3 *(U(1) center anomaly: closed side).* The class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ of Lemma 2.2 is the scalar-axis obstruction class to lifting the Lie algebra structure on $\bar{A}$ to the Lie algebra structure on the unreduced polynomial Hamiltonians $\mathfrak{h}_{\mathrm{poly}}$ along the abelian extension

$$0 \longrightarrow \mathbb{C} \cdot 1 \longrightarrow \mathfrak{h}_{\mathrm{poly}} \longrightarrow \bar{A} \longrightarrow 0.$$

Equivalently, $\mathfrak{h}_{\mathrm{poly}}$ is isomorphic to the central extension $\bar{A}_{[\bar{c}]} = \bar{A} \oplus \mathbb{C}K$ with bracket

$$[\,(f, aK), (g, bK)\,] = (\{f, g\}_{\bar{A}},\ \bar{c}(f, g)\,K),$$

and $[\bar{c}]$ spans the one-dimensional sub-line of $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ attached to this constant-Hamiltonian extension.

*Proof.* Central extensions of $\bar{A}$ by $\mathbb{C}$ are classified by $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ via the standard Chevalley-Eilenberg argument. The exact sequence above is a central extension because the constant Hamiltonian $1 \in \mathfrak{h}_{\mathrm{poly}}$ has zero Hamiltonian vector field, so $\mathbb{C} \cdot 1$ is central in $\mathfrak{h}_{\mathrm{poly}}$. The class of this extension is the obstruction to a Lie-algebra splitting; computing the class on monomial generators,

$$[z_1^k z_2^l, z_1^m z_2^n]_{\mathfrak{h}_{\mathrm{poly}}} = \{z_1^k z_2^l, z_1^m z_2^n\} = (kn - lm) z_1^{k+m-1} z_2^{l+n-1},$$

and projecting to $\mathbb{C} \cdot 1$ keeps only the $(0, 0)$-Taylor coefficient $\omega$ of Lemma 2.2. Thus the extension class is $[\bar{c}]$, and Lemma 2.2 shows it is nontrivial. The same lemma shows that the line $\mathbb{C} \cdot [\bar{c}]$ attached to this scalar extension is one-dimensional inside $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$, and that splitting this extension is exactly the problem of lifting back to $\mathfrak{h}_{\mathrm{poly}}$. □

Theorem 2.4 *(U(1) center anomaly: open side).* At finite $N$ the trace algebra Lie bracket $\{F, G\} = \mathrm{Tr}(\partial_{\phi_1} F\, \partial_{\phi_2} G - \partial_{\phi_2} F\, \partial_{\phi_1} G)$ on $\mathbb{C}[\mathfrak{gl}_N \oplus \mathfrak{gl}_N]^{GL_N}$ satisfies

$$\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N$$

in the unreduced ring, and zero in the scalar-reduced quotient $H^0_{\mathrm{red}} = H^0/\mathbb{C} \cdot \mathrm{Tr}(1)$. The scalar-axis central extension class of the unreduced trace algebra is the image of $[\bar{c}]$ under the boundary evaluation map

$$\partial^{\mathrm{full}}_{\mathrm{bb},N} : \mathbb{C}[z_1, z_2] \longrightarrow \mathbb{C}[\mathfrak{gl}^2_N]^{GL_N}, \qquad f \longmapsto \mathrm{Tr}\, f(\phi_1, \phi_2),$$

i.e. the same anomaly class is realised on the brane as the U(1) center of $\mathfrak{gl}_N$ with central element $I$.

*Proof.* The matrix Poisson bracket gives, on monomials of bidegree $(1, 0)$ and $(0, 1)$,

$$\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = \mathrm{Tr}\big((\partial_{\phi_1}\mathrm{Tr}\,\phi_1)(\partial_{\phi_2}\mathrm{Tr}\,\phi_2) - (\partial_{\phi_2}\mathrm{Tr}\,\phi_1)(\partial_{\phi_1}\mathrm{Tr}\,\phi_2)\big) = \mathrm{Tr}(I \cdot I) - 0 = N.$$

Hence the unreduced bracket realises the constant $N$ as a scalar element. In $H^0_{\mathrm{red}} = H^0/\mathbb{C}\cdot\mathrm{Tr}(1)$ this scalar is killed, matching the projection on the closed side. The boundary evaluation map $\partial^{\mathrm{full}}_{\mathrm{bb},N}$ sends $1 \in \mathbb{C}[z_1, z_2]$ to $\mathrm{Tr}(I) = N$ and satisfies, for monomials with $(k+m, l+n) = (1,1)$,

$$\partial^{\mathrm{full}}_{\mathrm{bb},N}(\{z_1^k z_2^l, z_1^m z_2^n\}) = (kn - lm)\cdot\mathrm{Tr}(I) = (kn-lm)\cdot N,$$

which equates the closed cocycle $\omega$ with the open trace bracket on the central direction. Hence the extension class on the brane is the image of $[\bar{c}]$. Since $\mathbb{C}\cdot I \subset \mathfrak{gl}_N$ is the U(1) centre with character Tr, the brane realisation of $[\bar{c}]$ is exactly the U(1) centre of $\mathfrak{gl}_N$. □

*Remark* 2.5. Every theorem that uses $\mathfrak{h}$, $\bar{A}$, or $\mathcal{A}_\partial = H^0_{\mathrm{red}}$ lives *after* killing the scalar. The unreduced statement requires keeping the central line $\mathbb{C}\cdot 1 \subset \mathfrak{h}_{\mathrm{poly}}$ on the closed side and $\mathbb{C}\cdot\mathrm{Tr}(1)$ on the open side, with the U(1) center anomaly class $[\bar{c}]$ recording the obstruction to a Lie-algebra splitting. Theorem 2.27, Theorem 3.161, Corollary 3.162, and Proposition 3.142 are all stated on the reduced side; their unreduced refinements are governed by $[\bar{c}]$ via the central extensions of Theorems 2.3 and 2.4.

### Quantum-classical equivalence of the U(1) center cocycle

THEOREM 2.6 *(Quantum-classical equivalence of the U(1) center class).* Let $\Phi = (\phi_1, \phi_2)$ be the classical brane field on the formal local disk and $\Phi_\hbar = (\phi_1^\hbar, \phi_2^\hbar)$ its Weyl quantization at finite $N$, with the Moyal product $f * g = m \circ \exp\big(\frac{\hbar}{2}P\big)(f\otimes g)$ for $P = \partial_{z_1}\otimes\partial_{z_2} - \partial_{z_2}\otimes\partial_{z_1}$. Then the normal-ordered quantum comoment imposes, in the degree-zero quantum Hamiltonian reduction, the Capelli/canonical relation

$$Y_\hbar X_\hbar - X_\hbar Y_\hbar + \hbar N I \equiv 0, \qquad X_\hbar = \phi_1^\hbar,\ Y_\hbar = \phi_2^\hbar,$$

modulo the quantum moment-map ideal. Its scalar trace part has semiclassical limit $\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N$ on the trace generators $X = \mathrm{Tr}\,\phi_1$ and $Y = \mathrm{Tr}\,\phi_2$. In the classical limit $\hbar \to 0$, the bracket $\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N$ recovers the central scalar of Theorem 2.4. The classical Poisson cocycle $\omega$ of Lemma 2.2 and the quantum Capelli shift $\hbar N$ represent the *same cohomology class* $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ under classical-to-quantum reduction, i.e. via the Rees deformation $U_\hbar(\bar{A}_{[\bar{c}]})$ specialising to $\widehat{\mathbf{S}}(\bar{A}_{[\bar{c}]})$ at $\hbar = 0$.

*Proof.* The Weyl convention gives $[X_\hbar, Y_\hbar]_* = \hbar\{X, Y\} + O(\hbar^3)$. On the trace generators, Proposition 3.142 at $(k,l,m,n) = (1,0,0,1)$ gives $\{X, Y\} = N$ before scalar reduction. Hence

$$X_\hbar Y_\hbar - Y_\hbar X_\hbar = \hbar N + O(\hbar^3),$$

and at finite $N$ the higher-order terms vanish on these linear generators because $P^r$ requires at least $r$ derivatives in each tensor factor. Rearranging gives the displayed normal-ordered moment relation $Y_\hbar X_\hbar - X_\hbar Y_\hbar + \hbar N\, I \equiv 0$ in the quantum Hamiltonian reduction. The Capelli identity [26, 27] of order one supplies the same relation as the leading nontrivial trace identity, and the Razmyslov–Procesi description [17, 18] identifies the full ideal of trace relations at level $N$ as a separate fact; the argument here uses only the order-one canonical commutator.

For the cohomology-class identification, the Rees algebra $U_\hbar(\bar{A}_{[\bar{c}]})$ has central fibre $\widehat{\mathbf{S}}(\bar{A}_{[\bar{c}]})$ at $\hbar = 0$ and generic fibre containing $\bar{A}_{[\bar{c}]}$ as Lie algebra at $\hbar = 1$. The PBW filtration's associated graded carries the Lie 2-cocycle as the primary obstruction to PBW-splitting the deformation. Both sides of the classical-to-quantum reduction therefore record the same class in $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$: classically $\omega$ is the obstruction to splitting the central extension $\mathfrak{h}_{\mathrm{poly}} \to \bar{A}$, quantum-mechanically the same $\omega$ is the leading-order coefficient of the Moyal star commutator on the central direction. □

*Remark* 2.7. The class $[\bar{c}]$ is a physical observable of the brane: the U(1) center of $\mathfrak{gl}_N$ acts on the brane Hilbert space with character Tr, and the Capelli shift $\hbar N$ is the charge of this central direction. Scalar reduction restricts to the kernel of this character.

### 2.5 *The two algebras*

The local model contains two algebras, both *derived* from the canonical mechanics above rather than postulated:

- The algebra $\mathcal{A}_N$ of $GL_N$-invariant local trace observables on the Dirac-reduced probe stack of (1); equivalently, the algebra of local operators of the open BV theory on the brane line $L = \mathbb{R}_{\text{brane}} \times \{0\} \times \{p\} \subset \mathbb{R}^2_{\text{top}} \times \mathbb{C}^2_{\text{hol}}$.
- The algebra $\mathcal{B}$ of local operators of the closed Hamiltonian BF sector (2); the operator-product expansion in the $\mathbb{R}$-direction equips $\mathcal{B}$ with an associative product, derived from the locally constant factorisation structure of the topological direction.

*Conjectural stable horizon.* The strategic large-$N$ expectation is the following: in the stable large-$N$ limit and within the Hamiltonian sector one should have

$$\lim_{N\to\infty} \mathcal{A}_N \simeq \mathcal{B}^!,$$

the Koszul dual of $\mathcal{B}$. This is conjectural and is not used to prove Theorem 2.27. The theorem below proves only the formal local Hamiltonian BF/Moyal shadow recorded in the claim-strength ledger; the stable bulk–boundary Koszul-duality statement remains a strategic open problem.

If proved, this stable horizon would realize the open/closed string duality philosophy of [4] in the present Hamiltonian sector by identifying the closed local theory with the derived Poisson centre of the open local theory. Here it functions only as motivation.

The provenance of this local model in higher-dimensional physics is motivational. Costello's protected sector of perturbative twisted M-theory on $\mathbb{R} \times X$, with $X$ a holomorphic-symplectic surface and a stack of $N$ topological branes on $\mathbb{R} \times p$, suggests a Hamiltonian formal sector at the symplectic disk $\widehat{X}_p$, but this manuscript does not prove such a sector for an arbitrary ambient background. If an ambient background supplies the admissible protected Hamiltonian restriction of Definition 5.2, then the local theorem of §2.14 applies to that restricted sector; §5.1 records the corresponding conditional protected-sector statement after the proved local core has been isolated by the claim-strength ledger.

### 2.6 *Three notions of locality*

Locality is not one notion in this model. It is three.

*(a) Support locality along the brane.* For an open interval $I \subset \mathbb{R}$ on the world-line, define the compactly supported Hamiltonian current algebra $\mathfrak{h}_I = \Omega^\bullet_c(I)\widehat{\otimes}\widehat{\mathfrak{h}}$ with bracket induced from the de Rham product and the Hamiltonian bracket. In the degree-zero test-function notation used for explicit measurements this reads

$$[a \otimes f,\, b \otimes g] = ab \otimes \{f, g\}_{\widehat{\mathfrak{h}}}.$$

The open observables on $I$ are $\mathcal{A}_\partial(I) = \mathbf{S}(\mathfrak{h}_I)$, generated by the smeared trace functionals

$$O_f(a) = \int_I a(t)\, \operatorname{Tr} f\big(\phi_1(t), \phi_2(t)\big)\, dt.$$

Their bracket is the canonical equal-time bracket:

$$\{O_f(a), O_g(b)\} = O_{\{f,g\}_{\widehat{\mathfrak{h}}}}(ab).$$

Disjoint supports multiply, since $ab = 0$ outside the intersection; overlapping supports give a delta-function contact term. This is the operational meaning of locality for the open factorisation algebra: disjoint independent measurements multiply, and Poisson brackets are supported where the measurements meet.

*(b) Topological locality.* In the topological direction (the $\mathbb{R}^2$ factor), translations are $Q$-exact:

$$L_v = [Q, \iota_v].$$

Equivalently, positive jets in the topological direction vanish in local-operator cohomology by the contraction homotopy $[d_{\mathbb{R}}, \iota_{\partial_t}] = \partial_t$ (Lemma 3.119). Therefore the precise position of an insertion is not physical; only the collision pattern of insertions matters. This is the structural reason the factorisation algebra is locally constant in the topological direction.

*(c) Formal holomorphic locality.* In the holomorphic direction (the $\mathbb{C}^2$ factor), the model has been completed at the brane point $p \in \mathbb{C}^2$. It is a theory of holomorphic jets at $p$, not a theory of finite propagation through $\mathbb{C}^2$. Antiholomorphic translations are $Q$-exact:

$$L_{\partial/\partial\bar{z}_i} = [Q, \iota_{\partial/\partial\bar{z}_i}].$$

The kinematic differential of the closed sector is $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$, and on the formal-jet model $\bar{\partial}$-cohomology is concentrated in degree zero. The cotangent boundary sector is *distributionally local, not polynomially local*: a coadjoint mode is a transverse principal part $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2$, paired with a brane Hamiltonian by residue,

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m}\delta_{b,n}.$$

This is exactly what one expects for a momentum dual field attached to a defect: support on the defect, transverse behaviour encoded by residues. The polynomial one-$\psi$ descendants $\Psi_{k,l}$ realise the PBW special fibre of $\mathfrak{h}^\vee[1]$ (Proposition 3.143); Corollary 3.144 records the deformation-level mismatch with the coadjoint module, exactly the ordinary distinction between polynomial and distributional realisations.

Under the support, topological, and formal-holomorphic locality hypotheses in (a)–(c), the factorisation-algebra structure is the resulting local algebra of measurements. Disjoint supports multiply along the brane; topological motion is $Q$-exact and changes nothing in cohomology; transverse holomorphic dependence is distributional. No external categorical input is imposed.

### 2.7 *Unitarity*

The model is a twisted BV sector. It is not a Lorentzian unitary quantum mechanics. There is no positive-definite Hilbert space, no Wightman positivity, and no reflection positivity. The fields include ghosts and antifields; topological time has been made cohomologically trivial.

The replacements for unitarity are the consistency conditions of the BV formalism: the classical master equation

$$QI + \tfrac{1}{2}\{I, I\} = 0$$

and the quantum master equation

$$QI + \tfrac{1}{2}\{I, I\} + \hbar\,\Delta I = 0,$$

together with associativity of the resulting factorisation/OPE algebra. If the local model arises as a protected sector of a larger unitary theory (for example, of a topological twist of a supersymmetric quantum field theory whose local observables for the local model arise as cohomology with respect to a supercharge), then unitarity is carried by the larger theory, and the local sector retains only the cohomological algebra of protected observables. No unitarity is imposed below, and no Lorentzian claims are made.

### *2.8 Thesis*

> A stack of branes is a system of probes for the formal symplectic normal disk. Dirac reduction of the probe phase space gives the open trace algebra. Hamiltonian BF theory is the shifted cotangent theory of canonical transformations. In the reduced formal Hamiltonian theorem proved below, the closed observables identify with derived Poisson central operations on the stable open Hamiltonian brane algebra.

### *2.9 Mixed holomorphic-topological strings*

By a local string theory we mean the portion of string field theory visible to local operators: closed fields forming a BV theory, local observables forming a factorization algebra, and a brane supplying a cyclic open field theory together with a bulk-to-boundary map. The words *topological* and *holomorphic* describe how those local operators vary in the spacetime directions. This is a local string-field-theoretic use of the word "string", not a definition of a global theory of worldsheet moduli, amplitudes on all backgrounds, or a completed quantum anomaly theory.

Gwilliam–Williams [5] are the precedent for formulating a holomorphic string in the Costello BV/factorization formalism. Their object is the holomorphic bosonic string on a Riemann surface, with Dolbeault fields $(\gamma, \beta, c, b)$, a one-loop quantization in the critical complex dimension, and a factorization algebra recovering the BRST cohomology. The object here is not that theory and not a replacement definition. It is a formal local target-space sector: de Rham/Floer directions, a holomorphic symplectic formal disk, Hamiltonian cotangent BF fields, and a brane large-$N$ operator algebra.

The topological word refers to locally constant factorization in the de Rham/Floer directions. The holomorphic word refers to the Dolbeault presentation: the kinetic operator contains $\bar{\partial}$, the fields are holomorphic jets after local formal completion, and anti-holomorphic translations act by cochain-homotopic zero. A holomorphic theory may acquire further topological descent after quantization; that extra descent is not part of the definition used here.

*Definition* 2.8 *(Working local topological factorization sector).* Let $M$ be a smooth manifold. A local topological closed-string sector on $M$ is a classical BV theory whose local observables form a locally constant factorization algebra on $M$: for inclusions of sufficiently small disks $U \subset V$, the factorization structure map

$$\mathrm{Obs}^{\mathrm{cl}}(U) \longrightarrow \mathrm{Obs}^{\mathrm{cl}}(V)$$

is a quasi-isomorphism. Attaching branes to such a sector means specifying a cyclic open BV theory and the closed-open operations it carries. In a de Rham model the closed fields carry the differential $d_M$ plus the BV linearized differential, and the locally constant factorization product records collision of operators up to isotopy.

*Remark* 2.9. The exact A-model factor used below is a local topological sector in this sense: the self-Floer model for $\mathbb{R} \subset T^*\mathbb{R}$ is represented by the de Rham complex $\Omega^*(\mathbb{R})$, and its local observables are insensitive to the metric geometry of the interval. This is the topological part of the mixed holomorphic-topological example.

*Definition* 2.10 *(Working local holomorphic factorization sector).* Let $Y$ be a complex manifold. A local holomorphic sector on $Y$ is a classical BV theory whose fields are Dolbeault complexes on $Y$, whose differential has the form $\bar{\partial}$ plus the BV linearized differential, whose interactions are holomorphic polydifferential operators, and whose local observables form a holomorphic factorization algebra. Here holomorphic factorization means that, on coordinate polydisks, the factorization products and restriction maps are represented by holomorphic polydifferential operations and every anti-holomorphic translation $\partial/\partial\bar{z}_i$ acts null-homotopically on observables. In the target-space Kodaira-Spencer examples the fields are Dolbeault-valued polyvectors; in the holomorphic symplectic examples below they are Hamiltonian polyvectors after removing constants. Branes are treated through cyclic open BV theory and the bulk-to-boundary operations constructed below.

*Remark* 2.11. BCOV/Kodaira-Spencer theory is the guiding threefold source convention [1]: its fields are Dolbeault-valued polyvectors, and the Calabi-Yau volume form supplies the divergence operator. In the holomorphic symplectic surface model used here, the analogous divergence-free vector fields are represented locally by Hamiltonians modulo constants. The surface model is not a truncation of threefold BCOV theory and no compact $CY_3$ BCOV theorem is used below.

*Remark* 2.12. The classical local observable theory underlying Gwilliam–Williams is an example of a pure holomorphic BV/factorization sector on a Riemann surface, with additional worldsheet ghost, BRST, and critical-dimension quantum structure. Their critical-dimension and BRST-cohomology theorems are not imported here. The model below is a different formal local target-space sector whose holomorphic sector is Hamiltonian BF, or the holomorphic symplectic Kodaira-Spencer analogue, and whose topological sector is de Rham/Floer. In particular, the critical complex dimension in Gwilliam–Williams is a worldsheet ghost-anomaly statement for the holomorphic bosonic string; it is not a constraint on the local $\mathbb{R}^2 \times \mathbb{C}^2$ Hamiltonian sector studied here.

*Definition* 2.13 *(Working classical local mixed holomorphic-topological sector).* Let $M$ be a smooth manifold and let $Y$ be a complex manifold. A classical local mixed holomorphic-topological closed sector on $M \times Y$ consists of:

(i) a sheaf of cochain complexes of fields $\mathcal{E}$ which, on product coordinate opens $D \times B$, has the form

$$\mathcal{E}(D \times B) = \Omega^*(D)\widehat{\otimes}\Omega^{0,*}(B, \mathcal{V})$$

for a graded holomorphic vector bundle or formal holomorphic vector space $\mathcal{V}$;

(ii) a degree $-1$ local BV pairing on $\mathcal{E}$ and a differential

$$Q = d_M + \bar{\partial}_Y + Q_0,$$

where $Q_0$ is holomorphic over $Y$ and is linear over de Rham forms on $M$;

(iii) a degree-zero local functional $I \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E})$, built from differential operators in the $M$ directions and holomorphic polydifferential operators in the $Y$ directions, satisfying the classical master equation

$$QI + \frac{1}{2}\{I, I\} = 0;$$

(iv) the associated classical observables

$$\mathrm{Obs}^{\mathrm{cl}}(U) = \left(\widehat{\mathrm{Sym}}(\mathcal{E}^!_c(U)), Q_{\mathrm{obs}}\right), \qquad Q_{\mathrm{obs}} = Q^! + \{I, -\},$$

with the standard factorization product for disjoint supports. Here $\mathcal{E}^!_c(U)$ denotes compactly supported continuous linear observables, i.e. the density-dual cosheaf of fields; extension by zero gives the covariant structure maps used by the factorization algebra.

These data must satisfy the mixed locality condition. For every inclusion of small disks $D \subset D'$ in $M$ and every holomorphic polydisk $B \subset Y$, the structure map

$$\mathrm{Obs}^{\mathrm{cl}}(D \times B) \longrightarrow \mathrm{Obs}^{\mathrm{cl}}(D' \times B)$$

is a quasi-isomorphism. In holomorphic coordinates on $B$, each anti-holomorphic translation $\partial/\partial\bar{z}_i$ acts null-homotopically on observables: there is a degree $-1$ derivation $\eta_i$ with

$$[Q_{\mathrm{obs}}, \eta_i] = \mathcal{L}_{\partial/\partial\bar{z}_i}.$$

A mixed brane for such a sector is supported on $N \times Z$, where $N \subset M$ is a smooth submanifold and $Z \subset Y$ is a holomorphic submanifold or a formal holomorphic neighbourhood of one, and is specified by an object $\mathcal{F}$ of the local holomorphic brane category on $Z$. Its open fields have the derived form

$$\Omega^*(N)\widehat{\otimes}\mathrm{RHom}_Y(\mathcal{F}, \mathcal{F}) \otimes \mathrm{End}(V)[1],$$

with a cyclic trace. For a smooth brane this recovers the Dolbeault model $\Omega^*(N)\widehat{\otimes}\Omega^{0,*}(Z, \mathcal{W})$; for a skyscraper brane it produces the Ext algebra. A full closed-open mixed datum would further require a morphism of factorization algebras

$$\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed}}\big|_{N\times Z} \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}}),$$

from closed observables to the Poisson/BV factorization-centre object of the open brane observables, compatible with the BV brackets and factorization products. The Hamiltonian model below constructs the degree-zero Hamiltonian current shadow of this map; the full closed CE/cotangent factorization-center lift is isolated in Problem 3.47 and treated there only with the reduced, chain-level, truncated, or weighted hypotheses stated in the local theorem.

*Remark* 2.14. A full quantum mixed holomorphic-topological string datum would be the corresponding $\hbar$-adic factorization algebra obtained from a renormalized BV action satisfying the quantum master equation. The manuscript constructs the formal Weyl/Moyal quantum Hamiltonian target and its degree-zero derived-center map; the full renormalized closed-open BV datum remains the analytic graph-realization problem of Problem 3.183.

Definition 2.13 separates three levels. A pure topological sector is locally constant in all real directions; in examples this includes the A-model local/Floer direction and the Chern-Simons-type open field theory on a topological brane. A pure holomorphic sector is Dolbeault in its complex directions and produces a holomorphic factorization algebra; Gwilliam–Williams' holomorphic bosonic string is a worldsheet example, while BCOV/Kodaira-Spencer gives the guiding target-space closed-string source convention. The Hamiltonian BF complex below is the formal local target-space sector used here. A mixed holomorphic-topological sector lives on $M \times Y$ with locally constant factorization in $M$ and holomorphic factorization in $Y$. The A/B terminology used below is one source of such examples: the A-side supplies the de Rham/Floer direction and the B-side supplies the Dolbeault or formal holomorphic direction.

### 2.10 *The local mixed model*

The paper studies the product local model

$$\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}.$$

The theorem is not a global statement about all twisted M-theory backgrounds. It is the formal local calculation obtained after completing at the brane $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ and taking the holomorphic symplectic normal disk at $p \in \mathbb{C}^2$, with

$$\omega = dz_1 \wedge dz_2.$$

The closed sector is the local Hamiltonian Kodaira-Spencer/BF analogue on this mixed space, not a pure A-model or pure B-model. The open sector is the cyclic open string field theory on the mixed brane: topological along $\mathbb{R}$ and formal holomorphic in the normal $\mathbb{C}^2$ directions.

*Remark* 2.15 *(Global obstruction).* The Hamiltonian identification used here is local. On a holomorphic symplectic surface $X$, a holomorphic vector field $v$ is symplectic when

$$\alpha_v = -\iota_v \omega$$

is a closed holomorphic one-form, and it is Hamiltonian only after choosing $f$ with $df = \alpha_v$, in the convention $\iota_{X_f}\omega = -df$. Thus globalizing the Hamiltonian sector requires either a sheaf of local Hamiltonians or vanishing of the corresponding holomorphic de Rham obstruction. The formal disk has no such obstruction by the holomorphic Poincare lemma; a general $X$ need not.

PROPOSITION 2.16 *(Global local-Hamiltonian restriction).* Let $(X, \omega)$ be a connected holomorphic symplectic surface and let $p \in X$. Put

$$\mathcal{H}_X = \mathcal{O}_X/\mathbb{C}_X, \qquad \widehat{\mathfrak{h}}_{X,p} = \widehat{\mathcal{O}}_{X,p}/\mathbb{C}.$$

The differential identifies $\mathcal{H}_X$ with the sheaf of closed holomorphic one-forms, hence with the sheaf of holomorphic symplectic vector fields by $\alpha = -\iota_v\omega$. For every $p \in X$ there is a Lie algebra restriction

$$H^0(X, \mathcal{H}_X) \longrightarrow \widehat{\mathfrak{h}}_{X,p}.$$

After choosing a formal Darboux coordinate system, the formal neighbourhood $(\widehat{X}_p, \omega)$ is identified with the stable Hamiltonian brane sector of the local model of Theorem 2.27. The resulting boundary map

$$\widehat{\mathfrak{h}}_{X,p} \longrightarrow H^0_{\mathrm{Ham}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{brane}})$$

is independent of this choice up to the natural action of the formal symplectomorphism group. The subspace coming from global Hamiltonian functions is the image of

$$H^0(X, \mathcal{O}_X)/\mathbb{C} \longrightarrow H^0(X, \mathcal{H}_X).$$

The obstruction for a locally Hamiltonian symplectic vector field to come from a global Hamiltonian function is its period class in $H^1(X, \mathbb{C}_X)$.

*Proof.* The holomorphic Poincare lemma gives an isomorphism

$$d : \mathcal{O}_X/\mathbb{C}_X \xrightarrow{\sim} \Omega^1_{X,\mathrm{cl}}.$$

The holomorphic symplectic form identifies a symplectic vector field $v$ with the closed one-form $\alpha_v = -\iota_v\omega$. Thus a section of $\mathcal{H}_X$ is exactly a local Hamiltonian primitive, modulo local constants, and the Poisson bracket is independent of the local constants. Taking the germ at $p$ and completing gives the displayed restriction to $\widehat{\mathfrak{h}}_{X,p}$.

The formal holomorphic Darboux theorem identifies the completed local symplectic algebra at $p$ with $\mathbb{C}[[z_1, z_2]]$ equipped with $dz_1 \wedge dz_2$. Under this identification Hamiltonians modulo constants become $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}$, and the point-brane Ext algebra is the coordinate-free exterior algebra $\mathbf{Ext}^*_X(\mathcal{O}_p, \mathcal{O}_p) \cong \Lambda^* T_p X$, with cyclic trace fixed by $\omega_p$. Hence the open BV reduction and the Hamiltonian boundary evaluation are exactly the formal local construction after choosing Darboux coordinates. A different Darboux coordinate system differs by a formal symplectomorphism, which acts on Hamiltonians and on the normal coordinate algebra; the substitution map $z_i \mapsto \phi_i$ is equivariant for this action.

The exact sequence

$$0 \to \mathbb{C}_X \to \mathcal{O}_X \to \mathcal{H}_X \to 0$$

gives

$$0 \to H^0(X, \mathcal{O}_X)/\mathbb{C} \to H^0(X, \mathcal{H}_X) \xrightarrow{\delta} H^1(X, \mathbb{C}_X) \to H^1(X, \mathcal{O}_X).$$

For a locally Hamiltonian vector field, $\delta$ is the period class of the closed one-form $-\iota_v\omega$. Vanishing of this class is exactly the existence of a single global holomorphic Hamiltonian primitive, unique modulo constants. □

*Remark* 2.17. Proposition 2.16 is the maximal global statement used below: it concerns local Hamiltonian restrictions only, not descent for factorization centres or quantum open-closed maps. If $H^1(X, \mathbb{C}) = 0$, global symplectic vector fields which are locally Hamiltonian have global Hamiltonian primitives. If $X$ is Stein, every finite jet needed for a finite local calculation can be lifted to a global holomorphic Hamiltonian, noncanonically: apply Cartan's theorem B to

$$0 \to \mathfrak{m}_p^{k+1} \to \mathcal{O}_X \to \mathcal{O}_X/\mathfrak{m}_p^{k+1} \to 0.$$

This does not imply surjectivity onto $\widehat{\mathcal{O}}_{X,p}$. If $X$ is compact connected, then $H^0(X, \mathcal{O}_X)/\mathbb{C} = 0$; abelian surfaces still have nonzero locally Hamiltonian translations, obstructed by their period classes. A global cochain-level Poisson factorization-center map or a global quantum deformation-quantization statement requires additional descent, anomaly, and background-choice data.

The closed Hamiltonian coefficient Lie algebra is the reduced Tate vector space

$$\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1.$$

We write a class by its unique zero-constant Taylor representative. The bracket is the projected Poisson bracket

$$\{f, g\} = \partial_{z_1} f \, \partial_{z_2} g - \partial_{z_2} f \, \partial_{z_1} g \mod \mathbb{C} \cdot 1.$$

Thus $\{z_1, z_2\} = 0$ in $\widehat{\mathfrak{h}}$, although the unreduced Poisson bracket of the chosen representatives is 1. The cotangent BF field contributes the continuous dual, so the residual local closed Lie algebra is

$$\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[1].$$

On product opens in $\mathbb{R}^2 \times \mathbb{C}^2$ the closed fields have the mixed form

$$\Omega^*(D)\widehat{\otimes}\Omega^{0,*}(B)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big),$$

with differential $d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}$ and BF interaction $\frac{1}{2}\int \langle \beta, \{\alpha, \alpha\}\rangle dz_1 dz_2$. Proposition 3.1 checks that this is a local mixed holomorphic-topological BV sector in the sense above.

The open brane is the product of the exact local A-brane $\mathbb{R} \subset T^*\mathbb{R}$ and the B-brane $\mathcal{O}_p$ at the formal holomorphic point. The Yoneda algebra of the skyscraper at the origin is computed in Proposition 3.2; for a cyclic Calabi-Yau $A_\infty$ category the open field of an object $\mathcal{F}$ is $\mathrm{RHom}(\mathcal{F}, \mathcal{F})[1]$, and the formal Chern-Simons action

$$S(a) = \sum_{n\geq 1} \frac{1}{n+1}\mathrm{Tr}\big(a\, m_n(a, \dots, a)\big), \qquad a \in \mathrm{RHom}(\mathcal{F}, \mathcal{F})[1],$$

is the algebraic core of Witten's open string field theory [4] and of the AKSZ packaging [6]. This is a source convention for the cyclic open BV form; the local proof below is the finite Dirac/Koszul matrix calculation. For the local model at hand,

$$\mathrm{RHom}_{\mathbb{C}[z_1,z_2]}(\mathcal{O}_p, \mathcal{O}_p) \cong \Lambda(\varepsilon_1, \varepsilon_2),$$

and the ghost-zero open field is

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2.$$

The ghost-zero action is the first-order gauged matrix model of (1), where $\phi_1, \phi_2$ are adjoint scalar normal coordinates and $A$ imposes the moment-map equation $[\phi_1, \phi_2] = 0$.

### 2.11 *Hamiltonian sector and stable large $N$*

Everything below is over $\mathbb{C}$. The phrase *formal local model* means the completed Taylor model at the chosen point of $\mathbb{C}^2$; equivalently, one may work polynomially degree by degree and then complete at the augmentation ideal. The symbol $\widehat{\mathbf{S}}$ denotes this completed symmetric algebra, and all Chevalley-Eilenberg cochains are continuous cochains for the same filtration.

Shifts follow the BV convention used in the field descriptions: an element of $V[1]$ has parity reversed and cohomological degree one lower than the corresponding element of $V$. The notation $\mathbf{S}(V)$ always means the free graded-commutative algebra on the graded vector space $V$.

Let

$$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1, \qquad \mathfrak{h} = \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1, \qquad \mathfrak{h}^\vee = \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}.$$

Constants are removed because a constant Hamiltonian has zero Hamiltonian vector field. We use zero-constant Taylor representatives for calculations, but the Lie bracket is always the Poisson bracket followed by projection modulo constants. On the brane side $\mathrm{Tr}_N(1) = N$ is the empty trace, a central scalar depending on the rank. We do not quotient the associative operator algebra by its unit. The Hamiltonian comparison uses the connected stable trace sector: the unit is retained in the symmetric algebra, but the empty trace is not a primitive generator and constants project to zero in the Hamiltonian

Lie/vector quotient. If $T$ is a positive word-degree trace expression, $\overline{T}$ denotes its connected trace class; constants have zero connected class.

At a fixed point of the brane line the reduced BV algebra is generated by two even matrices $\phi_1, \phi_2$ and the odd antifield $\psi = A^*$, with

$$Q\phi_1 = Q\phi_2 = 0, \qquad Q\psi = [\phi_1, \phi_2].$$

Thus the moment-map equation is imposed homologically. Boundary terms on $\mathbb{R}$ are discarded by taking compactly supported variations, or equivalently by working on an interval with fixed boundary values and passing to local operators in the interior.

The term *stable large-$N$* means the degreewise stable value of the invariant algebras under block restriction as $N \to \infty$. Inside a fixed finite matrix algebra all finite-$N$ trace identities remain present. In the stable limit, Procesi–Razmyslov invariant theory [17, 18] says that invariant functions of two matrices are freely generated by cyclic trace words. The differential $Q\psi = [\phi_1, \phi_2]$ then makes adjacent swaps exact, so ghost-zero cyclic words reduce to ordinary commutative monomials

$$\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \qquad k + l > 0.$$

The primitive one-$\psi$ Koszul homology supplies a second nonconstant family $\Psi_{k,l}$, obtained by inserting one $\psi$ and symmetrising over all words with $k$ copies of $\phi_1$ and $l$ copies of $\phi_2$. The scalar class $\mathrm{Tr}(\psi)$ is excluded from the Hamiltonian comparison because constants have been removed from $\bar{A}$; higher $\psi$-count homology lies outside this Hamiltonian sector. The collapse of cyclic words to commutative monomials on $H^0(Q)$ is the algebraic shadow of Dirac reduction: the constraint $Q\psi = [\phi_1, \phi_2]$ is the Koszul resolution of the moment-map equation.

The large-$N$ principle is the string-field-theoretic form of the Loday-Quillen-Tsygan principle [2, 3]; see also [24, Ch. 13] for the operadic packaging. Open strings on a stack produce matrix algebras; closed modes appear in the primitive stable part. LQT controls the stable invariants of the *unconstrained* matrix algebra; the moment-map equation $[\phi_1, \phi_2] = 0$ is the first-class constraint whose Koszul resolution is exactly the BV reduction $Q\psi = [\phi_1, \phi_2]$. Theorem 2.27 verifies this principle directly on the connected Hamiltonian trace sector.

Theorem 2.18 *(Loday-Quillen-Tsygan).* Let $A$ be a unital associative algebra over a field of characteristic zero. The stable Chevalley-Eilenberg homology of $\mathfrak{gl}_\infty(A)$ is the free graded-commutative algebra generated by shifted cyclic homology:

$$H_*^{\mathrm{CE}}\big(\mathfrak{gl}_\infty(A)\big) \cong \mathbf{S}\big(HC_{*-1}(A)\big).$$

LQT supplies the template; the reduced computation below uses Procesi–Razmyslov invariant theory together with the explicit Koszul differential. No global $HC_{*-1}$ convention is imported into Theorem 2.27: cyclic homology supplies the stable-trace analogy that organises the comparison, not the proof of the Hamiltonian sector.

### 2.12 *The concrete dictionary*

The dictionary uses three disjoint alphabets. Coincident bidegrees are only an indexing convention; they are not an identification of the underlying modules.

*Open PBW special fibre.* For $k, l \geq 0$ and $k + l > 0$ the open Hamiltonian trace label is

$$O_{k,l} = \overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}.$$

The primitive one-antifield PBW special-fibre label is the symmetrised trace

$$\Psi_{k,l} = \sum_{w \in \mathsf{W}_{k,l}} \mathrm{Tr}\big(\psi\, w(\phi_1, \phi_2)\big),$$

where $\mathsf{W}_{k,l}$ is the set of words with $k$ copies of $\phi_1$ and $l$ copies of $\phi_2$. The symbols $O_{k,l}$ and $\Psi_{k,l}$ are open brane classes in the connected stable PBW special fibre. They are not CE coordinates and they are not principal parts.

*Closed CE/PV coordinates.* For the reduced Hamiltonian Lie algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$, write $H_{k,l} = z_1^k z_2^l$ for the Hamiltonian basis element and $\eta_{k,l} = \delta_{k,l}[1] \in \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ for the shifted cotangent Lie generator, where

$$\delta_{k,l}(z_1^a z_2^b) = \delta_{k,a}\delta_{l,b}.$$

The CE/PV comparison is written in the coordinate alphabet

$$c^{k,l} \quad \text{dual to } H_{k,l}, \qquad u_{k,l} \quad \text{dual to } \eta_{k,l}.$$

The map in the CE/PV theorem sends $c^{k,l}$ to the corresponding Schouten coordinate $\theta^{k,l}$ and sends $u_{k,l}$ to the Hamiltonian central operation represented on the brane by $O_{k,l}$. Thus $u_{k,l}$ is a closed CE coordinate dual to the shifted cotangent generator; it is not the open one-$\psi$ class $\Psi_{k,l}$.

*Matlis principal parts.* The deformation-compatible cotangent module is the reduced Matlis principal-part module with basis

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2, \qquad a + b > 0,$$

paired with $H_{m,n}$ by residue:

$$\langle \rho_{a,b}, H_{m,n} \rangle = \delta_{a,m}\delta_{b,n}.$$

This realizes $\mathfrak{h}^\vee_{\mathrm{cont}}$ with the coadjoint lowering action of Proposition 3.147.

The vector-space pairing $e^{\mathrm{PBW}}_{k,l} \mapsto [\Psi_{k,l}]$ used in the PBW special fibre is therefore not the principal-part realization and not an $\mathfrak{h}$-module isomorphism to the Matlis module $\mathcal{P} = \bigoplus \mathbb{C}\rho_{a,b}$. A passage from $\Psi_{k,l}$ to $\rho_{k,l}$ would require an additional Rees/Fourier $D_\hbar$-module theorem; it is listed among the open obligations in Section 4.

Each later step has a concrete target: construct $O_{k,l}, \Psi_{k,l}$ from the open fields; construct $c^{k,l}, u_{k,l}$ in the CE/PV closed algebra; construct $\rho_{a,b}$ as the Matlis principal-part basis; prove that commutators of words reduce to commutative monomials in $H^0(Q)$; and verify that the Hamiltonian bracket coefficient is $kn - lm$.

### 2.13 *Status of assertions and the analytic problem*

Theorem 2.27 below is the precise local assertion; its constituents are constructed in the subsequent sections. The proof combines: the affine **Ext** calculation (Proposition 3.2); the open BV reduction (Lemma 3.5, Proposition 3.7); the Dirac reduction of the probe phase space (Lemma 3.6); the stable invariant-theory calculation of the Hamiltonian trace algebra (Proposition 3.120); the computation of

the primitive one-antifield Koszul homology (Proposition 3.125); the formal Hamiltonian BF model for the closed side (Proposition 3.13); the construction of the bulk-to-boundary Hamiltonian evaluation map (Construction 3.123, Theorem 3.161); the CE/Koszul-duality computation; and the first bracket calculation (5).

The formal Weyl/Moyal calculation below is an all-order algebraic deformation-quantisation target for the constant Poisson formal disk; its monomial coefficients are checked by finite exact arithmetic. A closed-string Feynman-integral derivation of those coefficients is a separate analytic problem, requiring the global BV/gauge-fixing data of Problem 3.183: a global elliptic complex extending the formal jet model; boundary conditions on $\mathbb{C}^2$; a gauge-fixing $Q^{\mathrm{GF}}$ and the heat-kernel propagator $P_{\epsilon,L} = \int_\epsilon^L Q^{\mathrm{GF}} K_t \, dt$; zero-mode treatment and the IR limit; UV renormalisation scheme and local counterterms; quantum master equation or anomaly class; bulk-to-boundary kernel for the brane; trace normalisation and large-$N$ weighting; and the connected/single-trace extraction convention. Costello's renormalized BV formalism supplies the general graph/RG package; Costello–Li prove the flat odd-dimensional BCOV/HCS case [13, 14]. The proof of Theorem 2.27 uses the formal local algebra alone and is independent of these analytic choices.

### 2.14 *The local theorem*

The theorem has three layers. The open calculation identifies the ghost-zero brane theory with the algebra of invariant functions on commuting matrix pairs together with the first Koszul descendants of the commuting-pair equation. The closed calculation constructs the CE/PV coordinates of the Hamiltonian cotangent Lie algebra and realizes its continuous cotangent module by Matlis principal parts. The Hamiltonian boundary map is the local algebraic operation of evaluating a Hamiltonian on the matrix-valued normal coordinates of the brane; the polynomial one-$\psi$ descendants remain PBW special-fibre classes, not principal-part currents.

The local theorem follows by combining the status index below. Its reduced core is the CE/PV derived-centre comparison, the brane-line factorization-current lift, the Matlis principal-part cotangent factor after de Rham-current contraction, the scalar-reduced degree-zero Moyal/Weyl deformation, and the U(1) scalar anomaly. The general conilpotent bar-cobar statement is only the template in its stated range. The unreduced Tate-coefficient cotangent lift, all-order one-antifield quantum lift, regulator independence, and mixed brane-defect QME are separated by claim strength in the claim-strength ledger and by the open obligations of Section 4.

#### 2.14.1 Theorem-lane index

This subsection is an index, not a second proof of the local theorem package. Each lane records the local hypotheses, the assertion whose proof is carried later in the manuscript, and the exact proof dependencies. The numbered entries keep stable cross-reference labels so that the summary package can refer to the same lanes, but the proof-bearing text is the cited theorem, proposition, lemma, or computation. No assertion below has claim strength beyond the cited proof-bearing block and the status vocabulary of the claim-strength ledger.

*Statement* 2.19 *(Index lane: Dirac brane probe). Status. proved in finite algebraic model.* The proof is confined to the formal local ghost-zero sector of the brane probe.

*Exact hypotheses.* The transverse formal disk is $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$; the stack size $N$ is finite; fields are $\phi_1, \phi_2, A \in \mathfrak{gl}_N$ on the brane line; variations are compactly supported; and the only residual gauge group retained in the lane is ordinary $GL_N$-conjugation.

*Local assertion.* For a stack of $N$ branes probing $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$, the ghost-zero open action is the first-order Dirac matrix system

$$S_\partial = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big),$$

and its reduced BV algebra is

$$\mathbb{C}[\phi_1{}^i{}_j, \phi_2{}^i{}_j] \otimes \Lambda(\psi^i{}_j), \qquad Q\psi = [\phi_1, \phi_2], \quad Q\phi_i = 0.$$

The moment map for conjugation is $-\mathrm{Tr}(\epsilon[\phi_1, \phi_2])$, so the BV differential is the Koszul differential for the commuting-pair equation.

*Proof dependency map.* Lemma 3.6 derives the ghost-zero action from the shifted normal directions and fixes the moment-map sign. Proposition 3.7 proves that the ghost-zero truncation is consistent and identifies the residual Koszul complex.

*Residuals.* The lane does not assert an untruncated open-string BV model beyond this ghost-zero finite algebraic truncation.

*Statement* 2.20 *(Index lane: stable trace and one-$\psi$ PBW shadow). Status. proved degreewise stable*. It is a PBW special-fibre trace statement, not a statement about the principal-part cotangent module.

*Exact hypotheses.* Work at fixed total polynomial degree and then in the stable connected large-$N$ range; pass to scalar-reduced connected traces; use the Koszul differential $Q\psi = [\phi_1, \phi_2]$; and take the PBW special fibre before any filtered or $D_\hbar$-interpolation is introduced.

*Local assertion.* The Hamiltonian open observables in ghost number zero are generated by scalar-reduced connected trace classes $\overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$. The primitive one-$\psi$ classes $\Psi_{a,b}$ are the primitive PBW special-fibre shadow classes. They do not realize the continuous $\mathfrak{h}$-coadjoint module and are not the principal-part currents $\rho_{a,b}$.

*Proof dependency map.* Procesi–Razmyslov stable invariant theory supplies the stable trace generators. The Koszul differential $Q\psi = [\phi_1, \phi_2]$ turns adjacent swaps into boundaries in the connected trace quotient. Proposition 3.125 computes the primitive one-$\psi$ homology. Theorem 3.101 identifies the chain-level primitive shadow, while Theorem 3.152 and Theorem 3.154 separate that PBW shadow from the coadjoint principal-part module. Theorem 3.104 proves the same separation for every $\hbar$-adic deformation whose leading term is the classical primitive label projection.

*Residuals.* Unstable finite-$N$ trace identities and any ordinary $\mathfrak{h}$-module identification of $\Psi_{a,b}$ with $\rho_{a,b}$ are outside the lane. The Fourier–Rees comparison of Theorem A.8 is only an associated-graded label bridge. The corrected quantum target, when needed, is the reduced distributional principal-part target of Construction 3.103, not the PBW one-$\psi$ span.

*Statement* 2.21 *(Index lane: CE/PV reduced Hamiltonian central-operation model). Status. proved degreewise stable*. The proof is at cochain level for the reduced formal Hamiltonian algebra and its continuous shifted cotangent extension. The bar-cobar consequence is used only under the stated Tate-conilpotent hypothesis.

*Exact hypotheses.* The Hamiltonian algebra is $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ with its $m$-adic topology; the cotangent direction is $\mathfrak{h}^\vee_{\mathrm{cont}}[1]$; CE cochains are completed and continuous; and every use of bar-cobar duality is restricted to the conilpotent Tate hypothesis named in the cited lemma.

*Local assertion.* Let

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1, \qquad \mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1].$$

The completed CE cochains of $\mathfrak{g}$ identify with the Schouten polyvector algebra of the Tate symmetric algebra:

$$\Phi : C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big) \simeq Z^{P_0}_{\mathrm{red,Ham}}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big),$$

with generator dictionary

$$c^I \mapsto \theta^I, \qquad u_I \mapsto O_I.$$

Here $|u_I| = 0$, $|c^I| = 1$, and $u_I$ is the CE coordinate dual to the shifted-cotangent generator, not to a Hamiltonian generator.

*Proof dependency map.* Proposition 3.135 identifies closed local operators as continuous CE cochains. Lemma 3.139 computes the Schouten differential. Theorem 3.27 checks $\Phi d_{\mathrm{CE}} = d_\pi \Phi$ on generators and extends by the completed Leibniz rule. Lemma 3.137 supplies the conilpotent bar-cobar template only in its own hypothesis; the untruncated formal Hamiltonian algebra with linear modes is not used as a lower-central conilpotent input.

*Residuals.* The lane does not prove an unrestricted factorization-centre theorem or a conilpotence statement for the untruncated formal Hamiltonian algebra. It also does not construct a BV heat-kernel propagator, a BV Laplacian, or a QME-compatible closed-open graph model for the unreduced cotangent sector.

*Statement 2.22 (Index lane: Matlis principal-part cotangent module). Status. proved degreewise stable.* The proof uses residue calculus and Matlis local-cohomology duality for the $m$-adic topology on $R = \mathbb{C}[[z_1, z_2]]$, after quotienting constants.

*Exact hypotheses.* Work over $R = \mathbb{C}[[z_1, z_2]]$ with maximal ideal $\mathfrak{m} = (z_1, z_2)$; put $\mathfrak{h} = R/\mathbb{C} \cdot 1$; take continuous duals for the $m$-adic topology; normalize the residue pairing only up to one nonzero scalar; and, for the Fourier–Rees comparison only, distinguish the Fourier delta Rees lattice $\mathcal{D}_\hbar/(z_1, z_2)$ from the full Cech local-cohomology lattice before $\hbar$ is inverted.

*Local assertion.* The continuous dual of the reduced Hamiltonian Lie algebra is the reduced Matlis-local-cohomology module

$$\mathfrak{h}^\vee_{\mathrm{cont}} \cong \mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1) \subset H^2_{\mathfrak{m}}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2,$$

with basis $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2$, $a + b > 0$, and coadjoint action

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1},$$

with negative-index targets equal to zero. The residue pairing is determined up to one nonzero scalar, and this scalar is the only normalization ambiguity in the lane.

*Proof dependency map.* Appendix A centralizes the reduced Matlis calculation in the notation of this lane. Theorem 3.148 proves the residue pairing and its scalar ambiguity. Proposition 3.147 computes the explicit coadjoint action. Theorem 3.150 gives the Matlis-form uniqueness statement. Theorem 3.152 and Theorem 3.154 state the no-go result for replacing $\mathcal{P}$ by polynomial descendants. Theorem A.8 supplies the limited Fourier–Rees label bridge through the Rees lattice $\bigoplus \hbar^{a+b}\mathbb{C}[[\hbar]]\rho_{a,b}$ and proves that it is not an $\mathfrak{h}$-module bridge.

*Residuals.* A same-action $D_\hbar$, Fourier, or Rees comparison identifying the PBW descendant module with the Matlis coadjoint module is ruled out on the generic fibre by local nilpotence unless the Hamiltonian action or the boundary target is changed. The Fourier-twisted label bridge does not supply such a same-action comparison.

*Statement 2.23 (Index lane: brane-line factorization currents). Status. proved degreewise stable.* The proof is for compactly supported de Rham Hamiltonian currents and compactly supported distributional principal-part currents on intervals. This lane is a strict $P_0$ current statement in the reduced Hamiltonian sector.

*Exact hypotheses.* Intervals lie in the brane line $\mathbb{R}$; $\mathfrak{h}_I = \Omega_c^\bullet(I)\widehat{\otimes}\mathfrak{h}$; $\mathcal{P}_I = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}$; extension by zero is the functoriality; and source coordinates in the displayed boundary operation are shifted-cotangent CE coordinates $u_{a(t)dt\otimes f}$, not Hamiltonian ghost coordinates.

*Local assertion.* For each interval $I \subset \mathbb{R}$,

$$\mathfrak{h}_I = \Omega_c^\bullet(I)\widehat{\otimes}\mathfrak{h}, \qquad \mathcal{P}_I = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}, \qquad \mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1].$$

The Hamiltonian factor admits an interval-wise locally constant $P_0$ factorization equivalence. On length-one shifted-cotangent CE coordinates the source is

$$u_{a(t)dt\otimes f} \longmapsto B_f(a) = \int_I a(t)\,\overline{\operatorname{Tr} f(\phi_1(t), \phi_2(t))}\,dt.$$

In the degree-zero notation used here $a \in \Omega_c^0(I)$, the paired density is $a(t)\,dt$, $|u_{a(t)dt\otimes f}| = 0$, and the de Rham-current pairing is

$$\langle B \otimes \rho,\ a \otimes f\rangle = B(a)\operatorname{Res}(\rho f).$$

For a regular current $B = b(t)\,dt$ this is $\int_I a(t)b(t)\,dt\ \operatorname{Res}(\rho f)$. Orientation reversal contributes only the ordinary sign of the one-form density $dt$; no extra sign is inserted in the lane.

*Proof dependency map.* Appendix B fixes the interval source, current dual, shifted-cotangent CE coordinate, and orientation convention used in this lane. Construction 3.18 fixes the de Rham-current objects and extension-by-zero functoriality. Proposition 3.19 proves disjoint-support locality. Theorem 3.24 supplies the smeared Hamiltonian central operation. Theorem 3.21 identifies the interval-wise factorization-centre map and its compatibility with factorization products.

*Residuals.* The strict current statement does not include the unreduced BV/factorization-centre morphism or its centrality homotopies against arbitrary open observables.

*Statement 2.24 (Index lane: reduced principal-part defect currents). Status. proved degreewise stable.* The proof applies only after the de Rham-current correction and only in the reduced $P_0$ model after contraction. It is not an unreduced BV/factorization theorem and does not supply centrality homotopies against arbitrary open observables.

*Exact hypotheses.* The algebra is the reduced defect-current model on intervals after de Rham-current contraction; the Hamiltonian factor is $\widehat{\mathbf{S}}(\mathfrak{h}_I)$; the cotangent factor is $\widehat{\mathbf{S}}(\mathcal{P}_I[1])$; brackets are strict $P_0$-brackets; and the primitive one-$\psi$ comparison is taken only after passing to indecomposables.

*Local assertion.* The principal-part cotangent factor extends the Hamiltonian brane-line algebra to the reduced defect-current model

$$A_\partial^{\mathrm{pp}}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{P}_I[1])$$

with

$$\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB), \qquad \{\Theta_\rho(B), \Theta_\sigma(C)\} = 0.$$

The chain-level primitive projection gives only the primitive indecomposable $P_0$ shadow of this reduced comparison. Its quantum analogue is the reduced Moyal principal-part target of Construction 3.103, where $f \cdot \rho$ is replaced by $\operatorname{ad}^\vee_{f,\hbar}\,\rho$.

*Proof dependency map.* Theorem 3.157 proves the reduced current model and its $P_0$ bracket. Theorem 3.159 records the boundary local-dual realization in the same reduced setting. Remark 3.158 and Remark 3.160 state the unreduced obligations. Theorem 3.101, Corollary 3.106, and Remark 3.107 supply only the primitive shadow status. Proposition 3.102 and Theorem 3.104 prove that the polynomial primitive label projection does not quantize to the Moyal coadjoint target.

*Residuals.* The unreduced cotangent factorization-centre morphism and its centrality homotopies remain outside the lane. The reduced Moyal principal-part target is constructed, but a quantum lift from polynomial one-$\psi$ primitives is ruled out under the leading-term assumptions above. The pointwise classical obstruction is the mismatch between the polynomial one-$\psi$ action and the principal-part coadjoint action:

$$O_{1,0} : \widetilde{\Psi}_{a,b} \mapsto b\,\widetilde{\Psi}_{a,b-1}, \qquad z_1 : \rho_{a,b} \mapsto -(b+1)\rho_{a,b+1},$$

and similarly

$$O_{0,1} : \widetilde{\Psi}_{a,b} \mapsto -a\,\widetilde{\Psi}_{a-1,b}, \qquad z_2 : \rho_{a,b} \mapsto (a+1)\rho_{a+1,b}.$$

Therefore the primitive projection records labels only; it does not supply centrality homotopies against arbitrary unreduced open observables.

*Statement* 2.25 *(Index lane: degree-zero Moyal/Weyl sector and graph boundary). Status. proved in finite algebraic model* for the formal Weyl/Moyal coefficient algebra; *conditional on named QME/radial-parts/unweighted-regulator theorem* for the finite-$N$ all-$N$ radial-parts reduction step. The descendant one-$\psi$ lift and the Costello brane-defect graph realization are not included in the proved algebraic lane.

*Exact hypotheses.* Work in the scalar-reduced degree-zero Hamiltonian sector at finite $N$, assume the radial-parts input of Statement D.5 for the all-$N$ reduction, then take the stable connected trace range degreewise; use the Capelli/Weyl normalization of the Weyl algebra and the formal Moyal product on $\bar{A}_\hbar$; and exclude one-antifield descendants and graph-theoretic BV data from the proved comparison.

*Local assertion.* On the scalar-reduced degree-zero Hamiltonian sector, conditional on the radial-parts input, the classical closed-open comparison deforms to all orders in $\hbar$. The commutator is the formal Moyal commutator on $\bar{A}_\hbar$ under the Capelli/Weyl normalization, and the stable connected trace map satisfies

$$\hbar^{-1}[\partial_{\mathrm{bb},\hbar}(f), \partial_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \partial_{\mathrm{bb},\hbar}(\{f,g\}_\hbar).$$

The first and third reduced open-line coefficients are $\hbar P(f,g)$ and $\hbar^3 P^3(f,g)/24$. A Costello-theoretic graph lift of those coefficients is conditional on the sector, propagator, counterterm, anomaly, and QME data named in the analytic graph problem. In the descendant direction the reduced target is not polynomial: Construction 3.103 gives the Moyal principal-part coadjoint complex, and Theorem 3.104 rules out any strict $\hbar$-adic deformation of the classical $\Psi_{a,b} \mapsto \rho_{a,b}$ projection with PBW linear Hamiltonian leading term.

*Proof dependency map.* Appendix D separates the formal Moyal/Capelli calculation from the external radial-parts input. Theorem 3.170 uses that input to obtain the finite-$N$ radial-parts identity and the stable degree-zero commutator. Corollary 3.171 and Theorem 3.187 assemble the stable Capelli-renormalized map. Proposition 3.186 proves the Moyal coefficient formula, with an independent finite exact arithmetic sweep as verification evidence. Proposition 3.191 and Proposition 3.192 compute the reduced open-line first and third coefficients. Theorem 3.194 is conditional on Problem 3.183, and Problem 3.68 carries the mixed brane-defect QME obstruction. Theorem 3.84 gives the finite nilpotent obstruction to an all-order descendant lift. Proposition 3.102, Construction 3.103, and Theorem 3.104 isolate the one-$\psi$ quantum target and the no-go for the polynomial primitive projection.

*Residuals.* Without the radial-parts source input the all-$N$ trace identity is conditional; the finite exact check covers only finite sweeps, the rank-two Cartan shadow, and the direct $N = 2, 3$ radial restrictions. The all-order Costello graph realization, the mixed brane-defect QME, the unweighted regulator problem, and an unreduced one-antifield Moyal lift are not consequences of this algebraic lane. The reduced principal-part Moyal target is only a target-side construction.

*Statement* 2.26 *(Index lane: scalar $U(1)$ anomaly). Status. proved in finite algebraic model.* The lane records the scalar-reduction obstruction to lifting the reduced Hamiltonian comparison back to the unreduced constant direction.

*Exact hypotheses.* The closed algebra is the unreduced polynomial Hamiltonian algebra $\mathfrak{h}_{\mathrm{poly}} = \mathbb{C}[z_1, z_2]$ and its quotient $\bar{A} = \mathfrak{h}_{\mathrm{poly}}/\mathbb{C} \cdot 1$; the open side is the finite-$N$ trace algebra before and after quotienting $\mathbb{C} \cdot \mathrm{Tr}(1)$; and the quantum comparison uses the same Capelli/Weyl normalization as the degree-zero Moyal lane.

*Local assertion.* The scalar-reduction axis carries the distinguished class

$$[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C}),$$

represented by the constant Taylor coefficient of the Poisson bracket. The same class appears on the open side as the centre-of-mass $\mathfrak{gl}_N$ anomaly and quantum mechanically as the Capelli shift $\hbar N$. All reduced Hamiltonian comparisons are made after killing constants; the unreduced scalar line is governed by this anomaly class.

*Proof dependency map.* Lemma 2.2 identifies the closed Poisson cocycle. Theorem 2.3 proves the closed-side central extension statement. Theorem 2.4 computes the open $U(1)$ centre. Theorem 2.6 identifies the classical cocycle with the quantum Capelli shift. Corollary 3.162 gives the explicit local central term in the smeared boundary bracket.

*Residuals.* The lane does not lift the reduced comparison to the unreduced scalar direction and does not imply compact Calabi–Yau, BKM, Igusa, or sister-volume anomaly statements.

COROLLARY 2.27 *(Main local summary corollary).* In the formal local mixed holomorphic-topological BV model on $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ with brane $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$, the status index 2.14.1 supplies the local proof dependencies: Dirac brane probe, stable trace and one-$\psi$ PBW shadow, CE/PV derived centre, Matlis principal parts, brane-line factorization currents, reduced principal-part defect currents, scalar-reduced degree-zero Moyal/Weyl deformation, and the U(1) scalar anomaly.

Consequently, after quotienting constants on the Hamiltonian side and the scalar trace on the brane side, the closed Hamiltonian BF observables $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1])$ identify with the reduced CE/PV Hamiltonian central-operation model attached to the stable open Hamiltonian brane algebra. This is not the full unreduced $P_0$-factorization-centre theorem. The cotangent direction is realized by the Matlis principal-part module, not by the polynomial one-$\psi$ descendants; the all-order quantum assertion is restricted to the degree-zero Hamiltonian Moyal/Weyl sector; and the unreduced scalar line is governed by the anomaly class $[\bar{c}]$.

The exact claim strengths are those of the claim-strength ledger. The open analytic, regulator, unreduced cotangent, one-antifield quantum, and cross-volume obligations are the obligations listed in Section 4.

*Proof.* This is a summary corollary, not a second proof. The listed theorem lanes give the proof dependencies and local hypotheses; the ledger records which parts are proved, conditional, conjectural, open, or not asserted. Taking their common reduced Hamiltonian consequence gives the displayed closed-open derived-centre statement. □

*Remark 2.28 (Range of the CE/PV lane).* The conilpotent CE/PV lane summarized in 2.27 holds for any pro-finite-dimensional Lie algebra $\mathfrak{h}$ in **TateVec** satisfying the hypothesis of Lemma 3.137. The full formal symplectic disk in dimension two is the explicit Hamiltonian specialization: its CE/PV cochain identity is checked on monomial structure constants, while the conilpotent bar-cobar consequences pass through the nilpotent or weighted replacement. The remaining proved lanes are dimension-two specific (they use the residue pairing on $\mathbb{C}^2$, the matrix Weyl algebra of pairs $(X, Y)$, and the U(1) center of $\mathfrak{gl}_N$). Higher-dimensional analogues require the same cochain identity together with the appropriate residue pairing, matrix Weyl data, and a separate verification of the required conilpotent, weighted, or completed coefficient hypotheses.

*Remark 2.29 (Dirac thesis, as a reduced summary corollary).* The summary corollary 2.27 makes precise the Dirac thesis sentence that organizes the manuscript:

> *A stack of branes is a system of probes for the formal symplectic normal disk. Dirac reduction of the probe phase space gives the open trace algebra. Hamiltonian BF theory is the shifted cotangent theory of canonical transformations. In the reduced formal Hamiltonian sector, the closed observables identify with derived Poisson central operations on the stable open Hamiltonian brane algebra.*

This is a corollary of 2.27: the brane stack is the open BV theory of $N$ matrix-valued probes $(\phi_1, \phi_2)$ for the formal symplectic disk; Dirac first-class reduction along the moment-map ideal $I_N = \mathcal{W}_N \widehat{\mu}(\mathfrak{gl}_N)$ produces the stable connected open trace algebra $H^{\circ}_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty})$; the closed Hamiltonian BF theory has fields valued in the shifted-cotangent semidirect product $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1]$ with $\mathfrak{h}$ the Hamiltonian Lie algebra of canonical transformations of the formal disk; and the closed observables $C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g})$ are identified, via $\Phi$, with the reduced CE/PV Hamiltonian central-operation model $Z^{P_\circ}_{\mathrm{red,Ham}}(A^{\mathrm{Ham}}_{\partial})$ on the brane line after de Rham-current contraction. The degree-zero Hamiltonian sector has the conditional quantum comparison stated later; the full unreduced factorization centre remains an open obligation.

The proof uses the formal local model alone. The unreduced Tate-coefficient cotangent lift is isolated in 3.47 and separated into its precise chain-level, nilpotent, weighted, and QME-obstruction components. What remains outside the proved formal core is the stronger analytic realization of the whole package as a literal Costello renormalized BV graph construction on an elliptic parent with brane-defect kernel, with the mixed brane-defect QME obstruction killed.

The protected twisted-M-theory origin is not used as a proof of the local theorem. It is recorded in §5.1 as a conditional outlook theorem: once an ambient background supplies the admissible protected Hamiltonian restriction, the local model applies to that protected sector and no further.

## 3 The Local Model

The theorem-status package extracts its content from a single local geometric setup: a topological line carrying a stack of probes, crossed with a holomorphic-symplectic disk to which they couple. The remainder of the manuscript constructs and quantizes that setup, and verifies that the theorem lanes hold on the resulting data.

### *3.1 Setup*

Let

$$M = \mathbb{R}^2, \qquad Y = \mathbb{C}^2, \qquad X = M \times Y,$$

with $M$ the topological factor, $Y$ the holomorphic factor, and all Hamiltonian calculations completed at the formal neighbourhood of $p = 0 \in Y$. The exact A-brane is the standard line $\mathbb{R}_{\text{brane}} \times \{0\} \subset M$ with trivial brane data; the B-brane is the skyscraper $\mathcal{O}_p \in \operatorname{Perf}(Y)$. Their product is the mixed brane

$$L = \mathbb{R}_{\text{brane}} \times \{0\} \times \{p\} \hookrightarrow \mathbb{R}^2_{\text{top}} \times \mathbb{C}^2_{\text{hol}}, \qquad p = 0.$$

The holomorphic factor carries the standard holomorphic symplectic form

$$\omega = dz_1 \wedge dz_2,$$

so the model is topological along $\mathbb{R}^2$ and holomorphic Hamiltonian along $\mathbb{C}^2$.

PROPOSITION 3.1 *(Local mixed model).* The formal local Hamiltonian data used here define a closed mixed holomorphic-topological BV sector and a compatible cyclic open brane BV theory. The constructed closed-open operation is the degree-zero Hamiltonian boundary evaluation. The full Poisson factorization-center morphism is Problem 3.47, not part of this proposition.

*Proof.* On product opens $D \times B$ in the completed formal neighbourhood of $\mathbb{R}^2 \times \{p\} \subset \mathbb{R}^2 \times \mathbb{C}^2$, the closed fields are

$$\mathcal{E}_{\text{cl}}(D \times B) = \Omega^*(D)\widehat{\otimes}\Omega^{0,*}(B)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\text{cont}}[2]\big).$$

Degreewise polynomial calculations use the dense subspace $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$. The differential is $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$, so $Q_0 = 0$ in Definition 2.13. The BV pairing is the local pairing on compactly supported fields, obtained by multiplying the form components, integrating over $D \times B$ against $dz_1\, dz_2$, and using the continuous evaluation pairing $\widehat{\mathfrak{h}}^\vee_{\text{cont}} \otimes \widehat{\mathfrak{h}} \to \mathbb{C}$. If $\alpha \in \Omega^p\widehat{\otimes}\widehat{\mathfrak{h}}[1]$ and $\beta \in \Omega^q\widehat{\otimes}\widehat{\mathfrak{h}}^\vee_{\text{cont}}[2]$, then $|\alpha| = p - 1$ and $|\beta| = q - 2$; the top-form condition $p + q = 4$ makes the pairing degree $-1$.

The free quadratic term

$$S_{\text{free}} = \int_{D\times B} \big\langle \beta, D\alpha \big\rangle dz_1\, dz_2$$

is the BV Hamiltonian for the differential $D$. The interaction in Definition 2.13 is only

$$I_{\text{BF}} = \frac{1}{2}\int_{D\times B} \big\langle \beta, \{\alpha, \alpha\} \big\rangle dz_1\, dz_2.$$

The total BF action is $S_{\text{free}} + I_{\text{BF}}$. The Poisson bracket on $\bar{A}$ is the holomorphic bracket determined by $dz_1 \wedge dz_2$. Since $D$ is a derivation of this bracket and the bracket satisfies Jacobi,

$$DI_{\text{BF}} + \frac{1}{2}\{I_{\text{BF}}, I_{\text{BF}}\} = 0,$$

so the total BF action satisfies the classical master equation without double-counting the free differential.

Mixed locality is the Cartan homotopy calculation on the interacting observables. For a constant vector field $v$ in the $\mathbb{R}^2$ direction, $[D, \iota_v] = \mathcal{L}_v$; for an anti-holomorphic coordinate vector field $\partial/\partial\bar{z}_i$, $[D, \iota_{\partial/\partial\bar{z}_i}] = \mathcal{L}_{\partial/\partial\bar{z}_i}$. The interaction is built from the de Rham wedge product and holomorphic polydifferential operators, so the contractions are derivations of the local observable algebra and commute with $\{I_{\text{BF}}, -\}$ up to the displayed Lie derivative. Thus $[Q_{\text{obs}}, \iota_v] = \mathcal{L}_v$ and $[Q_{\text{obs}}, \iota_{\partial/\partial\bar{z}_i}] = \mathcal{L}_{\partial/\partial\bar{z}_i}$ on local observables. The de Rham Poincare lemma in the $D$ directions gives the quasi-isomorphism for small disk inclusions, and the Dolbeault Poincare lemma gives the holomorphic dependence in the $B$ direction.

The brane is $L = \mathbb{R}_{\text{brane}} \times \{0\} \times \{p\}$, with $p \in Z$ understood as the formal holomorphic neighbourhood. Its open fields are

$$\Omega^*(\mathbb{R}) \otimes \mathbf{Ext}^*_{\mathbb{C}[z_1,z_2]}(\mathcal{O}_p, \mathcal{O}_p) \otimes \mathfrak{gl}_N[1],$$

and Proposition 3.2 identifies the Ext algebra with $\Lambda(\varepsilon_1, \varepsilon_2)$. This is the derived $\mathrm{RHom}_Y(\mathcal{O}_p, \mathcal{O}_p)$ allowed in Definition 2.13, not an ordinary Dolbeault algebra on the point. The trace on $\mathfrak{gl}_N$ and the Berezin trace on this exterior algebra give the cyclic open string field theory. Construction 3.123 and Theorem 3.161 construct the degree-zero Hamiltonian boundary component and prove its bracket compatibility. The full Poisson factorization-center closed-open map is Problem 3.47, not a theorem proved here. □

### 3.2 *Open mixed brane states*

The open string states are the tensor product of the topological brane direction and the formal holomorphic normal direction.

- *Topological factor.* In the exact self-Floer model for the graded, spin zero-section $\mathbb{R} \subset T^*\mathbb{R} \cong \mathbb{R}^2$, we use the local exact Fukaya model near a compact interval in the brane. A small compactly supported Hamiltonian perturbation identifies $CF^*(\mathbb{R}, \mathbb{R})$ with the Morse complex on that interval, and passage to local operators in the interior is represented by the de Rham complex $\Omega^*(\mathbb{R})$ with compactly supported variations. No wrapped generators at infinity are included in this local model.
- *Holomorphic factor.* Put affine coordinates at $p = 0$ and write $R = \mathbb{C}[z_1, z_2]$, $\mathfrak{m} = (z_1, z_2)$, and $\mathcal{O}_p = R/\mathfrak{m}$. The skyscraper sheaf has open string states

  $$\mathbf{Ext}^*_R(R/\mathfrak{m}, R/\mathfrak{m}),$$

  computed in Proposition 3.2.

Proposition 3.2 *(Affine skyscraper self-Ext).* Let $R = \mathbb{C}[z_1, z_2]$, $\mathfrak{m} = (z_1, z_2)$, and $k = R/\mathfrak{m}$. Then the Yoneda algebra of the skyscraper sheaf at the origin is

$$\mathbf{Ext}^*_R(k, k) \cong \Lambda_{\mathbb{C}}(\varepsilon_1, \varepsilon_2),$$

where $|\varepsilon_i| = 1$. In particular this is the free graded-commutative algebra on two odd generators.

*Proof.* The pair $(z_1, z_2)$ is a regular sequence in the polynomial ring $R$, so the affine Koszul complex

$$K = \big(0 \to R\, e_{12} \xrightarrow{d_2} R\, e_1 \oplus R\, e_2 \xrightarrow{d_1} R \to 0\big)$$

is a free resolution of $k$, with

$$d_1(ae_1 + be_2) = az_1 + bz_2, \qquad d_2(e_{12}) = z_2 e_1 - z_1 e_2.$$

Applying $\mathrm{Hom}_R(-, k)$ gives

$$0 \to k \xrightarrow{0} k\,\varepsilon_1 \oplus k\,\varepsilon_2 \xrightarrow{0} k\,\varepsilon_{12} \to 0,$$

because $z_1$ and $z_2$ act by zero on $k$. Hence the graded vector space of Ext groups is $k$ in degree 0, $k\varepsilon_1 \oplus k\varepsilon_2$ in degree 1, and $k\varepsilon_{12}$ in degree 2.

The product is the Yoneda product. The Koszul resolution is the exterior dg-algebra $R \otimes \Lambda(e_1, e_2)$ with $d(e_i) = z_i$; its diagonal sends $e_i$ to $e_i \otimes 1 + 1 \otimes e_i$. Dualising this diagonal gives $\varepsilon_i^2 = 0$ and $\varepsilon_1\varepsilon_2 = -\varepsilon_2\varepsilon_1$, while $\varepsilon_1\varepsilon_2$ spans the top Ext group. Thus the Yoneda algebra is $\Lambda_{\mathbb{C}}(\varepsilon_1, \varepsilon_2)$. □

### 3.3 *Open mixed brane field theory*

The open theory is the minimal local field theory which remembers the brane's two normal holomorphic directions. The two Ext generators are odd because they are derived normal directions of a point brane; after the BV shift their ghost-zero coefficients become the two ordinary matrix coordinates $\phi_1, \phi_2$. The one-form coefficient is the gauge field $A$ along the brane. Thus the formal Chern-Simons action is the mechanism which turns the Ext algebra into first-order matrix mechanics.

The total space of fields (in the Batalin-Vilkovisky formalism) is

$$\alpha \in \Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1]$$

with action

$$S(\alpha) = \int_{\mathbb{R}\times\mathbb{C}^{0|2}} \frac{1}{2}\operatorname{Tr}(\alpha\, d\alpha) + \frac{1}{3}\operatorname{Tr}\alpha^3.$$

Here $d = d_{\mathrm{dR}}$; the Ext algebra $\Lambda(\varepsilon_1, \varepsilon_2)$ has zero differential. The integral over $\mathbb{C}^{0|2}$ is the Berezin integral $\int d\varepsilon_1\, d\varepsilon_2$, which picks out the top component in the odd directions.

The cyclic trace is

$$\operatorname{Tr}_{\mathrm{open}}(\eta \otimes u \otimes M) = \int_{\mathbb{R}} \eta\, [\varepsilon_1\varepsilon_2]u\ \operatorname{Tr}_N(M),$$

where $[\varepsilon_1\varepsilon_2]u$ denotes the coefficient of the top Ext class. The top Ext class is the residue orientation dual to $dz_1 \wedge dz_2$. The de Rham integral has degree $-1$ and the Ext residue has degree $-2$; after the field shift by $[1]$, the induced BV pairing has degree $-1$.

*Construction* 3.3 *(The open field from the Ext basis).* Write $\varepsilon_{12} = \varepsilon_1\varepsilon_2$. Before imposing ghost number, an open field is a finite sum

$$\alpha = a_0 + a_1\varepsilon_1 + a_2\varepsilon_2 + a_{12}\varepsilon_{12},$$

where each $a_I$ is a de Rham form on $\mathbb{R}$ with values in $\mathfrak{gl}_N$, shifted by $[1]$. The ghost-number-zero components are exactly

$$A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N, \qquad \phi_1, \phi_2 \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N,$$

and they appear as

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2.$$

The Berezin integral kills every monomial except the coefficient of $\varepsilon_{12}$, so every term in the ghost-zero action is obtained by multiplying the displayed three summands and keeping exactly one real one-form and exactly the product $\varepsilon_1\varepsilon_2$.

*Remark* 3.4. The shift $[1]$ acts on the total cohomological degree: a form of de Rham degree $p$ and Grassmann degree $q$ has total degree $p + q$, which after the shift sits in ghost number $p + q - 1$. In particular, the 1-form component of $\alpha$ (de Rham degree 1, Grassmann degree 0) sits in ghost number 0 and is the bosonic gauge field. The scalar components proportional to $\varepsilon_i$ (de Rham degree 0, Grassmann degree 1) also sit in ghost number 0 and represent the motion of the brane in the two normal directions.

#### 3.3.1 GHOST-ZERO OPEN FIELD CONTENT

The ghost-zero truncation $\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2$ of 3.3 is the physical sector. Reducing the Chern-Simons action to it produces the first-order matrix mechanics governing the brane:

Lemma 3.5. With the Berezin convention $\int_{\mathbb{C}^{0|2}} \varepsilon_1\varepsilon_2 = 1$, the restriction of the BV action to $\alpha_0$ is, up to a boundary term on $\mathbb{R}$,

$$S(\alpha_0) = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

*Proof.* Only monomials containing one real one-form and the factor $\varepsilon_1\varepsilon_2$ survive integration over $\mathbb{R} \times \mathbb{C}^{0|2}$. In the quadratic term this leaves

$$\frac{1}{2}\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 - \phi_2\, d\phi_1\big),$$

with the sign produced by interchanging the two odd generators. The identity $d\,\mathrm{Tr}(\phi_1\phi_2) = \mathrm{Tr}(d\phi_1\,\phi_2 + \phi_1\, d\phi_2)$ rewrites the second summand as $\mathrm{Tr}(\phi_1\, d\phi_2)$ modulo an exact one-form. In the cubic term the six orderings of $A$, $\phi_1\varepsilon_1$, $\phi_2\varepsilon_2$ reduce, by cyclicity of the trace and $\varepsilon_2\varepsilon_1 = -\varepsilon_1\varepsilon_2$, to

$$3\,\mathrm{Tr}(A\phi_1\phi_2 - A\phi_2\phi_1).$$

The coefficient $\frac{1}{3}$ in the Chern-Simons action therefore produces $\mathrm{Tr}(A[\phi_1, \phi_2])$. □

Lemma 3.6 *(Dirac reduction of the probe phase space).* Let

$$\mathcal{P}_N = \mathfrak{gl}_N \oplus \mathfrak{gl}_N$$

with coordinates $(\phi_1, \phi_2)$ and symplectic form

$$\Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2).$$

With the convention $\iota_{X_f}\Omega_N = -df$, simultaneous conjugation by $\mathfrak{gl}_N$ is Hamiltonian with moment map components

$$\mu_\epsilon = -\mathrm{Tr}\big(\epsilon[\phi_1, \phi_2]\big), \qquad \epsilon \in \mathfrak{gl}_N.$$

They satisfy the first-class identity

$$\{\mu_\epsilon, \mu_\eta\} = \mu_{[\epsilon,\eta]}.$$

The action of Lemma 3.5 is the corresponding Dirac first-order action

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2) - \mu_A,$$

and varying $A$ imposes the constrained phase space

$$\mu^{-1}(0)/GL_N = \{[\phi_1, \phi_2] = 0\}/GL_N.$$

*Proof.* The symplectic form gives

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l\delta^k_j.$$

For $v_\epsilon(\phi_i) = [\epsilon, \phi_i]$, cyclicity of the trace gives

$$d\mu_\epsilon(\delta\phi_1, \delta\phi_2) = -\mathrm{Tr}\big(\epsilon([\delta\phi_1, \phi_2] + [\phi_1, \delta\phi_2])\big) = -\Omega_N(v_\epsilon, (\delta\phi_1, \delta\phi_2)).$$

Thus $X_{\mu_\epsilon} = v_\epsilon$. The moment-map bracket is therefore the infinitesimal equivariance identity $\{\mu_\epsilon, \mu_\eta\} = \mu_{[\epsilon,\eta]}$. Substituting $\epsilon = A(t)$ gives

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2) - \mu_A = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big),$$

and the $A$-equation is $[\phi_1, \phi_2] = 0$. □

PROPOSITION 3.7 *(Classical BV truncation).* The subspace of ghost-number-zero fields

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2$$

with all nonzero ghost-number components set to zero is a consistent classical truncation of the open BV theory. The induced classical action is the first-order gauged matrix model of Lemma 3.5.

*Proof.* Decompose the BV field as $\alpha = \sum_r \alpha^{(r)}$ by ghost number. The differential $d_{\mathbb{R}}$ and the product on $\Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N$ preserve total degree, and the BV action has ghost number zero. Hence every monomial in the Euler-Lagrange expression for a component of ghost number $r \neq 0$ contains at least one field component of nonzero ghost number when evaluated at $\alpha = \alpha_0$. Setting all such components to zero therefore makes the discarded equations vanish. The remaining equation is exactly the variation of $S(\alpha_0)$, computed in Lemma 3.5.

Concretely, the full Euler-Lagrange expression is the curvature

$$F(\alpha) = d\alpha + \alpha^2.$$

For $\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2$ on a line,

$$F(\alpha_0) = (d\phi_1 + [A, \phi_1])\varepsilon_1 + (d\phi_2 + [A, \phi_2])\varepsilon_2 + [\phi_1, \phi_2]\varepsilon_1\varepsilon_2.$$

The BV pairing keeps exactly real-form degree one and $\varepsilon_1\varepsilon_2$. Hence the displayed three components pair exactly with the variations of $A, \phi_2, \phi_1$ respectively, while variations of nonzero ghost number see complementary curvature components that are zero for $\alpha_0$ by real-form degree or $\varepsilon$-degree. This is a physical ghost-zero truncation; the residual ordinary $GL_N$ gauge symmetry remains. □

The truncated action is invariant under

$$\delta_\varepsilon\phi_i = [\varepsilon, \phi_i], \qquad \delta_\varepsilon A = d\varepsilon + [A, \varepsilon], \qquad \varepsilon \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N :$$

the $d\varepsilon$ variation of $\mathrm{Tr}(\phi_1\, d\phi_2)$ produces $-\mathrm{Tr}(d\varepsilon[\phi_1, \phi_2])$, which cancels the $d\varepsilon$ variation of $\mathrm{Tr}(A[\phi_1, \phi_2])$; the remaining terms cancel by invariance of the trace pairing.

*Remark* 3.8. The same local open model is obtained from the Calabi-Yau 2 point-brane residue calculation by tensoring the skyscraper object with $\mathbb{C}^N$ and passing to endomorphisms. This is a local $\mathrm{CY}_2$ pairing statement, not a global CY-to-chiral input. Global transfer would require the corresponding negative cyclic Calabi-Yau class, framing data, and descent; none of that is used in the proof below.

*Remark* 3.9. In AKSZ language this is the one-dimensional gauged sigma model with Hamiltonian target $\mathfrak{gl}_N \oplus \mathfrak{gl}_N$ and symplectic potential $\mathrm{Tr}(\phi_1\, d\phi_2)$. We use the resulting Poisson bivector

$$\Pi = \sum_{i,j} \frac{\partial}{\partial(\phi_1)^i{}_j} \wedge \frac{\partial}{\partial(\phi_2)^j{}_i},$$

so $\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l\delta^k_j$. Equivalently, with the convention $\iota_{X_f}\omega = -df$ used throughout, the symplectic form is $\mathrm{Tr}(d\phi_1 \wedge d\phi_2)$. For $\epsilon \in \mathfrak{gl}_N$, the Hamiltonian

$$\mu_\epsilon = -\mathrm{Tr}\big(\epsilon[\phi_1, \phi_2]\big)$$

generates $\delta_\epsilon\phi_i = [\epsilon, \phi_i]$. Its zero locus is the commuting-pair equation $[\phi_1, \phi_2] = 0$, and the classical reduced phase space is the stack of commuting pairs of matrices modulo conjugation.

*Remark* 3.10. Equivalently, take Chern-Simons theory on $\mathbb{R} \times T^2$ and restrict to the harmonic forms $1, \eta_1, \eta_2, \eta_1\eta_2$ on $T^2$. Writing $A_{3d} = A_t\, dt + \phi_1\eta_1 + \phi_2\eta_2$ and integrating over $T^2$ gives the same action; the odd variables $\varepsilon_i$ represent the degree-one cohomology classes $\eta_i$.

### 3.4 *Closed mixed Hamiltonian sector*

The closed field is the infinitesimal geometry of the formal normal surface. Because the surface is holomorphic symplectic, an infinitesimal canonical transformation is a Hamiltonian function modulo constants. The BF partner is the cotangent variable which makes the theory local and BV. This is the closed counterpart of the open first-order form: both sides are governed by the same symplectic normal plane.

The closed sector used here is a mixed field theory, not a tensor product of two independent closed-string state spaces. Its fields have a topological de Rham factor along $\mathbb{R}^2$ and a holomorphic Dolbeault factor along $\mathbb{C}^2$:

$$\Omega^*(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,*}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big).$$

The coefficient $\widehat{\mathfrak{h}}$ is the completed Lie algebra of nonconstant Hamiltonian functions on the formal holomorphic symplectic disk. The continuous dual copy $\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]$ is the cotangent/BF field. Thus the closed theory used here is the local Hamiltonian Kodaira-Spencer/BF analogue in the mixed holomorphic-topological model.

*Remark* 3.11. The original BCOV analysis [1] is written for Calabi-Yau threefolds; the higher-genus B-model on Calabi–Yau manifolds is taken up at the quantum level in [13]. On a holomorphic symplectic surface the natural local cotangent theory is the BF theory for the dg Lie algebra of Hamiltonian vector fields. The holomorphic symplectic form supplies the Poisson bracket, while the holomorphic volume form supplies the divergence operator on polyvectors.

We consider the space of polyvector fields

$$\mathbf{PV}^{*,*}(\mathbb{C}^2) \cong \Omega^{0,*}(\mathbb{C}^2)[\partial_{z_1}, \partial_{z_2}]$$

where $\partial_{z_i}$ are odd generators of poly-vector degree 1 (corresponding to the two holomorphic vector fields on $\mathbb{C}^2$). The operator

$$\partial : \mathbf{PV}^{i,*} \to \mathbf{PV}^{i-1,*}$$

is the divergence operator determined by the holomorphic volume form $\Omega = dz_1 \wedge dz_2$:

$$\partial_\Omega = \iota_\Omega^{-1} \circ \partial_{\mathrm{hol}} \circ \iota_\Omega, \qquad \iota_\Omega : \mathbf{PV}^{i,*} \xrightarrow{\sim} \Omega^{2-i,*}.$$

Thus for $X = a_1\partial_{z_1} + a_2\partial_{z_2}$,

$$\partial_\Omega X = (\partial_{z_1}a_1 + \partial_{z_2}a_2),$$

while $\partial_\Omega$ vanishes on functions.

We work in the formal polynomial local model at $0 \in \mathbb{C}^2$, so Dolbeault cohomology is represented by holomorphic Taylor polynomials. Here $\partial_\Omega$ is the divergence constraint selecting the Hamiltonian BF sector. The kinematic differential of the closed theory below is $D = d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}$. Thus the Poincare lemma identifies divergence-free vector fields with Hamiltonian potentials. The quotient by $\partial_\Omega$-exact Hamiltonians is absent. Analysing degree by degree:

- $\mathbf{PV}^{0,*}(\mathbb{C}^2) = \Omega^{0,*}(\mathbb{C}^2) \subset \mathbf{Ker}\,\partial$ trivially.
- $\mathbf{PV}^{1,*}(\mathbb{C}^2) \cong \Omega^{1,*}(\mathbb{C}^2)$ via $\iota_{(-)}\omega$. Under this identification, with the Hamiltonian convention $\iota_{X_f}\omega = -df$ used below, $\mathbf{Ker}\,\partial_\Omega$ is the space of closed holomorphic one-forms. Lemma 3.12 identifies these with $df$, uniquely modulo constants.

- The constant Hamiltonians act trivially, since their Hamiltonian vector fields vanish. We therefore use the reduced polynomial algebra
$$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$$
for nontrivial Hamiltonian fields.

LEMMA 3.12 *(Polynomial Poincare primitive).* Let
$$\theta = a(z_1, z_2)\, dz_1 + b(z_1, z_2)\, dz_2$$
be a polynomial one-form on $\mathbb{C}^2$. If $\partial_{z_2} a = \partial_{z_1} b$, then there is a polynomial $f \in \mathbb{C}[z_1, z_2]$, unique modulo constants, such that $\theta = df$.

*Proof.* Define
$$f(z_1, z_2) = \int_0^1 \big(z_1 a(tz_1, tz_2) + z_2 b(tz_1, tz_2)\big)\, dt.$$
Since $a$ and $b$ are polynomials, the integral is termwise division of each homogeneous term by its degree plus one, hence $f$ is polynomial. Differentiate under the integral sign. Closedness says $\partial_{z_2} a = \partial_{z_1} b$, so
$$\begin{aligned}
\partial_{z_1} f &= \int_0^1 \big(a(tz) + tz_1 \partial_{z_1} a(tz) + tz_2 \partial_{z_1} b(tz)\big)\, dt \\
&= \int_0^1 \big(a(tz) + tz_1 \partial_{z_1} a(tz) + tz_2 \partial_{z_2} a(tz)\big)\, dt \\
&= \int_0^1 \frac{d}{dt}\big(t\, a(tz)\big)\, dt = a(z).
\end{aligned}$$
The same calculation, with $a$ and $b$ interchanged, gives $\partial_{z_2} f = b(z)$. Therefore $df = \theta$. If $df = dg$, then $d(f - g) = 0$, hence $f - g$ is constant. □

PROPOSITION 3.13 *(Hamiltonian polyvector reduction).* In the formal polynomial model on $(\mathbb{C}^2, \omega = dz_1 \wedge dz_2)$, the divergence-closed polyvector-one fields reduce to Hamiltonian vector fields:
$$\mathbf{Ker}(\partial_\Omega : \mathbf{PV}^1 \to \mathbf{PV}^0) \cong \{df : f \in \mathbb{C}[z_1, z_2]\} \cong \bar{A}.$$
Under the convention
$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad \iota_{X_f}\omega = -df,$$
the commutator of vector fields is
$$[X_f, X_g] = X_{\{f,g\}_{\bar{A}}}.$$

*Proof.* Let $X = a_1 \partial_{z_1} + a_2 \partial_{z_2}$. The divergence operator is $\partial_\Omega X = \partial_{z_1} a_1 + \partial_{z_2} a_2$. Contracting with $\omega$ gives
$$\iota_X \omega = a_1\, dz_2 - a_2\, dz_1.$$
The equation $\partial_\Omega X = 0$ is exactly the closedness of this one-form:
$$d(\iota_X \omega) = (\partial_{z_1} a_1 + \partial_{z_2} a_2)\, dz_1 \wedge dz_2 = 0.$$

By Lemma 3.12, there is a polynomial $f$, unique modulo constants, such that $\iota_X \omega = -df$. Therefore

$$a_1 = -\partial_{z_2} f, \qquad a_2 = \partial_{z_1} f,$$

so $X = X_f$. Constants are exactly the kernel of $f \mapsto X_f$, giving $\bar{A}$.

Finally, for any polynomial $h$,

$$\begin{aligned}[X_f, X_g](h) &= \{f, \{g, h\}\} - \{g, \{f, h\}\} \\ &= \{\{f, g\}, h\}\end{aligned}$$

by the Jacobi identity for the constant Poisson bracket. Thus $[X_f, X_g] = X_{\{f,g\}}$, and projecting constants gives the bracket on $\bar{A}$. □

The Dolbeault-valued upgrade is immediate. The Hamiltonian sector of the closed mixed field theory replaces the divergence kernel $\mathbf{Ker}\,\partial_\Omega$ on poly-1-vectors with its Dolbeault enhancement: a Hamiltonian potential $f \in \Omega^{0,*}(\mathbb{C}^2) \widehat{\otimes} \bar{A}$ generates the Hamiltonian vector field

$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad \iota_{X_f}\omega = -df, \qquad \{f, g\}_\omega = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g.$$

Constants drop because $X_f = 0$ if and only if $f$ is constant; this is 3.13 read on Dolbeault-valued coefficients.

## 3.5 *Hamiltonian BF action*

### 3.5.1 The action functional

The closed BF action for the dg Lie algebra $\widehat{\mathfrak{h}}$ of formal Hamiltonians, with kinematic differential $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$, has fields

$$\alpha \in \Omega^*(\mathbb{R}^2) \widehat{\otimes} \Omega^{0,*}(\mathbb{C}^2) \widehat{\otimes} \widehat{\mathfrak{h}}[1], \qquad \beta \in \Omega^*(\mathbb{R}^2) \widehat{\otimes} \Omega^{0,*}(\mathbb{C}^2) \widehat{\otimes} \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2],$$

the Hamiltonian connection $\alpha$ and the cotangent partner $\beta$. The form-valued Hamiltonian bracket wedges the form factors and takes the Poisson bracket on the Hamiltonian factors with the tensor-product Koszul sign,

$$[\eta \otimes f, \theta \otimes g] = (-1)^{|f||\theta|}(\eta \wedge \theta) \otimes \{f, g\},$$

which equals +1 on the ghost-zero (even-coefficient) sector used below. The action is the BF Hamiltonian

$$S = \int_{\mathbb{R}^2 \times \mathbb{C}^2} \left\langle \beta,\ D\alpha + \tfrac{1}{2}\{\alpha, \alpha\} \right\rangle dz_1\, dz_2,$$

with the pairing the continuous evaluation $\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}} \otimes \widehat{\mathfrak{h}} \to \mathbb{C}$; the dense polynomial submodel pairs $\bar{A}^\vee$ with $\bar{A}$ in each finite degree. On ghost-zero fields, $\alpha$ has total real/Dolbeault form degree 1 and $\beta$ has total degree 2, so $\beta \wedge D\alpha$ and $\beta \wedge \{\alpha, \alpha\}$ have total degree 4; wedge with $dz_1 \wedge dz_2$ produces a top form on $\mathbb{R}^2 \times \mathbb{C}^2$.

*Remark* 3.14. The open mixed brane theory lives on the brane line; the closed Hamiltonian BF theory lives on the full mixed space-time $\mathbb{R}^2 \times \mathbb{C}^2$.

### 3.5.2 Classical field content

The BF action exhibits $\beta$ as a Lagrange multiplier whose equation of motion enforces flatness of the Hamiltonian connection,

$$F_\alpha = D\alpha + \tfrac{1}{2}\{\alpha, \alpha\} = 0,$$

where $\alpha$ is read as a connection on a bundle of holomorphic-symplectic surfaces over $\mathbb{R}^2$.

Consider the space-time $\mathbb{R}^2_{x_i} \times \mathbb{C}^2_{z_i}$. The ghost-number-zero components of $\alpha$ and $\beta$ are:

$$\alpha = \alpha_{x_i}\, dx_i + \alpha_{\bar z_j}\, d\bar z_j, \qquad \beta = \beta_{x_1 x_2}\, dx_1\, dx_2 + \beta_{x_i \bar z_j}\, dx_i\, d\bar z_j + \beta_{\bar z_1 \bar z_2}\, d\bar z_1\, d\bar z_2.$$

Here the coefficients of $\alpha$ are Hamiltonian functions modulo constants. The gauge symmetries are the BF-type transformations

$$\delta_\varepsilon \alpha = D\varepsilon + \{\alpha, \varepsilon\}, \qquad \delta_\varepsilon \beta = \mathrm{ad}^\vee_\varepsilon\, \beta,$$

and

$$\delta_\lambda \alpha = 0, \qquad \delta_\lambda \beta = D_\alpha \lambda,$$

where $D_\alpha = D + \{\alpha, -\}$ and $\lambda$ is a one-form valued in $\bar A^\vee$. The second symmetry is reducible on shell: if $\lambda = D_\alpha \mu$, then $D_\alpha \lambda = D_\alpha^2 \mu = \{F_\alpha, \mu\}$, which vanishes when $F_\alpha = 0$.

Analysing $\alpha$ as a Hamiltonian function on $\mathbb{C}^2$: the component $\alpha_{\bar z_j} d\bar z_j$ is the Hamiltonian potential whose vector field $X_{\alpha_{\bar z_j}} d\bar z_j$ is the Beltrami differential encoding a deformation of the complex structure of $\mathbb{C}^2$; the component $\alpha_{x_i} dx_i$ similarly defines a connection on $\mathbb{R}^2$ valued in Hamiltonian vector fields on $\mathbb{C}^2$, encoding a bundle of $\mathbb{C}^2$s over $\mathbb{R}^2$. Together, the equation of motion $F_\alpha = 0$ says this data constitutes an integrable deformation of complex structure with flat holonomy. The field $\beta$ appears linearly in the action and enforces exactly this equation.

*Remark* 3.15. This is the BF specialization to the dg Lie algebra of formal Hamiltonians: the multiplier $\beta$ enforces $F_\alpha = 0$, and the classical phase space is $T^*\mathcal{M}_{\mathrm{flat}}(\widehat{\mathfrak{h}})$.

### 3.5.3 Phase space

For an analytic model adequate to propagators, compactify one real direction to obtain $\mathbb{R} \times S^1 \times Y_{\mathrm{hol}}$ with $Y_{\mathrm{hol}}$ holomorphic symplectic, completed at the trivial solution: a compactified avatar of the formal local model, not a renaming of $X = M \times Y$. The equations of motion $F_\alpha = 0$ cut out the Maurer-Cartan problem for Hamiltonian vector fields on $S^1 \times Y_{\mathrm{hol}}$, with the analytic package of 3.183 fixed in this compact or formal setting (elliptic complex, brane-defect or boundary condition, gauge fixing, propagator, renormalization scheme, anomaly cancellation). The perturbative phase space is the derived cotangent stack

$$T^*_{\mathrm{der}}\mathrm{MC}\big(\Omega^*(S^1)\widehat{\otimes}\Omega^{0,*}(Y_{\mathrm{hol}}, \mathrm{Ham}(Y_{\mathrm{hol}})), D, \{-,-\}\big),$$

collapsing on the underived tangent at the trivial solution to the ordinary cotangent of the linearized deformation complex.

## *3.6 Closed-open brane coupling*

The brane sees a closed Hamiltonian by evaluating it on its matrix-valued normal coordinates and tracing over Chan-Paton indices. For a brane along $x_1$ and a closed field $dx_1 \otimes z_1^k z_2^l$, this gives the local boundary functional

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big) + \int_{\mathbb{R}} dx_1\, \mathrm{Tr}\, \phi_1^k \phi_2^l$$

inside the formal local algebraic model: the closed Hamiltonian is pulled back along the matrix-valued normal-coordinate map $z_i \mapsto \phi_i$ and traced. 3.123 is the algebraic form of this bulk-to-boundary prescription; 3.161 below proves that the resulting Hamiltonian evaluation preserves the classical bracket. This is the degree-zero Hamiltonian component of the closed-open map; the full Poisson factorization-center morphism is left to 3.47.

*Construction* 3.16 *(Reduced line-defect current kernel).* Let

$$X = \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}, \qquad L = \mathbb{R}_{x_1} \times \{x_2 = 0\} \times \{p\} \subset X,$$

and let $i : L \hookrightarrow X$ be the inclusion. This is a line defect, not a spacetime boundary. In the reduced formal local model, a test density $a(t)\,dt$ on $L$ defines the defect-supported current $i_*(a(t)\,dt)$, locally written as $a(x_1)\delta(x_2)\delta_p\,dx_1dx_2$ together with the formal holomorphic delta-current at $p$.

The degree-zero Hamiltonian part of the reduced PBW/Rees defect-current vertex is

$$K_L^{\mathrm{cl}}(a \otimes f) = \int_L a(t)\,\overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\,dt, \qquad f \in \bar{A}.$$

For the cotangent principal-part source, the reduced vertex sends

$$B \otimes \rho[1] \longmapsto \Theta_\rho(B), \qquad B \in \mathcal{D}'_c(L), \quad \rho \in \mathcal{P},$$

using Theorem 3.157. These formulas define a reduced PBW/Rees defect-current kernel. They are not a cochain-level map from closed CE observables, and they are not yet the quantum Costello brane-defect kernel: the Ward identity, anomaly cancellation, finite-scale propagator, and quantum moment-map compatibility remain separate obligations.

*Remark* 3.17. Geometrically, specifying a closed Hamiltonian field on $\mathbb{C}^2$ amounts to equipping $\mathbb{C}^2$ with a connection, so that the brane can parallel-transport its internal degrees of freedom as it moves along $\mathbb{R}$.

*Construction* 3.18 *(Interval Hamiltonian and factorization algebras).* For an interval $I \subset \mathbb{R}$, define

$$\begin{aligned} \mathfrak{h}_I &= \Omega^\bullet_c(I)\widehat{\otimes}\bar{A}, \\ \mathcal{P}_I &= \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}, \\ \mathfrak{g}_I &= \mathfrak{h}_I \ltimes \mathcal{P}_I[1], \end{aligned}$$

with brackets

$$\begin{aligned} [a \otimes f, b \otimes g]_{\mathfrak{h}_I} &= ab \otimes \{f, g\}_{\bar{A}}, \\ [a \otimes f, B \otimes \rho[1]]_{\mathfrak{g}_I} &= aB \otimes (f \cdot \rho)[1], \\ [B \otimes \rho[1], C \otimes \sigma[1]]_{\mathfrak{g}_I} &= 0, \end{aligned}$$

where products in $\Omega^\bullet_c(I)$ are de Rham products and, in the displayed degree-zero formulas, $aB$ is multiplication of a compactly supported current by a smooth compactly supported function and $f \cdot \rho = \pi_{\mathrm{pp}}\{f, \rho\}$ is the principal-part coadjoint action of 3.147. Inclusions $I' \subset I$ act by extension by zero on compactly supported de Rham forms and currents; disjoint unions act by sum. This makes $I \mapsto \mathfrak{h}_I, \mathfrak{g}_I$ cosheaves of (graded) Lie algebras on $\mathbb{R}$.

Define the boundary $P_0$ factorization algebras on $\mathbb{R}$ in the formal stable Hamiltonian sector by

$$\begin{aligned} A_\partial^{\mathrm{Ham}}(I) &= \widehat{\mathbf{S}}(\mathfrak{h}_I), \\ A_\partial^{\mathrm{pp}}(I) &= \widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{P}_I[1]), \end{aligned}$$

with $P_0$ bracket extended as a biderivation from

$$\begin{aligned} \{a \otimes f, b \otimes g\}_{A_\partial} &= ab \otimes \{f, g\}_{\bar{A}}, \\ \{a \otimes f, B \otimes \rho[1]\}_{A_\partial} &= aB \otimes (f \cdot \rho)[1], \\ \{B \otimes \rho[1], C \otimes \sigma[1]\}_{A_\partial} &= 0. \end{aligned}$$

The factorization product on disjoint intervals is the multiplication in $\widehat{\mathbf{S}}$ after extension by zero. These are the reduced targets after de Rham-current contraction for the closed-open factorization-center map. The Hamiltonian factor $A_\partial^{\mathrm{Ham}}$ is the connected component of the formal Hamiltonian sector of the open brane factorization algebra; the principal-part factor $A_\partial^{\mathrm{pp}}/A_\partial^{\mathrm{Ham}}$ is the reduced current object of 3.157.

PROPOSITION 3.19 *(Locality of the brane $P_0$ bracket).* Let $a \in \Omega_c^0(I_1), b \in \Omega_c^0(I_2)$ with $I_1 \cap I_2 = \emptyset$. For all $f, g \in \bar{A}$,

$$\{B_f(a), B_g(b)\}_{A_\partial^{\mathrm{Ham}}} = 0.$$

In particular, $A_\partial^{\mathrm{Ham}}$ has the strict factorization product property: the bracket between sections supported on disjoint intervals vanishes.

*Proof.* By 3.18, $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$. The pointwise product $ab \equiv 0$ since the supports are disjoint, so $B_{\{f,g\}}(ab) = 0$. □

*Remark* 3.20 *(Locality from the canonical equal-time bracket).* The locality input is the canonical equal-time bracket of the first-order open BV theory:

$$\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(s)\} = \delta^i_l \delta^k_j\, \delta(t-s), \qquad \{(\phi_a)^i{}_j(t), (\phi_a)^k{}_l(s)\} = 0 \ \ (a = 1, 2).$$

Tracing $f(\phi_1, \phi_2)(t)$ and $g(\phi_1, \phi_2)(s)$, applying the Leibniz rule, and integrating against $a(t)b(s)\delta(t-s)$ produces exactly the smeared identity $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$. This is the unsmeared content packaged in 3.162; the delta-function locality is what produces a strict $P_0$ factorization algebra rather than a homotopical one.

THEOREM 3.21 *(Open-closed reduced Hamiltonian central-operation equivalence, Hamiltonian sector).* Let $\mathrm{Obs}^{\mathrm{cl,red}}_{\mathrm{open},N}$ be the locally constant reduced open-brane factorization algebra of 3.24. Let $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H,\mathrm{Ham}}|_L$ denote the formal classical Hamiltonian-mode subfactorization algebra of the cotangent BF theory on the formal neighbourhood of $L$ obtained by restricting to the connected Hamiltonian central-operation sector. Define the interval-wise CE cochain factorization algebra by

$$\mathcal{F}_{\mathrm{cl}}(I) = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_I), \qquad \mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1],$$

with structure maps induced from extension by zero on $\Omega_c^\bullet$ and on compactly supported currents, cochain by cochain. Then:

(i) there is a continuous interval-wise morphism of $P_0$ factorization algebras

$$\Phi^{\mathrm{Ham}}_{\mathrm{fact}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H,\mathrm{Ham}}\big|_L \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{red},\mathrm{Ham}}\big(A^{\mathrm{Ham}}_{\partial}\big),$$

whose underlying CE cochain map on each interval $I$ is given on the shifted-cotangent length-one CE coordinates by

$$u_{a(t)dt\otimes f} \longmapsto B_f(a) := \int_I a(t)\,\overline{\mathrm{Tr}\, f(\phi_1(t),\phi_2(t))}\,dt;$$

(ii) this morphism is a quasi-isomorphism in each interval and is compatible with extension by zero and disjoint factorization products;

(iii) the induced equivalence is locally constant in the topological direction $\mathbb{R}$ in the sense of Lurie, *Higher Algebra*, §5.5; Costello–Gwilliam [15, Vol. I §6.4] supplies the published factorization-algebra terminology. Every inclusion $I' \hookrightarrow I$ of nonempty open intervals induces a quasi-isomorphism of the structure maps;

(iv) the factorization product $A^{\mathrm{Ham}}_{\partial}(I_1) \otimes A^{\mathrm{Ham}}_{\partial}(I_2) \to A^{\mathrm{Ham}}_{\partial}(V)$ for disjoint $I_1 \sqcup I_2 \subset V$ corresponds under $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$ to the CE coproduct on the closed side induced by extension by zero $\Omega^{\bullet}_c(I_j) \hookrightarrow \Omega^{\bullet}_c(V)$;

(v) at the stalk $I \to \{t_0\}$, $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$ restricts to the stalkwise central operation

$$f \longmapsto L_{B_f} \in \mathrm{Hoch}^0\big(A^{\mathrm{pt}}_N\big)$$

of 3.130.

*Proof. Step 1.* Identify the interval-wise central factorization algebra. In the formal stable Hamiltonian sector, the boundary factorization algebra $A^{\mathrm{Ham}}_{\partial}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$ is graded-commutative, with strict $P_0$ bracket inherited from $\mathfrak{h}_I$ as a biderivation. By the Hochschild–Kostant–Rosenberg theorem in the smooth/algebraic setting, applied here to the free graded-commutative algebra on the Tate vector space $\mathfrak{h}_I$, the Hochschild cochain complex $\mathrm{Hoch}^{\bullet}\big(\widehat{\mathbf{S}}(\mathfrak{h}_I)\big)$ is quasi-isomorphic to the polyvector fields $\mathrm{PV}^{\bullet}\big(\widehat{\mathbf{S}}(\mathfrak{h}_I)\big) = \widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathfrak{h}^{\vee}_{I,\mathrm{cont}}[-1])$ with the Schouten bracket. The locally constant factorization-center stalk on the line records exactly the Hochschild cochains in the corresponding $E_1$ algebra; for graded-commutative algebras this collapses to PV. Restricting to the polynomial subalgebra $\mathbf{S}(\mathfrak{h}_I) \subset \widehat{\mathbf{S}}(\mathfrak{h}_I)$ and to the constant polyvectors gives the central multiplication operations, which are recorded as the length-one shifted-cotangent CE coordinates $u_{a(t)dt\otimes f}$. The CE coordinates $c$ dual to Hamiltonian generators map instead to constant polyvectors. This is the interval version of the generator dictionary $c^I \mapsto \theta^I$, $u_I \mapsto O_I$ in 3.27.

*Step 2.* Construct the CE source. By 3.137, applied to $\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1]$ in the stated completed category,

$$C^{\bullet}_{\mathrm{CE},\mathrm{cont}}(\mathfrak{g}_I) = \widehat{\mathbf{S}}\big(\mathfrak{g}^{\vee}_{I,\mathrm{cont}}[-1]\big),$$

and its CE differential is the dual of the semidirect Lie bracket. The Hamiltonian part is controlled by the de Rham-current pairing on the $\Omega^{\bullet}_c(I)$ factor and the residue-Taylor pairing on $\bar{A}$. In degree-zero notation this is the familiar pairing between compactly supported test functions and compactly supported currents; the polynomial subspace $\bar{A} \subset \widehat{\bar{A}}$ pairs with the discrete Taylor dual generated by the $\delta_{a,b}$. The shifted-cotangent CE coordinate $u_{a(t)dt\otimes f}$ is the coordinate dual to the cotangent generator determined by the current $a(t)dt \otimes f$.

*Step 3.* Define $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}(I)$ on generators. By 3.135, the closed Hamiltonian cotangent BF observables on the formal neighbourhood of $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ in the Hamiltonian sector are $\widehat{\mathbf{S}}(\mathfrak{h}^{\vee}_{I,\mathrm{cont}}[-1])$, with CE differential dual to the Hamiltonian Lie bracket on $\mathfrak{g}_I$. The interval-wise map sends a length-one shifted-cotangent CE coordinate $u_{a(t)dt\otimes f} \in C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_I)$ to the element of the central operations factorization algebra given by the multiplication operator

$$L_{B_f(a)} \in \mathrm{Hoch}^0\big(A^{\mathrm{Ham}}_{\partial}(I)\big) \subset \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}\big(A^{\mathrm{Ham}}_{\partial}\big)(I),$$

with $B_f(a)$ as displayed. By 3.130, this operator is closed (Hochschild 0-cocycle on the graded-commutative open stalk). Extend $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}(I)$ as a continuous algebra morphism on $\widehat{\mathbf{S}}$.

*Step 4.* Verify CE differential compatibility. On length two, the CE differential of a shifted-cotangent generator $u_{a(t)dt\otimes f}$ is the coadjoint term dual to the Lie bracket on $\mathfrak{h}_I$. By 3.162 (its sharpened form below),

$$\{L_{B_f(a)}, L_{B_g(b)}\}_{A^{\mathrm{Ham}}_{\partial}} = L_{B_{\{f,g\}}(ab)}.$$

Pulling back through $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$ matches this with the shifted-cotangent CE differential dual to $[a \otimes f, b \otimes g]_{\mathfrak{h}_I} = ab \otimes \{f, g\}$. Inductive extension by length and the biderivation property propagates this to all CE lengths.

*Step 5.* Quasi-isomorphism on each interval. Both sides are free graded-commutative algebras on the same continuous Tate vector space after the generator dictionary $c \mapsto \theta, u \mapsto O$, with the same CE differential induced from the semidirect Lie bracket of $\mathfrak{g}_I$ (closed side: by definition; brane side: through the HKR identification of Step 1 and the Schouten bracket). The map respects the symmetric-algebra generators by construction. Thus it is an isomorphism of underlying graded vector spaces and intertwines the differentials. The map is therefore an isomorphism of $P_0$-cdg algebras on each interval.

*Step 6.* Factorization product compatibility. For disjoint $I_1, I_2 \subset V$, the CE source extension by zero gives $\mathfrak{h}_{I_1} \oplus \mathfrak{h}_{I_2} \hookrightarrow \mathfrak{h}_V$. The cobracket on the closed side is the dual of this: it is the splitting $\widehat{\mathbf{S}}(\mathfrak{g}^{\vee}_V[-1]) \to \widehat{\mathbf{S}}(\mathfrak{g}^{\vee}_{I_1}[-1])\widehat{\otimes}\widehat{\mathbf{S}}(\mathfrak{g}^{\vee}_{I_2}[-1])$ obtained from the dual surjection $\mathfrak{g}^{\vee}_V \twoheadrightarrow \mathfrak{g}^{\vee}_{I_1} \oplus \mathfrak{g}^{\vee}_{I_2}$. On the brane side, $A^{\mathrm{Ham}}_{\partial}(I_1) \otimes A^{\mathrm{Ham}}_{\partial}(I_2) \to A^{\mathrm{Ham}}_{\partial}(V)$ is the multiplication map under the inclusion of $\mathfrak{h}_{I_1} \oplus \mathfrak{h}_{I_2} \hookrightarrow \mathfrak{h}_V$. By 3.19, the bracket between the two factors vanishes on disjoint supports, so the product is strictly multiplicative. The induced map between central-operation factorization algebras is the corresponding tensor product of multiplication operators. This is exactly the dual of the CE-source extension by zero. Therefore the diagram

$$\begin{array}{ccc}
\mathcal{F}_{\mathrm{cl}}(I_1) \otimes \mathcal{F}_{\mathrm{cl}}(I_2) & \xrightarrow{\Phi(I_1)\otimes\Phi(I_2)} & \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}(A^{\mathrm{Ham}}_{\partial})(I_1) \otimes \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}(A^{\mathrm{Ham}}_{\partial})(I_2) \\
\uparrow & & \downarrow \\
\mathcal{F}_{\mathrm{cl}}(V) & \xrightarrow{\Phi(V)} & \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}(A^{\mathrm{Ham}}_{\partial})(V)
\end{array}$$

commutes after vertical reversal of the left arrow (CE coproduct = dual of inclusion).

*Step 7.* Locally constant structure. For each interval $I$, the integration map $p_I : \Omega^{\bullet}_c(I) \to \mathbb{C}[-1]$, $p_I(\alpha) = \int_I \alpha$, is a quasi-isomorphism: $H^0_c(I;\mathbb{C}) = 0$ and $H^1_c(I;\mathbb{C}) \cong \mathbb{C}$. The maps $p_I$ are natural for extension by zero. Choosing a compactly supported density $\lambda_I(t)\,dt$ with $\int_I \lambda_I(t)\,dt = 1$ and a homotopy $K_I$ gives an intervalwise contraction

$$\Omega^{\bullet}_c(I) \underset{i_I}{\overset{p_I}{\rightleftarrows}} \mathbb{C}[-1], \qquad i_I(1) = \lambda_I(t)\,dt, \quad p_I(\alpha) = \int_I \alpha,$$

with $dK_I + K_I d = \mathrm{id} - i_I p_I$; this choice is not asserted to be strictly natural in $I$. The degree-zero notation $u_{a(t)dt\otimes f}$ used above already includes the shifted-cotangent degree, so the density representative appears as a degree-zero CE coordinate after the BV shift. Translation is homotopically trivial by the same de Rham contraction, hence the interval-wise objects are locally constant in the topological direction after this density contraction.

*Step 8.* Stalk recovery. Under the density contraction, a length-one shifted-cotangent CE coordinate represented by $a(t)dt \otimes f$ maps to the multiplication operator $L_{B_f(a)}$. Stalk recovery is obtained by any cofinal family of compactly supported unit densities supported near $t_0$. This represents the compact-support cohomology generator under the locally constant factorization-algebra quasi-isomorphism; it is not a literal degree-zero point evaluation. □

*Remark* 3.22 *(Locally constant equivalence and $E_1$-algebra translation).* In the locally constant setting on $\mathbb{R}$, factorization algebras are equivalent to homotopy associative algebras up to translation invariance by Lurie, *Higher Algebra*, §5.5.4, in the Costello–Gwilliam vocabulary of locally constant factorization algebras [15, Vol. I §6.4]. The Hochschild cochain complex of the $E_1$ algebra computes the factorization-center stalk. For graded-commutative $E_1$ algebras (which the open Hamiltonian stalk is, in the reduced model), the Hochschild cochain complex is quasi-isomorphic to polyvector fields by HKR, and its strict $P_0$ subalgebra of multiplication operators with constant differential is the polynomial Hamiltonian sector $\widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathfrak{h}_I^\vee[-1])$. The CE source identification of 3.21 is exactly the closed dual of this PV picture in the connected Hamiltonian sector.

*Remark* 3.23 *(Quillen equivalence of presentations).* The two presentations of locally constant factorization algebras cited above are Quillen equivalent: the model-categorical presentation of [16, Cor. A.5.0.5] matches the $\infty$-categorical presentation of [35, Thm. A.3.7.5], so the $E_1$ translation used here is convention-independent.

For a factorization algebra $\mathcal{A}$ on the line, write $\mathrm{Z}_{\mathrm{fact}}(\mathcal{A})$ for the factorization algebra of $E_1$ central operations. In the locally constant one-dimensional case its stalk is the Hochschild cochain complex of the corresponding $E_1$ algebra. The notation $\mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathcal{A})$ means the same central operations with the additional compatibility homotopies for the classical Poisson/BV bracket. This is the local convention used below; no unrestricted Costello–Gwilliam central-operations theorem is imported into the reduced Hamiltonian proof. The proposition below records the degree-zero current shadow as a corollary of 3.21.

Theorem 3.24 *(Hamiltonian-current central operation, smeared).* 3.21 restricted to the length-one CE component on each interval recovers the degree-zero Hamiltonian current shadow: the assignment

$$\Phi_{I,N}(a \otimes f)(u) = B_f(a) \cdot u = \Big(\int_I a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt\Big)u$$

is a Hochschild 0-cocycle in $\mathrm{Z}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl,red}}_{\mathrm{open},N})(I)$, the assignments are compatible with interval inclusions and disjoint factorization products, and the induced operations satisfy

$$\{\Phi(a \otimes f), \Phi(b \otimes g)\}_{\mathrm{open}} = \Phi(ab \otimes \{f, g\}_{\bar{A}}).$$

The degree-zero current shadow is the length-one component of the quasi-isomorphism: its content is the strict $P_0$ identity, one of the structure equations defining $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$ of 3.21.

*Proof.* In the reduced open model $Q\phi_1 = Q\phi_2 = 0$. Hence $\overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ is $Q$-closed for every Hamiltonian $f$, and the compactly supported integral is $Q$-closed. Since the reduced open observable algebra is

graded-commutative, multiplication by this closed element is a Hochschild 0-cocycle; this is the interval-wise version of 3.130. Extension by zero of the test function is compatible with restriction to a larger interval. For disjoint intervals 3.19 gives vanishing of the $P_0$ bracket between the two factors, so the factorization product of two such operations is the strict multiplication by $I_\partial(a,f)I_\partial(b,g)$; this is the multiplicative extension from $\mathfrak{h}_I$ to $A_\partial^{\mathrm{Ham}}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$, not the linear current bracket.

The Poisson statement is local on the line. For compactly supported test functions $a, b$, 3.162 (the strict $P_0$ identity on the brane factorization algebra) gives

$$\{I_\partial(a,f), I_\partial(b,g)\}_{\mathrm{open}} = I_\partial(ab, \{f,g\}_{\bar{A}}).$$

Passing to the connected stable Hamiltonian quotient removes only the scalar Hamiltonian line, which is central for the bracket. Thus the current bracket is preserved. Taking $a$ and $b$ supported in a small coordinate interval and then passing to the locally constant stalk recovers 3.161. □

PROPOSITION 3.25 *(CE-source obstruction for perfect Hamiltonian Lie algebras).* Let $\mathfrak{h}$ be a perfect Lie algebra in **TateVec** (i.e. $[\mathfrak{h},\mathfrak{h}] = \mathfrak{h}$). Then no length-one CE coordinate $c^I$ dual to a Hamiltonian generator $H_I \in \mathfrak{h}$ is closed under $d_{\mathrm{CE}}$:

$$d_{\mathrm{CE}} c^K = -\tfrac{1}{2} C^K_{IJ} c^I c^J \neq 0.$$

Equivalently, $H^1(\mathfrak{h};\mathbb{C}) = 0$.

This obstruction is the structural reason that the boundary observable $O_I = J(H_I) \in A_\partial$ cannot be the image under $\Phi$ of $c^I$. Instead, under 3.27,

$$O_I = \Phi(u_I),$$

the image of the CE coordinate dual to the shifted cotangent generator $\eta_{\partial_I} \in \mathfrak{h}^\vee[1]$. The boundary observable lives in the cotangent direction of the CE cochain algebra, not in the Hamiltonian direction. Its CE differential is not zero but $d_{\mathrm{CE}} u_K = C^L_{KJ} u_L c^J$, mapping under $\Phi$ to the Hamiltonian vector field $d_\pi O_K = C^L_{KJ} O_L \theta^J$ of $O_K$ in the Schouten algebra.

This is exactly what a $P_0$-algebra map should produce: a function whose Schouten differential is its Hamiltonian vector field.

*Proof.* $\mathfrak{h}$ perfect means $C^K_{IJ} c^I c^J \neq 0$ for some triple $(I,J,K)$ realizing a non-trivial bracket; $H^1$ vanishing follows. Identification $O_I = \Phi(u_I)$ is the content of 3.42 combined with $\Phi(u_I) = O_I$ from 3.27. Differentials match on generators by 3.139. □

*Remark* 3.26. The reduced Hamiltonian Lie algebra $\mathfrak{h} = \widehat{A}/\mathbb{C}\cdot 1 = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$ of the formal symplectic disk in dimension two is perfect: for every monomial one has, before quotienting constants, $z_1^a z_2^b = \left\{z_1, \frac{1}{b+1} z_1^a z_2^{b+1}\right\}$, so $[\mathfrak{h},\mathfrak{h}] = \mathfrak{h}$. In particular $[H_{1,0}, H_{m,n}] = n\, H_{m,n-1}$ and $[H_{0,1}, H_{m,n}] = -m\, H_{m-1,n}$. The hypothesis of 3.25 is met. The CE-length-one cochain $\xi_{a,b} = c^{a,b}$ dual to $H_{a,b} = z_1^a z_2^b$ is therefore not closed, and no closed CE cochain has linear Hamiltonian part $c^{a,b}$, recovering the earlier source-obstruction observation.

THEOREM 3.27 *(Open-closed derived center, cochain level).* Let $\mathfrak{h}$ be any pro-finite-dimensional Lie algebra in **TateVec** over $\mathbb{C}$, concentrated in degree zero, satisfying the Tate-conilpotency hypothesis of 3.137. Let $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ be the shifted-cotangent extension. Let $A_\partial = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})$ be the Tate-completed Kirillov–Poisson algebra of 3.139.

Then there is a canonical isomorphism of completed dg graded-commutative $P_0$-algebras

$$\Phi : C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{PV}(A_\partial),$$

determined on generators by

$$\Phi(c^I) = \theta^I, \qquad \Phi(u_I) = O_I,$$

where $c^I$ is the CE coordinate dual to $H_I \in \mathfrak{h}$ and $u_I$ is the CE coordinate dual to the shifted cotangent generator $\eta_{\partial_I} \in \mathfrak{h}^\vee[1]$.

Identifying $\mathrm{PV}(A_\partial)$ with the reduced Hamiltonian central-operation object of the constant factorization algebra $A_\partial$ promotes $\Phi$ to

$$\Phi : C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \xrightarrow{\sim} Z^{P_0}_{\mathrm{red,Ham}}(A_\partial),$$

the cochain-level open-closed reduced Hamiltonian central-operation isomorphism in the formal/local sector.

*Proof. Generators and free extension.* By 3.137,

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}_{\mathrm{Tate}}\big((\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1])^\vee_{\mathrm{cont}}[-1]\big) \cong \widehat{\mathbf{S}}_{\mathrm{Tate}}\big(\{u_I\}\big)\widehat{\otimes}\widehat{\Lambda}\big(\{c^I\}\big),$$

with $|u_I| = 0$ and $|c^I| = 1$. On the open side, $\mathrm{PV}(A_\partial) = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\{O_I\})\widehat{\otimes}\widehat{\Lambda}(\{\theta^I\})$, with $|O_I| = 0$ and $|\theta^I| = 1$. The same generator data on both sides freely generate completed graded-commutative algebras in **TateVec** of the same shape; the assignment $c^I \mapsto \theta^I$, $u_I \mapsto O_I$ extends uniquely to a continuous bijection of completed dg-commutative algebras.

*Cochain identity on generators.* The CE differential is

$$d_{\mathrm{CE}} c^K = -\tfrac{1}{2} C^K_{IJ} c^I c^J, \qquad d_{\mathrm{CE}} u_K = C^L_{KJ} u_L c^J.$$

By 3.139,

$$d_\pi \theta^K = -\tfrac{1}{2} C^K_{IJ} \theta^I \theta^J, \qquad d_\pi O_K = C^L_{KJ} O_L \theta^J.$$

Therefore

$$\Phi(d_{\mathrm{CE}} c^K) = -\tfrac{1}{2} C^K_{IJ} \theta^I \theta^J = d_\pi \theta^K = d_\pi \Phi(c^K),$$
$$\Phi(d_{\mathrm{CE}} u_K) = C^L_{KJ} O_L \theta^J = d_\pi O_K = d_\pi \Phi(u_K).$$

*Cochain identity on the full completed algebra.* Both $d_{\mathrm{CE}}$ and $d_\pi$ are degree-1 derivations of the completed graded-commutative algebra structures on each side. They satisfy the graded Leibniz rule with the same Koszul sign convention. $\Phi$ is an algebra homomorphism, so

$$\Phi(d_{\mathrm{CE}}(xy)) = \Phi\big((d_{\mathrm{CE}} x)y + (-1)^{|x|} x\, d_{\mathrm{CE}} y\big) = \Phi(d_{\mathrm{CE}} x)\Phi(y) + (-1)^{|x|}\Phi(x)\Phi(d_{\mathrm{CE}} y),$$

while

$$d_\pi \Phi(xy) = d_\pi\big(\Phi(x)\Phi(y)\big) = (d_\pi \Phi(x))\Phi(y) + (-1)^{|x|}\Phi(x) d_\pi \Phi(y).$$

Generator-level agreement plus Leibniz/algebra-homomorphism gives agreement on every word in $\{c^I, u_J\}$ at fixed total length; Tate-completeness of both sides plus continuity of $d_{\mathrm{CE}}, d_\pi, \Phi$ in the Tate filtration extends the equality to the completed algebras.

*Inverse.* The assignment $\theta^I \mapsto c^I$, $O_I \mapsto u_I$ extends to a continuous algebra homomorphism in the opposite direction. By the same Leibniz argument, it intertwines $d_\pi$ and $d_{\mathrm{CE}}$. The two compositions fix generators, hence fix the completed algebras. So $\Phi$ is an isomorphism of complete dg graded-commutative algebras.

*$P_0$-bracket compatibility.* The CE side carries the canonical $(-1)$-shifted Poisson structure on $C^\bullet_{\mathrm{CE,cont}}(T^*[1]\mathfrak{h})$ with

$$\{c^I, u_J\}_{\mathrm{CE}} = \delta^I_J, \quad \{c^I, c^J\}_{\mathrm{CE}} = 0, \quad \{u_I, u_J\}_{\mathrm{CE}} = 0.$$

These are the standard Poisson brackets on the cochain algebra of a $(-1)$-shifted cotangent of a Lie algebra. The bracket compatibility used here is the displayed generator computation. The Schouten bracket on $\mathrm{PV}(A_\partial)$ satisfies

$$[\theta^I, O_J]_S = \delta^I_J, \quad [\theta^I, \theta^J]_S = 0, \quad [O_I, O_J]_S = 0,$$

the last because $O_I, O_J$ are functions on the formal affine space $\mathfrak{h}^\vee$ and the Schouten bracket of two functions is zero. Generator-level matching gives $\Phi\big(\{c^I, u_J\}_{\mathrm{CE}}\big) = \Phi(\delta^I_J) = \delta^I_J = [\theta^I, O_J]_S = [\Phi(c^I), \Phi(u_J)]_S$, and similarly for the other two pairs. Both brackets satisfy the graded Leibniz rule; bilinear extension on completed words gives compatibility on the full completed algebra.

*Identification with the reduced $P_0$ Hamiltonian central-operation model.* For the reduced constant-stalk convention used in this paper, the reduced $P_0$ Hamiltonian central-operation model of a Poisson algebra $A$ is by definition represented by the Poisson cochain complex $\mathrm{PV}(A)$ with the Lichnerowicz differential $[\pi, -]_S$. This is the local Poisson-CE model; it is not an import of an unreduced Costello–Gwilliam central-operations theorem. Therefore $\mathrm{PV}(A_\partial) \simeq Z^{P_0}_{\mathrm{red,Ham}}(A_\partial)$ as completed dg-commutative $P_0$-algebras, and $\Phi$ is the cochain-level open-closed derived center isomorphism in this local stalk sense. □

## 3.7 *Tate Quillen equivalence and the $\infty$-categorical upgrade of the cochain bar-cobar identity*

### 3.7.1 Categorical home

Let **TateVec** be the closed symmetric monoidal category of complete filtered Tate $\mathbb{C}$-vector spaces of 3.137. The filtration on every object is taken *complete and Hausdorff*: $V = \varprojlim_w V_{\leq w}$ and $\bigcap_w F^w V = 0$. Concrete morphisms are filtration-preserving continuous chain maps. The working ambient category is the bounded-below filtered subcategory $\mathbf{TateChain}_{\geq 0}$ of non-negatively filtered chain complexes in **TateVec**, large enough to contain every CE chain coalgebra and completed enveloping algebra used here. The model-categorical statements below are made in a fixed admissible envelope, not in the raw category of all Tate spaces.

*Statement* 3.28 *(Admissible Tate model envelope).* Fix a bounded-below, locally presentable symmetric monoidal model envelope $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$ containing the CE chains, completed enveloping algebras, completed symmetric algebras, and finite-window inverse systems used in this manuscript. Weak equivalences are admissible filtered quasi-isomorphisms: quasi-isomorphisms on every finite quotient and on associated graded. The tensor product is the completed Tate tensor product on Mittag-Leffler systems, and filtered inverse limits with surjective transition maps are exact. All Quillen statements below are relative to this envelope.

*Definition* 3.29 *(Tate cofibrantly generated model structure in the envelope).* Define generating cofibrations $I$ and acyclic cofibrations $J$ by

$$I = \big\{F^w S^{n-1} \hookrightarrow F^w D^n \mid n \in \mathbb{Z}_{\geq 0},\ w \in \mathbb{N}\big\}, \qquad J = \big\{0 \hookrightarrow F^w D^n \mid n \in \mathbb{Z}_{\geq 0},\ w \in \mathbb{N}\big\},$$

where $S^{n-1}$, $D^n$ are the standard sphere and disk chain complexes in **Vec** and the Tate window $F^w(-)$ takes the $w$-th piece of the filtration. A morphism $f : V \to W$ in $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$ is:

(i) a *weak equivalence* if it is an admissible filtered quasi-isomorphism in the sense of 3.137: a quasi-isomorphism on every quotient $V/F^wV \to W/F^wW$ and on the associated graded;

(ii) a *cofibration* if it is in the class generated by $I$ under transfinite composition, retracts, and pushouts;

(iii) a *fibration* if it has the right lifting property against $J$.

Theorem 3.30 *(Existence inside the admissible envelope).* Under Statement 3.28, the triple

$$\left(\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}, I, J\right)$$

is a cofibrantly generated model category. The class of weak equivalences is closed under retracts and satisfies the two-out-of-three property; cofibrations and fibrations form complementary lifting classes; the small-object argument terminates after countably many steps because every Tate window is finite-dimensional and every generating cofibration changes only one window.

*Proof.* Quillen's small-object argument ([36, §II.3]) applies in the admissible envelope once the following are verified:

(i) $I$ and $J$ permit the small-object argument: the domains and codomains of the generating maps are small with respect to relative $I$-cell complexes because each generator lives in one finite-dimensional Tate window $V_{\leq w}$, hence is $\aleph_0$-small in the admissible envelope.

(ii) Every map in $J$-cell is a weak equivalence: pushouts of acyclic generating cofibrations are levelwise acyclic by finite-dimensional standard chain-complex theory in each window, and the filtered colimit of windowwise acyclic cofibrations is acyclic by Mittag-Leffler exactness in the Tate filtration (3.137 item 1).

(iii) Every map with the right lifting property against $I$ is a trivial fibration: standard, levelwise.

The two-out-of-three property for filtered quasi-isomorphisms follows from the long exact sequence of the filtration: a composite $V \to W \to U$ has filtered $E_1$-page agreeing with the composite of $E_1$-pages of $V \to W$ and $W \to U$ in each window, so two-out-of-three on each window propagates to the global filtered class. This proves the model structure in the chosen envelope only. No assertion is made for arbitrary unbounded Tate complexes outside it. □

*Remark* 3.31 *(Boundedness restriction is harmless).* Restriction to $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$ avoids inverse-limit pathologies in unbounded filtrations. Every CE coalgebra $C^{\mathrm{CE}}_*(\mathfrak{g})$ and every completed enveloping algebra $\widehat{U}(\mathfrak{g})$ that appears here is non-negatively filtered (CE length and PBW length are $\geq 0$). A model structure on the raw unbounded category **TateChain** would require the Spaltenstein semi-model modification ([35, §7.5]) and is not needed for the application here.

### 3.7.2 Operad-algebra transfer

Let $\mathrm{Lie}_{\mathbf{TateVec}}$ be the Lie operad in **TateVec**, regarded as a $\Sigma$-cofibrant $\Sigma$-module (by the standard Koszul resolution of the $\Sigma$-cofibrant replacement), and similarly $\mathrm{Ass}^+_{\mathbf{TateVec}}$ for augmented associative. A $\mathrm{Lie}_{\mathbf{TateVec}}$-algebra is a Tate Lie algebra in **TateVec**; an $\mathrm{Ass}^+_{\mathbf{TateVec}}$-algebra is a Tate complete augmented associative algebra. Write

$$\mathrm{TateLie} = \mathrm{Lie}_{\mathbf{TateVec}}\text{-alg}_{\geq 0}, \qquad \mathrm{TateAssocAlg}^+ = \mathrm{Ass}^+_{\mathbf{TateVec}}\text{-alg}_{\geq 0}$$

in $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$.

LEMMA 3.32 *(Transferred model structures on Tate operad-algebras).* In the admissible envelope, the free-forget adjunction $\mathrm{Free}_{\mathcal{P}} \dashv \mathrm{Forget}$ for $\mathcal{P} \in \{\mathrm{Lie}_{\mathbf{TateVec}}, \mathrm{Ass}^{+}_{\mathbf{TateVec}}\}$ transfers the cofibrantly generated model structure of 3.30 to $\mathcal{P}\text{-alg}_{\geq 0}$ in the sense of Hinich [31, §2.2]: the weak equivalences and fibrations in $\mathcal{P}\text{-alg}_{\geq 0}$ are the maps whose underlying chain map in $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$ is one, and the cofibrations are determined by lifting.

*Proof.* Hinich's transfer theorem requires:

(1) a path object on every algebra of the form $\mathrm{Free}_{\mathcal{P}}(F^w D^n)$;

(2) that pushouts along generating acyclic cofibrations be weak equivalences.

Item (1) follows because $\mathcal{P} = \mathrm{Lie}$ or $\mathrm{Ass}^{+}$ is a $\Sigma$-cofibrant operad in **TateVec**; the path object is constructed from the cylinder object on chain complexes via the Berger-Moerdijk path-functor lift [29], valid whenever the operad is admissible and the ambient category is cofibrantly generated and proper. Item (2) follows because $\mathcal{P}$ acts levelwise on each Tate window, and the windowwise finite-dimensional pushout is a quasi-isomorphism by Schwede-Shipley [37, Lemma 2.3]. Filtered-colimit assembly preserves quasi-isomorphism by Mittag-Leffler exactness (3.137 item 1). □

The dual coalgebra side requires more care: Hinich's transfer applies to algebra categories, not coalgebra categories, because the cofree coalgebra functor is not a left adjoint in general. The substitute is the Aubry-Lemaire-Smirnov [24] §11.4 model structure on conilpotent dg coalgebras transferred through the bar-cobar adjunction itself.

*Definition* 3.33 *(Conilpotent Tate coalgebras).* Let

$$\begin{aligned}\mathrm{TateConilpCoAlg} = \big\{\, (C, \Delta_C, d_C) \;:\; & C \in \mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}, \\ & \Delta_C \colon C \to C \widehat{\otimes} C \text{ continuous coproduct}, \\ & \text{coaugmentation filtration exhaustive} \,\big\}.\end{aligned}$$

with morphisms continuous coalgebra maps. Conilpotency means the coaugmentation filtration $F^{(n)}_{\mathrm{coaug}} C = \ker(\overline{\Delta}^{(n)})$ is exhaustive: $C = \bigcup_n F^{(n)}_{\mathrm{coaug}} C$.

LEMMA 3.34 *(Transferred model structure on Tate conilpotent coalgebras).* TateConilpCoAlg admits, in the admissible envelope, a model structure with

- weak equivalences = maps $f \colon C \to C'$ such that $\Omega(f) \colon \Omega(C) \to \Omega(C')$ is a weak equivalence in $\mathrm{TateAssocAlg}^{+}$ (3.32);
- cofibrations = monomorphisms of underlying filtered chain complexes;
- fibrations = right lifting against acyclic cofibrations.

Equivalently, this is the Lefevre-Hasegawa [34, Th. 1.3.1.2] transferred model structure in the admissible Tate envelope.

*Proof.* Lefevre-Hasegawa, with proof sketched in Loday-Vallette §11.4.10, constructs the conilpotent-coalgebra transferred model structure in the symmetric monoidal category of $\mathbb{Q}$- or $\mathbb{C}$-vector spaces. The proof passes through the cobar-bar adjunction $\Omega \colon \mathrm{ConilpCoAlg} \rightleftarrows \mathrm{AugAssocAlg} \colon \mathrm{Bar}$ and uses that

(i) $\Omega$ takes coalgebra cofibrations (= mono with filtered cokernel) to algebra cofibrations;

(ii) $\Omega$ inverts coalgebra weak equivalences;

(iii) the unit $\eta_C\colon C \to \mathrm{Bar}\,\Omega(C)$ is a weak equivalence for every conilpotent dg coalgebra.

Items (i)-(iii) are checked windowwise (each Tate window is finite-dimensional, so the standard Lefevre-Hasegawa proof applies) and assemble by Mittag-Leffler exactness. The unit weak-equivalence (iii) is precisely 3.137 item 3 (admissible filtered qiso). Therefore the construction lifts verbatim to the admissible Mittag-Leffler Tate envelope. □

### 3.7.3 Bar-cobar Quillen equivalence in the admissible envelope

Theorem 3.35 *(Tate bar-cobar Quillen equivalence in the admissible envelope).* The bar-cobar adjunction

$$\Omega\colon \mathrm{TateConilpCoAlg} \rightleftarrows \mathrm{TateAssocAlg}^{+}\colon \mathrm{Bar}$$

is a Quillen equivalence with respect to the admissible-envelope model structures of 3.32 and 3.34. In particular:

(i) $\Omega$ is left Quillen: it preserves cofibrations and acyclic cofibrations.
(ii) Bar is right Quillen: it preserves fibrations and acyclic fibrations.
(iii) For every cofibrant conilpotent coalgebra $C$ and every fibrant augmented associative algebra $A$, the unit $\eta_C\colon C \to \mathrm{Bar}\,\Omega(C)$ and the counit $\varepsilon_A\colon \Omega\,\mathrm{Bar}(A) \to A$ are weak equivalences.
(iv) The induced derived adjunction $L\Omega\colon \mathrm{Ho}(\mathrm{TateConilpCoAlg}) \rightleftarrows \mathrm{Ho}(\mathrm{TateAssocAlg}^{+})\colon R\,\mathrm{Bar}$ is an equivalence of homotopy categories.

*Proof.* Items (1) and (2) are immediate from 3.34: weak equivalences in TateConilpCoAlg are defined as maps whose $\Omega$ is a weak equivalence, and fibrations on the coalgebra side are defined by lifting.

For item (3), the unit $\eta_C$: filter both sides by symmetric / tensor length on the underlying **TateVec** object, plus the Tate window filtration $F^w$. The associated graded gives, windowwise, a finite-dimensional dg coalgebra and the standard bar-cobar resolution $C^{\mathrm{gr}} \to \mathrm{Bar}\,\Omega(C^{\mathrm{gr}})$. By Loday-Vallette [24, §2.2.5, §11.1.5], this is a quasi-isomorphism on each window. Conilpotency (3.33) ensures the coaugmentation filtration on $C$ is exhaustive and Hausdorff, so the abutment of the spectral sequence of the filtration computes the global qiso. Mittag-Leffler exactness in the Tate filtration (3.137 item 1) lifts the windowwise qiso to a global filtered qiso.

For the counit $\varepsilon_A$: filter $\widehat{U}(\mathfrak{g})$ by PBW. 3.137 item 4 gives $\mathrm{gr}^{\bullet}_F \widehat{U}(\mathfrak{g}) \cong \widehat{S}_{\mathrm{Tate}}(\mathfrak{g})$ (PBW completed-symmetric associated graded), and item 3 gives the windowwise bar-cobar acyclicity for the Lie operad. The symmetric Koszul resolution of Loday-Vallette [24, §3.4.13] is qiso on associated graded. The same Mittag-Leffler argument propagates to the abutment.

Item (4) is formal from (1)-(3) by Quillen's adjoint-functor theorem ([33, Th. 1.3.10]). □

### 3.7.4 Promotion to the associated ∞-categories

Corollary 3.36 *(∞-categorical Tate bar-cobar equivalence).* Apply the Dwyer-Kan / Hammock-localization functor to the Quillen equivalence of 3.35:

$$N_{\Delta}(\mathrm{TateConilpCoAlg}^{\mathrm{cf}}, W) \simeq N_{\Delta}(\mathrm{TateAssocAlg}^{+,\mathrm{cf}}, W)$$

as the ∞-categories associated to the admissible model envelope in the sense of Lurie's *Higher Algebra* §A.3, where cf denotes the full subcategory of cofibrant-fibrant objects and $W$ is the class of weak equivalences. The equivalence is induced by $L\Omega$ and $R$ Bar at the cofibrant-fibrant level. Presentability is an attribute of the chosen locally presentable envelope, not of the raw Tate category.

*Proof.* Lurie [35, A.3.7.6, A.3.7.10]: a Quillen equivalence of model categories induces an equivalence of associated $\infty$-categories. The presentability of both sides, inside the envelope, is inherited from the cofibrant generation of the model structures (3.30, 3.32, 3.34) by Lurie A.2.6.8. □

### 3.7.5 Identification with the Lie operadic Koszul duality

The Quillen equivalence of 3.35 is *associative* bar-cobar. Specialization to the Lie operad gives the cochain-level equivalence used in the CE/PV theorem lane.

COROLLARY 3.37 *(Tate Lie bar-cobar / CE Koszul duality).* For $\mathfrak{g}$ a conilpotent Tate graded Lie algebra in the sense of 3.137, the bar-cobar Quillen equivalence applied to $A = \widehat{U}(\mathfrak{g}) \in \mathrm{TateAssocAlg}^{+}$ yields the equivalence in the homotopy category

$$L\Omega\big(C_*^{\mathrm{CE}}(\mathfrak{g})\big) \xrightarrow{\sim} \widehat{U}(\mathfrak{g}), \qquad R\,\mathrm{Bar}\big(\widehat{U}(\mathfrak{g})\big) \xrightarrow{\sim} C_*^{\mathrm{CE}}(\mathfrak{g}),$$

and hence a Lie-Koszul-dual equivalence of cochain complexes

$$C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}) \simeq \big(\widehat{U}(\mathfrak{g})\big)^{!}$$

in $\mathrm{Ho}(\mathrm{TateConilpCoAlg})$.

Inside the admissible envelope, by the symmetric-monoidal-functoriality of the bar-cobar Quillen equivalence [24, §13.2] and Lurie [35, §5.5], this lifts to an equivalence of symmetric monoidal $\infty$-categories,

$$N_\Delta\big(\mathrm{TateConilpCoAlg}^{\mathrm{cf}}_{\mathrm{Lie}}, W\big) \simeq N_\Delta\big(\mathrm{TateAssocAlg}^{+,\mathrm{cf}}_{\mathrm{Lie}}, W\big)$$

in the framework of Lurie *Higher Algebra* §5.5 (locally constant factorization algebras over $\mathbb{R}$, applied pointwise on the line; or directly the formal-disk version).

*Proof.* Apply 3.35 to the Lie-operadic Koszul-dual pair (Lie, CoComm) in **TateVec**. Loday-Vallette [24, Th. 13.4.6] identifies the operadic bar-cobar with the Lie-bar-cobar of CE chains and completed enveloping algebra. The symmetric-monoidal extension is direct because the operad Lie is a symmetric-monoidal Hopf operad in **Vec**, and base change to **TateVec** via the strong symmetric-monoidal functor **Vec** $\to$ **TateVec** preserves operadic Koszul duality ([24, §13.5]). □

### 3.7.6 Application to the local theorem lanes

THEOREM 3.38 *(Admissible-envelope promotion of the CE/PV cochain identity).* Let $\mathfrak{h}_{\mathrm{adm}}$ be a pro-finite-dimensional Lie algebra in **TateVec** whose shifted-cotangent extension $\mathfrak{g}_{\mathrm{adm}} = \mathfrak{h}_{\mathrm{adm}} \ltimes \mathfrak{h}^{\vee}_{\mathrm{adm,cont}}[1]$ satisfies the Tate-conilpotent admissible-envelope hypotheses of 3.35. The strict cochain map

$$\Phi\colon C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}) \longrightarrow \mathrm{PV}\big(\widehat{S}_{\mathrm{Tate}}(\mathfrak{h}_{\mathrm{adm}})\big)$$

of 3.27 promotes to an equivalence, inside that admissible Tate model envelope, of module $\infty$-categories over the two identified completed dg $P_0$-algebras

$$\widetilde{\Phi}\colon N_\Delta\big(C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}})\text{-mod}, W\big) \xrightarrow{\sim} N_\Delta\big(\mathrm{PV}\big(\widehat{S}_{\mathrm{Tate}}(\mathfrak{h}_{\mathrm{adm}})\big)\text{-mod}, W\big)$$

in the framework of Lurie's *Higher Algebra* §5.5. This is not a construction of a completed dg $P_0$ model structure or a factorization-centre theorem for the full untruncated formal Hamiltonian algebra. The equivalence itself is the admissible-envelope algebraic one.

*Proof.* By 3.37, the cochain-level identity

$$\Phi\colon C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}) \xrightarrow{\sim} \widehat{S}_{\mathrm{Tate}}(\mathfrak{h}_{\mathrm{adm}})\widehat{\otimes}\widehat{\Lambda}(\theta^I)$$

of 3.27 (which strictly intertwines $d_{\mathrm{CE}}$ and $d_\pi$ by 3.27) is the underlying 0-cell of an equivalence in $N_\Delta(\mathrm{TateConilpCoAlg}^{\mathrm{cf}}_{\mathrm{Lie}})$. The right-hand side, $\mathrm{PV}\big(\widehat{S}_{\mathrm{Tate}}(\mathfrak{h}_{\mathrm{adm}})\big)$, has the standard Schouten-Lichnerowicz $P_0$-algebra structure (3.139), and the closed-side cochain complex inherits a $P_0$-algebra structure by transport via $\Phi$.

The $\infty$-category structure on completed dg $P_0$-algebras follows from 3.35 applied to the operad $P_0 = \mathrm{Comm} \circ \Lambda^\bullet[\,\cdot\,,\,\cdot\,]_S$, which is Koszul self-dual up to suspension by Cattaneo-Felder [30] (cf. Costello-Gwilliam Vol. II §1.4 for the Tate version).

The equivalence on the module category follows from [35, Th. 7.1.4.6]: a Quillen equivalence of $E_n$-monoidal model categories induces an equivalence of module $\infty$-categories. Applied with $n = 1$ (associative monoidal in the bar-cobar; $P_0$ structure superimposed by the Lichnerowicz-Schouten differential) gives $\widetilde{\Phi}$.

Functoriality in the Lie algebra gives interval-wise compatibility for admissible interval objects $\mathfrak{h}_{\mathrm{adm},I} = \Omega^\bullet_c(I) \otimes \mathfrak{h}_{\mathrm{adm}}$ when these interval objects remain in the same admissible envelope. This is interval-wise categorical compatibility, not an independent Weiss descent theorem and not an unreduced open-BV centrality theorem. □

*Remark* 3.39 *(Relation to Lurie's higher-algebraic Koszul duality).* Lurie [35, §5.5.5] constructs a higher-algebraic Koszul duality between augmented $\infty$-operad-algebras and their Koszul-dual coalgebras. For the operad $\mathrm{Lie}_{\mathbf{TateVec}}$, Koszul self-duality (with shift) gives the equivalence of 3.37 in the Tate-conilpotent setting. The advantage of the model- categorical proof is concrete: weak equivalences are spelled out in terms of admissible filtered qiso of 3.137, not abstractly via the Hammock localization. The two constructions agree by Lurie A.3.7 plus the Hinich transfer of 3.32.

### 3.7.7 Compatibility with the descendant residual

The Tate Quillen equivalence of 3.35 operates on the bar-cobar duality between $\widehat{U}(\mathfrak{g})$ and $C^{\mathrm{CE}}_*(\mathfrak{g})$. It does not directly use the Casimir element $C_{\mathfrak{h}} = \sum_d e_{d,i} \otimes e^{i,\vee}_d$ of 3.137 or its convergence in the Tate completion.

Proposition 3.40 *(Quillen equivalence vs Casimir convergence).* The Quillen equivalence of 3.35 and the resulting $\infty$-categorical equivalence of 3.38, when their admissible-envelope hypotheses hold, are independent of the convergence of the Tate Casimir element. The cochain complexes $C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g})$ and $\mathrm{PV}\big(\widehat{S}_{\mathrm{Tate}}(\mathfrak{h})\big)$ are equivalent in the model category of 3.32 only after the same conilpotent admissibility hypotheses are imposed. Casimir convergence is not one of those hypotheses; conilpotence is.

*Proof.* The bar-cobar comparison uses $\widehat{S}^c(\mathfrak{g}[1])$ and $T(\mathfrak{g})$ as underlying graded objects. Both are well-defined and conilpotent in the admissible envelope under 3.137 items 1, 2 (continuous dual exact, conilpotent CE coalgebra). The Casimir $\sum_d e_{d,i} \otimes e^{i,\vee}_d$ is an element of $\mathfrak{h}\widehat{\otimes}\mathfrak{h}^\vee_{\mathrm{cont}}$; it is required for the Costello [12] heat-kernel propagator construction at the analytic / factorization-algebra-with-non-strict-bracket level. It is *not* required for the algebraic bar-cobar comparison.

Therefore the cochain-level $\Phi$ promotes, in the admissible model envelope and only there, to an $\infty$-categorical equivalence. What remains outside this algebraic promotion is the Weiss-cosheaf upgrade and the heat-kernel quantum lift, flagged in 3.41 below. □

PROPOSITION 3.41 *(Residual Weiss-cosheaf condition).* The $\infty$-categorical equivalence of 3.38 on stalks and on the locally-constant interval-factorization $\mathcal{F}_{\mathrm{cl}}$ of 3.21 does not, by itself, give the Weiss-cosheaf condition for the *unreduced* factorization algebra $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}$ on $\mathbb{R}^2 \times \mathbb{C}^2$. The Weiss / descent condition for an unreduced factorization algebra on a higher-dimensional manifold requires checking Čech descent for Weiss covers, which is a separate cosheaf-theoretic statement. Costello-Gwilliam Vol. I §§6.1, 6.5, 7.2 provide the published factorization, cosheaf, and basis-extension vocabulary; no theorem from those sections is used here to close the unreduced descent obligation. T3 closes the cochain-level $\infty$-categorical promotion; the Weiss upgrade is recorded as the standing residual on the unreduced-factorization side of the local theorem.

*Proof.* Lurie [35, §5.5.4], with the same terminology as Costello-Gwilliam [15, Vol. I §6.4]: a locally constant factorization algebra on $\mathbb{R}$ determines an $E_1$-algebra up to the standard contractible choice; the Quillen equivalence of 3.35 realizes this. On a higher-dimensional manifold, the Weiss cover $\{U_\alpha\}$ has structure-map descent independent of the algebraic bar-cobar; in particular the heat-kernel propagator on $\mathbb{R}^2 \times \mathbb{C}^2$ enters the analytic check. Without that propagator the bar-cobar identity is consistent but does not trivialize Weiss descent. □

### 3.7.8 SUMMARY DIAGRAM

$$\begin{array}{ccc}
\text{3.137 (filtered qiso of cochain complexes)} & \xrightarrow{\text{T3 promotion}} & \text{Quillen equivalence of model cats} \\
\downarrow & & \downarrow \\
\text{cochain identity } \Phi\, d_{\mathrm{CE}} = d_\pi\, \Phi & \xrightarrow{\text{Lurie A.3.7}} & \text{equivalence of } \infty\text{-cats} \\
 & & \downarrow \text{Lurie 5.5} \\
 & & \text{equivalence in admissible } \infty\text{-cats of} \\
 & & \text{completed dg } P_0\text{-algebras}
\end{array}$$

The chain on the left is $\Phi$ as a strict cochain map. The chain on the right is T3's promotion: model-categorical, then $\infty$-categorical, then $P_0$-algebra-categorical. The promotion does not depend on the Tate Casimir convergence (3.40); the Weiss-cosheaf descent on the unreduced factorization algebra is a separate residual (3.41).

COROLLARY 3.42 *(Moment-map shadow).* In the situation of 3.27, the degree-zero specialization of $\Phi$ on the cotangent direction is the moment-map / Kirillov assignment

$$J : \mathfrak{h} \longrightarrow A_\partial = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h}), \qquad J(f) = O_f.$$

The closed-side Lie bracket $\{f, g\}_{\mathfrak{h}}$ matches the Kirillov–Poisson bracket on $A_\partial$:

$$\{J(f), J(g)\}_\partial = J\big(\{f, g\}_{\mathfrak{h}}\big).$$

*Proof.* $J$ is the restriction of $\Phi$ to the cotangent direction $u_I \mapsto O_I$ followed by linear extension along $\mathfrak{h} \hookrightarrow A_\partial$ (the canonical embedding of $\mathfrak{h}$ as the linear functions on $\mathfrak{h}^\vee$). Bracket compatibility is the $(c^J, u_K)$-versus-$(\theta^J, O_K)$ identity from 3.27 read off in degree zero. Equivalently, this is the standard Kirillov–Kostant–Souriau theorem for linear Poisson structures. □

COROLLARY 3.43 *(CE/PV generator identity, formal symplectic disk).* Let $\mathfrak{h} = \widehat{A}/\mathbb{C} \cdot 1 = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$ be the reduced Hamiltonian Lie algebra of the formal symplectic disk on $\mathbb{C}^2$, with

bracket $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$ modulo constants. Choose the Taylor basis $H_{a,b} = z_1^a z_2^b$ for $a, b \geq 0$, $a + b > 0$. Structure constants:

$$[H_{a,b}, H_{c,d}] = (ad - bc)\, H_{a+c-1,b+d-1},$$

with the convention that $H_{m,n} = 0$ if $\min(m, n) < 0$ or $m + n = 0$.

Set $A_\partial = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})$, with linear coordinates $O_{a,b}$ on $\mathfrak{h}^\vee$ and Kirillov–Poisson bracket $\{O_{a,b}, O_{c,d}\}_\partial = (ad - bc) O_{a+c-1,b+d-1}$. The generator calculation underlying 3.27 gives the cochain-level CE/PV identity

$$\Phi : C^\bullet_{\mathrm{CE,cont}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big), \qquad c^{a,b} \mapsto \theta^{a,b}, \quad u_{a,b} \mapsto O_{a,b}.$$

Its moment-map shadow is the boundary evaluation map $J(f) = O_f = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ of 3.161 after identifying $A_\partial$ with the stable connected scalar-reduced Hamiltonian trace algebra of 3.131. The conilpotent bar-cobar and factorization derived-center isomorphism is read in the claim-strength ledger through the nilpotent or weighted replacements of §3.8.

*Proof.* The structure constants are an instance of 3.13 read on monomials. The CE/PV differential identity is checked directly on the monomial generators $c^{a,b}$, $u_{a,b}$ and then extended by the completed Leibniz rule; no lower-central conilpotence of the full Hamiltonian algebra is used. The moment-map shadow follows from 3.42 together with 3.161. □

*Example* 3.44 *(Low-degree monomial verification).* We verify $\Phi$ on small monomials of the formal symplectic disk.

- *Degree-1 generators.* Take $H_{1,0} = z_1$, $H_{0,1} = z_2$. The bracket $[H_{1,0}, H_{0,1}] = 1 \cdot 1 - 0 \cdot 0 = 1$ lifts to the constant $H_{0,0}$, which is set to zero in $\mathfrak{h}$. Hence $C^{0,0}_{(1,0),(0,1)} = 0$ in $\mathfrak{h}$. Consequently $d_{\mathrm{CE}} u_{0,0}$ is not present (no such generator); the $u_{0,0}$ degree of freedom does not appear, consistent with constants being removed.
- *Degree-2 calculation.* Take $H_{2,0} = z_1^2$, $H_{0,2} = z_2^2$. The bracket $\{z_1^2, z_2^2\} = 2z_1 \cdot 2z_2 - 0 \cdot 0 = 4z_1 z_2 = 4H_{1,1}$, so $C^{1,1}_{(2,0),(0,2)} = 4$ and the antisymmetric companion is $C^{1,1}_{(0,2),(2,0)} = -4$. The cochain identity reads

  $$d_{\mathrm{CE}} u_{1,1} = 4\, u_{2,0}\, c^{0,2} - 4\, u_{0,2}\, c^{2,0} + \cdots,$$

  where $\cdots$ denotes terms from other index triples $(K, J, L)$ with $K = (1, 1)$. Each such term $C^L_{(1,1),J}\, u_L\, c^J$ arises from $\{H_{1,1}, H_J\} = \sum_L C^L_{(1,1),J}\, H_L$. For example, $\{H_{1,1}, H_{1,0}\} = \partial_{z_1}(z_1 z_2) \cdot 0 - \partial_{z_2}(z_1 z_2) \cdot 1 = -z_1 = -H_{1,0}$, so $C^{1,0}_{(1,1),(1,0)} = -1$, contributing $-u_{1,0}\, c^{1,0}$ to $d_{\mathrm{CE}} u_{1,1}$. On the open/PV side,

  $$d_\pi O_{1,1} = 4\, O_{2,0}\, \theta^{0,2} - 4\, O_{0,2}\, \theta^{2,0} - O_{1,0}\, \theta^{1,0} + O_{0,1}\, \theta^{0,1} + \cdots,$$

  the same expression with $u \to O$, $c \to \theta$. Hence $\Phi(d_{\mathrm{CE}} u_{1,1}) = d_\pi \Phi(u_{1,1})$. An independent exact arithmetic check verifies the same structure-constant input on every monomial pair with exponents up to 6 via the first-order Moyal coefficient $kn - lm$ at $\hbar^1$, which equals $C^K_{IJ}$ for these indices.
- *Higher-order word.* For $K = (1, 1)$, $J = (1, 0)$, $J' = (0, 1)$, the cochain element $u_{1,1} c^{1,0} c^{0,1}$ has degree $0 + 1 + 1 = 2$. Its CE differential, by graded Leibniz, is

  $$d_{\mathrm{CE}}\big(u_{1,1} c^{1,0} c^{0,1}\big) = (d_{\mathrm{CE}} u_{1,1}) c^{1,0} c^{0,1} - u_{1,1} (d_{\mathrm{CE}} c^{1,0}) c^{0,1} + u_{1,1} c^{1,0} (d_{\mathrm{CE}} c^{0,1}),$$

with each summand applying the generator formulas above. The same word on the PV side is $O_{1,1}\theta^{1,0}\theta^{0,1}$, and $d_\pi$ applied with the same Leibniz convention produces the corresponding triple sum with $u \to O$ and $c \to \theta$. Term-by-term equality follows from the generator identities; the full word equality is the algebra-homomorphism Leibniz argument from 3.27.

*Construction* 3.45 *(Compact current PBW/Rees source).* Let $\mathcal{P}$ denote the principal-part module

$$\mathcal{P} = \bigoplus_{\substack{a,b\geq 0\\ a+b>0}} \mathbb{C}\,\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1}z_2^{-b-1}\,dz_1dz_2,$$

identified with $\mathfrak{h}^\vee_{\mathrm{cont}}$ by the residue pairing of Proposition 3.147. Put

$$\mathfrak{g}_{\mathrm{pp}} = \mathfrak{h} \ltimes \mathcal{P}[1],$$

with zero differential and bracket

$$[f,g] = \{f,g\}, \qquad [f,\rho[1]] = (f\cdot\rho)[1], \qquad [\rho[1],\sigma[1]] = 0,$$

where $f\cdot\rho = \pi_{\mathrm{pp}}\{f,\rho\}$. On monomials this action is

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1},$$

with negative-index targets zero.

For an interval $I \subset \mathbb{R}$, define the compactly supported current Lie algebra

$$\mathfrak{J}_{\mathfrak{g}}(I) = \Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{g}_{\mathrm{pp}}, \qquad [\alpha\otimes x, \beta\otimes y] = \alpha\beta\otimes[x,y].$$

Its PBW/Rees algebra is

$$U_\hbar\big(\mathfrak{J}_{\mathfrak{g}}(I)\big) = \widehat{T}\big(\mathfrak{J}_{\mathfrak{g}}(I)\big)[[\hbar]]\,/\,\langle xy - (-1)^{|x||y|}yx - \hbar[x,y]\rangle.$$

The PBW filtration is the tensor-word filtration. Its special fibre is

$$U_\hbar\big(\mathfrak{J}_{\mathfrak{g}}(I)\big)/(\hbar) \cong \widehat{\mathbf{S}}\big(\mathfrak{J}_{\mathfrak{g}}(I)\big),$$

equipped with the semiclassical $P_0$ bracket

$$\{\bar{X},\bar{Y}\}_{P_0} = \big(\hbar^{-1}(XY - (-1)^{|X||Y|}YX)\big) \bmod \hbar.$$

This is the PBW/Koszul-dual source for Hamiltonian and principal-part modes. It is not a closed CE cochain algebra map; the CE comparison must be made separately by bar-cobar twisting or by a completed pairing.

*Problem* 3.46 *(PBW-source $P_0$ center action and CE comparison).* Construct a filtered factorization-center action of the compactly supported current PBW/Rees source $U_\hbar(\mathfrak{J}_{\mathfrak{g}})$ of Construction 3.45. The target cannot be the polynomial open observable algebra alone if the cotangent generators are included; it must use the enlarged principal-part boundary object of Theorem 3.157. In the classical special fibre the action must send

$$f \longmapsto B_f = \overline{\mathrm{Tr}\, f(\phi_1,\phi_2)}, \qquad \rho_{a,b}[1] \longmapsto \Theta_{a,b},$$

and satisfy the stalkwise bracket identities

$$\{B_f, B_g\}_{\mathrm{open}} = B_{\{f,g\}}, \qquad \{B_f, \Theta_\rho\}_{\mathrm{open}} = \Theta_{f\cdot\rho}, \qquad \{\Theta_\rho, \Theta_\sigma\}_{\mathrm{open}} = 0.$$

This special-fibre statement is the natural $P_0$ theorem. A strict action of the full filtered Rees algebra on the same commutative reduced target is not a consequence of these identities: in the source

$$XY - (-1)^{|X||Y|}YX = \hbar[X, Y],$$

while multiplication operators on a commutative classical target commute. Thus the Rees-level theorem requires either a deformed boundary target or an explicit homotopical quantization of the central operations. The target data must include differential compatibility, factorization centrality homotopies, coherent $P_0$ bracket homotopies, and compact-support naturality for interval inclusions and disjoint unions before the de Rham contraction.

Separately compare this PBW/Koszul-dual source with the closed CE observable algebra by an explicit twisting or pairing construction. Without this comparison, the result is a PBW/Koszul-dual boundary action, not a cochain-level map from closed CE observables.

*Problem* 3.47 *(Unreduced Tate-coefficient cotangent lift).* Construct a Tate or weighted-coefficient extension of the BV heat-kernel calculus on $\mathbb{R}^2 \times \mathbb{C}^2$ in which:

(i) the formal Casimir $C_{\mathfrak{h}} = \sum_{d\geq 1}\sum_i e_{d,i} \otimes e^i_d$ converges as an element of $\widehat{\mathfrak{h}} \,\widehat{\otimes}\, \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$ [closed: 3.66(a) via the weighted Tate completion of 3.58];

(ii) the Costello propagator $P_{\epsilon,L}$ extends to that coefficient category [closed: 3.66(b)];

(iii) the primitive indecomposable shadow of the principal-part current sector

$$\widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I) = \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \,\widehat{\otimes}\, \widehat{\mathbf{S}}\big(\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I)\big)$$

admits a chain-level primitive projection $\pi_{\mathrm{prim}}$ with verified $Q$-compatibility and factorization compatibility on the indecomposable quotient [shadow closed: 3.101; the cotangent centrality homotopies remain part of item (4)];

(iv) the reduced principal-part defect-current lane of 2.27 lifts to the unreduced factorization-algebra model and the degree-zero Moyal/Weyl lane extends to the cotangent (one-antifield) sector at all orders in $\hbar$ [primitive shadow: 3.106; unreduced cotangent centrality and the one-antifield quantum lift remain conditional on the QME obstruction class of 3.68; the truncated nilpotent sector $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ closed classically by 3.83 and obstructed quantum-mechanically by 3.84].

Equivalently, lift the cochain-level 3.27 and the Hamiltonian-sector factorization-algebra 3.21 to a cochain-level morphism of factorization algebras

$$\Phi_{\mathrm{cl}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_L \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}\big(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}\big)$$

on the unreduced (cotangent-included) sector, with all $P_0$-center homotopies, factorization centrality homotopies, the stable connected quotient, and a boundary realization of the shifted cotangent generators. Any global version must also pass through the local-Hamiltonian descent and period obstruction of 2.16; a stalkwise formal disk map is not a Weiss/Ran descent theorem for factorization centers.

Four mathematical mechanisms analyze the residual (§3.8): Costello–Li 2012 BV-cohomology vanishing, refuted on the present geometry; weighted Tate completion, which repairs the coefficient kernel

and finite-scale propagator and gives graphwise RG control; nilpotent truncation to $\mathfrak{m}^3/\mathfrak{m}^{N+1}$, classical only, with an explicit Moyal obstruction at all orders; and direct chain-level primitive projection, closing the classical cosheaf and factorization compatibilities of the indecomposable shadow without analytic input.

### *3.8 Residual analysis of the unreduced Tate-coefficient cotangent lift*

The mechanisms in 3.8 analyzing 3.47 are separated below into proved partial results and the remaining obstruction. The BV-cohomology vanishing strategy of Costello–Li 2012 is recorded as a counter-example: it does not transfer to the geometry studied here, and the failure mode forces the replacement mechanisms below. The weighted Tate completion closes the Casimir convergence and the finite-scale heat-kernel propagator extension. The nilpotent truncation closes a conformal-symplectic-quotient sector classically and exhibits the all-order Moyal obstruction. The chain-level primitive projection closes the cosheaf and factorization compatibilities of the indecomposable shadow without analytic input. The weighted Hadamard/Mittag-Leffler theorem below reduces the remaining Costello-locality compatibility of 3.68 to the mixed brane-defect QME obstruction class.

## BV-cohomology vanishing and the unreduced cotangent lift

*Costello–Li 2012 hypotheses, restated*

*Statement* 3.48 *(Costello–Li 2012 BV-cohomology vanishing, restated).* Let $X$ be a compact Calabi–Yau manifold of complex dimension $d$, with holomorphic volume form $\Omega$. Let $\mathrm{PV}(X)[[t]]$ be the polyvector field complex with the Tate parameter $t$ (the worldsheet $S^1$ rotation), with differential $Q = \bar{\partial} + t\,\partial_\Omega$ and $(-1)$-shifted symplectic pairing

$$\langle \alpha, \beta \rangle = \int_X (\alpha \wedge \beta)\big|_{\mathrm{PV}^{d,d}} \cdot \Omega^2.$$

The classical BCOV interaction $I_{\mathrm{BCOV}}$ is the cubic vertex on $\mathrm{PV}(X)[[t]]$ supplied by [13]. Then:

(i) the cohomology of the local-functional obstruction complex $H^*\big(\mathrm{Loc}(X),\ Q + \{I_{\mathrm{BCOV}}, -\}\big)$ is concentrated in known degrees;

(ii) the one-loop quantization obstruction is a single explicit local class which vanishes by the Calabi–Yau condition ($\partial_\Omega$ is exact on $\mathrm{PV}(X)$ via $\Omega$);

(iii) all higher-loop obstructions vanish, giving a perturbative quantization compatible with the $S^1$ symmetry, unique up to the equivalences in the Costello–Li renormalization scheme.

The hypotheses are $X$ compact Calabi–Yau, $\Omega$ a holomorphic volume form, BCOV gauge fixing (Kähler metric, harmonic projection, $L^2$-regularized propagator), polyvector fields with the $t$-Tate completion, and pure closed BCOV without brane defects.

The 2015 sequel [14] extends to open-closed BCOV with open gauge data $\mathfrak{gl}(N \mid N)$, anomaly cancellation by the supertrace, on flat $\mathbb{C}^d$, $d$ odd. That is the precedent for defects in the Costello–Li line, but on $\mathbb{C}^d$, not on $\mathbb{R}^2 \times \mathbb{C}^2$ with topological factor.

*Applicability check on $\mathbb{R}^2 \times \mathbb{C}^2$ with brane $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ and $\mathfrak{gl}_N$*

PROPOSITION 3.49 *(Costello–Li 2012 hypotheses fail for the local model).* The hypotheses of 3.48 fail for the formal local model on $\mathbb{R}^2 \times \mathbb{C}^2$ with brane $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ and gauge data $\mathfrak{gl}_N$ for five independent reasons.

(F1) *Non-compactness.* $\mathbb{C}^2$ is not compact. Costello–Li's harmonic projection and $L^2$-regularized propagator depend on Hodge theory for $X$ compact. The formal local jet model is at most a stalk of the global theory and is not directly a Costello–Li target.

(F2) *Topological–holomorphic mixing.* The local theory has a topological factor $\mathbb{R}^2$ (de Rham) crossed with a holomorphic factor $\mathbb{C}^2$ (Dolbeault). Costello–Li 2012 is pure BCOV/HCS on a single holomorphic Calabi–Yau; there is no pure BCOV statement covering the mixed kinematics. The protected sector of twisted M-theory [25] has the right shape, but Costello–Li 2012 is not stated for it.

(F3) *Brane defect.* The brane $L$ is one-real-dimensional in the topological direction and point-supported in the holomorphic direction. Costello–Li 2012 has no brane sector. Costello–Li 2015 [14] has a brane sector but on flat $\mathbb{C}^d$ (not $\mathbb{R}^2 \times \mathbb{C}^2$) with $\mathfrak{gl}(N \mid N)$ open gauge data.

(F4) *Gauge data.* Ordinary $\mathfrak{gl}_N$ is not $\mathfrak{gl}(N \mid N)$. Costello–Li 2015 cancel the U(1) center anomaly by the supertrace, which fails for ordinary $\mathfrak{gl}_N$. The U(1) center class $[\bar{c}]$ in part (f) of 2.27 is precisely this anomaly.

(F5) *Tate direction.* Costello–Li's $t$-Tate completion is along the worldsheet $S^1$ rotation, not along the coefficient system. The Tate completion required by 3.47 is along the *coefficient* direction $\widehat{\mathfrak{h}} \widehat{\otimes} \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$ in which the Casimir $C_{\mathfrak{h}} = \sum_{d \geq 1} \sum_i e_{d,i} \otimes e^i_d$ must converge. By 3.182, the direct-sum dual fixed in the parent calculation obstructs convergence, and the coefficient category itself must change. Costello–Li provide no convergence theorem in this direction.

Hypothesis (F5) is the decisive obstruction: even if (F1)–(F4) were repaired by an Omega-background or twisted M-theory extension, the coefficient-system Tate completion would still be unprovided.

*Proof.* Each clause is a direct comparison of the hypotheses of 3.48 (and its 2015 sequel) with the data of 2.27 and the topology declared in the linear-heat / coefficient-Casimir analysis (3.181, 3.182). The non-compactness of $\mathbb{C}^2$ blocks the harmonic-projection step in Costello–Li's one-loop anomaly cancellation; the topological factor $\mathbb{R}^2$ is not BCOV kinematics; the brane $L$ is a topological line transverse to a holomorphic point, not an open boundary on a flat $\mathbb{C}^d$; ordinary $\mathfrak{gl}_N$ does not admit the Costello–Li supertrace cancellation (which is exactly the class $[\bar{c}]$ of part (f)); and the worldsheet-$t$ Tate completion is orthogonal to the coefficient-Casimir Tate direction. □

Corollary 3.50 *(Costello–Li 2012 does not close 3.47).* The Costello–Li 2012 BV-cohomology vanishing route does not by itself close the unreduced Tate-coefficient cotangent lift. The independent chain-level route 3.101 closes the classical primitive indecomposable shadow without analytic input. It does not construct the unreduced cotangent factorization-centre morphism. The quantum one-antifield extension of part (e) still requires the weighted coefficient category and the mixed brane-defect locality/QME input isolated in Problem 3.68 and 3.111.

*Proof.* By 3.49, the BV-cohomology vanishing of [13] does not transfer to the geometry studied here: it provides neither the coefficient-system Tate completion in which the Casimir converges, nor a brane-defect extension of the propagator. The closures cited in the statement supply the classical chain-level lift and the weighted finite-scale coefficient kernel by other routes. They do not make Costello–Li 2012 applicable to the mixed local geometry here. □

*What an extension of Costello–Li would have to supply*

*Remark* 3.51 *(Gap analysis: the four missing theorems).* An applicable analog of Costello–Li 2012 to the setup here would have to supply the following four theorems simultaneously:

(i) *Non-compact / Omega-background BV-cohomology vanishing.* A version of 3.48 on $\mathbb{R}^2 \times \mathbb{C}^2$, either via Costello's Omega-background [25] or via a compactification with Lagrangian boundary at infinity.

(ii) *Topological–holomorphic mixed kinematics.* The vanishing must hold on the kinematic differential $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$, with the linearized complex of 3.181, and on the Hamiltonian sector with cotangent [1] extension.

(iii) *Brane-defect extension.* The vanishing must allow the transverse one-real-dimensional defect $L$ on which open gauge data $\mathfrak{gl}_N$ live; not the flat-$\mathbb{C}^d$ $\mathfrak{gl}(N \mid N)$ setup of [14]. It must also decide the scalar contact class of Proposition C.3: closed exchange cancellation, supertrace removal, central-character normalization, or renormalized boundary generators.

(iv) *Coefficient-system Tate completion.* The coefficient direction $\widehat{\mathfrak{h}} \widehat{\otimes} \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$ must support the Casimir as a convergent kernel. By 3.182 this is a category-level change, not a damping factor.

The cited Costello–Li theorems do not supply these four inputs for this mixed brane-defect coefficient problem; together they constitute 3.47, and they remain a named open problem after T4.

*Remark* 3.52 *(Convention firewall, restated).* The local Hamiltonian BF sector here is not a compact Calabi–Yau output of any $\Phi_d$ functor, carries no $\kappa_{\mathrm{ch}}$ or $\kappa_{\mathrm{BKM}}$ consequence, and is not evidence for compact $\mathrm{CY}_3$ BCOV anomaly cancellation. Costello–Li 2012 is cited as motivation and convention template, not as a closing theorem for the unreduced cotangent lift. The Vol III CY-to-chiral frontier and the Igusa $\Delta_5$ / Borcherds / BKM frontier are independent obligations and are not affected by T4.

## Weighted Tate completion of the coefficient pair

*Status.* The weighted coefficient category proves the Casimir, finite-window Mittag-Leffler, heat-kernel, and classical cochain statements below. Renormalization-flow compatibility is asserted graph by graph at finite scale for polynomial-degree-bounded vertices. Any quantum descendant comparison is conditional on the weighted Costello-locality/QME obstruction in Problem 3.68. No regulator-independence theorem is asserted for the original unweighted product/direct-sum Tate pair. The only regulator-independence theorem proved below is independence of the chosen admissible weight inside the finite-window Mittag-Leffler sector; it is not an equivalence with the strict product/direct-sum theory.

*Diagnosis*

*Remark* 3.53 *(Why the unweighted Tate Casimir fails).* In the parent manuscript the Hamiltonian Lie algebra is the product/direct-sum Tate pair

$$\mathfrak{h} = \prod_{d \geq 1} H_d, \qquad \mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{d \geq 1} H_d^\vee, \tag{6}$$

with $H_d = \mathrm{Span}\{z_1^a z_2^b : a + b = d\}$ and Hamiltonian bracket satisfying $\{H_p, H_q\} \subset H_{p+q-2}$. By Lemma 3.137 item (1), the continuous dual of a Tate product is a strict direct sum, and *vice versa*. The formal Casimir

$$C_{\mathfrak{h}} = \sum_{d \geq 1} \sum_i e_{d,i} \otimes e_d^i$$

has a nonzero summand in every degree $d$, so its second tensor factor lies in $\prod_d H_d^\vee$, not in $\bigoplus_d H_d^\vee = \mathfrak{h}_{\mathrm{cont}}^\vee$. Thus $C_{\mathfrak{h}} \notin \mathfrak{h}\widehat{\otimes}\mathfrak{h}_{\mathrm{cont}}^\vee$. This is Lemma 3.182 of the parent.

The obstruction is categorical, not analytic: no rescaling of the bases $\{e_{d,i}\}$ or $\{e_d^i\}$ *within* the strict pair (6) can move the Casimir from the product $\prod_d (H_d \otimes H_d^\vee)$ to the strict sum-with-product $(\prod_d H_d)\widehat{\otimes}(\bigoplus_d H_d^\vee)$. The cure must change the coefficient *category*: the second factor must itself be a product, with the change of dualization inserted by an intermediate weight system that is itself a Mittag-Leffler inverse-limit object.

*Remark* 3.54 *(Costello's coefficient requirements).* In Costello's heat-kernel BV calculus, the BV propagator $P_{\epsilon,L}$ on a free elliptic complex $(\mathcal{E}, Q, \langle\,,\,\rangle)$ is the kernel of the operator $\int_\epsilon^L Q^{\mathrm{GF}} e^{-tD} dt$ in $\mathcal{E}\widehat{\otimes}\mathcal{E}$ [12, Ch. 5]. When the field complex factors as

$$\mathcal{E} = \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}_{\mathrm{cont}}^\vee[2]\big),$$

with the Tate coefficient pair acting passively on the spacetime, the propagator factors as

$$P_{\epsilon,L} = P_{\epsilon,L}^{\mathrm{base}}\widehat{\otimes}C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1]}, \tag{7}$$

where the spacetime piece $P_{\epsilon,L}^{\mathrm{base}}$ on $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)$ is the standard heat-kernel BV inverse on the bulk (Costello, op. cit., Ch. 5), and the coefficient piece is the BV inverse pairing on $\widehat{\mathfrak{h}} \oplus \widehat{\mathfrak{h}}_{\mathrm{cont}}^\vee[1]$. By Remark 3.181, the spacetime piece exists as an actual continuous operator. The whole kernel exists as an element of $\mathcal{E}\widehat{\otimes}\mathcal{E}$ *iff* the coefficient piece, the bare Casimir $C_{\mathfrak{h}}$ together with its dual coevaluation, exists as a coefficient kernel.

Costello's heat-kernel BV requires three properties of the coefficient module:

(C1) an actual distributional inverse pairing, i.e. a coefficient coevaluation $C_{\mathfrak{h}} \in \mathfrak{h}\widehat{\otimes}\mathfrak{h}_{\mathrm{cont}}^\vee$ realizing the bilinear form with kernel zero in the cohomology of the heat operator;

(C2) regularity in the cutoff parameters $\epsilon$ and $L$, in the sense that the heat-kernel propagator $P_{\epsilon,L}$ is a continuous family in the coefficient topology and converges as $L \to \infty, \epsilon \to 0^+$ to a distributional kernel after counterterm subtraction;

(C3) compatibility with the Costello RG flow $W(P_{\epsilon,L}, I)$: the weight or topology defining the coefficient module is preserved by the renormalization flow on effective interactions induced by varying $(\epsilon, L)$, in the precise sense that the source of the flow lies in the same completed coefficient category as its target.

The strict Tate pair (6) fails (C1); the subsection below repairs the coefficient kernel and the finite-scale propagator by a weighted Mittag-Leffler completion. Requirement (C3) is not a formal consequence of the coefficient change; it is the residual locality/QME problem isolated in Problem 3.68.

*Weighted Tate completion: definition*

*Definition* 3.55 *(Polynomial Hamiltonian degree weight).* A *degree weight* is a function $w\colon \mathbb{Z}_{\geq 1} \to \mathbb{R}_{>0}$ satisfying:

(W1) *Bracket-tame:* there exists a constant $C_w \geq 0$ such that $w(p+q-2) \leq C_w\, w(p)w(q)$ for every $p, q \geq 2$;

(W2) *Windowwise coadjoint-continuity:* for every Hamiltonian degree bound $R$ and every output Tate window $w_0$, the principal-part coadjoint shifts $\rho_{a,b} \mapsto \rho_{a-p+1,b-q+1}$ with $p+q \leq R$ and $a+b-p-q+2 \leq w_0$ involve only the finite input window $a+b \leq w_0 + R - 2$, and the finitely many corresponding weight ratios have a maximum;

(W3) *Mittag-Leffler coevaluation:* for each $w_0$ the finite-window Casimir $\sum_{d \le w_0} \sum_i w(d) e_{d,i} \otimes w(d)^{-1} e^i_d$ is compatible under the surjective truncation maps $w'_0 \to w_0$. Equivalently, the coefficient kernel is a strict Mittag-Leffler inverse system of finite-dimensional kernels.

LEMMA 3.56 *(Existence of weights).* Every exponential weight $w(d) = q^d$ with $q > 1$ satisfies (W1)–(W3). The proof uses only finite-window maxima for the coadjoint action; there is no uniform bound in the unbounded principal-part degree.

*Proof.* For $w(d) = q^d$, (W1) reads $q^{p+q-2} \le C_w q^{p+q}$, i.e. $C_w = q^{-2}$ suffices. For (W2), fix $R$ and $w_0$. If $p + q \le R$ and the output degree is $a + b - p - q + 2 \le w_0$, then $a + b \le w_0 + R - 2$. Hence the set of possible triples $(p, q, a + b)$ is finite, and the corresponding ratios of powers of $q$ admit a finite maximum. No bound independent of $a + b$ is asserted; the earlier uniform inequality would be false. Condition (W3) is the strict Mittag-Leffler compatibility of finite-window Casimirs under truncation. The transition maps are surjective, hence the system is Mittag-Leffler. □

*Remark* 3.57 *(Choice of exponential weight).* The exponential weight $w(d) = q^d$ is the working choice, with $q > 1$ formally an element of $\mathbb{R}_{>0}$ and concretely set to $q = 2$ for the proofs below. The choice is a grading weight on the coefficient module, not a symplectic automorphism of the formal disk. Thus the Poisson bracket and the counterterm representatives are compared between two weights only after transport by the diagonal map $R_{w,w'}$ below. In coordinates, the bracket constants in the basis $w(d)e_{d,i}$ are conjugated by the finite ratios $w(p)w(q)/w(p + q - 2)$, and condition (W1) is exactly the boundedness condition needed for this transported bracket on each finite graph. The weight is therefore a controlled coefficient regulator, not an assertion that rescaling the variables $z_1$, $z_2$ preserves the original Hamiltonian bracket.

*Definition* 3.58 *(Weighted Tate coefficient pair).* Given a degree weight $w$ as in Definition 3.55, write $H_d^{(w)}$ for the rescaled copy of the finite-dimensional degree piece $H_d$ with basis $w(d)e_{d,i}$, and $(H_d^\vee)^{(w^{-1})}$ for the rescaled dual basis $w(d)^{-1}e^i_d$. Define

$$\mathfrak{h}_w = \varprojlim_{w_0 \in \mathbb{Z}_{\ge 1}} \prod_{d=1}^{w_0} H_d^{(w)}$$

with the Tate filtration $F^{w_0}\mathfrak{h}_w = \ker(\mathfrak{h}_w \to \prod_{d \le w_0} H_d^{(w)})$, and define the *weighted coefficient dual*

$$\mathfrak{h}^{\vee,w}_{\mathrm{cont}} = \varprojlim_{w_0 \in \mathbb{Z}_{\ge 1}} \prod_{d=1}^{w_0} (H_d^\vee)^{(w^{-1})}.$$

The pairing is fixed degreewise by $\langle w(d)^{-1}e^i_d, w(d)e_{d,j}\rangle = \delta^i_j$. The weighted Tate coefficient pair is $(\mathfrak{h}_w, \mathfrak{h}^{\vee,w}_{\mathrm{cont}})$. The second factor is a product coefficient module, not the strict continuous dual of the first product Tate space.

*Remark* 3.59 *(Topological content).* Both $\mathfrak{h}_w$ and $\mathfrak{h}^{\vee,w}_{\mathrm{cont}}$ are Mittag-Leffler inverse limits of finite-dimensional Tate windows, and both are products in the degree direction. The two are no longer in strict Tate duality (one product, one direct sum); the point is precisely to replace the direct-sum dual by a symmetric Mittag-Leffler product coefficient module. This is an additional coefficient category for the BV kernel. It does not identify $\mathfrak{h}^{\vee,w}_{\mathrm{cont}}$ with the ordinary continuous dual of $\mathfrak{h}_w$.

Proposition 3.60 *(Casimir convergence).* In the weighted Tate coefficient pair $(\mathfrak{h}_w, \mathfrak{h}_{\mathrm{cont}}^{\vee,w})$ of Definition 3.58, the formal Casimir

$$C_{\mathfrak{h},w} = \sum_{d\geq 1}\sum_i \big(w(d)e_{d,i}\big)\otimes\big(w(d)^{-1}e_d^i\big) \tag{8}$$

converges as an element of $\mathfrak{h}_w\widehat{\otimes}\mathfrak{h}_{\mathrm{cont}}^{\vee,w}$. The pairing of $C_{\mathfrak{h},w}$ with any test functional in the dual is the identity on the corresponding finite-dimensional Tate window, so $C_{\mathfrak{h},w}$ is the BV coefficient inverse pairing in the weighted category.

*Proof.* The displayed Casimir, restricted to the Tate window $d \leq w_0$, is the finite sum $\sum_{d\leq w_0}\sum_i(w(d)e_{d,i}) \otimes (w(d)^{-1}e_d^i)$, an element of $\prod_{d\leq w_0} w(d)H_d \otimes \prod_{d\leq w_0} w(d)^{-1}H_d^\vee$. This is a tensor in a finite-dimensional vector space because each window is finite-dimensional. The tower of finite-window Casimirs is compatible: enlarging $w_0$ extends the sum by the terms in degrees $w_0+1,\dots,w_0'$. The inverse limit in **TateVec** exists by Definition 3.58 and (W3) of Definition 3.55.

Pairing the Casimir with a test functional $\phi \in \mathfrak{h}_{\mathrm{cont}}^{\vee,w}$, supported in finitely many Tate windows, gives a finite sum of identity pairings, so the result is the canonical evaluation. Hence $C_{\mathfrak{h},w}$ realizes the inverse pairing on the weighted Tate pair, satisfying Costello requirement (C1). □

*Remark* 3.61 *(Independence of basis).* Definition (8) is written in a basis $\{e_{d,i}\}$ of $H_d$, but does not depend on the choice. The dual basis $\{e_d^i\}$ is fixed by $\langle e_{d,i}, e_d^j\rangle = \delta_i^j$. An orthogonal change of basis $e_{d,i} \mapsto \sum_j O_i^j e_{d,j}$ with $O \in GL(H_d)$ shifts the dual by the inverse transpose, yielding the same Casimir. Thus $C_{\mathfrak{h},w} = \sum_d \mathrm{id}_{H_d\otimes H_d^\vee}$ is the basis-independent identity element of $\bigoplus_d H_d \otimes H_d^\vee$ extended to the weighted product.

*Heat-kernel propagator on the weighted pair*

Proposition 3.62 *(Heat-kernel propagator extension to the weighted pair).* Let $w$ be a degree weight as in Definition 3.55. The Costello heat-kernel propagator $P_{\epsilon,L}^{\mathrm{op}}$ of Remark 3.181 extends from the strict Tate field complex to the weighted field complex

$$\mathcal{E}_w = \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\mathfrak{h}_w[1]\oplus\mathfrak{h}_{\mathrm{cont}}^{\vee,w}[2]\big), \tag{9}$$

to give a continuous family of BV propagators $P_{\epsilon,L}^w \in \mathcal{E}_w\widehat{\otimes}\mathcal{E}_w$ for $0<\epsilon\leq L<\infty$, satisfying:

(1) the BV identity $[Q, P_{\epsilon,L}^w] = K_\epsilon^w - K_L^w$, where $K_t^w = K_t^{\mathrm{base}}\widehat{\otimes}C_{\mathfrak{h},w}$ is the weighted heat kernel of the BV inverse pairing;
(2) Costello regularity (C2): the family is continuous in $(\epsilon, L)$ in the weighted Tate topology;
(3) the limit $L\to\infty$, $\epsilon\to 0^+$ exists as a distributional kernel after the standard spacetime counterterm subtraction in Costello's heat-kernel RG package [12, Chs. 5–7]; the coefficient direction introduces no additional spacetime singular support. Locality and QME counterterm compatibility for the mixed brane-defect theory remain the content of Problem 3.68.

*Proof.* By the factorization (7), the propagator splits as a tensor product of the spacetime kernel $P_{\epsilon,L}^{\mathrm{base}}$ on $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)$ and the coefficient Casimir $C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1],w}$ on the weighted Tate pair. The spacetime piece is unchanged by the weight on the coefficient direction; it is the Costello heat-kernel inverse on the bulk free elliptic complex $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)$, constructed in [12, Ch. 5]. The BV identity for $P_{\epsilon,L}^{\mathrm{base}}$

is standard: with $Q^{\mathrm{base}} = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$, gauge-fixing $Q^{\mathrm{GF}} = d^*_{\mathbb{R}^2} + \bar{\partial}^*_{\mathbb{C}^2}$, and the associated Laplacian $\Delta = [Q^{\mathrm{base}}, Q^{\mathrm{GF}}]$, one has

$$[Q^{\mathrm{base}}, P^{\mathrm{base}}_{\epsilon,L}] = \int_\epsilon^L [Q^{\mathrm{base}}, Q^{\mathrm{GF}} e^{-t\Delta}]\, dt = \int_\epsilon^L \partial_t(-e^{-t\Delta})\, dt = e^{-\epsilon\Delta} - e^{-L\Delta} = K^{\mathrm{base}}_\epsilon - K^{\mathrm{base}}_L.$$

The coefficient piece is the Casimir of Proposition 3.60, which is annihilated by the trivial differential on $\mathfrak{h}_w[1] \oplus \mathfrak{h}^{\vee,w}_{\mathrm{cont}}[2]$. Therefore $[Q, P^w_{\epsilon,L}] = K^w_\epsilon - K^w_L$, and the family is continuous in $(\epsilon, L)$ because $C_{\mathfrak{h},w}$ is a fixed element of the weighted pair and the bulk family is continuous. Item (3), at the level asserted here, is the Costello bulk regularity package [12, Chs. 5–7] tensored with a fixed coefficient kernel. This proves finite-scale kernel regularity and absence of new coefficient wave-front directions. It does not by itself prove vanishing of the brane-defect QME obstruction classes. □

*Remark* 3.63 *(Brane defect).* The brane defect $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\} \subset \mathbb{R}^2 \times \mathbb{C}^2$ pulls the weighted field complex $\mathcal{E}_w$ to its restriction along the defect:

$$\mathcal{E}^\partial_w = \Omega^\bullet(\mathbb{R}) \widehat{\otimes} \big(\mathfrak{h}_w[1] \oplus \mathfrak{h}^{\vee,w}_{\mathrm{cont}}[2]\big).$$

The boundary propagator $P^{w,\partial}_{\epsilon,L} \in \mathcal{E}^\partial_w \widehat{\otimes} \mathcal{E}^\partial_w$ is the bulk-to-boundary restriction of $P^w_{\epsilon,L}$ followed by the zero-mode-removed midpoint propagator $G(t,s) = \frac{1}{2}\operatorname{sgn}(t-s)$ of Proposition 3.191. The weighted Casimir $C_{\mathfrak{h},w}$ is the same on the boundary as in the bulk; the weight $w$ does not interact with the coordinate $t$.

*Renormalization-flow compatibility*

The Costello RG flow on effective interactions is the assignment $I_L = W(P_{\epsilon,L}, I)$ that converts a local interaction $I$ at scale $\epsilon$ into the effective interaction at scale $L$, where $W$ is the homotopy renormalization group action on functionals [12, Ch. 6]. For the weighted mechanism to satisfy Costello requirement (C3), the RG flow on $\mathcal{E}_w$-valued interactions must preserve the weighted coefficient module: starting from a polynomial-degree-bounded interaction in $\mathcal{O}(\mathcal{E}_w)$, the effective interaction $I_L$ must remain in the same weighted category.

PROPOSITION 3.64 *(Graphwise RG-tame interactions).* Let $I \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ be a local interaction polynomial in the fields, of total polynomial-degree bounded by $D_0$ at each vertex in the coefficient direction. For every fixed Costello graph $\Gamma$ and every $0 < \epsilon \le L < \infty$, the graph contribution $W_\Gamma(P^w_{\epsilon,L}, I)$ has finite coefficient degree bounded by $D_0 V(\Gamma) - 2E(\Gamma)$ and lies in the weighted completed coefficient category. The bracket-tame estimate (W1), together with the finite-window coadjoint condition (W2), bounds this single graph by a finite constant times the corresponding spacetime graph weight.

*Proof.* The interaction $I$ has bounded polynomial degree $D_0$ at each vertex in the coefficient direction by hypothesis. A connected Costello graph $\Gamma$ with $V$ vertices, each of polynomial degree $\le D_0$, and $E$ internal edges, contributes a coefficient pairing of external length at most $V \cdot D_0 - 2E$ in the source factor and a matching length in the target. By the bracket-tame estimate (W1), each internal edge contraction $C_{\mathfrak{h},w} : \mathfrak{h}_w \widehat{\otimes} \mathfrak{h}^{\vee,w}_{\mathrm{cont}} \to \mathbb{C}$ changes the total weight by at most the finite constant attached to the degrees present in that graph. Coadjoint legs are controlled windowwise by (W2). Hence the graph contribution is a well-defined element of the weighted completion. No assertion is made that the full effective interaction preserves the original degree bound $D_0$; Costello's expansion is a completed sum over graph degrees. □

Theorem 3.65 *(Weighted RG flow, graphwise compatibility).* Let $w$ be a degree weight as in Definition 3.55 and let $I \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ be a local interaction satisfying a vertexwise polynomial-degree bound $D_0$. Each fixed graph contribution to the Costello RG flow

$$I_L = W(P^w_{\epsilon,L}, I)$$

lies in the weighted Tate completion and is continuous in $(\epsilon, L)$. The theorem is graphwise. The completed sum over counterterms and the QME obstruction class require the locality input of Problem 3.68.

*Proof.* By Proposition 3.64, every Feynman diagram considered individually remains in the weighted Tate category. Its finite coefficient-degree bound is $D_0 V(\Gamma) - 2E(\Gamma)$, and continuity in $(\epsilon, L)$ follows from Proposition 3.62 (2). Passing from graphwise well-definedness to a renormalized QME solution is precisely the residual analytic problem. □

*Cotangent-lift theorem*

Theorem 3.66 *(Cotangent-lift inside the weighted coefficient category).* Let $w(d) = q^d$ with $q > 1$ be the exponential degree weight of Remark 3.57. Then in the weighted Tate coefficient pair $(\mathfrak{h}_w, \mathfrak{h}^{\vee,w}_{\mathrm{cont}})$ of Definition 3.58 the following internal weighted-category statements hold:

(a) the Casimir $C_{\mathfrak{h},w} = \sum_d \sum_i w(d) e_{d,i} \otimes w(d)^{-1} e^i_d$ converges as an element of $\mathfrak{h}_w \widehat{\otimes} \mathfrak{h}^{\vee,w}_{\mathrm{cont}}$ (Proposition 3.60);

(b) the Costello heat-kernel propagator $P^w_{\epsilon,L}$ extends to the weighted field complex $\mathcal{E}_w$ as a continuous family in $(\epsilon, L)$ satisfying the BV identity (Proposition 3.62);

(c) each fixed Costello graph contribution preserves the weighted category on vertexwise polynomial-degree-bounded interactions (Theorem 3.65);

(d) the cochain-level $\Phi$ of Theorem 3.27 extends to the weighted cotangent extension $\mathfrak{g}_w = \mathfrak{h}_w \ltimes \mathfrak{h}^{\vee,w}_{\mathrm{cont}}[1]$ as a continuous $P_0$-algebra cochain map in the weighted coefficient envelope

$$\Phi^w \colon C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_w) \to \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h}_w)\big);$$

*Proof.* Items (a)–(c) are Propositions 3.60, 3.62, and Theorem 3.65. For item (d): the classical $\Phi$ of Theorem 3.27 acts on continuous CE cochains by the cochain identity $\Phi d_{\mathrm{CE}} = d_\pi \Phi$, which is a degree-by-degree algebraic identity on the polynomial Hamiltonian Lie algebra $\mathfrak{h}$ together with its dual $\mathfrak{h}^\vee[1]$. The identity is preserved under any rescaling of the bases by an invertible weight, so it descends to the weighted pair. The $P_0$-algebra structure on the source is the CE cochain $P_0$-algebra structure formed in the weighted coefficient envelope; this is not the strict continuous duality statement of Lemma 3.137 for a product/direct-sum Tate pair. The needed dual generators are inserted by the symmetric Mittag-Leffler coefficient pairing of Remark 3.59. On the target, the Schouten bracket descends to the weighted symmetric algebra by (W2).

No all-order QME or regulator-independent statement is part of this theorem; the quantum assertion is separated in Proposition 3.67. □

Proposition 3.67 *(Conditional weighted QME lift).* Assume the Costello-locality/QME compatibility identity of Problem 3.68: the finite-window counterterm system for the mixed brane-defect interaction has vanishing obstruction class and is compatible with the weighted coefficient category.

Under this assumption, the formal quantum derived-center map $\Phi_{\hbar}^{(0)}$ governed by Theorem 3.187 has a weighted-continuous cotangent extension

$$\Phi^{w}_{\hbar,\mathrm{cond}}\colon C^{\bullet}_{\mathrm{CE,cont}}\big(\mathfrak{h}_{w,\hbar} \ltimes \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}[1]\big)[[\hbar]] \xrightarrow[\text{conditional QME}]{\sim} Z^{P_{0,\hbar}}_{\mathrm{fact}}\big(\mathrm{Obs}^{\hbar}_{\mathrm{open}}\big)|_{\mathfrak{g}_w}.$$

The displayed isomorphism is a theorem only in this conditional weighted-continuous sector. It is not an unconditional all-order lift for the original unweighted Tate pair.

*Proof.* The algebraic cochain identity is supplied by Theorem 3.66. The BV graph realization decomposes the cotangent action $\mathrm{ad}^{\vee}$ into Costello graphs in $\mathcal{E}_w$, and Theorem 3.65 keeps each fixed graph inside the weighted category. What is not formal is the cancellation between Costello-locality counterterms and the bulk-boundary kernel at the brane defect. The proposition assumes exactly that weighted QME compatibility and the vanishing of the obstruction class named in Problem 3.68. With that assumption in force, the formal $\hbar$-adic construction of Theorem 3.187 applies in the weighted coefficient envelope. Without it, the statement stops at the graphwise finite-scale assertions above. □

*Problem* 3.68 *(Costello-locality compatibility for the weighted RG flow).* Verify that the Costello QME counterterms at the brane defect $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ in the weighted field complex $\mathcal{E}_w$ remain local-functional classes $O \in H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\})$, i.e. that the bulk Costello locality package [12, Chs. 5–7] extends to the weighted pair. The exact published theorem numbering is not an input to this problem. Concretely: a counterterm contribution $C_n \in \mathcal{O}^{\hbar^n}_{\mathrm{loc}}(\mathcal{E})$ at order $\hbar^n$ in the standard pair is a polynomial functional of bounded degree; one must check that the same counterterm, rescaled by the weight $w$ on each coefficient leg, remains a local functional of the same bounded polynomial degree in the weighted pair. The estimate (W1) bounds vertex weights, and (W2) bounds coadjoint legs graph by graph. The remaining question is whether the compatible finite-window counterterm system obtained from Costello's Hadamard locality theorem has vanishing obstruction class in $H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\})$ for the mixed brane-defect interaction. The Hadamard / Mittag-Leffler reduction of Theorem 3.111 controls the wave-front and finite-window locality estimates; it does not by itself compute that obstruction class.

COROLLARY 3.69 *(Conditional descendant cochain comparison).* Assume the obstruction class in Problem 3.68 vanishes, so that Proposition 3.67 applies. Then the Tate-coefficient lift of the descendant cochain comparison of Problem 3.189 exists in the weighted Tate completion only under this hypothesis: the deformation

$$\Phi^{(0,1)}_{\hbar,\mathrm{cond}}\colon C^{\bullet}_{\mathrm{CE,cont}}\big(\mathfrak{h}_{w,\hbar} \ltimes \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}[1]\big)^{(0)+(1)}[[\hbar]] \xrightarrow[\text{conditional QME}]{\sim} Z^{P_{0,\hbar}}_{\mathrm{fact}}\big(\mathrm{Obs}^{\hbar}_{\mathrm{open}}\big)^{(0)+(1)}|_{\mathfrak{g}_w}$$

is an isomorphism on the degree-zero plus one-antifield sector only after the conditional weighted QME input has been supplied. It does not assert an unconditional all-order descendant lift.

*Remark* 3.70 *(Sector restriction).* The weighted mechanism restricts the cotangent direction from the ambient polynomial Hamiltonian dual to the weighted dual $\mathfrak{h}^{\vee,w}_{\mathrm{cont}} = \prod_d w(d)^{-1} H^{\vee}_d$. The strict polynomial Hamiltonian dual $\mathfrak{h}^{\vee}_{\mathrm{cont}} = \bigoplus_d H^{\vee}_d$ is a *dense subspace* of $\mathfrak{h}^{\vee,w}_{\mathrm{cont}}$ for every $w$ with $w(d) \to \infty$, since the strict direct sum is finite-degree truncations and the weighted product is the weighted Mittag-Leffler inverse-limit completion. Hence the weighted lift extends the cochain map continuously from the strict polynomial sector to the weighted product completion. This recovers only the weighted-continuous Hamiltonian identities, not an unweighted regulator-independence statement and not an identification of the weighted product with the original strict continuous dual.

*Regulator independence and its boundary*

*Definition 3.71 (Regulator-admissible finite-window sector).* A local functional, interaction, obstruction cocycle, or graph coefficient in the weighted theory is *regulator-admissible* if it is represented by a strict Mittag-Leffler family on the finite Tate windows

$$\{\mathcal{E}_w^{w_0}\}_{w_0 \geq 1}$$

and satisfies the following four conditions.

(A1) The transition maps in the coefficient and local-functional directions are the truncations to degrees $d \leq w_0$, or the induced restriction obtained by setting higher-degree coefficient fields to zero. They are degreewise surjective maps of complexes.

(A2) At each fixed $\hbar$-order and for each fixed Costello graph, the coefficient expression uses only vertex-wise polynomial-degree-bounded Hamiltonian and cotangent legs.

(A3) If $w, w'$ are two degree weights satisfying Definition 3.55, the diagonal finite-window rescaling

$$R_{w,w'}^{w_0}(w(d)e_{d,i}) = w'(d)e_{d,i}, \qquad R_{w,w'}^{w_0}(w(d)^{-1}e_d^i) = w'(d)^{-1}e_d^i$$

transports the interaction, differential, bracket, propagator contraction, and counterterm representative from the $w$-window to the $w'$-window. The bracket is the transported bracket; the theorem does not compare two weights while holding the raw unweighted structure constants fixed.

(A4) The cohomology being computed is the admissible filtered cohomology detected on finite quotients and associated graded pieces in the envelope of Statement 3.28; no observable is allowed to detect an infinite product tail invisible in every finite quotient.

*Remark 3.72 (Closure of the admissible class under Costello RG flow).* Under the Costello RG-flow $W(P_{\epsilon,L}, I)$ of [12, §3–4], the regulator-admissible class 3.71 is closed: the flow preserves the strict Mittag-Leffler property, the polynomial-degree bound, the diagonal rescaling compatibility, and the admissible filtered cohomology. This closure is not a separate hypothesis here; it is implicit in the (CC3) compatibility recorded above and is named for referee orientation.

Theorem 3.73 *(Regulator independence inside the admissible finite-window sector).* Let $w, w'$ be degree weights satisfying Definition 3.55, and let $I_0^w$ be a regulator-admissible mixed brane-defect interaction. Put $I_0^{w'} = R_{w,w'} I_0^w$. Then the diagonal rescalings $R_{w,w'}^{w_0}$ assemble to an isomorphism of strict Mittag-Leffler local-functional complexes

$$R_{w,w'} : \varprojlim_{w_0} \big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w^{w_0}), Q + \{I_0^w, -\}\big) \xrightarrow{\cong} \varprojlim_{w_0} \big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_{w'}^{w_0}), Q + \{I_0^{w'}, -\}\big).$$

Under this isomorphism:

(i) truncated Casimirs, heat-kernel propagators, and fixed graphwise Costello contractions are identified;

(ii) the obstruction class

$$\mathrm{Ob}_w \in H^1_{\mathrm{adm}}(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0^w, -\})$$

maps to $\mathrm{Ob}_{w'}$;

(iii) vanishing or nonvanishing of the mixed brane-defect QME obstruction is independent of the admissible weight.

This theorem proves independence of the *weight* among admissible regulators. It does not identify the weighted product coefficient theory with the original strict product/direct-sum Tate pair, where the Costello coefficient Casimir does not exist.

*Proof.* On the finite window $d \le w_0$, the map $R^{w_0}_{w,w'}$ is an invertible linear map on the coefficient space and the identity on the spacetime factor. The bracket and the interaction in the $w'$-window are, by admissibility, the transported bracket and transported interaction. The map sends

$$\sum_{d \le w_0} \sum_i (w(d) e_{d,i}) \otimes (w(d)^{-1} e^i_d)$$

to the corresponding $w'$-weighted truncated Casimir. Hence it sends $P^{w,w_0}_{\epsilon,L}$ to $P^{w',w_0}_{\epsilon,L}$, commutes with the differential $Q$, and carries the transported BV bracket and every finite-window graph contraction to the $w'$-window contraction. Condition (A3) gives the same statement for $I_0$, the counterterm representatives, and the obstruction cocycle.

The maps $R^{w_0}_{w,w'}$ commute with truncation in $w_0$. The finite-window complexes therefore form an isomorphic pair of strict Mittag-Leffler inverse systems. By (A1), these are degreewise surjective towers of complexes. Hence the inverse limit is exact in the admissible envelope of Statement 3.28; equivalently $\varprojlim^1 = 0$ for this tower. Passing to the inverse limit gives the displayed isomorphism of local-functional complexes and identifies their admissible filtered cohomology classes. In particular the obstruction class vanishes for $w$ if and only if the transported obstruction class vanishes for $w'$.

The last sentence is the support calculation of Lemma 3.182: the strict continuous dual is a direct sum in degree, so the full Casimir has infinite support in a direct-sum factor. The weighted theorem compares only completed product coefficient theories for which the finite-window Casimirs form a strict Mittag-Leffler system. □

THEOREM 3.74 *(No strict unweighted finite-window endpoint).* Let

$$\mathfrak{h}_\Pi = \prod_{d \ge 1} H_d, \qquad D_\oplus = \bigoplus_{d \ge 1} H^\vee_d$$

be the original product/direct-sum Tate pair. There is no tensor $K \in \mathfrak{h}_\Pi \widehat{\otimes} D_\oplus$, and no tensor in $D_\oplus \widehat{\otimes} \mathfrak{h}_\Pi$, whose projection to every finite window $d \le w_0$ is the finite Casimir

$$C^{w_0}_{\mathfrak{h}} = \sum_{d \le w_0} \sum_i e_{d,i} \otimes e^i_d.$$

Consequently the strict unweighted pair cannot be the endpoint of a finite-window Costello regulator theory satisfying the coefficient kernel requirement (C1) and the windowwise compatibility used in Theorem 3.73.

Moreover, for any admissible weight $w$, the inclusion $D_\oplus \hookrightarrow D_{\Pi,w} := \prod_{d \ge 1} w(d)^{-1} H^\vee_d$ is only an admissible filtered weak equivalence. It is not an ordinary quasi-isomorphism of chain complexes with zero differential. Thus any observable or obstruction class detecting the quotient $D_{\Pi,w}/D_\oplus$ is outside the proved regulator-independent sector.

*Proof.* The direct-sum factor $D_\oplus$ is coefficient-finite by Lemma 3.137 item (1): every element of a completed tensor product with $D_\oplus$ has support in only finitely many $H^\vee_d$-degrees. If $K$ projected to $C^{w_0}_{\mathfrak{h}}$ for all $w_0$, then for every degree $d$ the $H^\vee_d$-component of $K$ would be nonzero. This is infinite support in the direct-sum direction, a contradiction. The same argument applies after reversing the tensor factors.

Costello's heat-kernel BV propagator for the cotangent coefficient field is $P^{\mathrm{base}}_{\epsilon,L}\widehat{\otimes}K$. Since no such $K$ exists in the strict product/direct-sum pair, the strict pair has an operator-level heat homotopy but no coefficient distributional inverse pairing and hence no graphwise regulator endpoint satisfying the displayed finite-window Casimir condition.

Finally, with zero differential,

$$H^0(D_{\Pi,w}/D_{\oplus}) = D_{\Pi,w}/D_{\oplus}.$$

The class of any sequence with one nonzero covector in every degree is nonzero in this quotient. Therefore the completion changes ordinary cohomology even though finite quotients and associated graded pieces cannot see the tail. This proves the asserted boundary of regulator independence. □

*Example* 3.75 *(The completion changes ordinary cohomology).* Let

$$D_{\oplus} = \bigoplus_{d\geq 1} H_d^{\vee}, \qquad D_{\Pi,w} = \prod_{d\geq 1} w(d)^{-1} H_d^{\vee}$$

with zero differential. The finite-window inclusion $D_{\oplus} \hookrightarrow D_{\Pi,w}$ induces an isomorphism on every quotient by the tail $d > w_0$, and therefore is a weak equivalence in the admissible filtered sense of Statement 3.28. It is not a quasi-isomorphism in the ordinary category of chain complexes:

$$H^0(D_{\Pi,w}/D_{\oplus}) = D_{\Pi,w}/D_{\oplus} \neq 0,$$

for example the class of any sequence with one nonzero covector in every degree is nonzero. Thus the weighted product contains tail states absent from the strict continuous dual. Any observable detecting such a tail is outside the regulator-admissible sector of Definition 3.71.

*Example* 3.76 *(Bad weights do not define the same regulator problem).* The super-exponential function $w(d) = \exp(d^2)$ is not a degree weight. Indeed

$$\frac{w(p+q-2)}{w(p)w(q)} = \exp\big((p+q-2)^2 - p^2 - q^2\big)$$

is unbounded, already along $p = q \to \infty$. Hence condition (W1) fails. The Poisson bracket is not bracket-tame in that coefficient topology, and the graphwise constants in Proposition 3.64 need not be finite uniformly in the degrees used by the interaction. The regulator-independence theorem above does not compare such a topology with the admissible weighted theory.

*Remark* 3.77 *(Exact status of the weighted mechanism).* The weighted Tate construction has four distinct statuses. It is a regulator for the finite-window graphwise sector; it is a filtered weak replacement of the strict direct-sum dual only inside the admissible Mittag-Leffler envelope; it is a genuine completion changing the ordinary chain complex once infinite product tails are admitted; and it is not a theorem about the original unweighted Costello graph calculus, because that calculus has no coefficient Casimir in the strict product/direct-sum topology.

*Convention firewall*

*Remark* 3.78 *(No transfer to Vol III).* The weighted Tate completion is a local-disk operation on the formal symplectic plane $\mathbb{C}^2$ with brane defect $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ and bulk $\mathbb{R}^2 \times \mathbb{C}^2$. It carries no compact Calabi-Yau datum, no $\Phi_d$ functor, no $\kappa_{\mathrm{ch}}$ or $\kappa_{\mathrm{BKM}}$ consequence, no $K3 \times E$ specialization, no Igusa cusp form, no Borcherds input form, no BKM denominator identity, and no Stage-1 or Stage-2 $\Phi_d$ specialization. It addresses exclusively the analytic obstruction named in Lemma 3.182 and Problem 3.189, and does so on the formal local model.

## NILPOTENT TRUNCATION OF THE COTANGENT SHADOW

The main theorem 2.27 is unconditional for the cochain-level derived-centre, principal-part, brane-line factorization, degree-zero Hamiltonian Moyal, and scalar-anomaly statements. Part (d) is proved at the reduced principal-part level after de Rham-current contraction; its unreduced primitive shadow and part (e) on the descendant (one-antifield) sector are governed by 3.47, the unreduced Tate-coefficient cotangent lift. The deciding obstructions are 3.182 (Casimir support fails in the direct-sum dual) and 3.180 (no naive finite-dimensional Lie truncation of the full sector). Two routes were named in 3.47: a weighted Tate completion or a nilpotent truncation. This section closes the classical nilpotent-truncation route on a precisely specified truncation, records what is lost, and records the quantum Moyal obstruction that prevents the same finite truncation from being an all-order descendant theorem.

*The truncation*

Let $\mathfrak{h} = (z_1, z_2)\mathbb{C}[[z_1, z_2]] = \mathfrak{m}$ be the reduced Hamiltonian Lie algebra of the formal symplectic disk on $\mathbb{C}^2$, with bracket $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$ modulo constants. Let $\mathfrak{m}^k = (z_1, z_2)^k \mathbb{C}[[z_1, z_2]]$ for $k \geq 1$ be the $k$-th power of the maximal ideal. Fix integers $N \geq k \geq 3$ and define the *nilpotent truncation*

$$\mathfrak{h}^{(k)}_{\leq N} := \mathfrak{m}^k / \mathfrak{m}^{N+1},$$

with $\mathfrak{m}$-adic Poisson bracket inherited from the ambient $\mathfrak{h}_{\mathrm{poly}}$. The standard choice is $k = 3$:

$$\mathfrak{h}_{\leq N} := \mathfrak{h}^{(3)}_{\leq N} = \mathfrak{m}^3 / \mathfrak{m}^{N+1}.$$

Concretely, $\mathfrak{h}_{\leq N}$ is the finite-dimensional vector space spanned by monomials $z_1^a z_2^b$ with $3 \leq a + b \leq N$.

*Remark* 3.79 *(Why $k = 3$ and not $k = 2$).* The naive choice $k = 2$ fails to give a nilpotent Lie algebra: the degree-2 piece $H_2 = \mathrm{span}\{z_1^2, z_1 z_2, z_2^2\}$ is closed under the Poisson bracket because $\{H_2, H_2\} \subset H_{2+2-2} = H_2$, and the structure is exactly $\mathfrak{sl}_2$:

$$\{z_1^2, z_2^2\} = 4 z_1 z_2, \quad \{z_1^2, z_1 z_2\} = 2 z_1^2, \quad \{z_1 z_2, z_2^2\} = 2 z_2^2.$$

Thus $\mathfrak{m}^2/\mathfrak{m}^{N+1}$ contains a nonabelian semisimple subalgebra and is not nilpotent, hence does not satisfy the Tate-conilpotency hypothesis of 3.137. The truncation $k = 3$ removes the $H_2$ direction; the bracket on $\mathfrak{m}^3$ raises $\mathfrak{m}$-adic degree strictly, making the truncation nilpotent.

LEMMA 3.80 *(Truncation is a finite-dimensional nilpotent Lie algebra).* For every $N \geq 3$:

(i) $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ is a finite-dimensional Lie algebra of dimension $\binom{N+2}{2} - 6$ (the number of monomials of degree between 3 and $N$ inclusive).

(ii) $\mathfrak{h}_{\leq N}$ is nilpotent of class at most $N - 2$ in the lower central series.

(iii) $\mathfrak{h}^\vee_{\leq N}$ is finite-dimensional, hence the formal Casimir $C_{\mathfrak{h}_{\leq N}} = \sum_{3 \leq d \leq N} \sum_i e_{d,i} \otimes e^i_d$ is a convergent element of $\mathfrak{h}_{\leq N} \otimes \mathfrak{h}^\vee_{\leq N}$ (a finite sum, no Tate completion required).

(iv) The linear Hamiltonians $z_1, z_2 \in \mathfrak{m} \setminus \mathfrak{m}^3$, the conformal-symplectic generators $H_2 = \mathrm{span}\{z_1^2, z_1 z_2, z_2^2\} \cong \mathfrak{sl}_2$, and the constant Hamiltonian $1 \notin \mathfrak{m}$ are absent from $\mathfrak{h}_{\leq N}$ by construction.

*Proof. (i)* Both $\mathfrak{m}^3$ and $\mathfrak{m}^{N+1}$ are Lie ideals of the polynomial Hamiltonian Lie algebra $\mathfrak{h}_{\mathrm{poly}}$: for $f \in \mathfrak{m}^p$ and $g \in \mathfrak{m}^q$ one has $\{f, g\} \in \mathfrak{m}^{p+q-2}$, since the Poisson bracket on $\mathbb{C}[z_1, z_2]$ lowers total polynomial degree by exactly two, and each partial derivative lowers $\mathfrak{m}$-adic degree by exactly one. For $p, q \geq 3$ the

bracket lands in $\mathfrak{m}^{p+q-2} \subset \mathfrak{m}^4 \subset \mathfrak{m}^3$, making $\mathfrak{m}^3$ a Lie subalgebra; for $p \geq 3$ and $q \geq N+1$ the bracket lands in $\mathfrak{m}^{p+q-2} \subset \mathfrak{m}^{N+2} \subset \mathfrak{m}^{N+1}$, making $\mathfrak{m}^{N+1}$ a Lie ideal of $\mathfrak{m}^3$. The quotient $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ inherits a well-defined Lie bracket. Dimension count: monomials $z_1^a z_2^b$ with $3 \leq a+b \leq N$ number $\binom{N+2}{2} - 6$ (the total number with $0 \leq a+b \leq N$ minus the six of degrees 0, 1, 2).

*(ii)* The Poisson bracket of two elements of $\mathfrak{m}^3$ raises $\mathfrak{m}$-adic degree by at least 1: $\{H_p, H_q\} \subset H_{p+q-2} \subset \mathfrak{m}^{p+q-2}$, and for $p, q \geq 3$, $p+q-2 \geq 4 > p$ (since $q \geq 3$). Thus the descending Lie series

$$\mathfrak{h}_{\leq N}^{(\geq m)} := \underbrace{[\mathfrak{h}_{\leq N}, [\mathfrak{h}_{\leq N}, \ldots]]}_{m \text{ brackets}} \subset \left(\mathfrak{m}^{m+3}/\mathfrak{m}^{N+1}\right)$$

satisfies $\mathfrak{h}_{\leq N}^{(\geq m)} = 0$ for $m+3 \geq N+1$, i.e. for $m \geq N-2$. Hence $\mathfrak{h}_{\leq N}$ is nilpotent of class at most $N-2$.

*(iii)* Immediate from finite-dimensionality.

*(iv)* Immediate from the $\mathfrak{m}^3$-adic cutoff and the exclusion $a+b \leq 2$. □

*Remark* 3.81 *(Why the $\mathfrak{m}$-adic obstruction does not apply).* 3.180 shows that $\mathfrak{h}/\mathfrak{m}^{N+1}$ is *not* a Lie quotient of the full Hamiltonian algebra $\mathfrak{h}_{\text{poly}}$: the linear Hamiltonian $z_1$ acting on $z_2^{N+1} \in \mathfrak{m}^{N+1}$ produces $\{z_1, z_2^{N+1}\} = (N+1)z_2^N \notin \mathfrak{m}^{N+1}$. The nilpotent truncation $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ *kills the linear Hamiltonian* and the conformal $\mathfrak{sl}_2$ direction from the algebra. The obstruction lemma is then irrelevant: no element of $\mathfrak{h}_{\leq N}$ has $\mathfrak{m}$-adic degree below 3, so the displayed counterexample $\{z_1,\ z_2^{N+1}\}$ never occurs in $\mathfrak{h}_{\leq N}$.

*Remark* 3.82 *(Why the Tate Casimir obstruction does not apply).* The product/direct-sum support obstruction of 3.182 is a feature of the unbounded expansion $\mathfrak{h} = \prod_{d\geq 1} H_d$: the Casimir $C_{\mathfrak{h}} = \sum_d \sum_i e_{d,i} \otimes e_d^i$ has nonzero component in every $H_d^\vee$, and $\bigoplus_d H_d^\vee$ admits no infinite-support element. On the truncation $\mathfrak{h}_{\leq N}$ the Casimir is the finite sum $C_{\mathfrak{h}_{\leq N}} = \sum_{d=3}^N \sum_i e_{d,i} \otimes e_d^i$ in the finite-dimensional vector space $\mathfrak{h}_{\leq N} \otimes \mathfrak{h}_{\leq N}^\vee$. There is no convergence question.

*The classical truncated cotangent CE/PV shadow*

Theorem 3.83 *(Classical truncated cotangent CE/PV shadow).* Set $\mathfrak{g}_{\leq N} = \mathfrak{h}_{\leq N} \ltimes \mathfrak{h}_{\leq N}^\vee[1]$. For every interval $I \subset \mathbb{R}$ let $\mathfrak{g}_{\leq N,I} = \Omega_c^\bullet(I) \widehat{\otimes} \mathfrak{h}_{\leq N} \ltimes \mathcal{D}_c'(I) \widehat{\otimes} \mathfrak{h}_{\leq N}^\vee[1]$. Then, after the Poisson, residue, and interval-smearing conventions fixed above, there is a filtered cochain isomorphism of completed dg graded-commutative $P_0$-algebras

$$\Phi_{\leq N} : C^\bullet_{\text{CE,cont}}(\mathfrak{g}_{\leq N,I}) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}(\mathfrak{h}_{\leq N,I})\big)$$

on every interval $I \subset \mathbb{R}$. This is the finite nilpotent CE/PV cotangent shadow of the reduced Hamiltonian model; it is not an unreduced open BV factorization-centre morphism. The isomorphism is determined on generators by $\Phi_{\leq N}(c^I) = \theta^I$, $\Phi_{\leq N}(u_I) = O_I$, with the same Schouten/Lichnerowicz convention as in 3.27. It is compatible with extension by zero and with disjoint-interval products in this reduced finite model, and on the Hamiltonian generators it recovers the stalkwise operation $f \mapsto L_{B_f}$. The cotangent generators are finite principal-part labels in the CE/PV shadow; their unreduced boundary centrality homotopies remain part of Problem 3.47.

*Proof.* The Lie algebra $\mathfrak{g}_{\leq N}$ is finite-dimensional ($\dim \mathfrak{g}_{\leq N} = 2 \dim \mathfrak{h}_{\leq N} = 2\binom{N+2}{2} - 12$), hence pro-finite-dimensional in the trivial constant filtration. It is Tate-conilpotent in the sense of 3.137: by 3.80(ii) the descending Lie powers of $\mathfrak{h}_{\leq N}$ vanish after at most $N-2$ brackets, and since $\mathfrak{h}_{\leq N}^\vee[1]$ is abelian,

the descending Lie powers of $\mathfrak{g}_{\leq N} = \mathfrak{h}_{\leq N} \ltimes \mathfrak{h}_{\leq N}^{\vee}[1]$ also terminate. Hence $\bigcap_m \mathfrak{g}_{\leq N}^{(\geq m)} = 0$ in any Tate filtration, and the conilpotency hypothesis of 3.137 is satisfied trivially.

Apply 3.27 (the universal cochain-level open-closed derived center) to $\mathfrak{h}_{\leq N}$. The conclusion is a canonical isomorphism of completed dg graded-commutative $P_0$-algebras

$$\Phi_{\leq N}^{\mathrm{stalk}} : C_{\mathrm{CE,cont}}^{\bullet}(\mathfrak{g}_{\leq N}) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}(\mathfrak{h}_{\leq N})\big),$$

with the explicit generator-level data of 3.27.

To pass from the stalk identification to the factorization-algebra identification on intervals $I \subset \mathbb{R}$, the Hamiltonian factor upgrade of 3.21 applies verbatim with $\mathfrak{h}$ replaced by $\mathfrak{h}_{\leq N}$: the locality input $\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(s)\} = \delta_l^i \delta_j^k \delta(t-s)$ is unchanged, the smearing $\mathfrak{h}_{\leq N,I} = \Omega_c^{\bullet}(I) \widehat{\otimes} \mathfrak{h}_{\leq N}$ is canonical because $\mathfrak{h}_{\leq N}$ is finite-dimensional (no Tate completion), and the strict $P_0$ bracket follows from the delta-function locality.

In this finite truncated classical sector, the cotangent factor has no Tate-Casimir convergence problem. The principal-part sub-factorization algebra

$$\widetilde{\mathrm{Obs}}_{\partial,\leq N}^{\mathrm{cl,pp}}(I) = \mathrm{Obs}_{\mathrm{open},\leq N}^{\mathrm{cl}}(I) \widehat{\otimes} \widehat{\mathbf{S}}\big(\mathfrak{h}_{\leq N,I}^{\vee}[1]\big)$$

admits a chain-level primitive projection $\pi_{\mathrm{prim}}^{\leq N}$ because the Tate Casimir $C_{\mathfrak{h}_{\leq N}} = \sum_{d=3}^{N} \sum_i e_{d,i} \otimes e_d^i$ is a finite sum, hence a literal element of the (uncompleted) tensor product $\mathfrak{h}_{\leq N} \otimes \mathfrak{h}_{\leq N}^{\vee}$. The $Q$-compatibility, factorization compatibility, and $P_0$-bracket homotopies are then formal, since at no step does one need an infinite-support tensor in the direct-sum dual.

The bracket compatibility and graded Leibniz extension to the completed finite CE/PV algebra is the same argument as in 3.27, with the additional simplification that all completions are trivial because $\mathfrak{h}_{\leq N}$ is finite-dimensional. Compatibility with extension by zero, disjoint-interval factorization, and the stalkwise limit follow as in 3.21. □

*Quantum status of the finite truncation*

THEOREM 3.84 *(Quantum status of the nilpotent truncation).* The classical finite truncation 3.83 is unconditional. The same finite Lie algebra $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ does *not* carry an all-order Moyal deformation for $N \geq 4$: it is not closed under the reduced Moyal bracket. Consequently the nilpotent truncation supplies no all-order quantum descendant theorem on the strict finite truncation unless an additional Moyal-flat replacement or a projected bracket with Jacobi and cochain compatibility is supplied.

What remains true at quantum level is the restriction of the full-sector degree-zero map 3.187 to the finite set of generators of $\mathfrak{h}_{\leq N}$, viewed inside the full $\mathfrak{h}_\hbar$. This is a restriction of a proved map on the full Hamiltonian source, not a closed CE theorem for a finite quantum Lie subalgebra.

*Proof.* The Moyal bracket is

$$\{f, g\}_\hbar = \sum_{j \geq 0} \frac{\hbar^{2j}}{(2j+1)!\, 2^{2j}} P^{2j+1}(f, g), \qquad P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}.$$

If $f \in \mathfrak{m}^p$ and $g \in \mathfrak{m}^q$, the order-$\hbar^{2j}$ summand lies in $\mathfrak{m}^{p+q-2(2j+1)}$ when nonzero. The Poisson term $j = 0$ preserves $\mathfrak{m}^3$ as a Lie subalgebra, but the higher Moyal terms need not. For $N \geq 4$,

$$P^3(z_1^4, z_2^4) = 24 z_1 \cdot 24 z_2 = 576 z_1 z_2,$$

so

$$\{z_1^4, z_2^4\}_\hbar = 16 z_1^3 z_2^3 + 24\hbar^2 z_1 z_2 + \cdots .$$

The term $24\hbar^2 z_1 z_2$ has $\mathfrak{m}$-adic degree 2. It is neither in $\mathfrak{m}^3$ nor a constant killed by the reduction modulo $\mathbb{C} \cdot 1$. Thus $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ is not a Moyal Lie subalgebra for $N \geq 4$.

Therefore the CE differential for a hypothetical finite quantum source $\mathfrak{h}_{\leq N,\hbar} \ltimes \mathfrak{h}^\vee_{\leq N,\hbar}[1]$ is not defined by restriction of the Moyal bracket. Any projected bracket would require a new Jacobi proof; the projected-degree warning of 3.180 shows that this is a real condition, not a formality. The all-order theorem 3.187 remains valid on the full Hamiltonian algebra $\mathfrak{h}_\hbar$, and one may evaluate its map on generators of degrees $3, \ldots, N$. That operation does not make those generators a finite quantum CE source. The descendant cotangent sector at $\hbar = 0$ is exactly 3.83; at $\hbar \neq 0$ it remains a named Moyal-flat truncation problem. □

*What is lost*

Proposition 3.85 *(Lost sectors).* The truncation $\mathfrak{h} \rightsquigarrow \mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ removes the following five data from the local theorem 2.27:

(i) The linear Hamiltonians $z_1, z_2 \in \mathfrak{m} \setminus \mathfrak{m}^2$, which generate translations in the symplectic plane and the center-of-mass mode of the brane.

(ii) The conformal-symplectic $\mathfrak{sl}_2$ generators $H_2 = \operatorname{span}\{z_1^2, z_1 z_2, z_2^2\}$, which generate rotations and dilations of the symplectic plane. This is the semisimple sector that prevents $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ from being nilpotent.

(iii) The constant Hamiltonian $1 \notin \mathfrak{m}$, on which the $\hbar$-normalization of the Moyal product depends. The central element $\mathbb{C} \cdot 1 \subset \mathfrak{h}_{\mathrm{poly}}$ is absent from $\mathfrak{m}^3$ a fortiori, so the $\hbar$-normalization in the truncated theorem is conventional.

(iv) The U(1) center anomaly cohomology class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ of 2.27(f), since the cocycle

$$\omega(z_1^k z_2^l, z_1^m z_2^n) = (kn - lm)\, \mathbf{1}[(k+m, l+n) = (1,1)]$$

is concentrated on multidegree $(1,1)$, with both arguments of total $\mathfrak{m}$-adic degree 1, hence the cocycle pairs only elements outside $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$. There is no central anomaly to anchor on the truncation.

(v) The stable connected limit $N \to \infty$ on the full $\mathfrak{h}$. The directed system $\{\mathfrak{h}_{\leq N}\}_{N \geq 3}$ has structure maps the natural surjections $\mathfrak{h}_{\leq N+1} \twoheadrightarrow \mathfrak{h}_{\leq N}$ induced by the $\mathfrak{m}$-adic projection $\mathfrak{m}^{N+1} \subset \mathfrak{m}^{N+2}$. The inverse limit is $\varprojlim_N \mathfrak{h}_{\leq N} = \mathfrak{m}^3$, the $\mathfrak{m}^3$-truncation of the Hamiltonian Lie algebra, not the full $\mathfrak{h} = \mathfrak{m}$.

What remains classically: the third-order and higher canonical transformations, which form a nilpotent Lie subalgebra $\mathfrak{m}^3 \subset \mathfrak{h}$, a Lie ideal in $\mathfrak{m}^2$, admitting the classical finite cotangent CE/PV shadow unconditionally. The unreduced open BV factorization-centre morphism is not constructed by this truncation. The all-order Moyal descendant lift is excluded by 3.84 unless extra quantum truncation data are added.

*Proof.* (i)–(iv) are immediate from the definitions and from the $\mathfrak{m}$-adic concentration of $\omega$. (v) is the explicit inverse limit calculation: a compatible family $\{f_N \in \mathfrak{m}^3/\mathfrak{m}^{N+1}\}_N$ assembles to a unique formal series in $\mathfrak{m}^3 \subset \mathbb{C}[[z_1, z_2]]$, and the $\mathfrak{m}$-adic Lie structure is preserved at every level. □

*Colim recovery question*

PROPOSITION 3.86 *(Inverse-limit recovery on $\mathfrak{m}^3$).* The classical truncated isomorphism $\Phi_{\leq N}$ of 3.83 is compatible with the directed system: for every $N \leq M$, the diagram

$$\begin{array}{ccc} C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\leq M,I}) & \xrightarrow{\Phi_{\leq M}} & Z^{P_0}_{\mathrm{red,Ham}}\big(\widehat{\mathbf{S}}(\mathfrak{h}_{\leq M,I})\big) \\ \downarrow \pi_{M\to N} & & \downarrow \pi_{M\to N} \\ C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\leq N,I}) & \xrightarrow{\Phi_{\leq N}} & Z^{P_0}_{\mathrm{red,Ham}}\big(\widehat{\mathbf{S}}(\mathfrak{h}_{\leq N,I})\big) \end{array}$$

commutes, where $\pi_{M\to N}$ is induced by the surjection $\mathfrak{h}_{\leq M} \twoheadrightarrow \mathfrak{h}_{\leq N}$. Passing to the inverse limit gives the cochain-level isomorphism

$$\Phi_{\mathfrak{m}^3} := \varprojlim_N \Phi_{\leq N} : C^\bullet_{\mathrm{CE,cont}}\big(\mathfrak{m}^3 \ltimes (\mathfrak{m}^3)^\vee_{\mathrm{cont}}[1]\big) \xrightarrow{\sim} Z^{P_0}_{\mathrm{red,Ham}}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{m}^3)\big).$$

*Proof.* Compatibility with the directed system follows from the naturality of 3.27 in the Lie source: a Lie surjection $\pi : \mathfrak{h}_M \to \mathfrak{h}_N$ induces a CE cochain map by pullback and a Schouten cochain map by pushforward, and the universal map $\Phi$ is built from the structure constants alone, so $\pi^*$ on the source and $\pi_*$ on the target intertwine $\Phi_M$ and $\Phi_N$. The inverse limit converges in **TateVec** because each truncation is finite-dimensional and the structure maps are surjective.

The Lie algebra $\mathfrak{m}^3$ is Tate-conilpotent in the $\mathfrak{m}$-adic Tate filtration: its descending Lie powers satisfy $(\mathfrak{m}^3)^{(\geq m)} \subset \mathfrak{m}^{m+3}$ because each iterated bracket raises the $\mathfrak{m}$-adic degree by at least 1 when both arguments lie in $\mathfrak{m}^3$ (since $\{H_p, H_q\} \subset \mathfrak{m}^{p+q-2}$ and $p, q \geq 3$ implies $p + q - 2 \geq p + 1$). Hence $\bigcap_m (\mathfrak{m}^3)^{(\geq m)} = 0$, and the conilpotency hypothesis of 3.137 holds. The Tate Casimir $C_{\mathfrak{m}^3} = \sum_{d\geq 3} \sum_i e_{d,i} \otimes e^i_d$ is governed by the same product/direct-sum convergence question as in 3.182, but on the smaller direct-sum dual $\bigoplus_{d\geq 3} H^\vee_d \subset \bigoplus_{d\geq 1} H^\vee_d$. The inverse-limit construction through finite-dimensional truncations bypasses the convergence question on $\mathfrak{m}^3$ specifically: the cochain isomorphism $\Phi_{\leq N}$ at every finite $N$ uses only the finite Casimir $\sum_{3\leq d\leq N} \sum_i e_{d,i} \otimes e^i_d$, and the isomorphism property is preserved at the inverse limit by the Mittag-Leffler argument because each $\pi_{M\to N}$ is a surjection of finite-dimensional spaces. □

*Remark* 3.87 *(The colim does NOT recover the full $\mathfrak{h}$).* The inverse limit $\Phi_{\mathfrak{m}^3}$ of 3.86 is the classical cotangent lift on $\mathfrak{m}^3$, not on $\mathfrak{h} = \mathfrak{m}$. To extend to all of $\mathfrak{h}$ one must include the low-degree sectors $\mathfrak{m}/\mathfrak{m}^3 = H_1 \oplus H_2 = \mathbb{C} \cdot z_1 \oplus \mathbb{C} \cdot z_2 \oplus \mathrm{span}\{z_1^2, z_1 z_2, z_2^2\}$, which by 3.180(iii) is not stable under any $\mathfrak{m}$-adic cutoff: the bracket $\{z_1, z_2^{N+1}\} = (N + 1) z_2^N$ is the explicit obstruction. Equivalently, the Tate Casimir $C_{\mathfrak{h}} = \sum_{d\geq 1} \sum_i e_{d,i} \otimes e^i_d$ has low-degree components $\sum_i e_{1,i} \otimes e^i_1 \in H_1 \otimes H_1^\vee$ and $\sum_i e_{2,i} \otimes e^i_2 \in H_2 \otimes H_2^\vee$ that the truncation removes; the degree-1 component is the source of the linear-Hamiltonian translation, and the degree-2 component is the source of the conformal-symplectic $\mathfrak{sl}_2$ direction on which the truncation fails to be nilpotent. Recovering the full $\mathfrak{h}$ requires a coefficient category in which $C_{\mathfrak{h}}$ converges *including* both low-degree components, which is exactly the Casimir support obstruction of 3.182. The truncation route does not solve that obstruction; it removes the sector on which the obstruction is felt.

*Problem* 3.88 *(Low-degree-sector lift to $\mathfrak{m}^3 \rightsquigarrow \mathfrak{m}$).* Extend the inverse-limit isomorphism $\Phi_{\mathfrak{m}^3}$ of 3.86 from $\mathfrak{m}^3$ to all of $\mathfrak{h} = \mathfrak{m}$. Equivalently, supply a Tate or weighted-coefficient extension in which the degree-1 and degree-2 Casimir components converge as kernels for the BV propagator, and in which the conformal-symplectic $\mathfrak{sl}_2$ direction $H_2$ is included without violating the Tate-conilpotency hypothesis

of 3.137. By 3.87 this is the low-degree-sector failure; by 3.182 this is the same coefficient-category problem as 3.47. Thus the nilpotent-truncation route closes the classical nilpotent cotangent shadow on $\mathfrak{m}^3$, and the residual moves from 3.47 as stated to its restricted form on the low-degree sector $\mathfrak{m}/\mathfrak{m}^3 = H_1 \oplus H_2$ alone.

*Remark* 3.89 *(The conilpotency obstruction is essential).* A second obstruction to extending $\Phi_{\mathfrak{m}^3}$ to all of $\mathfrak{h}$ is structural rather than analytic: the inclusion of $H_2 = \mathfrak{sl}_2$ violates Tate-conilpotency because $[H_2, H_2] = H_2$ is its own descending Lie series. Even if the degree-1 and degree-2 Casimir components converged in some weighted Tate completion, the universal cochain-level 3.27 would not apply directly to the resulting Lie algebra: a Koszul-bar/cobar duality argument for the semisimple $\mathfrak{sl}_2$ direction is non-trivial and is the same as the failure of the cobar formalism for non-conilpotent coalgebras ([24]). One either restricts to nilpotent radicals (the route taken here) or replaces the Lie source by a model that builds in semisimple subalgebras separately.

*Costello–Li precedent*

*Remark* 3.90 *(Costello–Li $\mathfrak{gl}(N \mid N)$ truncation as structural model).* Costello and Li [13] prove the quantum perturbative BCOV theorem on flat Calabi–Yau by an $N \to \infty$ stable connected limit through the open gauge algebra $\mathfrak{gl}(N \mid N)$, with the directed Lie embedding $\mathfrak{gl}(N \mid N) \hookrightarrow \mathfrak{gl}(N+k \mid N+k)$ and the BV-cohomology vanishing $H^*_{\mathrm{BV}}\big(\mathfrak{gl}(N \mid N)\big) = 0$ (supertrace identity $\mathrm{str}(I) = N - N = 0$ kills the would-be U(1)-anomaly). The structural analogy with the truncation here:

(i) Each window $\mathfrak{gl}(N \mid N)$ is finite-dimensional, BV-cohomology vanishes, the universal CE/Schouten identification and the BV-propagator coefficient kernel are well-defined on each window.

(ii) The inverse / direct system is the Lie embedding $\mathfrak{gl}(N \mid N) \hookrightarrow \mathfrak{gl}(N+1 \mid N+1)$ on the Costello–Li side, the Lie surjection $\mathfrak{h}_{\leq N+1} \twoheadrightarrow \mathfrak{h}_{\leq N}$ here.

(iii) The U(1)-anomaly is killed by the structure of each window on the Costello–Li side (supertrace), and is removed by the $\mathfrak{m}^3$-truncation here (the cocycle $\omega$ pairs only multidegree-$(1,0)$ with multidegree-$(0,1)$, both in $H_1 \subset \mathfrak{m} \setminus \mathfrak{m}^2$, hence outside $\mathfrak{m}^3$).

(iv) The colimit $N \to \infty$ converges on the Costello–Li side to the full $\mathfrak{gl}(\infty \mid \infty)$, providing the stable connected single-trace algebra. Here the inverse limit converges to $\mathfrak{m}^3 \subsetneq \mathfrak{h}$, not to all of $\mathfrak{h}$, by 3.87.

The structural difference: the $\mathfrak{gl}(N \mid N)$ system is graded by matrix size, with trivial action of one window on the next-window orthogonal complement; the $\mathfrak{h} = \mathfrak{m}$ system is graded by $\mathfrak{m}$-adic degree, with the linear and quadratic sectors acting non-trivially on every higher-degree window, which is precisely the finite-window obstruction of 3.180(iii) plus the $\mathfrak{sl}_2 \subset H_2$ semisimple obstruction to Tate-conilpotency.

*Summary*

The nilpotent truncation $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ closes the classical finite nilpotent cotangent CE/PV shadow on the truncated sector (degree-zero Hamiltonian plus finite cotangent labels). It does not construct the unreduced open BV factorization-centre morphism. The quantum all-order version on the same finite source is obstructed by 3.84: the Moyal bracket leaves $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ for $N \geq 4$. The inverse limit $\Phi_{\mathfrak{m}^3} = \varprojlim_N \Phi_{\leq N}$ of 3.86 is the classical cotangent lift on the full classical $\mathfrak{m}^3$ sector. What is lost: the low-degree sector $\mathfrak{m}/\mathfrak{m}^3 = H_1 \oplus H_2$ (linear translations and the conformal-symplectic $\mathfrak{sl}_2$), the constant-Hamiltonian $\hbar$-normalization, the U(1) center anomaly $[\bar{c}]$, and the stable connected limit on

the full $\mathfrak{h}$. Recovering the low-degree sector requires the coefficient-category change of 3.47 together with a non-conilpotent extension of 3.137 to accommodate the semisimple $\mathfrak{sl}_2 \subset H_2$. Thus the nilpotent truncation closes only the classical $\mathfrak{m}^3$ truncation; its all-order quantum version is the separate Moyal-flat truncation problem above.

This is the same structural truncation as Costello–Li $\mathfrak{gl}(N \mid N) \to \mathfrak{gl}(\infty \mid \infty)$, with the key difference that the symplectic-plane $\mathfrak{m}$-adic system does not admit a free directed exhaustion by nilpotent Lie subalgebras containing $H_1 \oplus H_2$; the truncation removes the low-degree sector by hypothesis.

### *3.9 Chain-level primitive projection*

The chain-level primitive projection

$$\pi_{\mathrm{prim}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \longrightarrow \Omega^\bullet_c(I)\widehat{\otimes}\bar{A}$$

required by Problem 3.47, item (3), is constructed below without recourse to any heat-kernel propagator or finite-scale regularization. The construction relies only on the chain structure of the reduced BV algebra (Construction 3.118), the de Rham contraction (Lemma 3.119), and the stable trace invariant theory of Proposition 3.120. The principal ingredient is the cumulant projection on the free symmetric algebra generated by primitive cyclic-word trace classes.

*Construction* 3.91 *(Trace-count filtration on the open observables).* Let $\mathcal{R}_N$ be as in Construction 3.118. In the stable connected limit $N \to \infty$, Proposition 3.120 identifies

$$\mathcal{R}_\infty^{GL_\infty,\mathrm{st}} \cong \mathbf{S}\big(\mathrm{Cyc}(F)_{\mathrm{nonempty}}\big)$$

with $F = \mathbb{C}\langle x_1, x_2\rangle$ and the cyclic words being primitive generators (Lemma 3.121 extends this to the super-cyclic case with one displayed odd $\psi$). The *trace-count filtration* on $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) = \Omega^\bullet_c(I)\widehat{\otimes}\mathcal{R}_N^{GL_N}$ is the symmetric-algebra filtration $F_k Obs$ generated by products of at most $k$ primitive trace classes (cyclic words):

$$F_0 \subset F_1 \subset F_2 \subset \cdots \subset \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I).$$

$F_0$ is the algebra of compactly supported de Rham forms on $I$ with scalar coefficients (the empty-trace stratum); $F_1/F_0$ is the *primitive* or *single-trace* component, isomorphic as a graded vector space to $\Omega^\bullet_c(I)\widehat{\otimes}\mathrm{Cyc}(F)_{\mathrm{nonempty}}$ in the stable limit.

LEMMA 3.92 *(Trace-count filtration is preserved by $Q$ and by the open $P_0$ bracket).* In the stable connected limit, the BV differential $Q$ and the open $P_0$ bracket $\{-,-\}_{\mathrm{open}}$ both preserve the trace-count filtration of Construction 3.91.

*Proof.* *$Q$ preserves $F_k$.* By Construction 3.118, $Q\psi = [\phi_1, \phi_2]$, $Q\phi_i = 0$. $Q$ is an interior derivation on $\mathcal{R}_N$, so on a product of $k$ traces it acts by the Leibniz rule and produces another sum of $k$-trace products (each application replaces a single $\psi$ inside a trace by $[\phi_1, \phi_2]$ inside the same trace; the total trace count is preserved). Smearing by a de Rham form on $I$ does not change the trace count. Hence $QF_k \subset F_k$.

*The open $P_0$ bracket preserves $F_k$.* By the canonical equal-time bracket of Remark 3.20, $\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(s)\} = \delta^i_l\delta^k_j\delta(t-s)$. On two traces $\mathrm{Tr}\, w_1$ and $\mathrm{Tr}\, w_2$ this bracket produces a single trace $\mathrm{Tr}\, w_3$ obtained by Wick-contracting one $\phi_1$ in $w_1$ with one $\phi_2$ in $w_2$ and joining the two cycles into one. This is the standard merging-of-cycles formula of Procesi–Razmyslov invariant theory. In particular, the bracket of two single-trace classes is a single-trace class, exactly Theorem 3.161. By the Leibniz rule for the Poisson bracket, the bracket of a $k_1$-trace product with a $k_2$-trace product is a sum

of products of total trace count exactly $(k_1 + k_2 - 1)$, that is, it stays in $F_{k_1+k_2-1} \subset F_{k_1+k_2}$. Hence the bracket preserves the filtration. □

*Construction 3.93 (Chain-level primitive projection).* In the stable connected limit, the symmetric-algebra structure $\mathcal{R}_\infty^{GL_\infty,\mathrm{st}} \cong \mathbf{S}(\mathrm{Cyc}(F)_{\mathrm{nonempty}})$ has the natural cumulant projection onto the linear span of generators (the Hopf-algebraic primitive). Tensored with the identity on $\Omega_c^\bullet(I)$, this gives

$$\pi_{\mathrm{prim}} : \Omega_c^\bullet(I)\widehat{\otimes}\mathbf{S}(\mathrm{Cyc}(F)_{\mathrm{nonempty}}) \twoheadrightarrow \Omega_c^\bullet(I)\widehat{\otimes}\mathrm{Cyc}(F)_{\mathrm{nonempty}}.$$

Composing with the identification $\mathrm{Cyc}(F)_{\mathrm{nonempty}} \simeq \bar{A}$ on the Hamiltonian-trace subspace (the abelianization of cyclic words: the trace of a cyclic word in the commuting variables $\phi_1, \phi_2$, taken modulo $Q$-exact reordering, is exactly the polynomial Hamiltonian $f \in \bar{A}$, a stable cyclic-word/polynomial dictionary recorded by Construction 3.123 and Lemma 3.141), gives

$$\pi_{\mathrm{prim}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \twoheadrightarrow \Omega_c^\bullet(I)\widehat{\otimes}\bar{A}.$$

At finite $N$ this is a degreewise stable construction, not a global map $\mathcal{R}_N \to \mathcal{R}_\infty^{\mathrm{st}}$. For a fixed word-degree $d$ and $N \geq d$, Proposition 3.120 identifies the degree-$d$ trace invariants with the stable cyclic-word span, and the above projection is applied on that stable range. Outside the stable range the finite-$N$ trace identities are not discarded; the statement is made after passing to the stable connected limit.

Lemma 3.94 *(Cumulant kernel is the closed ideal of decomposables).* The kernel of $\pi_{\mathrm{prim}}$ as a chain map is the closed ideal $F_{\geq 2}$ generated by products of at least two primitive trace classes. Equivalently, $\ker \pi_{\mathrm{prim}} = \Omega_c^\bullet(I)\widehat{\otimes}\mathbf{S}^{\geq 2}(\mathrm{Cyc}(F)_{\mathrm{nonempty}})$ on the multi-trace stratum, plus $\Omega_c^\bullet(I)\widehat{\otimes}\mathbb{C}\cdot 1$ on the empty-trace stratum (constants).

*Proof.* By construction $\pi_{\mathrm{prim}}$ is the projection onto generators of the symmetric algebra, which is the standard cumulant projection. Its kernel is the augmentation ideal squared $(\mathbf{S}^{\geq 1})^2 = \mathbf{S}^{\geq 2}$, complemented by the constants $\mathbb{C}\cdot 1 \subset F_0$. Tensoring with the de Rham cosheaf $\Omega_c^\bullet(I)$ preserves the kernel because the symmetric-algebra split is over the discrete coefficient ring on cyclic-word generators. □

Theorem 3.95 *(Chain-level primitive projection is a $Q$-chain map).* $\pi_{\mathrm{prim}}$ commutes strictly with $Q$:

$$\pi_{\mathrm{prim}} \circ Q = Q \circ \pi_{\mathrm{prim}}.$$

*Proof.* By Lemma 3.92, $Q$ preserves the trace-count filtration $F_\bullet$. In particular $Q$ restricts to endomorphisms of $F_0$, $F_1$, and the multi-trace ideal $F_{\geq 2}$. Hence $Q$ preserves the symmetric-algebra splitting

$$F_0 \oplus (F_1/F_0) \oplus F_{\geq 2} \,/\, F_{\geq 2}$$

in the associated graded sense. $\pi_{\mathrm{prim}}$ is by construction the projection onto $F_1/F_0$ in this splitting; the scalar empty-trace summand $F_0$ is killed by the tensor factor $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$. Therefore $\pi_{\mathrm{prim}} \circ Q$ and $Q \circ \pi_{\mathrm{prim}}$ both agree on each summand: on $F_1/F_0$ both equal $Q$ restricted to the single-trace component; on $F_0$ and on $F_{\geq 2}$ both equal zero. □

*Remark 3.96 (The cumulant target as the indecomposable part).* The target $\Omega_c^\bullet(I)\widehat{\otimes}\bar{A}$ is the *indecomposable* or *linear primitive* part of the brane $P_0$ factorization algebra $A_\partial^{\mathrm{Ham}}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$ of Construction 3.18: it is the generating Lie subspace $\mathfrak{h}_I = \Omega_c^\bullet(I)\widehat{\otimes}\bar{A}$. Equivalently, it is the quotient $\widehat{\mathbf{S}}(\mathfrak{h}_I)/\widehat{\mathbf{S}}^{\geq 2}(\mathfrak{h}_I) \cong \mathfrak{h}_I$ in the augmentation-ideal grading, after removal of constants. The factorization product on the cumulant

target side is therefore the linear (Lie) bracket of $\mathfrak{h}_I$, not a commutative algebra product; the multiplication of two distinct primitive elements lands in $\mathbf{S}^{\geq 2}$, which is the zero space at the indecomposable level.

THEOREM 3.97 *(Factorization compatibility).* For disjoint intervals $I_1 \sqcup I_2 \subset V$ in $\mathbb{R}$, the diagram

$$\begin{array}{ccc} \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I_1) \otimes \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I_2) & \xrightarrow{\mu} & \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(V) \\ \mathrm{aug}_{\oplus} \circ (\pi_{\mathrm{prim}} \otimes \pi_{\mathrm{prim}}) \downarrow & & \downarrow \pi_{\mathrm{prim}} \\ \big(\Omega^\bullet_c(I_1)\widehat{\otimes}\bar{A}\big) \oplus \big(\Omega^\bullet_c(I_2)\widehat{\otimes}\bar{A}\big) & \xrightarrow{\mu_{\bar{A}}} & \Omega^\bullet_c(V)\widehat{\otimes}\bar{A} \end{array}$$

commutes strictly modulo a homotopy $h_{\mathrm{fact}}$ that vanishes identically: $h_{\mathrm{fact}} \equiv 0$.

Here $\mu$ is the brane factorization product (multiplication in $\mathcal{R}^{GL_N}_N$ after extension by zero from $\Omega^\bullet_c(I_j)$ to $\Omega^\bullet_c(V)$), the bottom horizontal $\mu_{\bar{A}}$ is the cosheaf direct-sum extension by zero $\mathfrak{h}_{I_1} \oplus \mathfrak{h}_{I_2} \hookrightarrow \mathfrak{h}_V$ on the indecomposable (linear) part of the brane $P_0$ factorization algebra (cf. Remark 3.96), and the left vertical map factors through the augmentation

$$\big(\Omega^\bullet_c(I_1)\widehat{\otimes}\bar{A}\big) \otimes \big(\Omega^\bullet_c(I_2)\widehat{\otimes}\bar{A}\big) \twoheadrightarrow \big(\Omega^\bullet_c(I_1)\widehat{\otimes}\bar{A}\big) \oplus \big(\Omega^\bullet_c(I_2)\widehat{\otimes}\bar{A}\big)$$

obtained by killing the diagonal multi-trace component (a tensor product of two distinct linear primitives is multi-trace and is $\pi_{\mathrm{prim}}$-zero on the indecomposable-quotient target).

*Proof. Top-down route.* A typical class on $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I_1) \otimes \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I_2)$ is a tensor $\alpha \otimes \beta$ where each factor lies in some $F_{k_j}/F_{k_j-1}$. The brane factorization product $\mu$ extends $\alpha \otimes \beta$ by zero from $I_j$ to $V$ and multiplies; the result lies in $F_{k_1+k_2}/F_{k_1+k_2-1}$.

Apply $\pi_{\mathrm{prim}}$. The projection kills everything in $F_{\geq 2}$ (Lemma 3.94). Therefore the output is nonzero only when $k_1 + k_2 \leq 1$. The pairs $(k_1, k_2)$ satisfying $k_1 + k_2 \leq 1$ are $(0,0)$, $(1,0)$, $(0,1)$. The $(0,0)$ pair is killed by passing to $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$ (constants are excluded from the Hamiltonian quotient). The $(1,0)$ pair sends a single-trace $B_w(a)$ on $I_1$ tensored with a scalar de Rham form on $I_2$ to the single-trace $B_w(a)$ on $V$, multiplied by the same scalar form on $I_2$, and similarly for $(0,1)$. Modulo the scalar de Rham form, this is exactly the cosheaf extension-by-zero map $\mathfrak{h}_{I_j} \hookrightarrow \mathfrak{h}_V$ for $j = 1, 2$.

*Bottom-up route.* $\pi_{\mathrm{prim}} \otimes \pi_{\mathrm{prim}}$ applied to $\alpha \otimes \beta$ produces nonzero output only when each factor is in $F_1/F_0$. When both factors are nonzero linear primitives, the augmentation map kills the tensor (this is the multi-trace-vanishing in the indecomposable target, Remark 3.96); when only one is nonzero (and the other is the unit), the augmentation is the identity on that factor. The cosheaf extension by zero $\mu_{\bar{A}}$ then sends this to its image in $\Omega^\bullet_c(V)\widehat{\otimes}\bar{A}$.

Both routes give the same answer in every stratum: the linear cosheaf extension-by-zero map of the brane $P_0$ factorization algebra at the indecomposable level. Hence the diagram commutes strictly with $h_{\mathrm{fact}} \equiv 0$.

This is the chain-level statement of cosheaf compatibility. □

THEOREM 3.98 *($P_0$ bracket compatibility, with explicit homotopy).* For $F, G \in \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I)$ in the stable connected limit,

$$\pi_{\mathrm{prim}}\big(\{F, G\}_{\mathrm{open}}\big) - \big\{\pi_{\mathrm{prim}}F, \pi_{\mathrm{prim}}G\big\}_{\bar{A}} = (Q \circ h_{P_0} + h_{P_0} \circ Q)(F \otimes G),$$

with the explicit homotopy $h_{P_0} \equiv 0$ on every stratum $F_{k_1} \otimes F_{k_2}$. The chain-level $P_0$-bracket compatibility square commutes strictly.

*Proof.* We compute on each stratum separately.

*Stratum* $(k_1, k_2) = (1, 1)$. Both inputs are single traces $B_f(a)$ and $B_g(b)$ for $f, g \in \bar{A}$ and $a, b \in \Omega_c^\circ(I)$. By Lemma 3.92, the open $P_0$ bracket of two single traces is again a single trace (cycle merging in Procesi-Razmyslov), and by Theorem 3.161, in the connected Hamiltonian quotient, $\{B_f(a), B_g(b)\}_{\text{open}} = B_{\{f,g\}_{\bar{A}}}(ab)$. Apply $\pi_{\text{prim}}$: the result is the single-trace $ab \otimes \{f, g\}_{\bar{A}}$ in $\Omega_c^\bullet(I)\widehat{\otimes}\bar{A}$.

Independently, $\pi_{\text{prim}} B_f(a) = a \otimes f$ and $\pi_{\text{prim}} B_g(b) = b \otimes g$. The bracket on $\Omega_c^\bullet(I)\widehat{\otimes}\bar{A}$ is the $P_0$ bracket of Construction 3.18: $\{a \otimes f, b \otimes g\}_{\bar{A}} = ab \otimes \{f, g\}_{\bar{A}}$. This matches. Strict compatibility on this stratum.

*Stratum* $(k_1, k_2) = (1, k)$ *with* $k \geq 2$. $F = B_f(a)$ is a single trace; $G$ is a $k$-trace product $B_{w_1}(b_1) \cdots B_{w_k}(b_k)$. By the Leibniz rule,

$$\{F, G\}_{\text{open}} = \sum_{i=1}^{k} (\pm) B_{w_1}(b_1) \cdots \{B_f(a), B_{w_i}(b_i)\}_{\text{open}} \cdots B_{w_k}(b_k).$$

Each term is a $k$-trace product (cycle merging joined to a remaining $(k-1)$-trace tail). Hence the result lies in $F_k$, and by Lemma 3.94, $\pi_{\text{prim}}$ of this is zero whenever $k \geq 2$.

Independently, $\pi_{\text{prim}} G = 0$ because $G \in F_{\geq 2}$, hence $\{\pi_{\text{prim}} F, \pi_{\text{prim}} G\}_{\bar{A}} = 0$.

Both sides are zero on this stratum. Strict.

*Stratum* $(k_1, k_2)$ *with* $k_1 \geq 2$ *or* $k_2 \geq 2$. By symmetry the same argument: both sides vanish.

*Stratum* $(0, k)$ *or* $(k, 0)$. One factor is in $F_0$, that is, a scalar de Rham form. The bracket of a scalar de Rham form with anything is zero (the canonical equal-time bracket only acts on matrix fields). So $\{F, G\}_{\text{open}} = 0$ and $\{\pi_{\text{prim}} F, \pi_{\text{prim}} G\}_{\bar{A}} = 0$. Strict.

In every stratum the difference is zero; setting $h_{P_0} \equiv 0$ realizes the homotopy formula trivially. □

*Remark* 3.99 *(Why the homotopy is zero).* The strictness in Theorem 3.98 is not an accident of small word degree; it is a structural consequence of the trace-count filtration of Lemma 3.92. The bracket of two single traces is a single trace by Procesi–Razmyslov cycle merging; the bracket of a single trace with a multi-trace is a multi-trace by the Leibniz rule; cumulant projection projects to the single-trace component. Hence the failure of the bracket compatibility, if any, must lie in the multi-trace stratum, where both sides are zero.

*Construction* 3.100 *(Primitive one-$\psi$ shadow).* Extend $\pi_{\text{prim}}$ to the one-$\psi$ primitive shadow by recording the labels of the symmetrised traces $\Psi_{a,b}$ of Construction 3.123. Define

$$\pi_{\text{prim}}^{\text{cot}} : \text{Obs}_{\text{open},N}^{\text{cl,cot}}(I) \longrightarrow \Omega_c^\bullet(I)\widehat{\otimes}\big(\bar{A} \oplus \mathcal{P}[1]\big)$$

by:

- on the zero-$\psi$ Hamiltonian sector, $\pi_{\text{prim}}^{\text{cot}} = \pi_{\text{prim}}$ composed with the inclusion $\bar{A} \hookrightarrow \bar{A} \oplus \mathcal{P}[1]$;
- on a primitive one-$\psi$ class $\Psi_{a,b}$ smeared by a distributional test current $B \in \mathcal{D}_c'(I)$,

$$\pi_{\text{prim}}^{\text{cot}}\big(B \otimes \Psi_{a,b}\big) = B \otimes \rho_{a,b}[1] = \Theta_{a,b}(B);$$

- on multi-trace products of one-$\psi$ classes (or mixed products of zero-$\psi$ and one-$\psi$ traces), $\pi_{\text{prim}}^{\text{cot}} \equiv 0$ (cumulant on the multi-trace stratum).

This is a projection to the reduced principal-part label target, not a polynomial realization of the coadjoint module. The no-polynomial theorem 3.154 says precisely that the open polynomial one-$\psi$ descendants and the principal-part currents do not carry the same $\mathfrak{h}$-module structure. The map above is therefore an indecomposable shadow map after imposing the reduced current target $\mathcal{P}[1]$; it is not an unreduced boundary observable realizing $\mathcal{P}$ inside the polynomial BV/Koszul algebra.

THEOREM 3.101 *(Chain-level primitive projection and one-$\psi$ shadow).* In the stable connected limit, the chain-level primitive projection $\pi_{\mathrm{prim}}$ of Construction 3.93 (and its one-$\psi$ shadow $\pi^{\mathrm{cot}}_{\mathrm{prim}}$ of Construction 3.100) is a surjection to the indecomposable $P_0$ shadow of the boundary factorization algebra

$$\pi^{\mathrm{cot}}_{\mathrm{prim}} : \widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N} \twoheadrightarrow A^{\mathrm{pp}}_{\partial}$$

in the sense of Construction 3.18, with the following compatibilities:

(i) *(BV chain map.)* $\pi^{\mathrm{cot}}_{\mathrm{prim}} \circ Q = Q \circ \pi^{\mathrm{cot}}_{\mathrm{prim}}$ strictly at the chain level (Theorem 3.95 and Lemma 3.124 extending it to the cotangent sector via the closed cocycle property of $\Psi_{a,b}$);

(ii) *(Factorization product.)* The factorization-product compatibility square commutes strictly modulo a zero homotopy (Theorem 3.97), with the same trace-count argument on the one-$\psi$ label projection;

(iii) *($P_0$ bracket.)* The $P_0$-bracket compatibility square commutes strictly modulo a zero homotopy on the Hamiltonian zero-$\psi$ indecomposable quotient (Theorem 3.98). On the one-$\psi$ shadow the projection records labels only; it does not intertwine the open polynomial descendant raising action with the principal-part coadjoint lowering action. The reduced target bracket $\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB)$ is supplied separately by Theorem 3.157;

(iv) *(Cosheaf.)* $\pi^{\mathrm{cot}}_{\mathrm{prim}}$ is compatible with extension by zero in $\Omega^\bullet_c$, because both source and target are tensored over the de Rham cosheaf $\Omega^\bullet_c$ on $\mathbb{R}$.

The target is the nonunital indecomposable quotient: multiplication of two positive primitive generators is zero after projection, while the Lie/$P_0$ bracket remains the Hamiltonian bracket. Thus the statement is not an algebra map to $\bar{A}$ with its ordinary commutative multiplication.

Equivalently, $\pi^{\mathrm{cot}}_{\mathrm{prim}}$ is the chain-level realization of the unreduced-to-reduced primitive shadow required by Problem 3.47, item (3). It is not a proof of the unreduced cotangent factorization-center lift.

*Proof.* Items (1), (2), (3) on the Hamiltonian zero-$\psi$ sector are Theorems 3.95, 3.97, 3.98 above.

One-$\psi$ shadow. By Lemma 3.124, $Q\Psi_{a,b} = 0$ at the chain level. Hence $Q\,\pi^{\mathrm{cot}}_{\mathrm{prim}}(B \otimes \Psi_{a,b}) = \partial B \otimes \rho_{a,b}[1] = \pi^{\mathrm{cot}}_{\mathrm{prim}}(\partial B \otimes \Psi_{a,b} + B \otimes Q\Psi_{a,b}) = \pi^{\mathrm{cot}}_{\mathrm{prim}} \circ Q(B \otimes \Psi_{a,b})$ (the only contribution comes from the de Rham differential $\partial$ on the distributional test current $B$, since $Q$ kills $\Psi_{a,b}$). The factorization-product compatibility on one-$\psi$ generators uses the same trace-count filtration: a one-$\psi$ trace tensored with another trace is a multi-trace, killed by cumulant projection. The $P_0$-bracket compatibility asserted here is the zero-$\psi$ Hamiltonian statement already proved above. The one-$\psi$ shadow deliberately does not assert compatibility between the open descendant raising action and the principal-part coadjoint lowering action; their difference is Corollary 3.144. The bracket of two cotangent generators is zero on both sides because $\mathcal{P}[1]$ is an abelian odd ideal in $\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1]$.

Cosheaf compatibility is automatic: extension by zero on the $\Omega^\bullet_c$ factor commutes with the projection on the trace-algebra factor because the projection is of the form $\mathrm{id} \otimes p$ with $p$ a fixed linear map on the trace algebra.

The chain-level surjectivity is the surjectivity of the cumulant projection onto generators in the stable symmetric algebra (each generator is hit by itself), tensored with the identity on the compactly supported de Rham/current factor on the same interval.

This establishes all four items. □

Proposition 3.102 *(Quantum obstruction to the primitive one-$\psi$ label map).* Let $\mathfrak{h}_\hbar = \bar{A}_\hbar$ carry the Weyl/Moyal bracket

$$\{f, g\}_\hbar = \sum_{j \geq 0} \frac{\hbar^{2j}}{(2j+1)!\, 2^{2j}} P^{2j+1}(f, g), \qquad P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}.$$

The chain-level one-$\psi$ primitive label map $\Psi_{a,b} \mapsto \rho_{a,b}$ is not compatible with the $\mathfrak{h}_\hbar$-coadjoint action. The failure already occurs in the indecomposable target; it is not a decomposable multi-trace correction.

*Proof.* Compatibility would require the polynomial PBW descendant action of Proposition 3.143, after the label replacement $\widetilde{\Psi}_{a,b} \mapsto \rho_{a,b}$, to agree with the Moyal coadjoint action

$$(\mathrm{ad}^\vee_{f,\hbar}\, \delta)(g) = -\delta(\{f, g\}_\hbar).$$

Test this on the single Hamiltonian $f = z_1^4$ and the single descendant label $(a, b) = (1, 1)$.

On the polynomial one-$\psi$ side, Proposition 3.143 gives

$$z_1^4 : \widetilde{\Psi}_{1,1} \longmapsto 4\, \widetilde{\Psi}_{4,0}.$$

On the Moyal coadjoint side the classical Poisson term has negative first target index and vanishes, but the $P^3$ term does not:

$$P^3(z_1^4, z_2^4) = 576\, z_1 z_2, \qquad \frac{1}{3!\, 2^2} P^3(z_1^4, z_2^4) = 24\, z_1 z_2.$$

Hence

$$\mathrm{ad}^\vee_{z_1^4,\hbar}\, \delta_{1,1} = -24\, \hbar^2 \delta_{0,4} + \cdots,$$

equivalently $z_1^4 \cdot_\hbar \rho_{1,1} = -24\, \hbar^2 \rho_{0,4} + \cdots$ in the principal-part realization. The two answers have different $\hbar$-order and different primitive label. Since both sides have already been projected to the linear indecomposable summand, adding decomposable multi-trace terms cannot repair this square after applying $\pi^{\mathrm{cot}}_{\mathrm{prim}}$.

The finite exact-arithmetic attack verifies the same obstruction through exponents $\leq 5$ and Moyal order $\leq 5$: 1200 of 1225 nonzero test cases disagree under the naive label map, and 958 cases have genuine higher Moyal coadjoint terms. The same finite computation also checks the corrected principal-part target below in 3375 finite representation identities with exponents $\leq 3$, Moyal order $\leq 5$, and $\hbar$-power $\leq 4$, and checks the linear-generator leading-term obstruction to every $\hbar$-adic deformation of the classical label projection in 70 tests with labels $\leq 5$. The displayed $z_1^4$ calculation is already a proof that the classical primitive one-$\psi$ projection does not itself quantize to the descendant coadjoint sector. □

*Construction* 3.103 *(Reduced Moyal principal-part descendant target).* The corrected quantum descendant target is obtained by changing the target, not by correcting the polynomial one-$\psi$ representatives. For an interval $I \subset \mathbb{R}$ put

$$\mathfrak{h}_{\hbar,I} = \Omega^\bullet_c(I) \widehat{\otimes} \bar{A}_\hbar, \qquad \mathcal{P}_{\hbar,I} = \mathcal{D}'_c(I) \widehat{\otimes} \mathcal{P}[[\hbar]].$$

Define the reduced quantum principal-part descendant complex

$$A^{\mathrm{pp}}_{\partial,\hbar}(I) = \widehat{\mathbf{S}}_{\mathbb{C}[[\hbar]]}(\mathfrak{h}_{\hbar,I})\widehat{\otimes}\widehat{\mathbf{S}}_{\mathbb{C}[[\hbar]]}(\mathcal{P}_{\hbar,I}[1]).$$

The differential is the de Rham/current differential on the interval factors and zero on the coefficient modules $\bar{A}_\hbar$ and $\mathcal{P}[[\hbar]]$. The bracket is the Moyal semidirect bracket

$$\{B_f(a), B_g(b)\}_\hbar = B_{\{f,g\}_\hbar}(ab), \qquad \{B_f(a), \Theta_\rho(B)\}_\hbar = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB),$$

with

$$\{\Theta_\rho(B), \Theta_\sigma(C)\}_\hbar = 0.$$

Here $a, b \in \Omega^0_c(I)$, $B, C \in \mathcal{D}'_c(I)$, and $aB$ denotes multiplication of a compactly supported distribution by a smooth test function.

In monomial coordinates the full coadjoint action is as follows. Put

$$C_r(p,q;m,n) = \sum_{\alpha=0}^{r}(-1)^\alpha\binom{r}{\alpha}(p)_{r-\alpha}(q)_\alpha(m)_\alpha(n)_{r-\alpha}.$$

Then

$$z_1^p z_2^q \cdot_\hbar \rho_{a,b} = -\sum_{\substack{r\geq 1\\ r \text{ odd}}} \frac{\hbar^{r-1}}{2^{r-1}r!}\, C_r(p,q;a-p+r,b-q+r)\, \rho_{a-p+r,b-q+r},$$

with the convention that negative indices and the scalar target $(0,0)$ are zero. This is a finite sum for each monomial $z_1^p z_2^q$, because the falling factorials vanish for $r > p+q$. The $r = 1$ term is the classical Matlis coadjoint action of Theorem 3.150; the $r \geq 3$ terms are the genuine Moyal corrections.

This construction supplies a reduced distributional principal-part target for the quantum descendant sector. It does not make the polynomial PBW classes $\Psi_{a,b}$ into that target.

THEOREM 3.104 *(No $\hbar$-adic deformation of the primitive label projection).* Let $M_{\Psi,\hbar}$ be a $\mathbb{C}[[\hbar]]$-flat complex generated over $\mathbb{C}[[\hbar]]$ by lifts of the primitive polynomial one-$\psi$ classes $\widetilde{\Psi}_{a,b}$, $a+b>0$. Assume:

(i) the source action of the linear Hamiltonians reduces modulo $\hbar$ to the PBW descendant action

$$z_1 \cdot_0 \widetilde{\Psi}_{a,b} = b\,\widetilde{\Psi}_{a,b-1}, \qquad z_2 \cdot_0 \widetilde{\Psi}_{a,b} = -a\,\widetilde{\Psi}_{a-1,b};$$

(ii) the target coefficient module is $\mathcal{P}[[\hbar]][1]$, equivalently any fixed interval-current summand of $\mathcal{P}_{\hbar,I}[1]$ from Construction 3.103, with the full Moyal coadjoint action;

(iii) a candidate comparison $T_\hbar : M_{\Psi,\hbar} \to \mathcal{P}[[\hbar]][1]$ has the classical primitive label projection as leading term:

$$T_\hbar(\widetilde{\Psi}_{a,b}) = \rho_{a,b}[1] + O(\hbar).$$

Then $T_\hbar$ cannot intertwine the $\mathfrak{h}_\hbar$-action. The obstruction already appears after reduction modulo $\hbar$ and for the linear Hamiltonian $z_1$. Consequently no strict quantum descendant lift of the classical primitive projection exists under these assumptions. The only surviving quantum target in this lane is the reduced principal-part complex itself, not the polynomial one-$\psi$ source.

*Proof.* Reduce a hypothetical intertwiner modulo $\hbar$. For the class $\widetilde{\Psi}_{1,0}$, the PBW source action gives

$$z_1 \cdot_0 \widetilde{\Psi}_{1,0} = 0.$$

Hence the source-then-target route gives

$$T_0(z_1 \cdot_0 \widetilde{\Psi}_{1,0}) = 0.$$

The target-then-action route gives, by the $r = 1$ term of Construction 3.103,

$$z_1 \cdot_0 T_0(\widetilde{\Psi}_{1,0}) = z_1 \cdot \rho_{1,0}[1] = -\rho_{1,1}[1].$$

The principal-part basis vector $\rho_{1,1}$ is nonzero in $\mathcal{P}$. This contradiction is independent of all higher $\hbar$-corrections to $T_\hbar$, because those corrections vanish after taking the leading term. It is also independent of the higher Moyal terms: the linear Hamiltonian has no $r \geq 3$ Moyal correction.

The same computation with $z_2$ gives $z_2 \cdot_0 \widetilde{\Psi}_{0,1} = 0$ on the PBW side but $z_2 \cdot \rho_{0,1} = \rho_{1,1}$ on the principal-part side. Thus the mismatch is not a special feature of one coordinate. Since the coefficient differential on $\mathcal{P}[[\hbar]]$ is zero, this leading coefficient is not a boundary in the reduced target. A higher homotopy can remove it only after changing the target complex or the classical source action, which is outside the stated assumptions. □

*Remark* 3.105 *(Comparison with the heat-kernel propagator route).* The chain-level construction above avoids any heat-kernel propagator $P_{\epsilon,L}$ on $\mathbb{R}^2 \times \mathbb{C}^2$. In the Costello–Gwilliam BV propagator approach, the primitive projection is realized as a homotopy retraction onto the cohomology of the free open observable algebra, with the homotopy given by $G_L = \int_0^L K_t \, dt$ (the partial heat kernel). Here the role of $G_L$ is played by the symmetric-algebra splitting, and the key analytic fact (positive jets are $Q$-exact, Lemma 3.119) is needed only to identify the $E_1$-stalk of the brane factorization algebra with its $C^\infty$-smearing. The two routes agree on the classical cocycle level by uniqueness of the cumulant projection on a symmetric algebra.

The chain-level construction has the advantage of being independent of the analytic input of the heat kernel, and therefore independent of the Tate-completion convergence question recorded in Problem 3.47, item (1). It does not, however, by itself produce the renormalized BV homotopies required for the quantum (one-loop and higher) lift; that requires Costello renormalization of the brane-defect propagator, recorded separately in Problem 3.47, items (1)–(2).

Corollary 3.106 *(Primitive shadow of the reduced cotangent comparison).* Combining Theorem 3.101 with the cochain-level closed-open derived center (Theorem 3.27) and the Hamiltonian factorization-algebra equivalence (Theorem 3.21), one obtains a chain-level primitive shadow of the reduced cotangent comparison:

$$\begin{aligned} \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_L &\xrightarrow{\Phi_{\mathrm{cl}}} \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}\big(A^{\mathrm{pp}}_\partial\big) \\ &\xrightarrow{\pi^{\mathrm{cot}*}_{\mathrm{prim}}} \text{primitive indecomposable shadow of the open boundary sector.} \end{aligned}$$

This corollary does not construct the unreduced cotangent factorization-center morphism. The missing data are the centrality homotopies for principal-part currents against arbitrary unreduced open observables, together with the Tate/weighted coefficient kernel and the brane-defect propagator extension recorded in Problem 3.47.

*Proof.* $\Phi_{\mathrm{cl}}$ is constructed at the cochain level by Theorem 3.27 on the reduced sector after de Rham-current contraction. Theorem 3.21 promotes this to a $P_0$ factorization algebra equivalence in the Hamiltonian sector. Theorem 3.101 gives the chain-level primitive projection and one-$\psi$ label shadow. The composition therefore exists only at the indecomposable-shadow level. Since the one-$\psi$ projection does not intertwine the descendant raising action with the principal-part coadjoint lowering action, the asserted result is exactly the shadow statement and nothing stronger. □

*Remark* 3.107 *(Status of the primitive-shadow construction).* The chain-level primitive projection of Theorem 3.101 supplies the local primitive-projection component of Problem 3.47, item (3), and the primitive-shadow component of Theorem 2.27. It does not by itself close Problem 3.47, because the Tate-completion of the Casimir ($\bigoplus_{d\geq 1} H_d^\vee \to$ a topological coefficient in which the Casimir converges) and the propagator extension are separate analytic obligations not addressed here. Nor does it supply the missing centrality homotopies for principal-part currents against arbitrary unreduced open observables. Proposition 3.102 further shows that the one-$\psi$ label map is not compatible with the Moyal coadjoint action after quantization. The chain-level construction is therefore sufficient only for the classical primitive indecomposable shadow.

*Remark* 3.108 *(On the role of the de Rham contraction).* The de Rham contraction Lemma 3.119 enters Construction 3.93 only through the compactly supported density class on the brane line, not through any analytic propagator. Specifically, $H^\bullet(\Omega_c^\bullet(I)) = H_c^\bullet(I;\mathbb{C})$ is one-dimensional in degree 1 for an open interval $I$, represented by a compactly supported density $\lambda_I(t)\,dt$ with integral 1. The degree-zero notation in the primitive projection comes from the shifted BV/cotangent coordinate paired with this density; it is not ordinary point-evaluation in $H_c^0(I)$. The cosheaf structure is then transparent. No further analytic input is required; in particular no smoothing kernel, mollifier, or finite-scale BV gauge fix appears.

*Hadamard parametrix on the Mittag-Leffler coefficient module*

*Remark* 3.109 *(Spacetime/coefficient factorization of the Hadamard parametrix).* In Costello's heat-kernel BV calculus [12, Ch. 5], the Hadamard parametrix enters as the asymptotic expansion of the heat kernel $K_t^{\mathrm{base}}$ on the bulk free elliptic complex $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)$, combined with Hörmander wave-front-set tracking [32, Ch. 7] to control the singular support of the propagator $P_{\epsilon,L}^{\mathrm{base}}$ along the diagonal. Costello's locality package [12, Chs. 5–7] is used here only as the heat-kernel/RG vocabulary for free elliptic complexes: local functionals admit Hadamard-parametrix expansions whose counterterms stay in the local-functional class $\mathcal{O}_{\mathrm{loc}}$ under the hypotheses of that formalism. The exact published theorem number is not used as a local input.

In the weighted setting of 7, the propagator factors as $P_{\epsilon,L}^{w} = P_{\epsilon,L}^{\mathrm{base}}\widehat{\otimes}C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1],w}$. The bulk piece $P_{\epsilon,L}^{\mathrm{base}}$ is independent of the coefficient weight; the coefficient piece $C_{\mathfrak{h},w}$ is purely algebraic, with no scale parameter and no Laplacian. The Hadamard parametrix construction therefore lives entirely on the bulk factor; the coefficient factor enters only by tensor.

LEMMA 3.110 *(Windowwise parametrix compatibility).* For each $w_0 \in \mathbb{Z}_{\geq 1}$, let $C_{\mathfrak{h},w}^{w_0} = \sum_{d\leq w_0}\sum_i (w(d)e_{d,i}) \otimes (w(d)^{-1}e_d^i) \in \left(\prod_{d\leq w_0} w(d)H_d\right) \otimes \left(\prod_{d\leq w_0} w(d)^{-1}H_d^\vee\right)$ denote the truncated weighted Casimir. Define the windowwise field complex

$$\mathcal{E}_w^{w_0} = \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\left(\left(\prod_{d\leq w_0} w(d)H_d\right)[1] \oplus \left(\prod_{d\leq w_0} w(d)^{-1}H_d^\vee\right)[2]\right)$$

and the windowwise propagator $P^{w,w_0}_{\epsilon,L} = P^{\mathrm{base}}_{\epsilon,L} \widehat{\otimes} C^{w_0}_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1],w}$. Then:

(1) For each $w_0$, $\mathcal{E}^{w_0}_w$ is a free elliptic complex with finite-rank coefficient bundle, and Costello's Hadamard parametrix construction [12, Ch. 5] applies verbatim to $P^{w,w_0}_{\epsilon,L}$.

(2) The truncations $\{P^{w,w_0}_{\epsilon,L}\}_{w_0\geq 1}$ form a strict Mittag-Leffler inverse system in the coefficient direction, with surjective transition maps $P^{w,w'_0}_{\epsilon,L} \twoheadrightarrow P^{w,w_0}_{\epsilon,L}$ for $w'_0 \geq w_0$ given by truncation to degrees $d \leq w_0$.

(3) The wave-front set $\mathrm{WF}(P^{w,w_0}_{\epsilon,L})$ is contained in the bulk wave-front set $\mathrm{WF}(P^{\mathrm{base}}_{\epsilon,L}) \subset T^*(\mathbb{R}^2 \times \mathbb{C}^2) \setminus 0$, independently of $w_0$, since the coefficient factor $C^{w_0}_{\mathfrak{h},w}$ is a finite-rank algebraic tensor with no spacetime singularities.

*Proof.* (1) For fixed $w_0$, the coefficient bundle $\left(\prod_{d\leq w_0} w(d)H_d\right)[1] \oplus \left(\prod_{d\leq w_0} w(d)^{-1}H_d^\vee\right)[2]$ is a finite-rank $\mathbb{Z}$-graded vector space (finite product of finite-dimensional $H_d$'s). Costello's Hadamard parametrix construction [12, Ch. 5] applies verbatim to any free elliptic complex with finite-rank coefficient bundle, so it applies to $\mathcal{E}^{w_0}_w$. The bulk propagator $P^{\mathrm{base}}_{\epsilon,L}$ is unchanged; the coefficient factor $C^{w_0}_{\mathfrak{h},w}$ is a fixed finite-rank tensor.

(2) Surjectivity of the transition maps is the truncation $\prod_{d\leq w'_0} w(d)H_d \twoheadrightarrow \prod_{d\leq w_0} w(d)H_d$ on the coefficient direction; the bulk factor is unchanged. An inverse system with surjective transition maps satisfies the Mittag-Leffler condition trivially, so the inverse limit is exact in the admissible Mittag-Leffler envelope of Statement 3.28 and $\varprojlim^1 = 0$ for this tower. Compatibility with $P^{w,w'_0}_{\epsilon,L} \mapsto P^{w,w_0}_{\epsilon,L}$ follows from the tensor factorization and the truncation of the Casimir.

(3) The Hörmander wave-front set of a tensor product distribution $T_1 \otimes T_2$ on $X_1 \times X_2$ satisfies $\mathrm{WF}(T_1 \otimes T_2) \subseteq \left(\mathrm{WF}(T_1) \times \{(x_2, 0) : x_2 \in X_2\}\right) \cup \left(\{(x_1, 0) : x_1 \in X_1\} \times \mathrm{WF}(T_2)\right) \cup \left(\mathrm{WF}(T_1) \times \mathrm{WF}(T_2)\right)$ ([32, Thm. 8.2.9]). Here $T_1 = P^{\mathrm{base}}_{\epsilon,L}$ has wave-front set in the conormal of the diagonal of $\mathbb{R}^2 \times \mathbb{C}^2$, and $T_2 = C^{w_0}_{\mathfrak{h},w}$ is a finite-rank algebraic tensor with no point dependence, hence smooth, so $\mathrm{WF}(T_2) = \emptyset$. Hence $\mathrm{WF}(P^{w,w_0}_{\epsilon,L}) = \mathrm{WF}(P^{\mathrm{base}}_{\epsilon,L}) \times \{0\}$, which we identify with $\mathrm{WF}(P^{\mathrm{base}}_{\epsilon,L})$ on the spacetime, independent of $w_0$. □

Theorem 3.111 *(Graphwise Hadamard control on the Mittag-Leffler coefficient module).* Let $w(d) = q^d$ with $q > 1$ be the exponential degree weight of Remark 3.57, and let $\mathcal{E}_w$ be the weighted field complex (9). The Hadamard parametrix construction [12, Ch. 5] extends from each finite-window restriction $\mathcal{E}^{w_0}_w$ (Lemma 3.110) to the Mittag-Leffler limit $\mathcal{E}_w = \varprojlim_{w_0} \mathcal{E}^{w_0}_w$, preserving the Hadamard wave-front and finite-window locality estimates graph by graph. Specifically:

(i) The propagator $P^w_{\epsilon,L}$ admits a Hadamard parametrix expansion as the inverse limit $P^w_{\epsilon,L} = \varprojlim_{w_0} P^{w,w_0}_{\epsilon,L}$, with bulk wave-front set independent of $w_0$.

(ii) For every local interaction $I \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ of bounded polynomial-degree $D_0$ at each vertex in the coefficient direction, every fixed Costello graph $\Gamma$ of order $\hbar^n$ contributes a compatible finite-window counterterm system whose weighted coefficient size is bounded, on every finite output window, by a finite graph-dependent constant times $|C^{\mathrm{base}}_n[\Gamma]|$, where $|C^{\mathrm{base}}_n[\Gamma]|$ is the bulk spacetime counterterm of Costello.

(iii) For every fixed graph whose finite-window counterterms are degreewise surjective and compatible under the Mittag-Leffler transition maps, the corresponding inverse-limit representative lies in the local-functional class $\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$. This is graphwise finite-window Hadamard control, not a full Costello locality/RG theorem for the inverse-limit field theory. The remaining obstruction is the class in $H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\})$ for the mixed brane-defect interaction.

Thus the Hadamard and wave-front part of Problem 3.68 is reduced, graph by graph, to finite-window locality plus Mittag-Leffler compatibility. The theorem does not prove summability over graph topologies, a full Costello locality theorem for the inverse-limit field theory, or the vanishing of the mixed brane-defect QME obstruction class.

*Proof.* (i) Each truncation $P^{w,w_0}_{\epsilon,L}$ admits Costello's Hadamard parametrix expansion by Lemma 3.110 (1). The Mittag-Leffler inverse system of (2) of that lemma is strict (surjective transition maps), so the inverse limit $P^{w}_{\epsilon,L} = \varprojlim_{w_0} P^{w,w_0}_{\epsilon,L}$ exists in the weighted Tate category and inherits the wave-front bound (3) of that lemma uniformly in $w_0$. The inverse limit of the Hadamard expansions is itself a Hadamard expansion: the asymptotic Hadamard coefficients $a_k(x, y)$ on the bulk diagonal are independent of the coefficient direction, so the windowwise Hadamard expansion of $P^{w,w_0}_{\epsilon,L}$ is $\left(\sum_{k\geq 0} a_k(x, y)\, t^k\right)\widehat{\otimes} C^{w_0}_{\mathfrak{h},w}$, and the inverse limit in the coefficient direction commutes with the asymptotic expansion in the bulk parameter $t$.

(ii) Fix a graph $\Gamma$ with $V(\Gamma)$ vertices and $E(\Gamma)$ internal edges. Each vertex carries an interaction of polynomial degree $\leq D_0$ in the coefficient direction (hypothesis on $I$). Each internal edge contracts one source leg with one target leg via the Casimir $C_{\mathfrak{h},w}$. By (W1), each such contraction multiplies the weight by a factor $\leq C_w$ on Hamiltonian legs, and the cotangent legs are bounded on the finite windows appearing in the graph by (W2). The total coefficient size after $E(\Gamma)$ contractions is bounded by a finite graph-dependent constant times the bulk counterterm norm. The bulk counterterm norm $|C^{\mathrm{base}}_n[\Gamma]|$ is finite by Costello's bulk locality theorem [12, Chs. 5–7].

This graph-dependent constant is finite for the fixed graph $\Gamma$ and the fixed finite output window under consideration. No summability over all graph topologies at a fixed $\hbar$-order is asserted here. Such a summability statement belongs to the renormalized QME problem, together with the usual stability, counterterm, and locality hypotheses of Costello's graph expansion. The present theorem needs only the graphwise bound: for a fixed graph, the coefficient degrees and the corresponding weight ratios lie in a finite set, so the compatible finite-window counterterm system has a Mittag-Leffler limit when the windowwise representatives are themselves compatible.

(iii) The bulk Costello locality theorem at fixed window $w_0$ is used only as a finite-window input: the windowwise counterterm $C_n[\Gamma]^{w_0} \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}^{w_0}_w)$ is a local functional with kernel supported on the diagonal of $\mathbb{R}^2 \times \mathbb{C}^2$. By the wave-front bound of Lemma 3.110 (3), the support is the bulk diagonal independently of $w_0$. The transition map $\mathcal{O}_{\mathrm{loc}}(\mathcal{E}^{w'_0}_w) \to \mathcal{O}_{\mathrm{loc}}(\mathcal{E}^{w_0}_w)$ for $w'_0 \geq w_0$ is the restriction of polynomial functionals to truncated coefficient indices, equivalently setting higher coefficient fields to zero; in the regulator-admissible sector these maps are degreewise surjective. This preserves locality (the support of the truncated kernel is the same diagonal). The inverse limit $\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w) = \varprojlim_{w_0} \mathcal{O}_{\mathrm{loc}}(\mathcal{E}^{w_0}_w)$ exists in the weighted Tate category and is the candidate local-functional class on $\mathcal{E}_w$. The windowwise counterterms $\{C_n[\Gamma]^{w_0}\}_{w_0}$ form a compatible inverse system, and their inverse limit $C_n[\Gamma] = \varprojlim_{w_0} C_n[\Gamma]^{w_0}$ lies in $\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ with bound (ii).

The possible obstruction class $[C_n[\Gamma]] \in H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\})$ is represented by the inverse limit of the compatible windowwise classes when that compatibility holds. The theorem identifies its analytic support and coefficient topology; it does not assert that the resulting class vanishes for the mixed brane-defect interaction. □

*Remark* 3.112 *(Why the Hörmander commutation works).* The deciding technical point is whether the Mittag-Leffler limit in the coefficient direction commutes with Hörmander's wave-front-set analysis on the bulk. Three facts make this work:

(a) The propagator factorization (7) is exact (not asymptotic), so the bulk wave-front set is computed once and tensored.

(b) The coefficient factor $C_{\mathfrak{h},w}$ is not a distribution on any spacetime: it is an algebraic tensor in the weighted Tate category. Wave-front analysis on a smooth/algebraic tensor is trivial ($\mathrm{WF} = \emptyset$).

(c) The Hadamard parametrix coefficients $a_k(x, y)$ are universal in the bulk (depend only on the bulk Riemannian/Hermitian data of $\mathbb{R}^2 \times \mathbb{C}^2$); they do not see the coefficient direction.

Consequently, the wave-front bound of Lemma 3.110 (3) is exact and $w_0$-uniform; passing to the Mittag-Leffler limit preserves the local-functional class graph-by-graph at every order $\hbar^n$, conditional only on compatibility of the windowwise counterterm system.

Corollary 3.113 *(Hadamard reduction of the conditional weighted QME lift).* Proposition 3.67 is reduced to the remaining mixed brane-defect QME obstruction class. The candidate all-order quantum derived-center comparison

$$\Phi^w_\hbar : C^\bullet_{\mathrm{CE,cont}}\big(\mathfrak{h}_{w,\hbar} \ltimes \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}[1]\big)[[\hbar]] \longrightarrow Z^{P_{0,\hbar}}_{\mathrm{fact}}\big(\mathrm{Obs}^\hbar_{\mathrm{open}}\big)\big|_{\mathfrak{g}_w}$$

is not asserted unconditionally as an isomorphism. It extends to the weighted cotangent direction at all orders in $\hbar$ as a continuous $\mathbb{C}[[\hbar]]$-linear cochain map provided the obstruction class in $H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\})$ vanishes. Equivalently, Corollary 3.69 remains conditional on that obstruction-vanishing statement.

*Proof.* Proposition 3.67 is conditional on the Costello-locality/QME compatibility identity of Problem 3.68. By Theorem 3.111, the Hadamard wave-front-set and Mittag-Leffler coefficient-topology pieces of that identity are controlled. The remaining datum is the vanishing of the mixed brane-defect QME obstruction class. □

*Remark* 3.114 *(Status of Problem 3.68).* Problem 3.68 is reduced, not closed, by Theorem 3.111. The deciding technical input proved here is the spacetime/coefficient factorization of the Costello propagator together with the trivial Hörmander wave-front set of the algebraic Casimir factor. These reduce the Mittag-Leffler limit on the coefficient side to a passive tensor operation for Hadamard locality. They do not compute the brane-defect QME obstruction class.

*Remark* 3.115 *(Relation to regulator independence).* Theorem 3.73 proves that, after the finite-window Mittag-Leffler hypotheses of Definition 3.71 are imposed, the Hadamard reduction is independent of the admissible exponential weight. This is only weight-independence among regulated coefficient products. The unweighted product/direct-sum pair is not a further endpoint of this corollary, because Lemma 3.182 says that the required coefficient Casimir is absent there.

*Remark* 3.116 *(Closure status of the reduced core).* The summary corollary 2.27 is unconditional only in the proved Hamiltonian, factorization, principal-part, and anomaly statements stated above. The reduced cotangent factor after de Rham-current contraction is proved by 3.157; the chain-level primitive projection 3.101 supplies only its primitive indecomposable $P_0$ shadow on the unreduced open complex. It does not construct the unreduced cotangent factorization-centre morphism or its centrality homotopies. The weighted mechanism closes the coefficient kernel and graphwise analytic estimates, but the all-order one-antifield quantum extension of the degree-zero Moyal/Weyl lane remains conditional on the mixed brane-defect QME obstruction class in Problem 3.68. The nilpotent truncation closes only the classical $\mathfrak{m}^3$ sector.

*Remark* 3.117 *(Weiss/Ran descent firewall).* A global closed-open map is not obtained by choosing Darboux coordinates at each point and gluing the resulting stalk maps. The obstruction lives in the Weiss/Ran total complex

$$\operatorname{Tot} C^{\bullet}_{\mathrm{WR}}\Big(\operatorname{Ran}(L); \underline{\operatorname{RHom}}\Big(\operatorname{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}, Z^{P_0}_{\mathrm{fact}}(\operatorname{Obs}^{\mathrm{cl}}_{\mathrm{open}})\Big)\Big).$$

Vanishing of the local Hamiltonian period class is only the first condition. One must also prove source comparison, cotangent boundary realization, all $P_0$-center homotopies, and factorization descent. Until then the Hamiltonian boundary map, the $\Theta$-currents, and the PBW/Rees action are formal local or reduced-current statements.

### *3.10 Operators in the open brane gauge theory and the large $N$ limit*

The local-operator calculation is performed in the reduced BV model with fields $\phi_1, \phi_2$ and the antifield $\psi = A^*$ for the Lagrange multiplier $A$. We take $GL_N$-invariants in characteristic zero, so the Chevalley part of the gauge BRST differential acts trivially on trace functions by cyclicity.

The gauge field has one algebraic job: it imposes the moment-map equation $[\phi_1, \phi_2] = 0$. In BV language this equation is imposed by the Koszul differential $Q\psi = [\phi_1, \phi_2]$. This is why the stable trace algebra starts noncommutative, as cyclic words in two matrices, and then collapses in cohomology to commutative monomials on the formal symplectic plane.

*Construction* 3.118 *(The reduced BV algebra at the insertion point).* At a fixed point of the brane line, introduce three matrix variables

$$\phi_1, \phi_2 \in \mathfrak{gl}_N, \qquad \psi \in \mathfrak{gl}_N[1].$$

The reduced local polynomial algebra is

$$\mathcal{R}_N = \mathbb{C}[(\phi_1)^i{}_j, (\phi_2)^i{}_j] \otimes \Lambda\big((\psi)^i{}_j\big).$$

Its BRST differential is the derivation determined on generators by

$$Q(\phi_1)^i{}_j = 0, \qquad Q(\phi_2)^i{}_j = 0, \qquad Q(\psi)^i{}_j = ([\phi_1, \phi_2])^i{}_j.$$

This formula is the algebraic shadow of the $A$-equation of motion: varying

$$\int_{\mathbb{R}} \operatorname{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big)$$

with respect to $A$ gives $[\phi_1, \phi_2] = 0$, and the antifield $\psi = A^*$ records that equation as a Koszul differential.

LEMMA 3.119 *(De Rham jets).* Positive jets in the topological direction vanish in local operator cohomology. Equivalently, every cohomology class has a representative determined by the values of the fields and antifields at the insertion point.

*Proof.* For a coordinate $t$ on the line, the de Rham part of the BRST operator is $d_{\mathbb{R}}$. On the jet algebra one has the contraction homotopy $[d_{\mathbb{R}}, \iota_{\partial_t}] = \partial_t$. Hence a local expression containing a positive $t$-derivative is cohomologous to zero after applying the usual Euler homotopy on jet degree. This kills topological-direction jets and leaves the holomorphic $z_i$-derivatives used later on the closed side. □

Proposition 3.120 *(Stable trace invariants).* Let $F = \mathbb{C}\langle x_1, x_2\rangle$ and let $\mathrm{Cyc}(F) = F/[F,F]$ be the vector space of cyclic words. For each $N$, $GL_N$ acts on $\mathfrak{gl}_N^2$ by simultaneous conjugation. The degreewise stable invariant algebra

$$\mathbb{C}[\mathfrak{gl}_\infty^2]_{\mathrm{st}}^{GL_\infty} := \bigoplus_{d\geq 0} \varprojlim_N \big(\mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}\big)_d$$

with respect to block-restriction maps in each word-degree is the free commutative algebra generated by the nonempty cyclic words:

$$\mathbb{C}[\mathfrak{gl}_\infty^2]_{\mathrm{st}}^{GL_\infty} \cong \mathbf{S}\big(\mathrm{Cyc}(F)_{\mathrm{nonempty}}\big),$$

where a cyclic word $w$ maps to the trace function $(\phi_1, \phi_2) \mapsto \mathrm{Tr}\, w(\phi_1, \phi_2)$. More precisely, in total positive word-degree $d$ this map is an isomorphism for every $N \geq d$, after omitting the scalar empty trace as a primitive generator.

*Proof.* Procesi–Razmyslov invariant theory [17, 18] gives two inputs. First, $\mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}$ is generated by trace functions of words in the two generic matrices. Second, all polynomial relations among these trace functions are consequences of the fundamental trace identities of matrix size $N$; in particular every homogeneous trace relation has total word-degree at least $N+1$. Fix a total word-degree $d$. For every $N \geq d$ the degree-$d$ relations are exactly cyclicity of the trace. Passing degree by degree to the stable inverse limit removes all finite-$N$ trace identities. The surviving identification is $\mathrm{Tr}(uv) = \mathrm{Tr}(vu)$. This is exactly the symmetric algebra on cyclic words. □

Lemma 3.121 *(Stable marked trace invariants).* In the stable invariant theory used below, the primitive part linear or quadratic in an odd marked matrix $\psi$ is obtained from the corresponding even three-matrix trace invariant by polarization and skew-symmetrisation in the marked letters. The only stable single-trace relation is graded cyclicity: moving one $\psi$ through the trace has sign $+1$, while moving one $\psi$ past another $\psi$ gives a minus sign.

*Proof.* Apply Proposition 3.120 to three even generic matrices and take the homogeneous summand linear, respectively quadratic, in the third matrix. The Procesi–Razmyslov trace identities have no stable relations in the fixed total word-degree once $N$ is large enough. Declaring the marked matrix odd is the usual super-polarization: the linear part is unchanged, and the quadratic part is projected to the alternating component under exchange of the two marked occurrences. The trace relation therefore becomes exactly graded cyclicity. These are the signs used in the finite complex for one- and two-$\psi$ words. □

*Remark* 3.122. The empty trace is the finite-rank scalar $\mathrm{Tr}_N(1) = N$, which changes with $N$ and fails to define a stable primitive generator. The stable symmetric algebra still has its ordinary unit; what is omitted is the empty trace as a generator of the connected primitive part. The class $\mathrm{Tr}(\psi)$ is a genuine stable odd primitive class. It is excluded from the Hamiltonian comparison because constants have been removed from the closed Hamiltonian algebra.

Varying

$$S = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big)$$

with respect to $A$ gives the moment-map equation $[\phi_1, \phi_2] = 0$. In the reduced BV complex,

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0$$

on invariant trace functions.

*Construction* 3.123 *(Boundary evaluation of a closed Hamiltonian).* Let $f = \sum_{k,l} c_{k,l} z_1^k z_2^l \in \mathbb{C}[z_1, z_2]$. The brane replaces the two normal coordinates $z_1, z_2$ by the two adjoint scalar fields $\phi_1, \phi_2$. Choosing the ordered lift with all $\phi_1$'s to the left of all $\phi_2$'s gives the invariant function

$$\widetilde{\partial}_{\mathrm{bb},N}(f) = \sum_{k,l} c_{k,l} \operatorname{Tr}(\phi_1^k \phi_2^l) \in \mathcal{R}_N^{GL_N}.$$

Let

$$H^0_{\mathrm{Ham}}(\mathcal{R}_N^{GL_N}, Q) := H^0(\mathcal{R}_N^{GL_N}, Q)/\mathbb{C} \cdot \operatorname{Tr}(1)$$

as a vector space, and as a Lie algebra for the Poisson bracket. This is not an associative quotient of the operator algebra by its unit. Constants give the scalar $c_{0,0}N$ and therefore vanish in this Hamiltonian quotient. Thus the construction descends to a linear map

$$\partial_{\mathrm{bb},N} : \bar{A} \to H^0_{\mathrm{Ham}}(\mathcal{R}_N^{GL_N}, Q), \qquad [f] \longmapsto \overline{\widetilde{\partial}_{\mathrm{bb},N}(f)}.$$

Let $\mathsf{W}_{k,l}$ be the set of words in two letters containing $k$ copies of the first letter and $l$ copies of the second. The first descendant is obtained by inserting one Koszul variable and summing over all ordered lifts: For $k + l > 0$ the descendant part of the Hamiltonian comparison is

$$\partial^{(1)}_{\mathrm{bb},N}(e^{\mathrm{PBW}}_{k,l}) = \Psi_{k,l}, \qquad \Psi_{k,l} := \sum_{w \in \mathsf{W}_{k,l}} \operatorname{Tr}\big(\psi\, w(\phi_1, \phi_2)\big).$$

Here $e^{\mathrm{PBW}}_{k,l}$ is an abstract special-fibre PBW label in bidegree $(k, l)$, not the Taylor-dual coadjoint functional $\delta_{k,l}$ and not the Matlis principal part $\rho_{k,l}$. The same formula at $(k, l) = (0, 0)$ gives the decoupled scalar antifield $\operatorname{Tr}(\psi)$, which is excluded from the Hamiltonian pairing because constants have been removed from $\bar{A}$.

LEMMA 3.124 *(First descendants are cycles).* For every $k, l \geq 0$, the symmetrised trace $\Psi_{k,l}$ satisfies $Q\Psi_{k,l} = 0$.

*Proof.* Write $x = \phi_1$ and $y = \phi_2$. Since $x$ and $y$ are even and $Q\psi = [x, y]$, one has

$$Q\Psi_{k,l} = \sum_{w \in \mathsf{W}_{k,l}} \operatorname{Tr}\big((xy - yx)w(x, y)\big).$$

Group the resulting traces by cyclic word of multidegree $(k + 1, l + 1)$. The coefficient of a cyclic word in the positive sum is the number of cyclic adjacent occurrences of the block $xy$; the coefficient in the negative sum is the number of cyclic adjacent occurrences of the block $yx$. On a circle the number of transitions from an $x$-run to a $y$-run equals the number of transitions from a $y$-run to an $x$-run. The two grouped sums are therefore equal, so $Q\Psi_{k,l} = 0$. □

PROPOSITION 3.125 *(Primitive one-antifield Koszul homology).* Fix $k, l \geq 0$. The middle homology of the primitive single-trace complex with one displayed $\psi$ and $k, l$ displayed $x, y$ letters is one-dimensional, generated by the class of $\Psi_{k,l}$. The nonconstant part relevant to the Hamiltonian comparison is the direct sum over $k + l > 0$.

*Proof.* Write $x = \phi_1$ and $y = \phi_2$. Let $\mathsf{W}_{k,l}$ be the set of ordinary words with $k$ letters $x$ and $l$ letters $y$, and let $\mathsf{N}_{a,b}$ be the set of cyclic words with $a$ letters $x$ and $b$ letters $y$. By Lemma 3.121, primitive cyclic words are taken modulo the graded cyclic rule

$$\operatorname{Tr}(ab) = (-1)^{|a||b|}\operatorname{Tr}(ba), \qquad |x| = |y| = 0, \quad |\psi| = 1.$$

Graded cyclicity gives a unique representative $\mathrm{Tr}(\psi w)$ for every one-$\psi$ cyclic word. Thus the relevant one-$\psi$ chain group is

$$C_1(k,l) = \mathbb{C}\{\mathsf{W}_{k,l}\}.$$

The zero-$\psi$ target is

$$C_0(k,l) = \mathbb{C}\{\mathsf{N}_{k+1,l+1}\},$$

and the differential is the incidence map

$$\partial_1(w) = [xyw]_{\mathrm{cyc}} - [yxw]_{\mathrm{cyc}}.$$

The two-$\psi$ source is the alternating quotient generated by traces $\mathrm{Tr}(\psi u \psi v)$ with $u$ and $v$ ordinary words whose total bidegree is $(k-1, l-1)$:

$$\mathrm{Tr}(\psi u \psi v) = -\mathrm{Tr}(\psi v \psi u).$$

Hence diagonal pairs vanish, and if $k = 0$ or $l = 0$ this source is zero. Applying $Q\psi = xy - yx$ gives

$$\begin{aligned}\partial_2(u,v) &= Q\,\mathrm{Tr}(\psi u \psi v)\\ &= \mathrm{Tr}\big(\psi vxyu - \psi vyxu - \psi uxyv + \psi uyxv\big),\end{aligned}$$

so, after removing the common front $\psi$,

$$\partial_2(u,v) = vxyu - vyxu - uxyv + uyxv.$$

The displayed formula changes sign under $(u,v) \leftrightarrow (v,u)$, so it is well defined on the alternating quotient. A direct substitution shows $\partial_1\partial_2 = 0$.

If $k = 0$ or $l = 0$, then $C_1(k,l)$ is spanned by $x^k$ or $y^l$, $\partial_1 = 0$, and $C_2(k,l) = 0$. Hence the middle homology is already the line spanned by $\Psi_{k,l}$. Assume from now on that $k, l > 0$.

We compute the first homology of this finite complex. Form a graph $\Gamma_{k,l}$ whose vertices are the necklaces in $\mathsf{N}_{k+1,l+1}$ and whose oriented edges are the words $w \in \mathsf{W}_{k,l}$, with

$$w : [yxw]_{\mathrm{cyc}} \longrightarrow [xyw]_{\mathrm{cyc}}.$$

Then $\partial_1$ is the cellular boundary of this graph. For each pair $(u,v)$ as above attach the square whose oriented boundary is $vxyu - vyxu - uxyv + uyxv$. Denote the resulting two-complex by $X_{k,l}$.

Equivalently, $X_{k,l}$ is the two-skeleton of the chip-moving state complex $Y_{k,l}$. The full complex $Y_{k,l}$ has an $r$-cell for $r$ mutually chosen chip moves on a cycle with $l+1$ oriented edges and $k+1$ indistinguishable chips, modulo cyclic relabelling of the edge labels. A vertex is a cyclic distribution of the chips among the vertices of this cycle, hence a necklace in $\mathsf{N}_{k+1,l+1}$. A one-cell moves one chosen chip across one oriented edge; cutting the cycle at this moving edge records exactly a word $w \in \mathsf{W}_{k,l}$, and its two endpoints are $[yxw]_{\mathrm{cyc}}$ and $[xyw]_{\mathrm{cyc}}$. A two-cell records two chip moves performed independently, and its oriented boundary is precisely $vxyu - vyxu - uxyv + uyxv$. Therefore the cellular chain complex of $Y_{k,l}$ through degree two is exactly

$$C_2(k,l) \longrightarrow C_1(k,l) \longrightarrow C_0(k,l),$$

and the inclusion $X_{k,l} = Y_{k,l}^{(2)} \hookrightarrow Y_{k,l}$ induces an isomorphism on $H_1$, since cells of dimension at least three do not affect first homology.

The homology of $Y_{k,l}$ is the homology of the cycle. For completeness we recall the elementary Abel-map argument. Before quotienting by cyclic relabelling, send a chip configuration to its Abel coordinate, the sum of the chip positions in $\mathbb{R}/(l+1)\mathbb{Z}$, and extend linearly over each cell. In the universal cover of the target circle, a fibre is described by weakly ordered lifted chip positions with fixed total sum; it is a convex polytope, subdivided by the same chip-crossing cells. Hence the Abel map is a cellular homotopy equivalence to $S^1$. The finite cyclic relabelling of the $y$-gaps is equivariant for this Abel map: it rotates the target circle, and every stabilizer acts affinely on a convex fibre, whose quotient is again contractible. Thus the quotient target is still a circle and the quotient complex $Y_{k,l}$ has the homology of $S^1$. Consequently

$$H_1(X_{k,l};\mathbb{C}) \cong H_1(S^1;\mathbb{C}) \cong \mathbb{C}.$$

Lemma 3.124 identifies this representative as a cycle in the original trace complex. The functional

$$\epsilon\big(\mathrm{Tr}(\psi w)\big) = 1, \qquad w \in \mathsf{W}_{k,l},$$

kills every displayed two-$\psi$ boundary because each boundary has two positive and two negative summands, while

$$\epsilon(\Psi_{k,l}) = \binom{k+l}{k} \neq 0.$$

Thus the one-$\psi$ homology in bidegree $(k,l)$ is the one-dimensional line generated by $[\Psi_{k,l}]$. □

*Remark* 3.126. A finite exact homology check constructs the finite complex $C_2(k,l) \to C_1(k,l) \to C_0(k,l)$ over $\mathbb{Q}$, with the alternating two-$\psi$ source above, and confirms $\dim\ker\partial_1/\mathrm{im}\,\partial_2 = 1$ for $0 \le k,l \le 5$. It also checks the open Hamiltonian descendant action of Proposition 3.143 (below) for exponents at most 3, and the closed coadjoint Taylor-dual formula together with the principal-part realization of Proposition 3.147 (below) for exponents at most 5. The finite exact check verifies the displayed finite models without adding assumptions.

COROLLARY 3.127 *(PBW special-fibre one-antifield pairing).* Let $\bar{A}_{\mathrm{desc}}[1]$ be the abstract shifted PBW label copy with basis $e^{\mathrm{PBW}}_{k,l}$ in nonconstant bidegree $(k,l)$. The assignment

$$e^{\mathrm{PBW}}_{k,l} \longmapsto [\Psi_{k,l}]$$

extends to an isomorphism from $\bar{A}_{\mathrm{desc}}[1]$ to the nonconstant primitive one-$\psi$ brane homology.

*Proof.* The monomials $z_1^k z_2^l$ with $k+l>0$ form the polynomial basis of $\bar{A}$ and therefore index the PBW special-fibre label copy $\bar{A}_{\mathrm{desc}}$. By Proposition 3.125, the primitive one-$\psi$ homology has one generator in each bidegree, namely $[\Psi_{k,l}]$. Discarding the scalar bidegree $(0,0)$ and sending the abstract PBW label in every nonconstant bidegree to that generator gives a filtered vector-space isomorphism. This is the boundary pairing used in the PBW special fibre. It is not the continuous Taylor-dual/coadjoint module and not the Matlis principal-part realization. □

LEMMA 3.128 *(Adjacent swaps).* In the invariant reduced BV cohomology,

$$\mathrm{Tr}(u\phi_1\phi_2 v) - \mathrm{Tr}(u\phi_2\phi_1 v)$$

is $Q$-exact for all words $u, v$.

*Proof.* By cyclicity of the trace,

$$\begin{aligned} Q\,\mathrm{Tr}(\psi vu) &= \mathrm{Tr}([\phi_1, \phi_2]vu) \\ &= \mathrm{Tr}(\phi_1\phi_2 vu - \phi_2\phi_1 vu) \\ &= \mathrm{Tr}(u\phi_1\phi_2 v) - \mathrm{Tr}(u\phi_2\phi_1 v). \end{aligned}$$

□

Corollary 3.129 *(Matrix evaluation is commutative in cohomology).* If two words in $\phi_1, \phi_2$ have the same numbers of $\phi_1$ and $\phi_2$, then their traces define the same class in the reduced BV cohomology. Consequently $\partial_{\mathrm{bb},N}(f)$ is determined by the commutative polynomial $f \in \mathbb{C}[z_1, z_2]$ and is independent of the chosen ordered lift.

*Proof.* Adjacent transpositions generate all reorderings of a finite word. Each adjacent transposition of the adjacent letters $\phi_1, \phi_2$ changes the trace by a $Q$-exact term by Lemma 3.128; cyclicity moves the chosen adjacent pair into the displayed position of that lemma. Summing the resulting exact differences proves the claim. □

Proposition 3.130 *(Stalk central multiplication).* Let

$$A_N^{\mathrm{pt}} = (\mathcal{R}_N^{GL_N}, Q)$$

be the invariant reduced open Koszul algebra at a brane insertion point. For $f \in \mathbb{C}[z_1, z_2]$, set

$$B_f = \widetilde{\partial}_{\mathrm{bb},N}(f) = \sum_{k,l} c_{k,l}\mathrm{Tr}(\phi_1^k \phi_2^l).$$

Then $QB_f = 0$. Regard $B_f$ as a Hochschild 0-cochain; its associated central operation is multiplication

$$L_{B_f} : A_N^{\mathrm{pt}} \to A_N^{\mathrm{pt}}, \qquad a \longmapsto B_f a.$$

This Hochschild 0-cochain is closed. The same statement holds for multiplication by a closed one-antifield class $\Psi_{k,l}$. After projection to the Hamiltonian quotient and stabilization in $N$, the degree-zero classes satisfy

$$\{B_f, B_g\}_{\mathrm{open}} = B_{\{f,g\}_{\bar{A}}}.$$

*Proof.* Since $Q\phi_1 = Q\phi_2 = 0$, every $B_f$ is $Q$-closed. The algebra $A_N^{\mathrm{pt}}$ is graded-commutative, so the Hochschild differential of a 0-cochain $b$ is

$$d_{\mathrm{Hoch}} b(a) = Qb\, a + ba - (-1)^{|b||a|} ab.$$

This vanishes for every closed graded-central element $b$. The classes $\Psi_{k,l}$ are closed by Lemma 3.124 and are central for the same graded-commutative reason. Ordered-lift independence for $B_f$ is cohomological and is exactly Corollary 3.129. The Poisson identity follows from Theorem 3.161 (below). □

Let $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$. The removed constant corresponds on the gauge side to the scalar $\mathrm{Tr}(1) = N$ and has zero Hamiltonian vector field.

PROPOSITION 3.131 *(Connected brane Hamiltonian sector).* In the stable large-$N$ open BV model, the connected Hamiltonian descendant sector freely generated by the nonempty primitive ghost-zero traces and the nonconstant primitive one-$\psi$ Koszul homology is

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane},H} \cong \mathbf{S}\big(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]\big).$$

The class of $z_1^k z_2^l$ with $k + l > 0$ corresponds to $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$. The corresponding descendant basis vector corresponds to the symmetrised trace with one $\psi$ and $k, l$ copies of $\phi_1, \phi_2$.

*Proof.* Proposition 3.120 gives the free commutative algebra on cyclic words. Lemma 3.128 identifies any two cyclic words with the same numbers of $\phi_1$ and $\phi_2$, because adjacent transpositions generate all reorderings. Thus the connected ghost-zero cyclic words reduce to the monomials $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$ with $k + l > 0$. Proposition 3.125 computes the primitive one-$\psi$ homology in each multidegree; the nonconstant generators are $\Psi_{k,l}$ with $k + l > 0$. Products of the connected ghost-zero primitive classes and these one-$\psi$ primitive classes give the free graded-commutative multi-trace algebra in the Hamiltonian sector. By definition of the Hamiltonian sector, single traces containing two or more occurrences of $\psi$ are excluded from this algebra. □

*Remark* 3.132. The word "sector" is literal. The calculation above presents the stable algebra generated by the primitive no-$\psi$ Hamiltonians and the nonconstant primitive one-$\psi$ homology generated by $\Psi_{k,l}$ for $k + l > 0$. The full invariant Koszul homology in higher $\psi$-count lies outside this sector, as does the decoupled scalar class $\mathrm{Tr}(\psi)$.

*Problem* 3.133 *(Full primitive Koszul homology).* Let

$$A_\psi = \mathbb{C}\langle x, y, \psi\rangle, \qquad |x| = |y| = 0, \quad |\psi| = 1,$$

with differential $d\psi = xy - yx$ and $dx = dy = 0$. Put

$$\mathrm{Cyc}(A_\psi) = A_\psi / [A_\psi, A_\psi]_{\mathrm{gr}}, \qquad [uv] = (-1)^{|u||v|}[vu].$$

Give $x, y, \psi$ weights

$$\mathrm{wt}(x) = (1, 0), \qquad \mathrm{wt}(y) = (0, 1), \qquad \mathrm{wt}(\psi) = (1, 1).$$

For fixed $(R, S)$ define

$$K^p_{R,S} = \mathbb{C}\{[w]_{\mathrm{cyc}} : \#\psi(w) = p,\ \#x(w) = R - p,\ \#y(w) = S - p\},$$

with $0 \le p \le \min(R, S)$. The differential is

$$\partial[w] = \sum_{i : a_i = \psi} (-1)^{\#\{\psi \text{ before } i\}} [a_1 \cdots a_{i-1} x y\, a_{i+1} \cdots a_n - a_1 \cdots a_{i-1} y x\, a_{i+1} \cdots a_n]_{\mathrm{cyc}}.$$

Prove the primitive homology formula

$$H_p(K_{R,S}) \cong \begin{cases} \mathbb{C} \cdot [x^R y^S], & p = 0, \\ \mathbb{C} \cdot [\Psi_{R-1,S-1}], & p = 1,\ R, S \ge 1, \\ 0, & p \ge 2. \end{cases}$$

Equivalently, prove that the primitive stable Koszul homology is $\mathbb{C}[x, y] \oplus \mathbb{C}[x, y]\theta$ with $|\theta| = 1$ and $\mathrm{wt}(\theta) = (1, 1)$.

The required new ingredient is the all-$p$ orientation statement identifying this graded cyclic complex with the cellular chains of the chip-moving necklace state complex, together with the Abel-map proof that the relevant state complex has the homology of $S^1$. Proposition 3.125 proves exactly the $p \leq 2$ skeleton needed for the Hamiltonian sector. The complete theorem also requires the all-$\psi$ stable trace presentation and an exact computational check of the finite complex.

### 3.11 *Operators in the closed Hamiltonian sector*

The brane side has produced a stable algebra of trace classes generated by the primitive ghost-zero monomials $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$ and the primitive one-$\psi$ descendants $\Psi_{k,l}$. The closed Hamiltonian sector must produce the same data as a Lie algebra of local modes whose Chevalley–Eilenberg cochains carry the residual BRST differential.

Let $\mathfrak{h} = \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ with the projected Poisson bracket, and let $\mathfrak{h}^\vee$ be its continuous Taylor dual. The cotangent field is a coadjoint module, so the local Hamiltonian Lie algebra is

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1].$$

For $f, g \in \mathfrak{h}$ and $\lambda, \mu \in \mathfrak{h}^\vee$, the brackets are

$$[f, g] = \{f, g\}, \qquad [f, \eta_\lambda] = \eta_{\mathrm{ad}_f^\vee \lambda}, \qquad [\eta_\lambda, \eta_\mu] = 0,$$

where $(\mathrm{ad}_f^\vee \lambda)(g) = -\lambda(\{f, g\})$. On polynomial monomials in the dense subalgebra $\bar{A}$ this reads

$$[z_1^k z_2^l, z_1^m z_2^n] = \overline{(kn - lm) z_1^{k+m-1} z_2^{l+n-1}},$$

where the bar means projection to $\bar{A}$; in particular the constant bracket of $z_1$ and $z_2$ is zero in the reduced Hamiltonian Lie algebra.

Lemma 3.134 *(Shifted coadjoint signs).* Let $\eta_\lambda$ denote the suspension of $\lambda \in \mathfrak{h}^\vee$ into $\mathfrak{h}^\vee[1]$. Since the Hamiltonians in $\mathfrak{h}$ are even,

$$[f, \eta_\lambda] = \eta_{\mathrm{ad}_f^\vee \lambda}, \qquad [\eta_\lambda, \eta_\mu] = 0.$$

Consequently the enveloping algebra relations involving two shifted coadjoint generators are odd-commutative:

$$\eta_\lambda \eta_\mu + \eta_\mu \eta_\lambda = 0.$$

*Proof.* The semidirect product bracket before shifting is the ordinary coadjoint action of the even Lie algebra $\mathfrak{h}$ on its continuous dual. Suspending the module raises the degree of every $\lambda$ by one. The mixed-bracket sign is $+1$ because $|f| = 0$. The shifted module is abelian as a Lie ideal, so its bracket with itself is zero; in the associative enveloping algebra this zero Lie bracket gives $\eta_\lambda \eta_\mu - (-1)^{1 \cdot 1} \eta_\mu \eta_\lambda = 0$. □

The kinematic BRST operator is

$$Q_0 = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}.$$

The same jet contraction as in Lemma 3.119, together with the Dolbeault Poincare lemma in the formal local model, leaves precisely the holomorphic Taylor coefficients. Thus the linear local coordinates are the functionals

$$\partial_{z_1}^k \partial_{z_2}^l \alpha|_0, \qquad \partial_{z_1}^k \partial_{z_2}^l \beta|_0.$$

With the coefficient-dual convention fixed above, these are represented by $(k!l!)^{-1}\partial_{z_1}^k \partial_{z_2}^l|_0$. Since zero-form components of $\alpha$ and $\beta$ have negative field ghost numbers, the corresponding linear functions have positive ghost numbers.

PROPOSITION 3.135 *(Closed local operators are continuous CE cochains).* The classical local operators of the formal Hamiltonian cotangent BF sector are the continuous Chevalley-Eilenberg cochains

$$O_{\mathrm{grav}}^{\mathrm{cl}} \cong C_{\mathrm{CE,cont}}^{\bullet}(\mathfrak{g}) = \widehat{\mathbf{S}}(\mathfrak{g}_{\mathrm{cont}}^{\vee}[-1]),$$

with CE differential dual to the Lie brackets above.

*Proof.* Work at the formal insertion point. The jet algebra of the fields is the completed free graded-commutative algebra on the Taylor coefficients of $\alpha$ and $\beta$. The kinematic differential $Q_0 = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$ contracts all positive real and anti-holomorphic jets: in the real directions the homotopy is $[d_{\mathbb{R}^2}, \iota_{\partial_{x_i}}] = \partial_{x_i}$, and in the anti-holomorphic directions it is the formal Dolbeault Poincare homotopy. The $Q_0$-cohomology of the jet algebra is therefore generated by the holomorphic Taylor coefficients alone.

The remaining $\alpha$-coefficients are Taylor Hamiltonians modulo constants, hence elements of $\mathfrak{h} = \widehat{\bar{\mathfrak{h}}}$; degreewise polynomial coefficients form the dense subalgebra $\bar{A}$. The cotangent field $\beta$ supplies the odd shifted continuous dual $\mathfrak{h}^{\vee}[1]$. Thus the linear space of residual local modes is $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^{\vee}[1]$.

The nonlinear BRST differential is determined by the BV Hamiltonian

$$S = \int \left\langle \beta, D\alpha + \frac{1}{2}\{\alpha, \alpha\} \right\rangle.$$

The linear part is exactly $Q_0$, already killed in cohomology. The quadratic term sends two Hamiltonian $\alpha$-modes $f, g$ to their reduced Poisson bracket $\{f, g\}$, and the action on the cotangent variable is the coadjoint action displayed above. The quadratic term has zero component with two cotangent inputs, so $[\eta_\lambda, \eta_\mu] = 0$. Hence the residual BRST differential on functions is precisely the derivation dual to the Lie bracket on $\mathfrak{g}$, which is the Chevalley-Eilenberg differential. Continuity records the completion over polynomial degree. □

### *3.12 Koszul Duality*

The closed local operators have been identified with continuous Chevalley–Eilenberg cochains of $\mathfrak{g}$. The brane Hamiltonian sector is a free graded-commutative algebra on $\bar{A}$ and $\bar{A}_{\mathrm{desc}}[1]$. To compare these as $E_1$-algebras one passes to the operation that exchanges commutative cochains and associative enveloping algebras: Koszul duality.

*Definition* 3.136. For an augmented complete dg algebra $B$, its Koszul dual is

$$B^! = \mathrm{RHom}_B(\mathbb{C}, \mathbb{C})$$

[23]. We regard commutative Chevalley-Eilenberg cochains as $E_1$ algebras by restriction. All bar and cobar constructions below are taken in the category of complete filtered Tate vector spaces, with continuous maps. The completions are by symmetric or tensor length together with the coefficient topology coming

from Taylor degree. For a Tate space $V$, the notation $V^\vee_{\mathrm{cont}}$ means the continuous dual with its dual Tate topology, $\widehat{\otimes}$ means the completed Tate tensor product, and

$$\widehat{\mathbf{S}}^n(V) = (V^{\widehat{\otimes} n})_{\mathbb{S}_n}.$$

Chevalley-Eilenberg chains below are direct sums in CE length with completed coefficients in each length; CE cochains, completed cobar algebras, and completed enveloping algebras are products in the corresponding length filtration.

Lemma 3.137 *(Tate CE chains, completed PBW, and Koszul duality).* Work in the symmetric monoidal category **TateVec** of complete filtered Tate $\mathbb{C}$-vector spaces:

$$V = \varprojlim_{w\in\mathbb{N}} V_{\leq w}, \qquad V_{\leq w} \text{ finite-dimensional}, \qquad F^w V = \mathbf{Ker}\big(V \to V_{\leq w-1}\big),$$

with $\widehat{\otimes}$ the completed tensor product $V\widehat{\otimes}W = \varprojlim_w (V_{\leq w} \otimes W_{\leq w})$ and $V^\vee_{\mathrm{cont}} = \varinjlim_w V^*_{\leq w}$ the continuous dual. Admissible quasi-isomorphisms are the filtered ones: chain maps inducing a quasi-isomorphism on every quotient $V/F^wV$ and on the associated graded.

Let $\mathfrak{g}$ be a Tate graded Lie algebra, conilpotent in the sense that the descending Lie powers $\mathfrak{g}^{(\geq m)} = [\mathfrak{g}, \mathfrak{g}^{(\geq m-1)}]$ satisfy $\bigcap_m \big(\mathfrak{g}^{(\geq m)} + F^w\mathfrak{g}\big) = F^w\mathfrak{g}$ for every $w$. Then:

(i) For $V = \prod_d V_d$ with all $V_d$ finite-dimensional and $W = \bigoplus_d W_d$ with all $W_d$ finite-dimensional,

$$(V \oplus W)^\vee_{\mathrm{cont}} = \bigoplus_d V_d^\vee \oplus \prod_d W_d^\vee,$$

and the dualization $V \mapsto V^\vee_{\mathrm{cont}}$ is exact and continuous in the filtration $F^w$.

(ii) The continuous Chevalley–Eilenberg cochains

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \mathrm{Hom}_{\mathrm{cont}}\big(\widehat{\mathbf{S}}^c(\mathfrak{g}[1]), \mathbb{C}\big)$$

identify with $\widehat{\mathbf{S}}_{\mathrm{Tate}}\big(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]\big)$, and the coaugmentation filtration on the CE coalgebra $C^{\mathrm{CE}}_*(\mathfrak{g}) = \widehat{\mathbf{S}}^c(\mathfrak{g}[1])$ is conilpotent in **TateVec**: every element of any finite Tate window lies in a finite power of the coaugmentation coideal.

(iii) The continuous bar resolution of the augmentation $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \to \mathbb{C}$ has dual cobar algebra $\Omega C^{\mathrm{CE}}_*(\mathfrak{g})$, and the canonical Lie bar–cobar map

$$\Omega C^{\mathrm{CE}}_*(\mathfrak{g}) \longrightarrow \widehat{U}(\mathfrak{g})$$

is an admissible (filtered) quasi-isomorphism in **TateVec**.

(iv) The PBW filtration on $\widehat{U}(\mathfrak{g})$ by tensor length is complete and separated. Its associated graded is the completed symmetric algebra

$$\mathrm{gr}^\bullet_F \widehat{U}(\mathfrak{g}) \cong \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{g}).$$

*Proof. Item (1).* A continuous functional on $V = \prod V_d$ depends on finitely many coordinates: the topology on $\mathbb{C}$ is discrete, so the kernel of any continuous map contains some $F^wV$. Therefore $V^\vee_{\mathrm{cont}} = \bigoplus_d V_d^*$. Conversely, a continuous functional on the discrete sum $W = \bigoplus W_d$ is given by an arbitrary product of degreewise functionals, hence $W^\vee_{\mathrm{cont}} = \prod_d W_d^*$. Direct sum of these gives the displayed

formula. Exactness of dualization in **TateVec** is preservation of finite limits; continuity in the Tate filtration is direct.

*Item (2).* The completed CE coalgebra $\widehat{\mathbf{S}}^c(\mathfrak{g}[1]) = \bigoplus_n \widehat{\mathbf{S}}^n(\mathfrak{g}[1])$ has coaugmentation coideal $\widehat{\mathbf{S}}^{\geq 1}(\mathfrak{g}[1])$. An element $x$ in the Tate window $V_{\leq w}$ projects to $\mathfrak{g}_{\leq w}$, whose finite-dimensional symmetric powers are conilpotent because the Lie bracket on $\mathfrak{g}_{\leq w}$ has finite descending central series in any finite-dimensional approximation. Pass to the inverse limit using the Tate-conilpotency hypothesis on $\mathfrak{g}$: the descending powers $(\widehat{\mathbf{S}}^c)^{(\geq m)}$ have empty Tate intersection modulo every finite filtration step. The displayed equality identifies cochains as continuous functionals on the CE coalgebra, using item (1) windowwise.

*Item (3).* Filter the CE coalgebra by symmetric coalgebra length and by Tate window. Project to a fixed window $\mathfrak{g}_{\leq w}$ and bound symmetric/tensor length to a fixed $n$. The resulting finite-dimensional diagram is the ordinary Lie bar–cobar comparison [24]: the bar resolution computes $\mathrm{RHom}_{C^\bullet_{\mathrm{CE}}(\mathfrak{g}_{\leq w}, \leq n)}(\mathbb{C}, \mathbb{C})$, and the cobar algebra maps to the universal enveloping algebra of the same window. These finite comparisons are compatible with enlargement of the Tate window and increase of the length cap: the enlargement-of-window map is the dual of the surjection $\mathfrak{g} \to \mathfrak{g}_{\leq w}$, and Lie-algebra surjections induce maps of universal enveloping algebras. Passing to the Tate limit and completing in tensor length gives the asserted filtered quasi-isomorphism in **TateVec**. The PBW filtration is, windowwise, the standard PBW filtration on $U(\mathfrak{g}_{\leq w})$, hence preserved under inverse limits.

*Item (4).* Tensor length is a separated, descending filtration on $\widehat{U}(\mathfrak{g})$ because the Lie relations $xy - (-1)^{|x||y|} yx - [x, y] = 0$ have leading tensor-length term $xy - (-1)^{|x||y|} yx$ of length 2 and lower-order term $-[x, y]$ of length 1; so the associated graded retains only the symmetric (length-fixed) part. Completeness is clear from the Tate construction. The associated graded is therefore the completed symmetric algebra $\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{g})$. □

*Remark* 3.138. The conilpotency hypothesis in item (2) is a genuine hypothesis. It is satisfied by the nilpotent truncations and by the controlled weighted replacements used later; it is not asserted for the full reduced formal Hamiltonian algebra with linear modes, which is perfect in the sense of Remark 3.26. The Beilinson–Feigin–Mazur explicit product/direct-sum Tate convention [22] is the special case $\mathfrak{g} = \left(\prod P_d\right) \oplus \left(\bigoplus Q_d\right)$ with the bracket bilinear in the two displayed pieces.

LEMMA 3.139 *(Linear Poisson Schouten differential).* Let $\mathfrak{h}$ be a pro-finite-dimensional Lie algebra in **TateVec**. The completed symmetric algebra $\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})$ is naturally a Poisson algebra with Kirillov–Kostant–Souriau bracket $\{O_I, O_J\}_\partial = O_{[H_I, H_J]} = C^K_{IJ}\, O_K$ on a chosen homogeneous basis $\{H_I\}$ of $\mathfrak{h}$, where $O_I \in \mathfrak{h}^*$ is the linear coordinate. The associated Poisson bivector is the linear bivector

$$\pi_\partial = \tfrac{1}{2}\, C^K_{IJ}\, O_K\, \theta^I \wedge \theta^J, \qquad \theta^I = \partial/\partial O_I \in \mathrm{Der}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big),$$

and the Lichnerowicz/Schouten differential $d_\pi = [\pi_\partial, -]_S$ on

$$\mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big) = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h}) \widehat{\otimes} \widehat{\Lambda}(\theta^I)$$

acts on generators by

$$d_\pi \theta^K = -\tfrac{1}{2}\, C^K_{IJ}\, \theta^I \theta^J, \qquad d_\pi O_K = C^L_{KJ}\, O_L\, \theta^J.$$

*Proof.* Direct calculation. The Schouten Leibniz rule with $[\theta^I, O_J]_S = \delta^I_J$, $[\theta^I, \theta^J]_S = 0$, $[O_I, O_J]_S = 0$ gives

$$\begin{aligned}[\pi_\partial, O_K]_S &= \tfrac{1}{2} C^L_{IJ}\, O_L\, [\theta^I\theta^J, O_K]_S \\ &= \tfrac{1}{2} C^L_{IJ}\, O_L\, \left(\delta^I_K \theta^J - \delta^J_K \theta^I\right) \\ &= \tfrac{1}{2}\left(C^L_{KJ} - C^L_{JK}\right) O_L \theta^J = C^L_{KJ} O_L \theta^J,\end{aligned}$$

using the antisymmetry $C^L_{IJ} = -C^L_{JI}$ in the lower indices. For $[\pi_\partial, \theta^K]_S$, the Schouten bracket with $\theta^K$ uses $[\theta^I\theta^J, \theta^K]_S = 0$ on the bivector factor and $[O_L, \theta^K]_S = -\delta^K_L$; the result is $-\frac{1}{2} C^K_{IJ}\theta^I\theta^J$. □

Proposition 3.140 *(Koszul dual of continuous CE cochains in the admissible envelope).* Let $\mathfrak{g}_{\mathrm{adm}}$ be a Tate graded Lie algebra satisfying the conilpotency hypothesis of Lemma 3.137; in this paper this means one of the nilpotent finite-window replacements, the weighted admissible envelope where the hypothesis is explicitly imposed, or another stated admissible replacement. Put

$$B_{\mathrm{adm}} = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee_{\mathrm{adm,cont}}[-1]\big).$$

Then

$$B^!_{\mathrm{adm}} \simeq \widehat{U}(\mathfrak{g}_{\mathrm{adm}})$$

in the completed Tate category. No statement of this proposition is made for the full formal Hamiltonian algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$ with its linear and conformal-symplectic low-degree modes.

*Proof.* This is exactly Lemma 3.137 applied to $\mathfrak{g}_{\mathrm{adm}}$. The conilpotency condition is the load-bearing hypothesis: it ensures that the coefficient-completed CE coalgebra is conilpotent in every finite Tate window, that the continuous bar resolution is admissible, and that the completed Lie bar-cobar map

$$\Omega C^{\mathrm{CE}}_*(\mathfrak{g}_{\mathrm{adm}}) \longrightarrow \widehat{U}(\mathfrak{g}_{\mathrm{adm}})$$

is a filtered quasi-isomorphism.

The full formal Hamiltonian algebra is handled elsewhere in this manuscript by the direct CE/PV generator calculation and by the nilpotent or weighted replacements. The proposition is therefore a theorem about the admissible envelope, not a theorem that the untruncated formal Hamiltonian algebra is conilpotent. □

The completed PBW/order Rees enveloping algebra

$$U_\hbar(\mathfrak{g}) = \widehat{T}(\mathfrak{g})[[\hbar]] \Big/ \overline{\left\langle xy - (-1)^{|x||y|} yx - \hbar[x, y]\right\rangle},$$

with closure in tensor length and the Tate coefficient topology, makes the PBW special-fibre bookkeeping precise in every admissible replacement to which Proposition 3.140 applies. The Rees parameter $\hbar$ records word length in the enveloping algebra and is independent of Taylor degree. At $\hbar = 0$ the completed PBW special fibre is

$$U_\hbar(\mathfrak{g})/(\hbar) \cong \widehat{\mathbf{S}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\widehat{\mathfrak{h}} \oplus \mathfrak{h}^\vee[1]\big).$$

Under the monomial-dual Taylor basis $\mathfrak{h}^\vee \cong \bar{A}_{\mathrm{desc}}$, the degreewise polynomial subalgebra $\mathbf{S}(\bar{A} \oplus \mathfrak{h}^\vee[1])$ is the stable classical brane Hamiltonian descendant algebra of Proposition 3.131: the degreewise polynomial part of the PBW associated graded gives the brane Hamiltonian descendant labels. For the full

formal Hamiltonian algebra this is a PBW/Rees label comparison, not a deformation-level Koszul-duality theorem. The interacting closed Hamiltonian bracket deforms the special fibre inside the admissible enveloping algebra; on the brane side the matching deformation at order one is the Hamiltonian trace bracket proved below.

### 3.12.1 First-order corrections, order hbar one

The PBW special-fibre label correspondence used in the reduced comparison reads

$$z_1^k z_2^l \leftrightarrow \overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}.$$

The reduced Poisson bracket on the gravity side is:

$$\{z_1^k z_2^l,\ z_1^m z_2^n\}_{\bar{A}} = \overline{(kn - lm)\, z_1^{k+m-1} z_2^{l+n-1}}.$$

If an exponent in the displayed monomial is negative then the coefficient $kn - lm$ is already zero; if the displayed monomial is constant, its class in $\bar{A}$ is zero. The matching bracket on the open side is the canonical bracket induced by the first-order kinetic term $\int \mathrm{Tr}(\phi_1\, d\phi_2)$, with the sign convention $\iota_{X_f}\omega = -df$ fixed in the AKSZ remark above:

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j, \qquad \{(\phi_1)^i{}_j, (\phi_1)^k{}_l\} = 0 = \{(\phi_2)^i{}_j, (\phi_2)^k{}_l\}.$$

LEMMA 3.141 *(Hamiltonian trace bracket).* The canonical matrix Poisson bracket descends to the invariant reduced BV cohomology and then to the Hamiltonian Lie quotient by constant trace classes.

*Proof.* For $\epsilon \in \mathfrak{gl}_N$, the component $\mu_\epsilon = -\mathrm{Tr}(\epsilon[\phi_1, \phi_2])$ of the moment map generates the infinitesimal conjugation action, with the sign fixed by the convention for the bracket. Hence $\{F, \mu_\epsilon\} = 0$ for every invariant function $F$. If $I_\mu$ is the ideal generated by all $\mu_\epsilon$, then $\{F, \mu_\epsilon K\} = \mu_\epsilon\{F, K\}$ lies again in $I_\mu$. Therefore the bracket of invariant functions descends to $(\mathcal{R}_N/I_\mu)^{GL_N}$, equivalently to the degree-zero reduced BV cohomology. The central line $\mathbb{C} \cdot \mathrm{Tr}(1)$ is central for this Lie bracket, so the Hamiltonian vector-space quotient by constants is compatible with the bracket. No associative quotient by the unit is taken. □

PROPOSITION 3.142 *(First-order trace bracket).* In the stable connected Hamiltonian trace cohomology,

$$\{\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}\} = \overline{(kn - lm)\mathrm{Tr}(\phi_1^{k+m-1} \phi_2^{l+n-1})}.$$

*Proof.* Write $F_{k,l} = \mathrm{Tr}(\phi_1^k \phi_2^l)$. The matrix Poisson bracket induced by

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j$$

is

$$\{F, G\} = \mathrm{Tr}\left(\frac{\partial F}{\partial \phi_1}\frac{\partial G}{\partial \phi_2} - \frac{\partial F}{\partial \phi_2}\frac{\partial G}{\partial \phi_1}\right),$$

where the derivatives are cyclic matrix derivatives. Varying one occurrence at a time gives

$$\frac{\partial F_{k,l}}{\partial \phi_1} = \sum_{a=0}^{k-1} \phi_1^{k-1-a} \phi_2^l \phi_1^a,$$

$$\frac{\partial F_{k,l}}{\partial \phi_2} = \sum_{b=0}^{l-1} \phi_2^{l-1-b} \phi_1^k \phi_2^b.$$

Therefore the first summand in the bracket is a sum of $kn$ traces, each obtained by removing one $\phi_1$ from the first word and one $\phi_2$ from the second word. The second summand is a sum of $lm$ traces obtained by removing one $\phi_2$ from the first word and one $\phi_1$ from the second. Every trace appearing in either nonzero sum has multidegree $(k+m-1, l+n-1)$. Lemma 3.128 identifies all cyclic words with this multidegree in reduced BRST cohomology. The coefficient is therefore $kn - lm$, and the common class is $\overline{\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}$. When the common word has multidegree $(0,0)$, this class is the scalar trace and is zero in the Hamiltonian quotient by constants. □

Proposition 3.143 *(Open action on primitive descendants).* Let

$$\widetilde{\Psi}_{a,b} = \binom{a+b}{a}^{-1} \Psi_{a,b}$$

be the averaged primitive one-$\psi$ class. In the stable primitive one-$\psi$ homology,

$$\{\overline{\mathrm{Tr}(\phi_1^p \phi_2^q)}, \widetilde{\Psi}_{a,b}\} = (pb - qa)\, \widetilde{\Psi}_{a+p-1,b+q-1},$$

with the convention that a term with a negative exponent is zero. Thus the unreduced polynomial descendant classes carry the adjoint polynomial Hamiltonian action. The Hamiltonian descendant sector is the quotient by the scalar line $\mathbb{C}\widetilde{\Psi}_{0,0}$, so the displayed action is projected further by killing the scalar target.

*Proof.* Put $x = \phi_1$ and $y = \phi_2$. The cyclic derivatives of $F_{p,q} = \mathrm{Tr}(x^p y^q)$ are

$$\frac{\partial F_{p,q}}{\partial x} = \sum_{i=0}^{p-1} x^{p-1-i} y^q x^i, \qquad \frac{\partial F_{p,q}}{\partial y} = \sum_{j=0}^{q-1} y^{q-1-j} x^p y^j.$$

For a one-$\psi$ cyclic word $\mathrm{Tr}(\psi w)$ of bidegree $(a,b)$, differentiating with respect to $y$ gives one summand for each of the $b$ occurrences of $y$ in $w$, and differentiating with respect to $x$ gives one summand for each of the $a$ occurrences of $x$. Multiplying by the displayed cyclic derivatives of $F_{p,q}$ therefore produces $pb$ positive one-$\psi$ words and $qa$ negative one-$\psi$ words for each $w \in \mathsf{W}_{a,b}$, all of total bidegree $(a+p-1, b+q-1)$.

The bracket is a cycle because $F_{p,q}$ is invariant, hence $\{F_{p,q}, \mu_\epsilon\} = 0$ for every moment-map component $\mu_\epsilon = -\mathrm{Tr}(\epsilon[x,y])$. Equivalently, the Hamiltonian vector field of $F_{p,q}$ preserves the Koszul differential $Q\psi = [x,y]$ and acts on Koszul homology. By Proposition 3.125, the target primitive one-$\psi$ homology in that bidegree is one-dimensional. The functional $\epsilon(\mathrm{Tr}(\psi w)) = 1$ detects its coefficient. Since $\epsilon(\widetilde{\Psi}_{a,b}) = 1$, the averaged input gives

$$\epsilon\big(\{F_{p,q}, \widetilde{\Psi}_{a,b}\}\big) = pb - qa.$$

Also $\epsilon(\widetilde{\Psi}_{a+p-1,b+q-1}) = 1$. Hence the displayed formula follows. If the target bidegree is $(0,0)$, the same formula is still the unreduced primitive one-$\psi$ action; the Hamiltonian descendant quotient sends that scalar class to zero. □

Corollary 3.144 *(Descendant action and coadjoint action: structural separation).* Under the PBW special-fibre pairing of Corollary 3.127, the open Hamiltonian descendant action on the polynomial $\Psi_{a,b}$-basis is the symmetric raising action

$$F_{p,q} : \widetilde{\Psi}_{a,b} \longmapsto (pb - qa)\, \widetilde{\Psi}_{a+p-1,b+q-1},$$

while the closed coadjoint action on the principal-part basis $\rho_{a,b} \leftrightarrow \delta_{a,b} \in \mathfrak{h}^\vee_{\mathrm{cont}}$ is the lowering action

$$\mathrm{ad}^\vee_{z_1^p z_2^q}\, \delta_{a,b} = -(pb - qa + p - q)\, \delta_{a-p+1,b-q+1},$$

with negative-index targets zero. The two actions agree only on the trivial Hamiltonian $p = q = 0$. The deformation quantisation of the cotangent factor sees the second action; the special-fibre PBW symmetric algebra sees the first. The structural reason is the PBW shadow versus coadjoint-module separation of Theorem 3.152 below: the $\widetilde{\Psi}_{a,b}$ and the $\rho_{a,b}$ have the same index set but different $\mathfrak{h}$-module structures.

*Proof.* Proposition 3.143 gives the symmetric raising formula on $\widetilde{\Psi}_{a,b}$. The coadjoint action is dual to the Poisson bracket: for $f = z_1^p z_2^q$ and the Taylor-dual basis vector $\delta_{a,b}$,

$$(\mathrm{ad}^\vee_f\, \delta_{a,b})(z_1^m z_2^n) = -\delta_{a,b}\big((pn - qm) z_1^{p+m-1} z_2^{q+n-1}\big),$$

giving the displayed lowering formula. For example, $z_1$ acts on open descendants by $\widetilde{\Psi}_{a,b} \mapsto b\, \widetilde{\Psi}_{a,b-1}$, whereas it acts coadjointly by $\delta_{a,b} \mapsto -(b+1)\, \delta_{a,b+1}$. The scalar edge $\{\, \mathrm{Tr}(\phi_1), \widetilde{\Psi}_{0,1} \,\} = \Psi_{0,0}$ is removed with the scalar descendant. The two formulas have opposite index shift directions and differ by $-(pb - qa + p - q)\rho_{a-p+1,b-q+1} - (pb - qa)\rho_{a+p-1,b+q-1}$ after any bidegree-only vector-space relabelling between the two alphabets, which is non-zero unless both targets vanish identically; this happens exactly on the trivial $p = q = 0$ action. The PBW special-fibre product identification is Corollary 3.127; the coadjoint half is Theorem 3.150; the Laurent/distributional principal-part realization is Theorem 3.147 and the separation Theorem 3.152. □

Three notions of "local" on the brane line $\mathbb{R}$ coexist; the coadjoint identification on the principal-part side requires that they remain distinct.

LEMMA 3.145 *(Three notions of locality).* On the brane line $\mathbb{R}$ with formal holomorphic transverse coordinates $z_1, z_2$ at the brane point $p \in \mathbb{C}^2$, the following three notions of locality are independent.

(i) *Support locality*:distributional support on the brane: a section $T \otimes m$ of $\mathcal{D}'_c(I) \widehat{\otimes} M$ has support locality whenever $\mathrm{supp}\, T \subset I \subset \mathbb{R}$.

(ii) *Topological locality*:locally constant in the $\mathbb{R}$-direction up to $Q$-exact translations: every section is identified with its translate by a parameter on $\mathbb{R}$ modulo the open BV differential $Q_{\mathrm{open}}$.

(iii) *Formal holomorphic locality*:polynomial, or Taylor-series, dependence on the transverse coordinates $z_1, z_2$.

The polynomial trace observables $O_f(a) = \int_{\mathbb{R}} a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt$ have all three. The shifted cotangent currents $\Theta_\rho(B)$ of Construction 3.156 have (1) and (2) but not (3): they are momentum modes attached to the brane and live in the $\mathbb{C}$-linear residue annihilator of scalar Hamiltonians inside the top local-cohomology principal-part module, $\mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1)$.

*Proof.* Independence is a category-theoretic statement: a polynomial in $z_1, z_2$ with point support is constant in $\mathbb{R}$, so (1) does not imply (3); a delta-current at one point of $\mathbb{R}$ has compact support but only trivial $Q$-exact translation, so (1) does not imply (2). The cotangent currents $\Theta_\rho(B)$ are smooth or distributional sections of the line $\mathbb{R}$ (hence (1)) and translation-invariant up to $Q_{\mathrm{open}}$-exact cocycles (hence (2)); their transverse content $\rho \in \mathcal{P}$ is a top-local-cohomology principal part, hence a residue-dual distributional mode rather than a Taylor-polynomial mode (hence not (3)). □

*Remark* 3.146. In canonical quantization of a constrained system, the configuration variables are locally polynomial in spatial coordinates and the conjugate momenta are dual distributions. This is exactly the separation Lemma 3.145 records: configuration observables on a brane have formal holomorphic locality (3), and momentum observables have only (1) and (2). The principal-part sector is the algebraic image of this separation in the BCOV Hamiltonian model.

Proposition 3.147 *(Principal-part realization of the coadjoint module).* Let

$$\mathcal{P} = \bigoplus_{\substack{a,b\geq 0\\ a+b>0}} \mathbb{C}\,\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2,$$

with residue pairing

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \mathrm{Res}_{z_1=z_2=0}\, z_1^{m-a-1} z_2^{n-b-1}\, dz_1 dz_2 = \delta_{a,m}\delta_{b,n}.$$

Let $\pi_{\mathrm{pp}}$ denote projection of Laurent two-forms to the span of the principal parts $\rho_{a,b}$. The action

$$f \cdot \rho = \pi_{\mathrm{pp}}\big(\{f, \rho\}\big)$$

identifies $\mathcal{P}$ with the continuous coadjoint module $\mathfrak{h}^\vee_{\mathrm{cont}}$. On monomials,

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1},$$

with the convention that a term with a negative index is zero.

*Proof.* The residue pairing identifies $\rho_{a,b}$ with the Taylor-dual functional $\delta_{a,b}$. Hamiltonian vector fields preserve the holomorphic volume form $dz_1 dz_2$, so integration by parts for the residue pairing gives

$$\langle \{f, \rho\}, g \rangle = -\langle \rho, \{f, g\} \rangle.$$

Hence the Laurent principal-part action realizes the coadjoint action. Directly, for $f = z_1^p z_2^q$ and $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$,

$$\begin{aligned}\{f, z_1^{-a-1} z_2^{-b-1}\} &= p z_1^{p-1} z_2^q\,(-(b+1)) z_1^{-a-1} z_2^{-b-2} \\ &\quad - q z_1^p z_2^{q-1}\,(-(a+1)) z_1^{-a-2} z_2^{-b-1} \\ &= -(pb - qa + p - q) z_1^{p-a-2} z_2^{q-b-2}.\end{aligned}$$

Since $z_1^{p-a-2} z_2^{q-b-2}\, dz_1 dz_2 = \rho_{a-p+1,b-q+1}$ when both indices are nonnegative, principal-part projection gives the displayed formula. □

Two rigidity statements pin the principal-part identification after the residue scalar is fixed: the residue pairing is unique up to a single nonzero scalar, and the coadjoint identification $\mathcal{P} \cong \mathfrak{h}^\vee_{\mathrm{cont}}$ is the unique continuous $\mathfrak{h}$-equivariant isomorphism modulo that scalar.

Theorem 3.148 *(Residue pairing, normalized up to scalar).* Up to a nonzero scalar, the residue pairing

$$\langle\,\cdot\,,\,\cdot\,\rangle : \mathcal{P} \times (\mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1) \longrightarrow \mathbb{C}, \qquad \langle \rho_{a,b}, [z_1^m z_2^n] \rangle = \delta_{a,m}\delta_{b,n},$$

is the unique non-degenerate continuous $\mathfrak{h}$-equivariant pairing graded of total degree zero with the property that every element of $\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)$ annihilates the constant line.

*Proof. Existence.* The residue pairing is non-degenerate by the displayed Kronecker relation; it is continuous because $\mathcal{P}$ is graded discrete and $\mathbb{C}[[z_1, z_2]]$ carries the Taylor filtration; equivariance is integration by parts for the Poisson bracket, since Hamiltonian vector fields preserve the holomorphic two-form $dz_1 dz_2$:

$$\langle \{f, \rho\}, g\rangle = \operatorname{Res}(\{f, \rho\} g) = -\operatorname{Res}(\rho\, \{f, g\}) = -\langle \rho, \{f, g\}\rangle.$$

*Uniqueness.* Let $\beta : \mathcal{P} \times (\mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1) \to \mathbb{C}$ be a second pairing satisfying the same conditions, and write $\beta_{a,b;m,n} = \beta(\rho_{a,b}, z_1^m z_2^n)$. Since $\beta$ is graded of total degree zero, $\beta_{a,b;m,n} = 0$ unless $a + b = m + n$.

*Diagonal weight constraint.* The Hamiltonian $z_1 z_2$ acts diagonally on monomials by $\{z_1 z_2, z_1^m z_2^n\} = (n - m) z_1^m z_2^n$. Direct Poisson computation on principal parts uses the Hamiltonian vector field $X_{z_1 z_2} = z_2\, \partial_{z_2} - z_1\, \partial_{z_1}$ (with $X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}$), giving $X_{z_1 z_2}(z_1^{-a-1} z_2^{-b-1}) = (a - b)\, z_1^{-a-1} z_2^{-b-1}$, so $\{z_1 z_2, \rho_{a,b}\} = (a - b)\rho_{a,b} = -(b - a)\rho_{a,b}$. Equivariance $\beta(\{z_1 z_2, \rho_{a,b}\}, z_1^m z_2^n) = -\beta(\rho_{a,b}, \{z_1 z_2, z_1^m z_2^n\})$ gives $-(b - a)\beta_{a,b;m,n} = -(n - m)\beta_{a,b;m,n}$, hence $\beta_{a,b;m,n} = 0$ unless $a - b = m - n$. Combined with the weight constraint $a + b = m + n$, this gives $\beta_{a,b;m,n} = 0$ unless $(a, b) = (m, n)$. So $\beta$ is diagonal in the residue basis: $\beta_{a,b;m,n} = c_{a,b}\, \delta_{a,m} \delta_{b,n}$ for some scalars $c_{a,b} \in \mathbb{C}$.

*Coefficients are constant.* The Hamiltonian generator $f = z_1$ has $X_{z_1} = \partial_{z_2}$, so $\{z_1, z_1^m z_2^n\} = n\, z_1^m z_2^{n-1}$ and a direct Poisson computation gives $\{z_1, \rho_{a,b}\} = -(b + 1)\rho_{a,b+1}$. Equivariance reads $\beta(\{z_1, \rho_{a,b}\}, z_1^m z_2^n) = -\beta(\rho_{a,b}, \{z_1, z_1^m z_2^n\})$, i.e., $-(b + 1)\beta_{a,b+1;m,n} = -n\, \beta_{a,b;m,n-1}$. Both sides vanish identically unless the diagonality condition is met on each pair. The right-hand side is non-zero only if $(a, b) = (m, n - 1)$, i.e., $m = a$, $n = b + 1$. The left-hand side is non-zero only if $(a, b + 1) = (m, n)$, i.e., $m = a$, $n = b + 1$. Both indices coincide, and we obtain $(b + 1)\, c_{a,b+1} = (b + 1)\, c_{a,b}$, hence $c_{a,b+1} = c_{a,b}$ for all $b \geq 0$ with $a + b + 1 \geq 1$. Symmetrically from $f = z_2$ (using $X_{z_2} = -\partial_{z_1}$, so $\{z_2, z_1^m z_2^n\} = -m\, z_1^{m-1} z_2^n$ and $\{z_2, \rho_{a,b}\} = (a + 1)\rho_{a+1,b}$) we get $c_{a+1,b} = c_{a,b}$ for all $a \geq 0$. Connecting these chains, all $c_{a,b}$ for $a + b \geq 1$ equal a single constant $c \in \mathbb{C}$. Non-degeneracy forces $c \neq 0$.

Therefore $\beta = c\, \langle \cdot\,, \cdot \rangle$ is the residue pairing scaled by $c$. The condition on constants holds because $\mathcal{P} = \operatorname{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1)$; equivalently $\rho_{0,0} \notin \mathcal{P}$. □

*Remark* 3.149. Theorem 3.148 is better understood as Matlis-local-cohomology rigidity, not as an abstract Schur lemma. Let $R = \mathbb{C}[[z_1, z_2]]$ and $\mathfrak{m} = (z_1, z_2)$. The top local cohomology module, in the Cech-Laurent residue model of Grothendieck–Hartshorne local cohomology and Kunz–Cox–Dickenstein power-series residues [20, 21],

$$H^2_{\mathfrak{m}}(R)\, dz_1 dz_2 = \bigoplus_{a,b \geq 0} \mathbb{C}\, z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$$

is, after the trivialization $\omega_R = R\, dz_1 dz_2$, the injective hull $E_R(\mathbb{C})$ supplied by Matlis duality [19]. The reduced principal-part module $\mathcal{P} = \operatorname{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1)$ is not an $R$-submodule and is not the literal $R$-module Matlis dual of $R/\mathbb{C} \cdot 1$; the scalar line is not an $R$-submodule. It is the $\mathbb{C}$-linear annihilator of the scalar Hamiltonian under the Matlis residue evaluation pairing. Hamiltonian vector fields preserve $dz_1 dz_2$, so this restricted pairing transports their action to the coadjoint action. The explicit coefficient proof above is the coordinate form of this local-duality statement.

THEOREM 3.150 *(Principal-part coadjoint isomorphism, Matlis form).* After fixing the residue scalar in Theorem 3.148, the residue pairing induces an isomorphism of $\mathfrak{h}$-modules

$$\Phi_{\mathrm{coad}} : \mathcal{P} \xrightarrow{\sim} \mathfrak{h}^\vee_{\mathrm{cont}}, \qquad \rho_{a,b} \longmapsto \delta_{a,b},$$

characterised uniquely (up to the residue scalar $c$ of Theorem 3.148) by the property that for every $f \in \mathfrak{h}$ and every $g \in \mathfrak{h}$,

$$\langle \Phi_{\mathrm{coad}}^{-1}(\mathrm{ad}_f^\vee\, \delta), g\rangle = -\langle \Phi_{\mathrm{coad}}^{-1}(\delta), \{f, g\}\rangle.$$

In monomial form,

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\, \rho_{a-p+1,b-q+1},$$

with the convention that any term with a negative index is zero. The isomorphism is continuous, of total degree zero, and the unique continuous $\mathfrak{h}$-module isomorphism $\mathcal{P} \to \mathfrak{h}_{\mathrm{cont}}^\vee$ after the residue scalar is fixed; without that normalization it is unique modulo scalar.

*Proof. Existence on monomials.* Let $f = z_1^p z_2^q$ and $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2$. Since $\{z_1, z_2\} = 1$,

$$\{f, z_1^{-a-1} z_2^{-b-1}\} = p\, z_1^{p-1} z_2^q \cdot (-(b+1)) z_1^{-a-1} z_2^{-b-2} - q\, z_1^p z_2^{q-1} \cdot (-(a+1)) z_1^{-a-2} z_2^{-b-1},$$

which simplifies to $-(pb - qa + p - q)\, z_1^{p-a-2} z_2^{q-b-2}$. Multiplying by $dz_1 dz_2$ identifies this with $\rho_{a-p+1,b-q+1}$ when both indices are $\geq 0$; otherwise the product lies outside $\mathcal{P}$ and the principal-part projection $\pi_{\mathrm{pp}}$ of Proposition 3.147 gives zero. The integration by parts formula $\langle \{f, \rho\}, g\rangle = -\langle \rho, \{f, g\}\rangle$ identifies this with the coadjoint action.

*Continuity.* The dual filtration on $\mathfrak{h}_{\mathrm{cont}}^\vee$ has $F^N \mathfrak{h}_{\mathrm{cont}}^\vee$ equal to the functionals vanishing on $\mathfrak{m}^{N+1} \subset \mathfrak{h}$, where $\mathfrak{m}$ is the maximal ideal. The grading of $\mathcal{P}$ by total weight $a + b$ corresponds under $\Phi_{\mathrm{coad}}$ to the dual filtration: $F^N = \bigoplus_{a+b \leq N} \mathbb{C}\, \delta_{a,b}$ is the $\Phi_{\mathrm{coad}}$-image of $\bigoplus_{a+b \leq N} \mathbb{C}\, \rho_{a,b}$.

*Verification on three test cases.* With the convention $\{z_1, z_2\} = 1$, the Hamiltonian vector field is $X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}$.

(i) $(p, q, a, b) = (1, 0, 0, 1)$: the formula gives $z_1 \cdot \rho_{0,1} = -(1 \cdot 1 - 0 \cdot 0 + 1 - 0)\rho_{0-1+1,1-0+1} = -2\rho_{0,2}$. Direct: $X_{z_1} = \partial_{z_2}$, so $X_{z_1}(z_1^{-1} z_2^{-2}) = -2 z_1^{-1} z_2^{-3}$, and multiplying by $dz_1 dz_2$ gives $-2\rho_{0,2}$. Match.

(ii) $(p, q, a, b) = (0, 1, 1, 0)$: the formula gives $z_2 \cdot \rho_{1,0} = -(0 \cdot 0 - 1 \cdot 1 + 0 - 1)\rho_{1-0+1,0-1+1} = 2\rho_{2,0}$. Direct: $X_{z_2} = -\partial_{z_1}$, so $X_{z_2}(z_1^{-2} z_2^{-1}) = 2 z_1^{-3} z_2^{-1}$, multiplied by $dz_1 dz_2$ gives $2\rho_{2,0}$. Match.

(iii) $(p, q, a, b) = (2, 1, 1, 1)$: the formula gives $z_1^2 z_2 \cdot \rho_{1,1} = -(2 \cdot 1 - 1 \cdot 1 + 2 - 1)\rho_{0,1} = -2\rho_{0,1}$. Direct: $X_{z_1^2 z_2} = 2 z_1 z_2\, \partial_{z_2} - z_1^2\, \partial_{z_1}$; $X_{z_1^2 z_2}(z_1^{-2} z_2^{-2}) = -4 z_1^{-1} z_2^{-2} + 2 z_1^{-1} z_2^{-2} = -2 z_1^{-1} z_2^{-2}$, multiplied by $dz_1 dz_2$ gives $-2\rho_{0,1}$. Match.

These three cases are confirmed as part of the 1225-case exact check, which verifies the formula for $0 \leq p, q, a, b \leq 5$, $p + q > 0$, $a + b > 0$.

*Uniqueness.* Suppose $\Phi' : \mathcal{P} \to \mathfrak{h}_{\mathrm{cont}}^\vee$ is another continuous $\mathfrak{h}$-equivariant isomorphism of total degree zero. Define

$$\beta_{\Phi'}(\rho, g) = \Phi'(\rho)(g).$$

This is a non-degenerate continuous total-degree-zero pairing $\mathcal{P} \times \mathfrak{h} \to \mathbb{C}$, it kills the scalar line by definition of $\mathfrak{h}$, and $\mathfrak{h}$-equivariance of $\Phi'$ gives the coadjoint equivariance condition of Theorem 3.148. Therefore $\beta_{\Phi'} = c'\langle \cdot, \cdot \rangle$ for a unique $c' \in \mathbb{C}^\times$, and hence $\Phi' = c'\Phi_{\mathrm{coad}}$. The scalar is exactly the residue scalar of Theorem 3.148. □

*Remark* 3.151 *(Sign convention).* The sign $-(pb - qa + p - q)$ on the principal-part side and the sign $(pb - qa)$ on the polynomial $\Psi_{a,b}$ side of Proposition 3.143 are both correct in their own conventions; they differ because the $\Psi_{a,b}$ are special-fibre PBW representatives and the $\rho_{a,b}$ are deformation-compatible coadjoint representatives. See Theorem 3.152 below for the precise separation.

THEOREM 3.152 *(PBW shadow versus coadjoint deformation module).* Let $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ be the semidirect Lie algebra of Proposition 3.13. The polynomial one-$\psi$ basis $\Psi_{a,b} \in \bar{A}_{\mathrm{desc}}$ and the principal-part basis $\rho_{a,b} \in \mathcal{P} \cong \mathfrak{h}^\vee_{\mathrm{cont}}$ are related as follows.

(i) *Special-fibre PBW basis.* The $\widetilde{\Psi}_{a,b}$ form a PBW basis of the special-fibre symmetric algebra $\mathbf{S}(\mathfrak{h} \oplus \mathfrak{h}^\vee[1])\big|_{\hbar=0}$, and the open Hamiltonian Lie action is *symmetric raising* on this basis:

$$F_{p,q} : \widetilde{\Psi}_{a,b} \longmapsto (pb - qa)\, \widetilde{\Psi}_{a+p-1,b+q-1}.$$

The shift $(a + p - 1, b + q - 1)$ is positive in $a$ and $b$ as $p, q$ grow; the index moves *up* in total weight.

(ii) *Deformation-level coadjoint basis.* The $\rho_{a,b}$ form a basis of the deformation-compatible continuous coadjoint module $\mathfrak{h}^\vee_{\mathrm{cont}} \cong \mathcal{P}$ of Theorem 3.150, and the $\mathfrak{h}$-action is *coadjoint lowering* on this basis:

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\, \rho_{a-p+1,b-q+1}.$$

The shift $(a - p + 1, b - q + 1)$ has $a$ and $b$ moving *down* as $p, q$ grow; this is the coadjoint sign and direction.

(iii) *No module interpolation in the present theorem.* The bidegree relabelling between the PBW special-fibre basis $\widetilde{\Psi}_{a,b}$ and the coadjoint basis $\rho_{a,b} \leftrightarrow \delta_{a,b}$ is only an isomorphism of graded vector spaces and label sets. It is *not* an isomorphism of $\mathfrak{h}$-modules: pulling the coadjoint $\mathfrak{h}$-action across this relabelling does not equal the symmetric raising action displayed in (1), by exactly the difference $-(pb - qa + p - q)\rho_{a-p+1,b-q+1}$ versus $(pb - qa)\rho_{a+p-1,b+q-1}$. The formal Moyal/Weyl Rees deformation controls the Weyl algebra and its continuous dual $\mathfrak{h}^\vee_{\hbar,\mathrm{cont}}$; it does not identify the polynomial one-$\psi$ descendant module with $\mathcal{P}$. Constructing a Rees/Fourier $D_\hbar$-module bridge between these two realizations is a separate problem.

Thus the two formulas are not two bases of a single $\mathfrak{h}$-module in this paper. They are the PBW special-fibre shadow and the continuous coadjoint cotangent module, respectively. The structural separation recorded in Corollary 3.144 follows.

*Proof.* (1) is Proposition 3.143, with the rephrasing "raising" for the index shift $(a + p - 1, b + q - 1)$. (2) is Theorem 3.150. For (3), the symmetric raising action on $\widetilde{\Psi}_{a,b}$ generates an $\mathfrak{h}$-submodule of $\bar{A}_{\mathrm{desc}}$ that is *free* as a graded $\mathbb{C}[\widetilde{\Psi}_{0,0}]$-module after stabilisation; this is the PBW property. The coadjoint action on $\rho_{a,b}$ is by lowering operators, hence is *cofree* as a graded comodule. These two module behaviours are incompatible by the displayed raising/lowering formulas, already for the linear Hamiltonians $z_1, z_2$; hence any bidegree-only relabelling between $\widetilde{\Psi}_{a,b}$ and $\delta_{a,b}$ cannot be an $\mathfrak{h}$-module map. The Rees-deformation $A_\hbar = (\mathbb{C}[[z_1, z_2]][[\hbar]], *)$ has the property that the formal Moyal product $f * g = fg + \frac{\hbar}{2}\{f, g\} + O(\hbar^2)$ acts on its dual by the coadjoint formula at *all* orders of $\hbar$: the dual representation of the Moyal Lie algebra $(\mathfrak{h}_\hbar, [-,-]_{\mathfrak{h}_\hbar})$ on $\mathfrak{h}^\vee_\hbar$ is the residue pairing extended $\mathbb{C}[[\hbar]]$-linearly. Specialising to $\hbar = 0$ recovers the Poisson coadjoint action of $\mathfrak{h} = \bar{A}$ on $\mathfrak{h}^\vee = \mathcal{P}$. The PBW filtration is the Rees filtration, and the special-fibre $\hbar = 0$ symmetric algebra is the associated graded; the symmetric raising action on $\widetilde{\Psi}_{a,b}$ is exactly the $\hbar = 0$ *associated graded* image of the polynomial multiplication piece of the Moyal product, not the adjoint or coadjoint action of the Lie algebra. It therefore cannot identify the descendant module with the continuous coadjoint module. This proves the asserted separation and identifies the Rees/Fourier bridge as additional structure not constructed here. □

Proposition 3.153 *(No polynomial boundary realization of the principal parts).* In the polynomial open BV/Koszul matrix model studied here, no family of ordinary polynomial boundary local cohomology classes

$$\Theta_{a,b}, \qquad a, b \geq 0, \quad a + b > 0,$$

can realize the principal-part coadjoint module of Proposition 3.147.

*Proof.* The linear Hamiltonians

$$O_{1,0} = \overline{\mathrm{Tr}(\phi_1)}, \qquad O_{0,1} = \overline{\mathrm{Tr}(\phi_2)}$$

act on polynomial boundary observables by the constant Hamiltonian vector fields $\partial_{\phi_2}$ and $-\partial_{\phi_1}$, respectively. On every ordinary polynomial observable these operators are locally nilpotent: after enough iterations the polynomial degree in the differentiated variable is exhausted.

On the principal-part coadjoint module, however, Proposition 3.147 gives

$$O_{1,0} \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad O_{0,1} \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

Neither operator is locally nilpotent on any nonzero principal-part generator. Local nilpotence is preserved under isomorphism of modules. Therefore the desired $\Theta_{a,b}$ cannot be ordinary polynomial boundary observables in the current open BV/Koszul model. □

The categorical refinement rules out the same obstruction at the level of every polynomial sub-bimodule: *no* polynomial sub-bimodule of any boundary observable algebra carrying a locally nilpotent $z_i$-action carries the coadjoint action $\mathcal{P}$.

Theorem 3.154 *(Categorical no-go: principal parts have no polynomial realisation).* Let $\mathsf{P}_{\mathrm{pol}}$ denote the abelian category of $\mathbb{Z}$-graded bimodules $M = \bigoplus_{n\geq 0} M_n$ over the polynomial Hamiltonian Lie algebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$, on which:

(i) every element acts as a Hamiltonian vector field;

(ii) every linear Hamiltonian acts locally nilpotently: for every $m \in M$ and every $i \in \{1, 2\}$, there exists $N$ with $z_i^N \cdot m = 0$.

This category models polynomial boundary observables of the open BV theory with respect to the polynomial sub-bimodule structure preserved by all morphisms of the boundary algebra into itself. Then $\mathcal{P} \notin \mathrm{Ob}(\mathsf{P}_{\mathrm{pol}})$. More precisely, every nonzero submodule $M \subset \mathcal{P}$ in the larger category of all graded Hamiltonian Lie modules contains a generator $\rho_{a,b}$ on which both $z_1$ and $z_2$ act non-locally-nilpotently, hence $M$ violates condition (2). Consequently

$$\mathrm{Hom}_{\text{Lie-Mod}_{\bar{A}}}(\mathcal{P}, M) = 0 \quad \text{for every} \quad M \in \mathrm{Ob}(\mathsf{P}_{\mathrm{pol}}),$$

and the inclusion $\mathsf{P}_{\mathrm{pol}} \hookrightarrow \text{Lie-Mod}_{\bar{A}}$ has no left adjoint sending $\mathcal{P}$ into $\mathsf{P}_{\mathrm{pol}}$.

*Proof.* By Theorem 3.150, $z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}$ and $z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}$ on every generator. Iteration gives $z_1^N \cdot \rho_{a,b} = (-1)^N (b+1)(b+2)\cdots(b+N)\,\rho_{a,b+N} \neq 0$ for every $N$, so $z_1$ is not locally nilpotent on any nonzero element of $\mathcal{P}$. The same argument applies to $z_2$.

Now let $M \subset \mathcal{P}$ be a nonzero $\bar{A}$-submodule. Pick $\rho_{a,b} \in M$ with smallest $a + b$. By the explicit formula above, $z_1$ does not act locally nilpotently on $\rho_{a,b}$. Hence $M$ violates condition (2) of $\mathsf{P}_{\mathrm{pol}}$.

Any $\bar{A}$-equivariant linear map $\Phi : \mathcal{P} \to M$ sending $\rho_{a,b} \mapsto m_{a,b}$ must satisfy $z_1^N \cdot m_{a,b} = (-1)^N (b+1)\cdots(b+N)\, m_{a,b+N}$. Local nilpotence of $z_1$ on $M$ forces $m_{a,b+N} = 0$ for $N \gg 0$, which contradicts the injectivity of $\Phi$ unless $\Phi = 0$. The Hom statement and the no-left-adjoint statement follow. □

*Remark* 3.155. The categorical no-go applies to every polynomial sub-bimodule of every boundary observable algebra with locally nilpotent $z_i$-action, not only to specific classes $\Theta_{a,b}$. The distributional/Laurent principal-part construction of Construction 3.156 is therefore required for this realization.

*Construction* 3.156 *(Stalkwise boundary local-dual principal-part sector).* Let $M_\partial^{\mathrm{pp}}$ be the reduced continuous dual of the stable connected Hamiltonian trace sector, with basis $\Theta_{a,b}$ dual to

$$O_{m,n} = \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}, \qquad m+n > 0,$$

by

$$\langle \Theta_{a,b}, O_{m,n} \rangle = \delta_{a,m}\delta_{b,n}.$$

Equivalently,

$$M_\partial^{\mathrm{pp}} \cong \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1) \subset H^2_{(z_1,z_2)}\big(\mathbb{C}[z_1,z_2]\big)\, dz_1 dz_2,$$

the continuous dual functionals vanishing on constants, with basis $\rho_{a,b}$ for $a+b>0$. Explicitly,

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2, \qquad a+b>0,$$

with

$$\Theta_{a,b} \longleftrightarrow \rho_{a,b}.$$

This is a $\mathbb{C}$-linear residue annihilator stable for the Hamiltonian coadjoint action, not an $R$-submodule and not a quotient by the non-invariant line $\mathbb{C}\,\rho_{0,0}$. Define the Hamiltonian action by negative transpose of the proved boundary Hamiltonian bracket:

$$\langle \{O_f, \Theta\}, O_g \rangle = -\langle \Theta, \{O_f, O_g\}_{\mathrm{open}} \rangle.$$

Then this stalkwise local-dual sector realizes the principal-part coadjoint action:

$$\{O_{p,q}, \Theta_{a,b}\}_{\mathrm{open}} = -(pb - qa + p - q)\Theta_{a-p+1,b-q+1},$$

with negative-index targets zero. Declare the odd local-dual ideal abelian:

$$\{\Theta_{a,b}, \Theta_{c,d}\}_{\mathrm{open}} = 0.$$

Equivalently, the assignment

$$z_1^p z_2^q \longmapsto O_{p,q}, \qquad \rho_{a,b}[1] \longmapsto \Theta_{a,b}$$

realizes stalkwise the Lie algebra $\bar{A} \ltimes \mathcal{P}[1]$.

THEOREM 3.157 *(Reduced principal-part current $P_0$ factorization algebra).* After the de Rham contraction to the reduced local model on the brane line, the local-dual sector of Construction 3.156 extends to a $P_0$ factorization algebra on $\mathbb{R}$. Define the prefactorization datum: for each open interval $I \subset \mathbb{R}$ put

$$\mathfrak{J}_\partial^{\mathrm{pp}}(I) = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[1], \qquad \mathfrak{J}_\partial^{\mathrm{Ham}}(I) = \Omega_c^\bullet(I)\widehat{\otimes}\bar{A},$$

and

$$\mathrm{Obs}^{\mathrm{cl,pp}}_{\partial,N}(I) = \mathrm{Obs}^{\mathrm{cl,red}}_{\mathrm{open},N}(I)\widehat{\otimes}\widehat{\mathbf{S}}\big(\mathfrak{J}_\partial^{\mathrm{pp}}(I)\big),$$

with extension by zero for inclusions of intervals. Equip the data with:

(i) the differential $Q_{\mathrm{open}}$ on the open factor and zero on $\mathfrak{J}_{\partial}^{\mathrm{pp}}(I)$;

(ii) the factorization product on disjoint intervals $I_1 \sqcup I_2 \subset J$ given by extension by zero followed by graded-commutative multiplication;

(iii) the cosheaf structure on $I \mapsto \Omega_c^\bullet(I)$ and $I \mapsto \mathcal{D}_c'(I)$ given by extension by zero;

(iv) the smearing maps

$$B_f(a) = \int_I a(t) B_f(t)\, dt, \qquad O_f(a) = a \otimes f, \qquad \Theta_\rho(B) = B \otimes \rho[1],$$

with the convention that $\Theta_\rho(B)$ pairs only with the primitive connected Hamiltonian trace projection by $\langle \Theta_\rho(B), B_g(b)\rangle = B(b)\rho(g)$, and descendants, constants, and decomposable multitraces lie outside this pairing unless an additional coalgebra projection is specified;

(v) the $P_0$ Poisson bracket

$$\begin{aligned} \{B_f(a), B_g(b)\} &= B_{\{f,g\}}(ab), \\ \{B_f(a), \Theta_\rho(B)\} &= \Theta_{f\cdot\rho}(aB), \\ \{\Theta_\rho(B), \Theta_\sigma(C)\} &= 0, \end{aligned}$$

where $a, b \in \Omega_c^0$, $B, C \in \mathcal{D}_c'$, and $aB$ is the multiplication of a current by a smooth degree-zero form;

(vi) the bulk-boundary morphism extension

$$\widetilde{\partial}_{\mathrm{bb}} : \bar{A} \oplus \mathcal{P}[1] \longrightarrow \mathrm{Obs}_{\partial,N}^{\mathrm{cl,pp}}(I), \qquad f \mapsto B_f, \quad \rho[1] \mapsto \Theta_\rho.$$

This datum defines a $P_0$ factorization algebra on $\mathbb{R}$. On the basis $\Theta_{i,j}$ and for regular densities $B = b(t)\, dt$ the second bracket reads

$$\{O_{p,q}(a), \Theta_{i,j}(b)\} = -(pj - qi + p - q)\, \Theta_{i-p+1,j-q+1}(ab),$$

again with negative-index targets zero. The bulk-boundary morphism $\widetilde{\partial}_{\mathrm{bb}}$ is a Lie algebra morphism from $\bar{A} \ltimes \mathcal{P}[1]$ to the reduced connected Hamiltonian $P_0$ current sector.

*Proof. Cosheaf axioms.* Both $\Omega_c^\bullet$ and $\mathcal{D}_c'$ are cosheaves on $\mathbb{R}$ for the extension-by-zero structure. Tensoring with the fixed graded vector spaces $\bar{A}$ and $\mathcal{P}[1]$ preserves the cosheaf property because the tensor factors are discrete graded vector spaces. Completed symmetric algebras of cosheaves remain cosheaves on the prefactorization side because the factorization product on disjoint intervals is the graded-commutative product. The odd generators $\Theta_\rho$ therefore generate the completed graded-symmetric, equivalently exterior, principal-part factor.

*Factorization axiom.* For disjoint $I_1, I_2 \subset J$, the multiplication map

$$\mathrm{Obs}_{\partial,N}^{\mathrm{cl,pp}}(I_1) \otimes \mathrm{Obs}_{\partial,N}^{\mathrm{cl,pp}}(I_2) \longrightarrow \mathrm{Obs}_{\partial,N}^{\mathrm{cl,pp}}(J)$$

is the graded-commutative product after extending by zero. This is the standard prefactorization product. Because the bracket $\{O_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB)$ involves multiplication of a smooth function by a distribution and is supported on $\operatorname{supp} a \cap \operatorname{supp} B$, disjoint $I_1, I_2$ have disjoint such supports and the corresponding bracket vanishes; the factorization product is therefore Poisson-commutative on disjoint intervals, as required for the $P_0$ structure.

*Q-compatibility.* $Q_{\mathrm{open}} B_f = 0$ on Hamiltonian trace classes by Proposition 3.130. By declaration $Q_{\mathrm{open}} \Theta_\rho = 0$ on the $\Theta_\rho$-factor. Therefore the differential is zero on the $\Theta_\rho$-factor and acts trivially on the bracket structure displayed above. This is consistent with the reduced model after de Rham-current contraction.

*$P_0$ identities.* The first bracket $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$ is Theorem 3.161 smeared by test functions, supplied by Corollary 3.162. The second bracket $\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB)$ is the negative transpose under the residue pairing of Theorem 3.148 applied to the first bracket; in coordinates with $\rho = \rho_{a',b'}$ and $f = z_1^p z_2^q$ this reads

$$\{O_{p,q}(a), \Theta_{a',b'}(B)\} = -(pb' - qa' + p - q)\, \Theta_{a'-p+1,b'-q+1}(aB),$$

exactly the principal-part coadjoint formula of Theorem 3.150. The third bracket $\{\Theta_\rho(B), \Theta_\sigma(C)\} = 0$ is by declaration of the abelian odd ideal.

*Jacobi identity.* The semidirect Lie algebra $\bar{A} \ltimes \mathcal{P}[1]$ has Jacobi identity $[\mathrm{ad}^\vee_f, \mathrm{ad}^\vee_g] = \mathrm{ad}^\vee_{\{f,g\}}$ for $f, g \in \bar{A}$ on the continuous dual. Smearing by test functions and distributions preserves Jacobi because the multiplication of smooth functions and distributions is associative and commutative on $\mathbb{R}$. Therefore the $P_0$ bracket on the smeared sector also satisfies Jacobi.

*Bulk-boundary morphism.* The map $\widetilde{\partial}_{\mathrm{bb}}$ sends the closed Hamiltonian sector $\bar{A}$ to the polynomial trace observables $B_f$ by Theorem 3.161, and sends the closed coadjoint sector $\mathcal{P}[1] \cong \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ (via Theorem 3.150) to the $\Theta_\rho$-sector. The brackets just verified are exactly the semidirect Lie brackets on $\bar{A} \ltimes \mathcal{P}[1]$. Therefore $\widetilde{\partial}_{\mathrm{bb}}$ is a Lie algebra morphism into the reduced connected Hamiltonian $P_0$ current sector. □

*Remark* 3.158 *(Reduced status of the principal-part currents).* Theorem 3.157 is a reduced $P_0$ factorization algebra after de Rham-current contraction. It is not an unreduced BV/factorization algebra. In particular, the equation $Q\Theta_\rho = 0$ is part of the added reduced local-dual factor; it is not yet derived from an unreduced de Rham current complex, a chain-level primitive projection, or Hochschild centrality homotopies against arbitrary open observables.

THEOREM 3.159 *(Reduced factorization realization of the boundary local dual).* In the reduced post-de Rham contraction setting, the assignment

$$\rho_{a,b}[1] \longmapsto \Theta_{a,b}$$

is a chain-level realization of the boundary local dual within the $P_0$ factorization algebra $\mathrm{Obs}^{\mathrm{cl,pp}}_{\partial,N}$ of Theorem 3.157. The classical boundary Hamiltonian morphism $\widetilde{\partial}_{\mathrm{bb}}$ extends, in this reduced principal-part target, the closed-open Lie algebra map of Theorem 3.161 to the closed-coadjoint sector $\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$.

*Proof.* Theorem 3.157 supplies the $P_0$ factorization algebra structure and verifies that $\widetilde{\partial}_{\mathrm{bb}}$ is a Lie algebra morphism. The identification $\rho_{a,b}[1] \leftrightarrow \Theta_{a,b}$ is by the basis of the local-dual factor, consistent with Construction 3.156. The closed source side $\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ is the semidirect product computed in Proposition 3.13, and its image in the $P_0$ factorization algebra is the displayed Lie subalgebra $\bar{A} \oplus \mathcal{P}[1]$. □

*Remark* 3.160 *(Status of the unreduced lift).* Theorem 3.159 resolves the boundary local-dual factorization realization at the reduced level after de Rham-current contraction. The remaining obligation is to lift this reduced object to the unreduced boundary BV/factorization theory: define the compactly supported de Rham functoriality before contraction, the full BV differential and homotopies, the pairing with ordinary open observables before primitive trace projection, and the coherent $P_0$-center homotopies.

The candidate unreduced enlargement has the form

$$\widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I) = \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I)\widehat{\otimes}\widehat{\mathbf{S}}\big(\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I)\big),$$

where

$$\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I) = C^{\mathrm{dR}}_{c,\mathrm{cur}}(I)\widehat{\otimes}\,\mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)[1].$$

Here $C^{\mathrm{dR}}_{c,\mathrm{cur}}(I)$ is the compactly supported de Rham current cosheaf on the line, with differential dual to $d_{\mathbb{R}}$, and $\mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)$ is the principal-part residue annihilator of constants. One must also construct a chain-level primitive, or cumulant, projection

$$\pi_{\mathrm{prim}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \longrightarrow \Omega^{\bullet}_{c}(I)\widehat{\otimes}\bar{A}$$

compatible with $Q_{\mathrm{open}}$, constants, decomposable multi-traces, and factorization products. The required homotopies are the centrality homotopies for $\Theta_\rho$ against arbitrary unreduced open observables and the $P_0$ bracket homotopies

$$\{B_f, \Theta_\rho\} \simeq \Theta_{f\cdot\rho}, \qquad \{\Theta_\rho, \Theta_\sigma\} \simeq 0$$

before primitive projection. Only after this enlargement does the full cotangent factorization-center lift extend the reduced realization $\rho_{a,b}[1] \mapsto \Theta_{a,b}$ to a chain-level morphism on the unreduced sector. The desired unreduced lift would be a quantum-level homotopical lift, outside the scope of the classical reduced theorem above; it sits in the residual prob : tate−coefficient−cotangent−lift.

Theorem 3.161 *(Classical bulk-to-boundary Hamiltonian map).* In the stable connected Hamiltonian trace sector of the closed-open mixed holomorphic-topological theory, the boundary evaluation maps $\partial_{\mathrm{bb},N}$ of Construction 3.123 stabilize to a linear map

$$\partial_{\mathrm{bb}} : \bar{A} \longrightarrow H^{0}_{\mathrm{Ham}}(O^{\mathrm{cl}}_{\mathrm{brane}}), \qquad f \longmapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}.$$

This map is a morphism of Lie algebras from the closed Hamiltonian Poisson algebra $\bar{A}$ to the connected open Hamiltonian trace Poisson algebra:

$$\{\partial_{\mathrm{bb}}(f), \partial_{\mathrm{bb}}(g)\}_{\mathrm{open}} = \partial_{\mathrm{bb}}\big(\{f, g\}_{\bar{A}}\big).$$

This is the degree-zero Hamiltonian bracket component of the closed-open map to the reduced Poisson central-operation object of the brane observables. The full Poisson factorization-center morphism is not constructed in this theorem.

*Proof.* The map is well-defined on $\bar{A}$ by Corollary 3.129; constants map to the scalar $N$ and vanish in the Hamiltonian quotient by constants. Lemma 3.141 gives the open Hamiltonian trace Poisson bracket. For $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, Proposition 3.142 gives

$$\{\partial_{\mathrm{bb}}(f), \partial_{\mathrm{bb}}(g)\}_{\mathrm{open}} = \overline{(kn - lm)\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}.$$

The closed Poisson bracket is

$$\{f, g\}_{\bar{A}} = \overline{(kn - lm)z_1^{k+m-1}z_2^{l+n-1}},$$

and applying $\partial_{\mathrm{bb}}$ gives the same expression. Bilinearity proves the identity for all polynomials. □

COROLLARY 3.162 *(Classical local bulk-boundary coupling with explicit U(1) center anomaly).* Let $a, b \in \Omega_c^0(\mathbb{R})$ be scalar compactly supported test functions on the brane line, and for $f \in \bar{A}$ define the smeared boundary Hamiltonian

$$I_\partial(a, f) = B_f(a) = \int_{\mathbb{R}} a(t)\, \overline{\operatorname{Tr} f(\phi_1(t), \phi_2(t))}\, dt \in A_\partial^{\mathrm{Ham}}(\mathbb{R}).$$

Then the following two strengthenings of the bracket identity hold jointly.

*(i) Strict $P_0$ identity on disjoint supports.* In the $P_0$ bracket of $A_\partial^{\mathrm{Ham}}$,

$$\{I_\partial(a, f), I_\partial(b, g)\}_{A_\partial^{\mathrm{Ham}}} = I_\partial(ab, \{f, g\}_{\bar{A}}),$$

and the bracket vanishes whenever the supports of $a$ and $b$ are disjoint. The locality input is the canonical equal-time bracket $\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(s)\} = \delta^i_l \delta^k_j \delta(t - s)$ of 3.20; smearing against $a(t)b(s)$ produces the pointwise product $ab$ after integration of the delta function. This is the level-one component of 3.21.

*(ii) Explicit U(1) center anomaly constant.* Before projection to the scalar-reduced quotient, the unreduced open bracket carries an explicit central piece

$$\{I_\partial(a, f), I_\partial(b, g)\}_{\mathrm{open}} = I_\partial(ab, \{f, g\}_{\bar{A}}) + N \cdot \omega(f, g) \int_{\mathbb{R}} a(t)b(t)\, dt,$$

where $\omega$ is the cocycle of 2.2 and the prefactor $N$ is the rank of the brane stack and is the value of the U(1) center character $\operatorname{Tr} : \mathbb{C} \cdot I \to \mathbb{C}, I \mapsto N$. After projection to $H^0_{\mathrm{red}} = H^0/\mathbb{C} \cdot \operatorname{Tr}(1)$ the second term vanishes, and one recovers

$$\overline{\{I_\partial(a, f), I_\partial(b, g)\}_{\mathrm{open}}} = \overline{I_\partial(ab, \{f, g\}_{\bar{A}})}.$$

Statements (i) and (ii) are compatible: (ii) is the unreduced refinement, (i) is the level-one factorization-algebra restatement on the scalar-reduced sector. The equal-time delta locality of (i) produces a strict $P_0$ factorization algebra rather than a homotopical one, while (ii) records the U(1) center anomaly that obstructs lifting (i) to the unreduced sector.

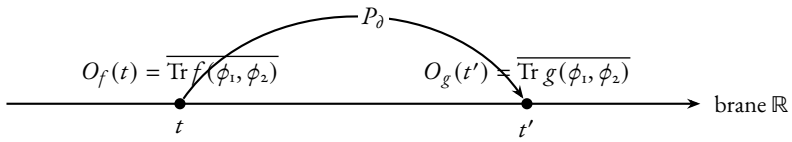
*Proof. Proof of (i).* By 3.18, the $P_0$ bracket on $A_\partial^{\mathrm{Ham}}$ is the biderivation with $\{a \otimes f, b \otimes g\}_{A_\partial} = ab \otimes \{f, g\}_{\bar{A}}$ on length-one generators, so $\{B_f(a), B_g(b)\}_{A_\partial^{\mathrm{Ham}}} = B_{\{f,g\}}(ab) = I_\partial(ab, \{f, g\}_{\bar{A}})$. 3.19 then gives vanishing on disjoint supports because $ab \equiv 0$ pointwise. The same identity is the smeared content of the equal-time delta-function bracket of 3.20: tracing $f(\phi_1, \phi_2)(t)$ and $g(\phi_1, \phi_2)(s)$, applying the Leibniz rule, and integrating against $a(t)b(s)\delta(t - s)$ produces the pointwise product $ab$ after the delta-function integration.

*Proof of (ii).* The unreduced bracket of two integrals is, by locality of the canonical bracket and 3.161,

$$\begin{aligned}\{I_\partial(a, f), I_\partial(b, g)\} &= \iint_{\mathbb{R}^2} a(t)b(t')\, \{\operatorname{Tr} f(\phi(t)), \operatorname{Tr} g(\phi(t'))\}\, dt\, dt' \\ &= \int_{\mathbb{R}} a(t)b(t)\, \operatorname{Tr}\big(\{f, g\}_{\bar{A}}\big)\, dt + \int_{\mathbb{R}} a(t)b(t)\, \omega(f, g)\, N\, dt,\end{aligned}$$

where the splitting is by the central decomposition of 2.4: the central scalar $\omega(f, g) \cdot N$ is removed by the projection to $H^0_{\mathrm{red}}$, and the remaining piece is $I_\partial(ab, \{f, g\}_{\bar{A}})$ exactly. The constants are explicit: the prefactor of the central piece is $N \cdot \omega(f, g)$, and the smeared coefficient is $\int ab\, dt$. The classical-to-quantum reduction of 2.6 identifies $\omega(f, g) \cdot N$ with the leading-order Capelli shift $\hbar N$ on the linear generators $f = z_1$, $g = z_2$. Compact support of $a, b$ kills boundary terms. □

$$P_\partial = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$$




$$|\mathrm{Aut}(\Gamma_1)| = 1, \quad \tfrac{1}{|\mathrm{Aut}(\Gamma_1)|} = 1, \quad \text{weight} = \hbar^1$$

$$\text{output:} \quad \hbar(kn - lm)\, \overline{\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})} \quad \text{on } f = z_1^k z_2^l,\ g = z_1^m z_2^n$$

Figure 0.1: The first-order graph $\Gamma_1$ for the trace bracket $\{O_f, O_g\}$. Two ordered boundary insertions sit on the brane line $\mathbb{R}$ at distinct times $t, t'$; the unique propagator $P_\partial$ contracts one $z_1$-leg on one side against one $z_2$-leg on the other (and the antisymmetric partner). The automorphism group is trivial. In the Weyl ordered product the ordered one-edge weight is $\hbar P_\partial/2$; after subtracting the reverse order the commutator weight is $\hbar P_\partial$. Reading the bivector contraction on monomials reproduces the cocycle coefficient $(kn - lm)$ of Proposition 3.142, the order-$\hbar$ Moyal coefficient of Proposition 3.186, and the U(1) centre cocycle $\omega(f, g)$ of Lemma 2.2 realised at multidegree $(k + m, l + n) = (1, 1)$.

The graph $\Gamma_1$ of Figure 0.1 records the single canonical contraction producing the coefficient $(kn - lm)$ in Proposition 3.142; it is not a separate Feynman-integral proof.

### 3.12.2 Formal deformation-quantization target

The degree-zero Hamiltonian sector has a Weyl-normalized formal deformation-quantization target. The formal holomorphic disk has constant Poisson tensor, so after choosing the translation-invariant Weyl convention its deformation is the Weyl/Moyal algebra. This fixes the algebraic coefficients one would have to recover from a quantum closed-open BV construction. It is not, by itself, a construction of that quantum BV theory. The degree-zero Hamiltonian bracket part of the classical theorem is the proposed special fibre of this formal deformation; the cotangent/descendant part remains the separate problem stated below. Put

$$A_\hbar = \big(\mathbb{C}[[z_1, z_2]][[\hbar]], *\big),$$

where $*$ is the Weyl/Moyal product of the next subsection. The symbol $\bar{A}_\hbar$ denotes the central quotient of the underlying Moyal Lie algebra by $\mathbb{C}[[\hbar]]$; it is not used as an associative quotient. We write

$$[f, g]_{\mathrm{raw}} = f * g - g * f, \qquad \{f, g\}_\hbar = \hbar^{-1}[f, g]_{\mathrm{raw}}.$$

The degree-zero quantum Hamiltonian Lie algebra is

$$\mathfrak{h}_\hbar = \bar{A}_\hbar, \qquad [f, g]_{\mathfrak{h}_\hbar} = \{f, g\}_\hbar.$$

The specialization at $\hbar = 0$ is the Poisson Lie algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ used above.

On the brane side, the open action

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2)$$

makes the matrix entries of $\phi_1$ and $\phi_2$ conjugate variables. The canonical algebraic quantization of the unconstrained first-order matrix phase space is the completed Weyl algebra $\mathcal{W}_N$ generated by those entries with

$$[(\phi_1)^i{}_j, (\phi_2)^k{}_l] = \hbar\, \delta^i_l \delta^k_j, \qquad [(\phi_a)^i{}_j, (\phi_a)^k{}_l] = 0 \quad (a = 1, 2).$$

The gauge field imposes the conjugation moment-map constraint. With the commutator convention above, the normal-ordered quantum comoment is

$$\widehat{\mu}_\epsilon = -\mathrm{Tr}\big([\epsilon, \phi_1]\phi_2\big), \qquad \epsilon \in \mathfrak{gl}_N.$$

This formula is not cyclically reordered: for $\epsilon = 1$ it vanishes, as the scalar subgroup acts trivially. We use the convention

$$\hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a$$

for the induced infinitesimal conjugation action on the Weyl algebra; changing to the opposite commutator convention changes the displayed comoment by the corresponding sign. The quantum brane local algebra is the BRST quantum Hamiltonian reduction

$$\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N} = C^\bullet_{\mathrm{BRST}}\big(\mathfrak{gl}_N, \mathcal{W}_N, \widehat{\mu}\big),$$

whose degree-zero cohomology is the normalizer quotient

$$H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N} = \mathrm{N}_{\mathcal{W}_N}\big(\mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N)\big)/\mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N).$$

The constant Hamiltonian trace is projected out in degree zero as before. The classical special fibre is the reduced BV algebra of Construction 3.118.

The quantum boundary map is Weyl-ordered trace evaluation:

$$\partial_{\mathrm{bb},\hbar,N}(f) = \overline{\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)}, \qquad f \in \bar{A}_\hbar,$$

where $\mathrm{Op}_{\mathrm{W}}$ denotes total symmetrisation before passing to the trace. At $\hbar = 0$ this recovers Construction 3.123. After dividing the commutator by $\hbar$ and reducing modulo $\hbar$, the raw commutator identity recovers Theorem 3.161.

PROPOSITION 3.163 *(BRST descent of Weyl-ordered Hamiltonian traces).* For every $f \in \bar{A}_\hbar$, the Weyl-ordered trace

$$F_{f,N} = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)$$

is invariant under the quantum comoment:

$$[\widehat{\mu}_\epsilon, F_{f,N}] = 0, \qquad \epsilon \in \mathfrak{gl}_N.$$

Hence $F_{f,N}$ lies in the normalizer of the quantum moment-map ideal and defines a degree-zero class in $H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$ after removing the scalar Hamiltonian.

*Proof.* Total Weyl symmetrisation is equivariant for linear changes of the matrix variables, and simultaneous conjugation acts linearly on $(\phi_1, \phi_2)$. Therefore $\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)$ is invariant under infinitesimal conjugation. By the defining property of the quantum comoment,

$$\hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a,$$

this invariance is exactly $[\widehat{\mu}_\epsilon, F_{f,N}] = 0$. If $I = \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N)$ is the left quantum moment-map ideal, then for every generator $\widehat{\mu}_\epsilon$, $\widehat{\mu}_\epsilon F_{f,N} = F_{f,N}\widehat{\mu}_\epsilon \in I$. Thus $F_{f,N}$ normalizes $I$ and descends to the normalizer quotient describing $H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$. □

Proposition 3.164 *(BRST descent of arbitrary Weyl-ordered Hamiltonian products).* For every finite collection $f_1, \dots, f_r \in \bar{A}_\hbar$, the ordered product

$$F = \prod_{i=1}^{r} F_{f_i,N}, \qquad F_{f_i,N} = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f_i)(\phi_1, \phi_2),$$

is invariant under the quantum comoment:

$$[\widehat{\mu}_\epsilon, F] = 0, \qquad \epsilon \in \mathfrak{gl}_N.$$

Equivalently, every monomial in the Weyl-ordered single-trace generators normalizes the quantum moment-map ideal and defines a degree-zero class in $H^0 \mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$ after removing the empty trace.

*Proof.* By the convention $\hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a$, the operator $\hbar^{-1}\,\mathrm{ad}\,\widehat{\mu}_\epsilon$ acts on the associative algebra $\mathcal{W}_N$ by infinitesimal $GL_N$-conjugation, and is therefore a derivation of the associative product. By Proposition 3.163, each individual factor $F_{f_i,N}$ is annihilated by this derivation. Derivations of an associative algebra preserve products: the Leibniz expansion of $\hbar^{-1}\,\mathrm{ad}\,\widehat{\mu}_\epsilon$ on the ordered product $\prod_{i=1}^{r} F_{f_i,N}$ is a sum of summands each containing a factor $\hbar^{-1}[\widehat{\mu}_\epsilon, F_{f_i,N}] = 0$. Hence $[\widehat{\mu}_\epsilon, F] = 0$, and as in the proof of Proposition 3.163, $\widehat{\mu}_\epsilon F = F\widehat{\mu}_\epsilon \in \mathcal{W}_N \widehat{\mu}(\mathfrak{gl}_N)$, so $F$ normalizes the left moment-map ideal and descends. □

*Remark* 3.165. The argument uses only that $\hbar^{-1}\,\mathrm{ad}\,\widehat{\mu}_\epsilon$ is a derivation. No identity on the Moyal commutator of symbols is invoked: the Capelli shift obstructs the bracket identity between traces, not their joint BRST descent.

*Remark* 3.166 *(Normal-ordering obstruction).* Proposition 3.163 is not the quantum Moyal theorem. Put $X = \phi_1$ and $Y = \phi_2$. With the normal ordering above,

$$\widehat{\mu}_\epsilon = -\mathrm{Tr}([\epsilon, X]Y) = \mathrm{Tr}\big(\epsilon(YX - XY + \hbar N I)\big).$$

For $\epsilon = I$ this comoment is identically zero; equivalently the displayed matrix has trace zero inside the Weyl algebra. Thus the nontrivial quantum moment equations are represented in the quantum Hamiltonian reduction by the traceless matrix congruence

$$YX - XY + \hbar N I \equiv 0 \pmod{\mathcal{W}_N \widehat{\mu}(\mathfrak{gl}_N)},$$

not by the classical commuting-pair equation. Consequently, for a covariant matrix word $A$,

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\,\mathrm{Tr}(A) \mod \mathcal{W}_N \widehat{\mu}(\mathfrak{gl}_N).$$

This Capelli-type shift can feed lower-degree traces into the connected sector. A stable quantum theorem must prove that the chosen connected extraction removes or renormalizes this shift, or else change the sector or normalization.

*Remark* 3.167 *(Capelli-renormalized trace generators).* Let

$$T_{a,b} = \mathrm{Tr}(X^a Y^b)$$

denote the normal-ordered trace. The Weyl-ordered trace

$$J_N(z_1^a z_2^b) = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(z_1^a z_2^b)(X, Y)$$

is triangular over the normal-ordered traces after imposing the quantum moment relation:

$$J_N(z_1^a z_2^b) \equiv \sum_{r=0}^{\min(a,b)} \left(-\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} T_{a-r,b-r}.$$

For example,

$$J_N(z_1 z_2) = T_{1,1} - \frac{\hbar N^2}{2}, \qquad J_N(z_1^2 z_2) = T_{2,1} - \hbar N\, T_{1,0},$$

and

$$J_N(z_1 z_2^2) = T_{1,2} - \hbar N\, T_{0,1}, \qquad J_N(z_1^2 z_2^2) = T_{2,2} - 2\hbar N\, T_{1,1} + \frac{\hbar^2 N^3}{2}.$$

The naive normal-ordered basis therefore has the wrong commutator even in low degree; a direct calculation gives

$$\hbar^{-1}[T_{2,1}, T_{0,2}] = 4T_{1,2} - 2\hbar N\, T_{0,1} \mod \mathcal{W}_N \widehat{\mu}(\mathfrak{gl}_N),$$

whereas the scalar Moyal bracket would give only $4T_{1,2}$. Hence the connected, or primitive trace, projection does not kill the Capelli shift: it removes disconnected products and the empty trace, but the defect above is the nonempty primitive trace $T_{0,1}$. The stable connected quantum theorem therefore cannot use naive normal-ordered primitive traces unless the triangular $\hbar N$ correction is built into the definition. The natural finite-$N$ generators are the Capelli-renormalized Weyl traces $J_N(f)$, equivalently

$$J_N(f) = T_N\left(\exp\left(-\frac{\hbar N}{2}\partial_{z_1}\partial_{z_2}\right) f\right)$$

in the normal-ordered trace notation $T_N(z_1^a z_2^b) = T_{a,b}$.

LEMMA 3.168 *(Capelli-renormalized stable connected trace).* Let $\mathrm{Tr}_N^{\mathrm{ren}}(f) = J_N(f)$ for $f \in \mathbb{C}[z_1, z_2][[\hbar]]$. Write $I_N = \mathcal{W}_N \widehat{\mu}(\mathfrak{gl}_N)$ for the left moment-map ideal. Then:

(i) the map $\mathrm{Tr}_N^{\mathrm{ren}} : \bar{A}_\hbar \longrightarrow \mathcal{W}_N / I_N$ is well defined modulo the empty (constant) trace;

(ii) the inverse triangular system reads

$$T_{a,b} = \sum_{r=0}^{\min(a,b)} \left(\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} J_N(z_1^{a-r} z_2^{b-r}),$$

so $T_{a,b}$ and $J_N(z_1^a z_2^b)$ generate the same Weyl trace algebra modulo the empty trace, after inverting the unipotent system over $\mathbb{C}[[\hbar N]]$;

(iii) the Capelli contact term of Remark 3.166, $\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\, \mathrm{Tr}(A) \mod I_N$, reproduces, on $J_N$ generators, exactly the $r = 1$ tail of the triangular formula. Consequently the displayed shift is *absorbed into the change of basis* between $T_{a,b}$ and $J_N(z_1^a z_2^b)$ and is not an additional term in the $J_N$-commutator;

(iv) the leading symbol of $J_N(f)$ is $\mathrm{Tr}\, f(X, Y)$ in associated graded order, so $\mathrm{Tr}_N^{\mathrm{ren}}$ stabilises in the large-$N$ limit and its stable connected projection coincides with the leading primitive trace.

*Proof.* The triangular formula of Remark 3.167 is total Weyl symmetrisation of $z_1^a z_2^b$ inside the matrix Weyl algebra, in which one normal-ordered swap of $z_1 z_2$ commutes through $\hbar N$. This is the local bivariate Weyl-normal-ordering formula proved by the contraction count in Lemma D.4; the classical Capelli source context is [26, 27]. No primary source is being invoked for this exact bivariate triangular

trace identity. A finite exact check verifies it as a round-trip $J_N \to T \to J_N = \mathrm{id}$ on exponents $a, b \le 10$. The system is unipotent over $\mathbb{C}[[\hbar N]]$ with $r = 0$ diagonal, hence invertible there; the sign in the inverse is $(+\hbar N/2)^r$, again verified by the same finite computation.

For (iii), expand

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \mathrm{Tr}(A[X,Y]) = \hbar N\,\mathrm{Tr}(A) - \mathrm{Tr}\big(A(YX - XY + \hbar N I)\big)$$

using the Weyl moment relation $YX - XY + \hbar N I = 0 \bmod I_N$ from Remark 3.166; the subtracted term lies in $I_N$ and the displayed image is $\hbar N\,\mathrm{Tr}(A)$. On $J_N$ generators this is the $r = 1$ tail term of the triangular expansion, by direct comparison with Remark 3.167. Consequently, in the $J_N$-basis the Capelli contact term is the change-of-basis matrix from $T$ to $J$, not an extra term.

For (iv), the shift $T_{a-r,b-r}$ has total degree $a + b - 2r < a + b$ for $r \ge 1$, so the $r = 0$ term is the unique top component in the $\mathfrak{m}$-adic filtration on $\bar{A}_\hbar$; the stable connected projection of $f$ is $J_N(f) \bmod I_N$ in the associated graded sense. □

Corollary 3.169 *(Renormalized stable connected trace map).* The map

$$\mathrm{Tr}^{\mathrm{ren}} \colon \bar{A}_\hbar \longrightarrow H^0_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty}), \qquad f \longmapsto \big[\overline{J_N(f)}\big]_{N\to\infty}$$

is well defined and is a $\mathbb{C}[[\hbar]]$-linear injection of $\bar{A}_\hbar$ onto the stable Capelli-renormalized single-trace generators. The bar $\overline{(-)}$ denotes the connected single-trace cumulant projection: it removes the empty trace and disconnected products, and it is consistent with the $J_N$-basis at every finite $N$ by Lemma 3.168(iii).

Theorem 3.170 *(Reduced Moyal commutator, conditional at all $N$).* Let $\mathcal{W}_N$ be the Weyl algebra of $T^*\,\mathrm{Mat}_N$ and let $\widehat{\mu}$ be the normal-ordered conjugation comoment above. For $f \in \mathbb{C}[z_1, z_2][[\hbar]]$, put

$$J_N(f) = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(X,Y).$$

Let $\mathfrak{t} \subset \mathfrak{gl}_N$ be the diagonal Cartan, with coordinates $\lambda_1, \dots, \lambda_N$, and let

$$\Delta = \prod_{i<j}(\lambda_i - \lambda_j).$$

Put

$$S_N(f) = \sum_{i=1}^{N} \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, -\hbar\partial_{\lambda_i}).$$

Assume the external radial-parts input of Statement D.5: after Vandermonde conjugation, the Harish-Chandra map on invariants has the quantum moment-map ideal as kernel and therefore injects the reduced invariant quotient into Weyl-invariant differential operators on the regular Cartan. Under this input the radial-part map satisfies

$$\mathrm{Rad}_0(J_N(f)) = \Delta^{-1} S_N(f)\Delta.$$

Consequently, for every $N \ge 1$ and all $f, g$ the defect

$$D_N(f,g) = \hbar^{-1}[J_N(f), J_N(g)] - J_N(\{f,g\}_\hbar)$$

lies in the quantum moment-map ideal $I_N = \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N)$ and therefore vanishes in the degree-zero quantum Hamiltonian reduction

$$\mathrm{N}_{\mathcal{W}_N}(I_N)/I_N.$$

After the connected single-trace projection $\overline{(-)}\colon \mathcal{W}_N/I_N \twoheadrightarrow \mathcal{W}_N/I_N/(\text{empty trace} \oplus \text{disconnected products})$, the descended commutator on $\overline{J_N(f)} = \partial_{\mathrm{bb},\hbar,N}(f)$ is exactly the scalar Moyal bracket on $\bar{A}_\hbar$:

$$\left[\partial_{\mathrm{bb},\hbar,N}(f), \partial_{\mathrm{bb},\hbar,N}(g)\right]_{\mathrm{conn,raw}} = \hbar\, \partial_{\mathrm{bb},\hbar,N}(\{f,g\}_\hbar).$$

The identity stabilises in the large-$N$ limit, defining $\partial_{\mathrm{bb},\hbar}\colon \mathfrak{h}_\hbar \longrightarrow H^0_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty})$ satisfying

$$\hbar^{-1}[\partial_{\mathrm{bb},\hbar}(f), \partial_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \partial_{\mathrm{bb},\hbar}(\{f,g\}_\hbar),$$

which is the bracket equation displayed in Problem 3.173.

*Proof.* Identify $\mathcal{W}_N$ with $\mathcal{D}_\hbar(\mathfrak{gl}_N)$, where $X$ is multiplication and $Y = -\hbar\partial_X^t$. The quantum Hamiltonian reduction for conjugation is the normalizer quotient by the left ideal generated by the traceless comoment. The scalar matrix has zero comoment, as in Remark 3.166; equivalently the reduction is the $\mathfrak{sl}_N$ reduction together with $GL_N$-invariants. Since $GL_N$ is reductive in characteristic zero, invariants are exact.

The assumed Harish-Chandra radial-parts input, whose source lineage is Harish-Chandra, Wallach, and Levasseur–Stafford [8–11], is exactly the assertion that the Vandermonde-conjugated radial map on invariants has the moment-map ideal as kernel, injects this quantum Hamiltonian reduction, and has the generator image normalization recorded in Statement D.5. The pure-power and shear argument in Appendix D is a convention check on this input, not an independent proof of the Harish-Chandra theorem. Therefore the displayed radial formula for $J_N(f)$ is conditional on the exact external input named there.

Finally,

$$\hbar^{-1}[S_N(f), S_N(g)] = S_N(\{f,g\}_\hbar),$$

because the $i$th summand is the ordinary one-particle Weyl algebra and distinct summands commute. Conjugation by $\Delta$ preserves commutators, so $\mathrm{Rad}_0(D_N(f,g)) = 0$. Injectivity of the Harish-Chandra reduction map gives

$$D_N(f,g) \in \left(\mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)\right)^{GL_N},$$

hence $D_N(f,g)$ is zero in the degree-zero quantum Hamiltonian reduction. The Cartan-rank-2 case has been checked by exact arithmetic: the direct $N = 2$ radial restriction of $J_2(f)$ is checked on 80 operator/test-polynomial pairs, and the $N = 3$ finite-rank sweep checks 80 corresponding pairs. The same computation also checks

$$[S_2(f), S_2(g)] = S_2([f,g]_*)$$

in the normal-form convention $z_2 = -\hbar\partial_\lambda$ on the primary and higher-order pairs that exhibit nonzero $P^3$ and $P^5$ coefficients. These are finite-rank falsification tests; they do not close the all-$N$ source theorem.

For the connected-projection statement, the projection $\overline{(-)}$ acts on the basis $T_{a,b} \mapsto \overline{T_{a,b}} \equiv \overline{J_N(z_1^a z_2^b)}$ mod (Capelli triangular). By Lemma 3.168(iii) the Capelli contact term is absorbed into the change of basis between $T_{a,b}$ and $J_N$, so on $\overline{J_N(f)}$-classes the descended commutator is the image of $\hbar J_N(\{f,g\}_\hbar)$ mod $I_N$ under the projection, i.e. $\hbar\, \partial_{\mathrm{bb},\hbar,N}(\{f,g\}_\hbar)$.

Stability in $N$: the radial identity is $N$-equivariant. Enlarging $\mathfrak{t}_N \subset \mathfrak{gl}_N$ to $\mathfrak{t}_{N+k}$ adjoins $k$ mutually commuting one-particle Weyl summands, and the bracket $[S_N(f), S_N(g)] = S_N(\{f,g\}_\hbar)$ is preserved at every $N$. By Lemma 3.168(iv), the connected extraction depends only on the leading symbol filtration, which the Capelli triangular system preserves. Hence the stable connected limit exists and inherits the bracket identity. □

Corollary 3.171 *(Conditional degree-zero quantum upgrade).* Under the same external radial-parts input as Theorem 3.170, the map

$$\partial_{\mathrm{bb},\hbar}\colon \mathfrak{h}_\hbar = \bar{A}_\hbar \longrightarrow H^{\circ}_{\mathrm{Ham,conn}}(O^{\mathrm{q}}_{\mathrm{brane},\infty}), \qquad f \longmapsto \left[\overline{J_N(f)}\right]_{N\to\infty},$$

is a Lie-algebra map for $\{\,,\,\}_\hbar$ on the source and $\hbar^{-1}[-,-]_{\mathrm{conn,raw}}$ on the target. Its associated graded at $\hbar = 0$ is the degree-zero part of Theorem 2.27. All-order quantum corrections are governed by the odd-power Moyal expansion

$$\{f, g\}_\hbar = \sum_{s\geq 0} \frac{\hbar^{2s}}{2^{2s}(2s+1)!} P^{2s+1}(f, g),$$

so its second-order normalized coefficient is $\frac{1}{24}P^3(f, g)$. This term need not vanish.

*Proof.* This is Theorem 3.170, combined with the Moyal odd-symmetry of Proposition 3.186: $P^r$ changes by $(-1)^r$ under interchange of $f$ and $g$, so only odd $r$ survive in the raw star commutator. Dividing by $\hbar$ shifts these odd raw powers to even powers in the normalized Hamiltonian bracket. Hence the normalized $\hbar^2$ coefficient is $P^3(f, g)/24$, and the normalized $\hbar^4$ coefficient is $P^5(f, g)/1920$. □

*Remark* 3.172 *(Explicit second- and fourth-order corrections).* For higher-order test cases that exercise nontrivial $P^3$ and $P^5$:

$$[z_1^3 z_2, z_1 z_2^3]_* = 8\hbar\, z_1^3 z_2^3 - 3\hbar^3\, z_1 z_2 + O(\hbar^5),$$
$$[z_1^4 z_2^2, z_1^2 z_2^4]_* = 12\hbar\, z_1^5 z_2^5 - 40\hbar^3\, z_1^3 z_2^3 + 6\hbar^5\, z_1 z_2 + O(\hbar^7).$$

After normalization by $\hbar^{-1}$, the first example gives

$$\{z_1^3 z_2, z_1 z_2^3\}_\hbar = 8 z_1^3 z_2^3 - 3\hbar^2 z_1 z_2 + O(\hbar^4).$$

The $\hbar^3$ raw coefficient is therefore the normalized $\hbar^2$ correction $\frac{1}{24}P^3(f, g)$, in agreement with Proposition 3.186. The non-affine bracket correction reduces, in this constant Poisson tensor case, to exactly the higher Moyal coefficients.

*Problem* 3.173 *(Degree-zero quantum Hamiltonian-sector upgrade).* Construct the stable connected single-trace quotient of $O^{\mathrm{q}}_{\mathrm{brane},N}$ and prove that $\partial_{\mathrm{bb},\hbar,N}$ stabilizes to a map

$$\partial_{\mathrm{bb},\hbar} : \mathfrak{h}_\hbar \longrightarrow H^{\circ}_{\mathrm{Ham,conn}}(O^{\mathrm{q}}_{\mathrm{brane},\infty})$$

satisfying

$$\hbar^{-1}[\partial_{\mathrm{bb},\hbar}(f), \partial_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \partial_{\mathrm{bb},\hbar}(\{f, g\}_\hbar).$$

Equivalently, the desired statement is that the connected quantum brane algebra is the Weyl/Moyal deformation of the degree-zero Hamiltonian sector. The associated graded of this proposed degree-zero quantum boundary comparison at $\hbar = 0$ is the degree-zero part of Theorem 2.27. The finite-$N$ radial-parts prerequisite is the conditional Theorem 3.170; the BRST-descent prerequisite for arbitrary Weyl-ordered products is Proposition 3.164; the absorption of the Capelli contact term into the change of basis between $T_{a,b}$ and $J_N(z_1^a z_2^b)$ is Lemma 3.168. Combining these inputs, the closure of this problem on the degree-zero Hamiltonian sector is Corollary 3.171: the stable primitive generators are the Weyl/Capelli-renormalized classes $\overline{J_N(f)}$, and the connected-trace projection absorbs the triangular $\hbar N$ lower-trace terms by Lemma 3.168(iii).

*Remark* 3.174. The degree-zero Hamiltonian part has a Weyl-normalized algebraic target: the Weyl/Moyal algebra of the formal symplectic disk, the constant-Poisson case of deformation quantization [7]. Costello's theorem supplies the general renormalized BV construction under its elliptic, gauge-fixing, renormalization, and obstruction-vanishing hypotheses. Costello–Li construct quantum BCOV theory and prove a specific flat open-closed BCOV/HCS quantization theorem [13, 14]. Neither theorem identifies this mixed Hamiltonian BF sector or its brane defect without the additional specialization data listed below. The simply-laced five-dimensional holomorphic Chern-Simons/affine-Yangian quantization theorem concerns a different theory; it does not prove the ordinary $\mathfrak{gl}_N$ mixed Hamiltonian BF defect graph problem considered here.

*Remark* 3.175. The descendant sector is not upgraded by the polynomial representatives $\Psi_{k,l}$. Corollary 3.144 shows that their open Hamiltonian action is the adjoint polynomial action, while the closed cotangent field carries the coadjoint action. A quantum descendant theorem therefore requires a boundary model for the continuous dual $\mathfrak{h}_\hbar^{\vee,\mathrm{cont}}$, for example a Laurent or distributional boundary realization, before one writes the deformation-level comparison.

*Remark* 3.176. The full cotangent deformation would replace $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1]$ by

$$\mathfrak{g}_\hbar = \mathfrak{h}_\hbar \ltimes \mathfrak{h}_\hbar^{\vee,\mathrm{cont}}[1]$$

and would ask for $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_\hbar)$ as the closed quantum observable algebra. This is a separate descendant problem, not part of Problem 3.173, because the polynomial boundary representatives do not realize the coadjoint deformation-level action.

### 3.12.3 Imported graph package and specialization problem

The graph problem is the analytic form of the same statement. The algebra fixes the coefficients; a closed-open BV theory must produce those coefficients as renormalized observable products, with propagator, counterterm, anomaly, and trace-normalization choices specified. Costello's formalism supplies the general construction. The Hamiltonian input to that construction is the open task below.

*Statement* 3.177 *(Imported Costello perturbative BV package).* For an elliptic BV theory with a gauge fixing, heat kernel $K_t$, propagator

$$P_{\epsilon,L} = \int_\epsilon^L Q^{\mathrm{GF}} K_t \, dt,$$

and a renormalization scheme, Costello constructs finite-scale effective interactions by the RG operator

$$W(P_{\epsilon,L}, I[\epsilon]) = \hbar \log\bigl(\exp(\hbar \partial_{P_{\epsilon,L}}) \exp(I[\epsilon]/\hbar)\bigr).$$

The renormalized interaction is obtained by adding local counterterms and taking the $\epsilon \to 0$ limit. It has the connected-graph expansion

$$I[L] = \sum_{\Gamma \text{ connected}} \frac{\hbar^{b_1(\Gamma)+\sum_v g_v}}{|\operatorname{Aut}\Gamma|} W_\Gamma(P_{\epsilon,L}; I_{g_v}[\epsilon] + I^{\mathrm{CT}}_{g_v}(\epsilon)),$$

satisfies RG flow and locality, and satisfies the quantum master equation when the obstruction classes vanish [12]. The resulting quantum observables form a factorization algebra [15, 16].

*Statement* 3.178 *(Imported Costello–Li flat open-closed BCOV theorem).* Costello–Li prove a perturbative quantization of open-closed BCOV theory on flat $\mathbb{C}^d$, for $d$ odd, unique up to the equivalences in their

renormalization scheme, with open gauge algebra $\mathfrak{gl}(N \mid N)$ and compatibility under $\mathfrak{gl}(N \mid N) \hookrightarrow \mathfrak{gl}(N+k \mid N+k)$. In that flat open-closed BCOV/HCS theory they compute the anomaly cancellation needed for the graph construction [14]. Ordinary $\mathfrak{gl}_N$ and the mixed $\mathbb{R}^2 \times \mathbb{C}^2$ Hamiltonian brane model are not covered by this theorem without an additional reduction or summand argument.

*Remark* 3.179. The two imported statements supply the universal renormalized BV framework and one major flat BCOV/HCS instance, subject to their hypotheses. They do not by themselves prove the graph realization for the Hamiltonian sector of this paper. Here the brane $L = \mathbb{R}_{\text{brane}} \times \{0\} \times \{p\}$ is a defect in the mixed space, not a spacetime boundary unless a half-space model is separately chosen.

Lemma 3.180 *(Finite-window obstruction for the full Hamiltonian sector).* The polynomial Hamiltonian Lie algebra of the formal symplectic plane admits no naive finite-dimensional Lie truncation compatible with the full Hamiltonian sector. More precisely:

(i) finite degree windows are not Lie subalgebras;
(ii) projected degree-window brackets need not satisfy Jacobi;
(iii) $\mathfrak{m}$-adic jet quotients are not Lie quotients in the presence of linear Hamiltonians;
(iv) there is no directed exhaustion by finite-dimensional Lie subalgebras containing the full polynomial Hamiltonian algebra.

*Proof.* Let $H_d$ be the span of homogeneous polynomials of degree $d$, and let $\mathfrak{m} = (z_1, z_2)$. The Poisson bracket satisfies $\{H_p, H_q\} \subset H_{p+q-2}$. Thus

$$\{z_1^N, z_2^N\} = N^2 z_1^{N-1} z_2^{N-1},$$

which has degree $2N - 2 > N$ for $N > 2$, so degree $\leq N$ windows are not Lie subalgebras.

If one instead projects the bracket back to a degree window, Jacobi can fail. For the bracket $\pi_{\leq 3}\{-,-\}$, take

$$f = z_2, \qquad g = z_2^3, \qquad h = z_1^2 z_2.$$

A direct calculation gives the projected Jacobiator

$$[[f,g]_3, h]_3 + [[g,h]_3, f]_3 + [[h,f]_3, g]_3 = 6z_2^3.$$

The $\mathfrak{m}$-adic quotients are not stable under translations generated by linear Hamiltonians:

$$\{z_1^{N+1}, z_2\} = (N+1) z_1^N \notin \mathfrak{m}^{N+1}.$$

Hence $\mathfrak{h}/\mathfrak{m}^{N+1}$ is not a Lie quotient for the full Hamiltonian algebra.

Finally, any finite-dimensional Lie subalgebra containing

$$a = z_1^2 z_2^2, \qquad b = z_1^3 z_2$$

contains all iterated brackets

$$(\mathrm{ad}_a)^n(b) = (-4)^n z_1^{n+3} z_2^{n+1}, \qquad n \geq 0,$$

an infinite linearly independent set. Thus no directed finite-dimensional Lie-subalgebra exhaustion can contain all polynomial Hamiltonians. □

*Remark* 3.181 *(Linear heat operator versus BV kernel).* On the linear field complex

$$\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1]\oplus\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big),$$

the flat gauge fixing

$$Q^{\mathrm{GF}} = d^*_{\mathbb{R}^2} + \bar\partial^*_{\mathbb{C}^2}, \qquad \Delta = [d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}, Q^{\mathrm{GF}}]$$

gives finite-scale heat operators

$$K_t = K_t^{\mathrm{base}}\widehat{\otimes}\mathrm{id}, \qquad P^{\mathrm{op}}_{\epsilon,L} = \int_\epsilon^L Q^{\mathrm{GF}} K_t^{\mathrm{base}}\, dt\widehat{\otimes}\mathrm{id}$$

satisfying

$$[d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}, P^{\mathrm{op}}_{\epsilon,L}] = K_\epsilon - K_L$$

as continuous operators on the completed coefficient spaces. This is only an operator-level homotopy. To obtain Costello's BV propagator one also needs a coefficient coevaluation kernel for the Tate pair $\widehat{\mathfrak{h}}\oplus\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$. In the product/direct-sum topology used here, the formal sum of homogeneous coefficient coevaluations is not a convergent element of the completed tensor product. Thus the analytic graph problem requires a genuine Tate or weighted coefficient extension of the heat-kernel calculus, or a declared restriction to a smaller sector.

LEMMA 3.182 *(Tate Casimir obstruction for the full Hamiltonian sector).* In the product/direct-sum Tate topology

$$\mathfrak{h} = \prod_{d\geq 1} H_d, \qquad \mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{d\geq 1} H_d^\vee,$$

the coefficient coevaluation

$$C_{\mathfrak{h}} = \sum_{d\geq 1}\sum_i e_{d,i}\otimes e^i_d$$

is not an element of $\mathfrak{h}\widehat{\otimes}\mathfrak{h}^\vee_{\mathrm{cont}}$, nor of $\mathfrak{h}^\vee_{\mathrm{cont}}\widehat{\otimes}\mathfrak{h}$. Consequently the full Hamiltonian sector has a BV pairing but no Costello coefficient kernel in this topology. The operator-level homotopy of Remark 3.181 therefore does not define the BV propagator, the BV Laplacian, or renormalized graph weights for the full sector.

*Proof.* For each $d$, choose a basis $\{e_{d,i}\}_i$ of $H_d$ and the dual basis $\{e^i_d\}_i$ of $H_d^\vee$. An element of a completed tensor product with the direct-sum factor $\bigoplus_d H_d^\vee$ has finite support in that direct-sum direction in the Tate convention used here. The displayed formal Casimir has a nonzero component in every $H_d^\vee$. Hence it is not a convergent tensor. The same support obstruction holds with the tensor factors reversed. Costello's heat-kernel BV formalism requires the inverse pairing as an actual distributional kernel, not merely an abstract continuous operator on the field complex. The weighted Tate completion of Theorem 3.66 resolves this Casimir obstruction by changing the coefficient category; the unweighted diagnosis here remains the structural reason that change is required. □

*Problem* 3.183 *(Hamiltonian specialization of Costello perturbative BV).* Costello supplies the universal renormalized BV graph calculus. For the Hamiltonian sector here, the remaining theorem is conditional on exhibiting an elliptic closed-open parent theory, or a stated dimensional reduction of one, whose restriction is this Hamiltonian BF sector and whose obstruction class vanishes. The required data are:

(i) choose the mechanism: either prove that the Hamiltonian cotangent BF complex

$$\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big)$$

is literally a closed sector of a Costello-type open-closed BV theory, or prove that it is obtained by a specified dimensional reduction. The chosen mechanism must determine the field topology, degree $-1$ BV pairing, differential, local interaction, bracket degree, and a homotopy retract closed under the propagator and RG flow. Since $\widehat{\mathfrak{h}}$ is a completed Tate coefficient system, the proof cannot use a naive finite-degree or $\mathfrak{m}$-adic truncation for the full sector, by Lemma 3.180. It must instead construct a genuine Tate or weighted-coefficient extension of Costello's heat-kernel calculus [12] compatible with the Poisson bracket and RG flow, or restrict to a smaller Lie sector such as the classical $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ truncation and explicitly declare that translations, the conformal $\mathfrak{sl}_2$ sector, and the all-order Moyal closure have been removed. By Lemma 3.182, a damping factor inside the direct-sum dual fixed above is not enough; the coefficient category itself must be replaced by one in which the Casimir converges, or the sector must be restricted;

(ii) specify the asymptotic conditions on $\mathbb{C}^2$ and the brane-defect condition on $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$;

(iii) choose the induced finite-scale gauge fixing, heat-kernel propagator, and BV Laplacian in this sector. The infrared limit $L \to \infty$ may be used only after the zero modes and harmonic projection have been fixed;

(iv) compute the obstruction/anomaly class for this mixed Hamiltonian sector. Concretely, the obstruction should be represented as a local functional class

$$\mathrm{Ob}_n \in H^1\big(\mathrm{Loc}_{\mathrm{bulk+defect}}, Q + \{I_0, -\}\big)$$

coming from the $n$-loop failure of the quantum master equation after counterterms. Costello–Li's flat $\mathfrak{gl}(N \mid N)$ anomaly cancellation supplies a source theorem and a model; transfer is not automatic. To execute it one must prove that constant-Hamiltonian projection and connected trace extraction kill the ordinary $\mathfrak{gl}_N$ anomaly, or else replace the open sector by the appropriate super or special-linear variant. The first defect-local obstruction is the Capelli contact term

$$\omega_{\mathrm{Cap}}(A) = N\,\overline{\mathrm{Tr}(A)},$$

represented by

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\,\mathrm{Tr}(A) \mod \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N).$$

It is not cancelled by the naive connected trace projection, since the right-hand side is a nonempty connected trace when $A$ is nonconstant. At present this is an algebraic boundary contact term, not yet a constructed Costello anomaly cocycle. Known candidate QME repairs first lift it to a degree-one local class and then cancel this class by an explicit closed exchange graph and counterterm,

$$\omega_{\mathrm{Cap}} + W_{\mathrm{closed\,exch}} + (Q + \{I_0, -\})C = 0,$$

remove it by a specified supertrace or central-character normalization, or build it into the renormalized boundary generators. Exhaustiveness is not asserted here;

(v) construct the bulk-to-defect kernel as a cochain map to the Poisson central-operation object of the brane observables. On degree-zero Hamiltonians it must send $f(z_1, z_2)$ to the Weyl-ordered boundary observable $\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)$, and it must satisfy the Ward identity and the quantum moment-map constraint;

(vi) define the relevant boundary ribbon graphs: cyclic order, boundary components, automorphism factors, propagator labels, and the relation between Costello's loop-counting $\hbar$ and the Weyl contraction parameter. Equivalently, one must prove that a single boundary cross-contraction is $P/2$ in the Weyl convention;

(vii) fix the one-propagator boundary graph normalization. For monomials $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, the raw first boundary commutator must have coefficient

$$\hbar(kn - lm) z_1^{k+m-1} z_2^{l+n-1},$$

equivalently the normalized commutator must have coefficient

$$(kn - lm) z_1^{k+m-1} z_2^{l+n-1};$$

(viii) after the first normalization, compute the three-propagator boundary graph normalization and prove that the connected order-$\hbar^3$ commutator contributes exactly

$$\frac{\hbar^3}{24} P^3(f, g)$$

with $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$;

(ix) give the convention for extracting connected single-trace and multi-trace coefficients as cumulants or primitive trace classes, and prove that this extraction cancels, renormalizes, or otherwise accounts for the normal-ordering shift of Remark 3.166.

The computations here settle the formal polynomial Hamiltonian algebra and the formal Moyal coefficients. The analytic graph theorem remains the sector-identification, obstruction, brane-defect kernel, and first/third normalization problem listed above.

*Remark* 3.184 *(Convention firewall).* This local Hamiltonian BF sector is not a compact Calabi-Yau output of a $\Phi_d$ functor, carries no $\kappa_{\mathrm{ch}}$ or $\kappa_{\mathrm{BKM}}$ consequence, and is not evidence for compact $\mathrm{CY}_3$ BCOV anomaly cancellation. The Moyal computation below is local deformation quantization of a formal symplectic disk. A closed-open factorization source is extra structure: descent, QME/anomaly cancellation, completion, and the specialization or defect map must be supplied before one may pass from a formal coefficient calculation to a Costello graph realization. The Weyl algebra fixes the target coefficients; it does not reconstruct the elliptic BV source. In particular the formal disk carries no compact Calabi-Yau datum, no $K3 \times E$ specialization, no chosen $(\Sigma, C)$ specialization, no Igusa cusp form, no Borcherds input form, no BKM denominator identity, and no Stage-1 or Stage-2 $\Phi_d$ specialization.

*Remark* 3.185 *(Parabolic functoriality of the local model).* The m-adic completion at $z_1$ used here breaks the formal symplectomorphism group $\mathrm{Symp}_{\mathrm{form}}(\mathbb{C}^2, 0)$ to its parabolic stabiliser $P_{(z_1)} = \{\varphi \in \mathrm{Symp}_{\mathrm{form}} : \varphi^* \mathfrak{m} \subset \mathfrak{m}\}$ of the maximal ideal. Any cross-volume transfer must therefore supply a parabolic-equivariant comparison, not a full $\mathrm{Symp}_{\mathrm{form}}$-equivariant one; this is one operational meaning of the no-formal-disk-transfer firewall.

### 3.12.4 Formal Weyl/Moyal target, all orders

We use the Weyl/Moyal Rees convention for the formal quantization of the constant Poisson tensor [7]:

$$f * g = m \circ \exp\left(\frac{\hbar}{2}(\partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1})\right)(f \otimes g).$$

The factor $1/2$ is already determined by linear Hamiltonians in this Weyl convention: if $u = az_1 + bz_2$ and $v = cz_1 + dz_2$, then $[\widehat{u}, \widehat{v}] = \hbar(ad - bc)$, so the Baker-Campbell-Hausdorff formula for Weyl-ordered exponentials contributes $\exp(\hbar P(u, v)/2)$. Differentiating these exponentials gives the cross-contraction $P/2$ in the formal Weyl convention. Then the star commutator is odd in $\hbar$:

$$[f, g]_* = \hbar\{f, g\} + \frac{\hbar^3}{24} P^3(f, g) + O(\hbar^5),$$

where $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$. Thus the $\hbar^2$ coefficient of the raw star commutator vanishes in this quantization convention. The normalized quantum Hamiltonian bracket is instead

$$\{f, g\}_\hbar = \hbar^{-1}[f, g]_* = \sum_{s \geq 0} \frac{\hbar^{2s}}{2^{2s}(2s+1)!} P^{2s+1}(f, g).$$

Proposition 3.186 *(Moyal coefficients on monomials).* Let $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$. For every $r \geq 0$,

$$P^r(f, g) = \sum_{a=0}^{r} (-1)^a \binom{r}{a} (k)_{r-a} (l)_a (m)_a (n)_{r-a} z_1^{k+m-r} z_2^{l+n-r},$$

where $(s)_q = s(s-1)\cdots(s-q+1)$ and $(s)_0 = 1$. Terms with a negative displayed exponent are understood to be zero; equivalently the corresponding falling factorial already vanishes. Consequently

$$[f, g]_* = \sum_{\substack{r \geq 1 \\ r \text{ odd}}} \frac{2}{2^r r!} \hbar^r P^r(f, g),$$

so the second-order coefficient of the raw commutator vanishes. The first four odd raw coefficients are

$$\hbar P(f, g), \qquad \frac{\hbar^3}{24} P^3(f, g), \qquad \frac{\hbar^5}{1920} P^5(f, g), \qquad \frac{\hbar^7}{322560} P^7(f, g).$$

*Proof.* Since

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1},$$

the two summands commute as differential operators on the tensor product. The binomial theorem gives

$$P^r = \sum_{a=0}^{r} (-1)^a \binom{r}{a} \partial_{z_1}^{r-a} \partial_{z_2}^a \otimes \partial_{z_1}^a \partial_{z_2}^{r-a}.$$

Applying this to $z_1^k z_2^l \otimes z_1^m z_2^n$ gives the displayed falling factorial formula. Interchanging $f$ and $g$ changes $P^r(f, g)$ by the sign $(-1)^r$, so the star commutator keeps exactly odd $r$ and contributes the factor $2/(2^r r!)$ in order $r$. □

The all-order Moyal expansion, the reduced commutator, and the Capelli-renormalized stable trace assemble into a $\mathbb{C}[[\hbar]]$-linear comparison on degree-zero Hamiltonian generators after the external radial-parts input of Statement D.5 is matched to this Hamiltonian reduction. This is the conditional algebraic part of the quantum derived-center story. It is not a construction of the descendant sector and it is not the Costello graph realization.

THEOREM 3.187 *(Conditional all-order quantum Hamiltonian comparison, degree-zero sector).* Assume the external radial-parts input of Statement D.5, with the Weyl/Capelli image convention matched to the reduced Hamiltonian trace model used here. Let $\mathfrak{h}_\hbar = \bar{A}_\hbar$ be the Moyal Hamiltonian Lie algebra, and let

$$H^{\circ}_{\mathrm{Ham,conn}}(O^{\mathrm{q}}_{\mathrm{brane},\infty})_{\mathrm{ren}}$$

denote the stable connected Hamiltonian trace sector generated by the Capelli-renormalized classes $\left[\overline{J_N(f)}\right]_{N\to\infty}$. Then

$$\Phi^{(\circ)}_\hbar \colon \mathfrak{h}_\hbar \xrightarrow{\sim} H^{\circ}_{\mathrm{Ham,conn}}(O^{\mathrm{q}}_{\mathrm{brane},\infty})_{\mathrm{ren}}, \qquad f \longmapsto \partial_{\mathrm{bb},\hbar}(f) = [\overline{J_N(f)}]_{N\to\infty}$$

is a $\mathbb{C}[[\hbar]]$-linear isomorphism onto the renormalized degree-zero Hamiltonian sector, and it is a Lie-algebra map:

$$\Phi^{(\circ)}_\hbar(\{f,g\}_\hbar) = \hbar^{-1}[\Phi^{(\circ)}_\hbar(f), \Phi^{(\circ)}_\hbar(g)]_{\mathrm{conn,raw}}.$$

Under this assumption, the associated graded at $\hbar = 0$ is the degree-zero Hamiltonian restriction of $\Phi$ in Theorem 2.27. The descendant ($\mathfrak{h}^\vee$, one-antifield Koszul) sector remains governed by Problem 3.47; see Problem 3.189 below for the pinned obstruction.

*Proof.* Corollary 3.169 identifies the target with the stable Capelli-renormalized single-trace generators, hence gives the displayed $\mathbb{C}[[\hbar]]$-linear bijection. Flatness over $\mathbb{C}[[\hbar]]$ follows from the unipotent Capelli triangular system over $\mathbb{C}[[\hbar N]]$ (Lemma 3.168(ii)); the associated graded is the classical trace map because the $r = 0$ triangular term is the leading symbol (Lemma 3.168(iv)). The Lie identity is precisely the conditional content of Theorem 3.170, passed to the connected stable limit under the same radial-parts input. No statement about the cotangent cochain summand is used here. □

COROLLARY 3.188 *(Quantum derived-center comparison, restricted status).* The candidate full quantum derived-center comparison

$$\Phi^{\mathrm{cand}}_\hbar \colon C^\bullet_{\mathrm{CE,cont}}(\mathfrak{h}_\hbar \ltimes \mathfrak{h}^\vee_{\hbar,\mathrm{cont}}[1])[[\hbar]] \longrightarrow Z^{P_{\circ,\hbar}}_{\mathrm{fact}}(\mathrm{Obs}^\hbar_{\mathrm{open}})$$

is not asserted here as an isomorphism. Under the external radial-parts input of Statement D.5, it is proved only after restriction to the degree-zero Hamiltonian generator sector, where it is Theorem 3.187. The descendant sector is conditional on Problem 3.47 and Problem 3.189. A realization of this comparison by Costello renormalized closed-open graphs is the separate analytic problem of Problem 3.183.

*Problem* 3.189 *(Tate-coefficient lift of the descendant cochain isomorphism).* This is the quantum specialization of 3.47. The classical Tate obstruction is closed by the weighted Tate completion of Theorem 3.66 (parts (a)–(d)); the all-order quantum lift on the descendant sector is available only under the hypothesis of 3.68. The weighted-completion and nilpotent-truncation mechanisms are recorded in 3.8. Lemma 3.182 remains the diagnosis of the obstruction in the unweighted Tate pair; the resolution requires a category-level change of coefficient module, supplied by the weighted completion.

*Remark* 3.190. After the chain Proposition 3.163–Proposition 3.164–Lemma 3.168–Theorem 3.170–Corollary 3.171–Proposition 3.191–Theorem 3.187, and assuming the external radial-parts input of Statement D.5, there is no remaining obstruction in the formal Moyal coefficient algebra, the Capelli renormalization, the conditional finite-$N$ radial-parts identity, or the reduced open-line boundary product. There are still two different obligations outside that algebraic package: the Tate completion needed for the descendant cochain summand, and the analytic Costello specialization of Problem 3.183. The classical analogue, Problem 3.47, is already pinned to the Tate obstruction at $\hbar = 0$.

Proposition 3.191 *(Conditional all-order boundary-product normalization in the reduced open-line Weyl model).* For Weyl-ordered Hamiltonian boundary observables placed on the open line $\mathbb{R}$ in the reduced first-order open model $S_0 = \int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2)$, with the zero-mode-removed midpoint propagator

$$G(t,s) = \tfrac{1}{2}\,\mathrm{sgn}(t-s), \qquad \langle \phi_1(t)\phi_2(s)\rangle = \hbar G(t,s), \qquad \langle \phi_2(t)\phi_1(s)\rangle = -\hbar G(t,s),$$

the boundary product on $\bar{A}_\hbar$ is

$$F \star G = m \circ \exp\left(\frac{\hbar}{2} P_\partial\right)(F \otimes G), \qquad P_\partial = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}.$$

Equivalently, every odd raw commutator coefficient is $\frac{2}{2^r r!}\hbar^r P^r(f,g)$. In particular the first and third coefficients are $\hbar P(f,g)$ and $\frac{\hbar^3}{24} P^3(f,g)$. This reduced open-line computation uses no Costello specialization. With these conventions, the Weyl normalization is determined by:

(i) the matrix Weyl algebra commutator $[(\phi_1)^i{}_j, (\phi_2)^k{}_l] = \hbar \delta^i_l \delta^k_j$;
(ii) the zero-mode-removed midpoint propagator $G(\pm,\mp) = \pm\frac{1}{2}$;
(iii) the Baker–Campbell–Hausdorff determination of the cross-contraction weight $P/2$ on linear Hamiltonians;
(iv) the linear-Hamiltonian commutator $[\widehat{u},\widehat{v}] = \hbar(ad - bc)$ for $u = az_1 + bz_2$, $v = cz_1 + dz_2$.

*Proof.* Items (i)–(iv) form a normalization system. By (i)–(ii) the midpoint propagator at $t \neq s$ has weight $\pm\hbar/2$. By (iv) the linear-Hamiltonian Weyl commutator is $\hbar(ad - bc)$, so the cross-contraction weight on linear Hamiltonians is exactly $1/2$, giving $P/2$. Hence by (iii) the BCH formula on the Heisenberg algebra of linear Hamiltonians fixes the cross-contraction kernel on the boundary as $P_\partial = P$ with weight $\hbar/2$.

The all-order boundary product on $\bar{A}_\hbar$ is then determined by: (a) the universal property of the Weyl quantization of the constant Poisson tensor; among associative deformations of the symmetric algebra on $(z_1, z_2)$ extending the Heisenberg commutator, the symmetrised Moyal product is the unique translation-invariant deformation [7, Theorem 6.1]; (b) Proposition 3.186 together with item (iii) above; (c) the antisymmetrisation argument of Proposition 3.192.

The "no additional factor of $N$, sign, or scalar moment-map term" clause is the content of Lemma 3.168(iii): the Capelli contact term is absorbed into the $T \leftrightarrow J_N$ change of basis, and the connected single-trace projection is the standard cumulant projection. Consequently all odd raw commutator coefficients are $\frac{2}{2^r r!}\hbar^r P^r(f,g)$, and for $r = 1, 3$ they are $\hbar P(f,g)$ and $\frac{\hbar^3}{24} P^3(f,g)$. The four primary cases (a)–(d) and a sweep of 14641 monomial pairs through order $\hbar^{11}$ are exact-arithmetic verification checks for this coefficient formula. They are not a substitute for the external finite-$N$ radial-parts input. □

Proposition 3.192 *(Open-line midpoint graph weights).* In the reduced open first-order line model

$$S_0 = \int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2),$$

choose the zero-mode-removed midpoint propagator

$$G(t,s) = \frac{1}{2}\,\mathrm{sgn}(t-s), \qquad \langle \phi_1(t)\phi_2(s)\rangle = \hbar G(t,s), \quad \langle \phi_2(t)\phi_1(s)\rangle = -\hbar G(t,s).$$

For the ordered observable product with the $f$-insertion placed to the right of the $g$-insertion, the induced boundary cross-contraction is $P/2$. Hence the one-edge ordered graph has weight $\hbar P(f,g)/2$, and the three-parallel-edge ordered graph has weight $\hbar^3 P^3(f,g)/48$. After subtracting the reverse ordered product, the commutator contributions are

$$\hbar P(f,g), \qquad \frac{\hbar^3}{24} P^3(f,g).$$

*Proof.* The midpoint value $G(+,-) = 1/2$ assigns one copy of

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$$

to each cross-contraction, with the Weyl factor $1/2$. For one edge there is no nontrivial edge automorphism, so the ordered product contributes $\hbar P(f,g)/2$. The reverse ordered product contributes $\hbar P(g,f)/2 = -\hbar P(f,g)/2$, and the commutator gives $\hbar P(f,g)$.

For three parallel edges between the same two labelled observable vertices, the automorphism group permutes the three identical edges and has order $3!$. The ordered product contribution is therefore

$$\hbar^3 \frac{1}{3!}\left(\frac{1}{2}\right)^3 P^3(f,g) = \frac{\hbar^3}{48} P^3(f,g).$$

Since $P^3(g,f) = -P^3(f,g)$, subtracting the reverse ordered product doubles this coefficient and gives $\hbar^3 P^3(f,g)/24$. □

*Remark* 3.193. Proposition 3.192 is a computation in the reduced open-line Weyl model. It is not the Costello closed-open graph theorem. The latter still requires an elliptic parent or dimensional reduction, a brane-defect condition, finite-scale gauge fixing, the induced bulk-to-defect propagator, orientations and measures on configuration spaces, counterterms, anomaly cancellation, and compatibility with the quantum Hamiltonian reduction.

### 3.12.5 Third-order target for Costello's graph problem

For monomials $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, the third-order Moyal coefficient from Proposition 3.186 is

$$P^3(f,g) = \sum_{a=0}^{3} (-1)^a \binom{3}{a} (k)_{3-a} (l)_a (m)_a (n)_{3-a} z_1^{k+m-3} z_2^{l+n-3},$$

where $(r)_s = r(r-1)\cdots(r-s+1)$ and $(r)_0 = 1$. Therefore

$$[f,g]_* = \hbar(kn - lm) z_1^{k+m-1} z_2^{l+n-1} + \frac{\hbar^3}{24} P^3(f,g) + O(\hbar^5).$$

The exact-arithmetic verification checks this first-order coefficient, the vanishing of the second-order coefficient in the raw commutator, the displayed third-order formula, and the star-commutator normalization on every monomial pair with exponents between 0 and 10 and commutator orders through 11.

This algebraic computation is an observable-product target, not an effective action computation. In Costello's formalism the connected effective action and the product/OPE of quantum observables are related but distinct. For two boundary observables $F, G$, the finite-scale free-field model has the schematic product

$$F \star_L G = m \circ \exp(\hbar \partial_{P_{\partial,L}})(F \otimes G),$$

where $P_{\partial,L}$ is the boundary or brane-defect propagator induced by the chosen kernel. The connected single-trace part is the cumulant, or primitive trace, extracted from this product. In this convention the order in $\hbar$ is the number of boundary cross-contractions; it is not the loop order of the connected effective interaction.

Thus there are two equivalent normalizations to keep separate. In Costello's observable-product notation one asks for the boundary contraction kernel

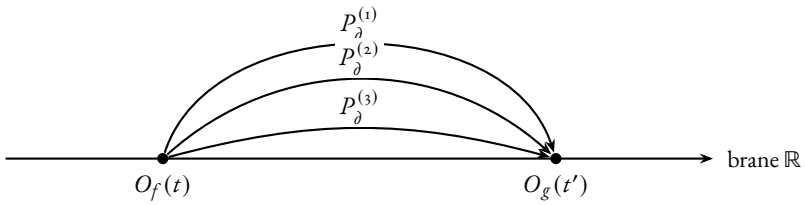
$$P_{\partial,L} \rightsquigarrow \frac{1}{2}P$$

on degree-zero Hamiltonians, so that

$$m \circ \exp(\hbar\partial_{P_{\partial,L}}) \rightsquigarrow m \circ \exp\left(\frac{\hbar}{2}P\right),$$

the Weyl/Moyal product used above. The ordered product has labeled inputs $(f, g)$; its $r$ identical cross-contractions contribute the denominator $r!$. No additional division by interchanging the two boundary vertices is made in the ordered product. The reverse order is the separate product $g * f$ whose subtraction forms the commutator.

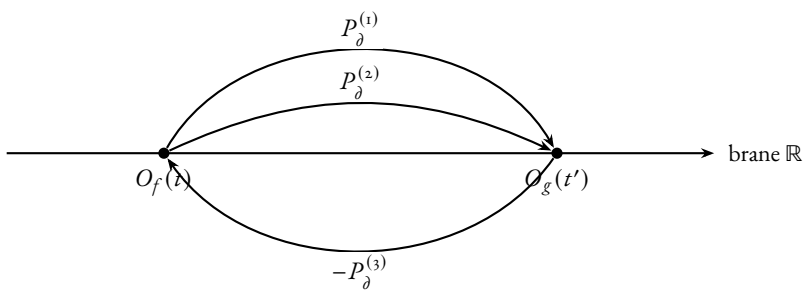
$\Gamma_{3a}$: three propagators, common orientation




$$\mathrm{Aut}(\Gamma_{3a}) = S_3, \quad |\mathrm{Aut}(\Gamma_{3a})| = 3! = 6, \quad \tfrac{1}{|\mathrm{Aut}(\Gamma_{3a})|} = \tfrac{1}{6}$$

orientations $a = 0, 3$: contributions $(k)_3(n)_3$ and $-(l)_3(m)_3$

$\Gamma_{3b}$: two propagators forward, one reversed




$$\mathrm{Aut}(\Gamma_{3b}) = S_2, \quad |\mathrm{Aut}(\Gamma_{3b})| = 2! = 2, \quad \tfrac{1}{|\mathrm{Aut}(\Gamma_{3b})|} = \tfrac{1}{2}$$

orientations $a = 1, 2$: contributions $-3(k)_2 lm(n)_2$ and $+3k(l)_2(m)_2 n$

FIGURE 0.2: The third-order graphs $\Gamma_{3a}$ (top) and $\Gamma_{3b}$ (bottom) for the order-$\hbar^3$ boundary contractions whose tensor target is $P^3(f, g)$. Each cross-contraction contributes one copy of the Poisson bivector $P_\partial$; reversing the orientation of a propagator contributes a relative sign. The orientation multiplicities are the binomial coefficients $\binom{3}{a}$ of Proposition 3.186: $\Gamma_{3a}$ realises $a \in \{0, 3\}$ with automorphism group $S_3$ permuting the three identical propagators (symmetry factor 1/6), and $\Gamma_{3b}$ realises $a \in \{1, 2\}$ with automorphism group $S_2$ permuting the two same-orientation propagators (symmetry factor 1/2). Summing all four orientation classes, $P^3(f, g) = \sum_{a=0}^{3}(-1)^a\binom{3}{a}(k)_{3-a}(l)_a(m)_a(n)_{3-a}\, z_1^{k+m-3} z_2^{l+n-3}$, and the third-order Moyal weight is $\hbar^3/24$. The reduced open-line weights are computed in Proposition 3.192; the Costello counterterm and anomaly checks remain the analytic obligations of Problem 3.196.

The graphs $\Gamma_{3a}$ and $\Gamma_{3b}$ of Figure 0.2 record the displayed order-$\hbar^3$ boundary contraction types whose tensor target is $P^3(f, g)$. Each cross-contraction contributes one copy of the constant Poisson bivector $P$. Their reduced open-line weights and automorphism factors are computed above. Their realization as Costello brane-defect graphs still requires the sector specialization, propagator, counterterm, and anomaly checks in the problem below.

THEOREM 3.194 *(Conditional first and third Costello graph normalization test).* Assume the Hamiltonian specialization data of Problem 3.183 have been supplied for a Costello-type closed-open BV theory, and assume that the induced brane-defect observable product restricts on degree-zero Hamiltonians to the midpoint kernel $P/2$ of Proposition 3.192. Assume further that at order $\hbar^3$ every connected local term not represented by the three-parallel-edge cross-contraction has been shown to vanish, to

be counterterm-exact, or to land in the discarded scalar trace. Then the first- and third-order Costello boundary commutator coefficients are determined by the Weyl/Moyal coefficients:

(i) the connected one-propagator brane-defect ribbon graph carries coefficient $\hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1}$ for monomial inputs $f = z_1^k z_2^l$, $g = z_1^m z_2^n$;
(ii) the surviving three-propagator contribution carries coefficient $\frac{\hbar^3}{24}P^3(f, g)$;
(iii) any finite counterterm ambiguity that changes the first or third coefficient is fixed by matching the Moyal coefficients 1 and 1/24 at these orders. No all-order Costello uniqueness is asserted here.

*Proof.* Under the stated hypotheses, the degree-zero boundary observable product is the reduced open-line Weyl product of Proposition 3.191. Thus each boundary cross-contraction contributes the operator $P$ with weight $\hbar/2$.

Item (i): one edge, no edge automorphisms, weight $\hbar P/2$ for the ordered product (Proposition 3.192); antisymmetrisation gives

$$\hbar P(f, g) = \hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1}$$

on monomials.

Item (ii): for three parallel edges between the two labelled boundary observables, the edge automorphism group is $S_3$. The ordered weight is

$$\hbar^3 \frac{1}{3!}\left(\frac{1}{2}\right)^3 P^3(f, g) = \frac{\hbar^3}{48}P^3(f, g),$$

and antisymmetrisation doubles it to $\hbar^3 P^3(f, g)/24$. The hypotheses exclude the remaining connected local terms at the same order.

Item (iii): a finite local counterterm that changes the one- or three-edge boundary product changes the coefficient of $P$ or $P^3$. The linear-Hamiltonian Heisenberg commutator fixes the coefficient of $P$ to 1, and Proposition 3.186 fixes the coefficient of $P^3$ to 1/24 in the raw commutator. Thus matching these two coefficients removes precisely the counterterm freedom visible at these orders. □

*Remark* 3.195. In the reduced open-line model the Weyl parameter $\hbar$ counts boundary cross-contractions. If a Costello specialization restricts to this free boundary kernel, then its observable-product expansion must be read with this edge-counting convention on degree-zero Hamiltonians. Without the specialization data of Problem 3.183, the statement is only the formal Weyl/Moyal normalization test above.

*Problem* 3.196 *(First and third graph normalizations inside Costello theory).* Work inside the observable product supplied by Statement 3.177, after solving the Hamiltonian specialization data of Problem 3.183. The first graph fixes the normalization; the third graph tests whether the Costello specialization really lands on the Weyl/Moyal product rather than on another deformation quantization convention. The analytic problem is reduced to the following checks:

(i) identify the connected one-propagator brane-defect ribbon graph whose tensor contraction is $P(f, g)$, and prove that its antisymmetrised boundary product gives the raw coefficient $\hbar P(f, g)$;
(ii) fix the sign, automorphism factor, boundary orientation, and trace normalization in this first-order graph so that the coefficient is $\hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1}$ on monomials;
(iii) identify the connected three-propagator brane-defect ribbon graphs whose tensor contraction is $P^3(f, g)$;
(iv) prove that every other connected order-$\hbar^3$ contribution in the Hamiltonian sector vanishes, is counterterm-exact, or lands in the discarded scalar trace;

(v) compute the graph weight and automorphism factor after the first-order normalization and show that the coefficient is $\frac{1}{24}$;

(vi) check that connected large-$N$ extraction introduces no additional factor of $N$, sign, or scalar moment-map term.

The desired output is the pair of algebraic coefficients

$$\hbar P(f,g), \qquad \frac{\hbar^3}{24} P^3(f,g)$$

in the Weyl/Moyal commutator. The formal coefficients are proved above; Theorem 3.194 records the conditional implication once the Costello specialization supplies these checks. Outside that hypothesis, items (i)–(vi) remain the open analytic specialization and normalization checks.

*Remark* 3.197. In characteristic zero, $GL_N$ is linearly reductive on rational representations. Therefore taking $GL_N$-invariants is exact, and

$$H\big((\mathcal{R}_N)^{GL_N}, Q\big) \cong H(\mathcal{R}_N, Q)^{GL_N}.$$

This justifies the invariant calculation used above. A Chevalley BRST model for the quotient stack introduces Lie algebra ghosts as presentation data. Linearly reductive exactness identifies derived algebraic invariants with ordinary invariants in this setting.

## 4 Open obligations

The proved local core is confined to the formal Hamiltonian, residue, factorization-current, scalar-reduced Moyal, and scalar-anomaly claims stated above. The following statements are outside those theorems.

(i) *Proof ownership behind theorem-lane entries.* Each index lane requires a cited theorem, proposition, lemma, or computation whose hypotheses and conclusion match the lane. The index entry itself is not a proof-bearing theorem.

(ii) *Sign and convention appendix.* A convention block must record the BV degrees, Ext shifts, CE/PV shifts, current-density pairings, interval-orientation sign, scalar quotient, and Capelli/Weyl normalization. It must distinguish convention signs from computed signs.

(iii) *Matlis/local-cohomology source boundary.* Appendix A contains the topology on $\mathfrak{h}$, the continuous dual calculation, the residue action, the local-cohomology uniqueness statement, and the no-polynomial-descendant theorem. It also contains the limited Fourier–Rees label bridge: the bridge is a Weyl-module Fourier-normalization and associated-graded label comparison through the Fourier delta Rees lattice, not an $\mathfrak{h}$-module identification and not an equality with the full Cech lattice before $\hbar$ is inverted. The remaining obligation is any theorem that changes the Hamiltonian action or the boundary target enough to evade the same-action local-nilpotence obstruction.

(iv) *Radial-parts source matching.* Appendix D fixes the Weyl algebra conventions, Capelli triangular signs, scalar $U(1)$ quotient, and relation to the stated finite exact ranges. The finite-$N$ all-$N$ radial-parts identity remains conditional on matching the Harish-Chandra–Wallach–Levasseur–Stafford input to this quantum Hamiltonian reduction. The finite exact check is independent finite-range evidence, not a substitute for that source match.

(v) *Unreduced cotangent factorization centre.* An unreduced open BV model whose cotangent sector carries principal-part currents before de Rham-current contraction remains open. The corresponding centrality homotopies against arbitrary open observables are also open. The obstruction is visible before any analytic graph integral is chosen. In the grading used here, $|\phi_i| = 0$, $\psi = A^*$ is odd of BV degree $-1$, a Hamiltonian generator $H_f$ has degree 0, a shifted cotangent generator $\eta_\rho = \rho[1]$ has degree $-1$, and the CE coordinates have $|c_f| = 1$, $|u_\rho| = 0$. The polynomial one-$\psi$ descendants cannot serve as unreduced cotangent representatives: for the linear Hamiltonians,

$$\{O_{1,0}, \widetilde{\Psi}_{a,b}\} = b\,\widetilde{\Psi}_{a,b-1}, \qquad \{O_{0,1}, \widetilde{\Psi}_{a,b}\} = -a\,\widetilde{\Psi}_{a-1,b},$$

while the principal-part coadjoint action is

$$z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

The index shifts go in opposite directions. Hence the primitive one-$\psi$ projection is only the indecomposable $P_0$ shadow of this problem; it is not a centrality homotopy and not an unreduced factorization-centre morphism.

(vi) *One-antifield quantum lift.* The extension of the degree-zero Moyal/Weyl comparison to the descendant sector is open at the unreduced open-BV and factorization-centre level. The finite nilpotent source $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ is not stable under the Moyal bracket for $N \geq 4$, so the lift requires a different Moyal-flat source or a weighted coefficient construction with its own hypotheses. There is, however, a target-side reduced complex: Construction 3.103 defines the distributional principal-part target with full Moyal coadjoint action

$$\{B_f(a), \Theta_\rho(B)\}_\hbar = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB), \qquad \{\Theta_\rho(B), \Theta_\sigma(C)\}_\hbar = 0.$$

This target does not come from the primitive polynomial one-$\psi$ classes. Proposition 3.102 computes

$$\mathrm{ad}^\vee_{z_1^4,\hbar}\,\delta_{1,1} = -24\hbar^2\delta_{0,4} + \cdots,$$

whereas the PBW one-$\psi$ action sends $\widetilde{\Psi}_{1,1}$ to $4\widetilde{\Psi}_{4,0}$. Thus the obstruction survives after primitive projection and is not a multi-trace artifact. Theorem 3.104 strengthens this: no $\hbar$-adic deformation whose leading term is $\widetilde{\Psi}_{a,b} \mapsto \rho_{a,b}$ and whose source linear Hamiltonians have the PBW leading action can intertwine the Moyal coadjoint action. The remaining obligation is therefore not to add higher corrections to the old label map, but to construct the unreduced boundary model, centrality homotopies, and analytic Costello/QME lift landing in the principal-part target.

(vii) *Regulator independence and the strict endpoint.* The weighted Tate regulator is independent of the chosen admissible degree weight inside the finite-window Mittag-Leffler sector, by Theorem 3.73. This is not a theorem that the original product/direct-sum Tate pair carries the same closed-open graph theory: in that strict topology the Costello coefficient Casimir is absent by Lemma 3.182. The stronger strict endpoint is ruled out, under the Costello coefficient-kernel and finite-window compatibility hypotheses, by Theorem 3.74. Tail-sensitive observables in the weighted product completion remain outside the proved regulator-admissible sector, and ordinary cohomology of the weighted product is not identified with the strict direct-sum dual.

(viii) *Mixed brane-defect QME.* The quantum master equation for the brane-defect Costello graph model, with the required boundary kernel and scalar-anomaly normalization, is open. The

Hadamard/Mittag-Leffler theorem supplies the analytic reduction, not the vanishing of the obstruction class. The residual class is the brane-defect local functional

$$\mathrm{Ob} \in H^1\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\}\big),$$

together with the scalar contact normalization. Classically, before scalar reduction,

$$\{I_\partial(a,f), I_\partial(b,g)\}_{\mathrm{open}} = I_\partial(ab, \{f,g\}_{\bar{A}}) + N\,\omega(f,g)\int_{\mathbb{R}} a(t)b(t)\,dt.$$

Quantum mechanically the same boundary contact appears as

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\ \mathrm{Tr}(A) \quad \mod\ \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N).$$

Here $X, Y$ are the Weyl generators corresponding to $\phi_1, \phi_2$. Known candidate QME lifts include cancellation by a closed exchange graph and counterterm, removal by an explicitly stated supertrace or central-character normalization, or absorption into renormalized boundary generators. Exhaustiveness would require a computation of the relevant degree-one obstruction complex.

(ix) *All-order Costello graph realization.* An all-order renormalized BV graph model whose boundary product is the Weyl/Moyal product in the normalization used here remains open. The first and third graph normalizations remain computations to stated order.

(x) *Rees/Fourier bridge between $\Psi$ and $\rho$.* The constructed Fourier–Rees comparison relates the PBW polynomial labels and the principal-part labels only after Fourier twist and degreewise Rees normalization:

$$z_1^a z_2^b \mapsto \partial_1^a \partial_2^b e_0 \mapsto (-1)^{a+b}\hbar^{a+b} a!b!\rho_{a,b}.$$

Before $\hbar$ is inverted this is the Fourier delta Rees lattice $\mathcal{D}_\hbar/(z_1, z_2)$, not the full Cech lattice $\bigoplus \mathbb{C}[[\hbar]]\rho_{a,b}$. It does not intertwine the original Hamiltonian action: Fourier sends the Rees-normalized operators $\hbar X_{z_1}, \hbar X_{z_2}$ to multiplication operators, while the Matlis cotangent module uses the original coadjoint vector fields on principal parts. A same-action bridge from $\Psi_{a,b}$ to $\rho_{a,b}$ is ruled out on the generic fibre by local nilpotence unless the Hamiltonian action or the boundary target is changed. Theorem 3.104 records the corresponding $\hbar$-adic no-go for deformations of the classical primitive projection.

(xi) *Claim-status vocabulary.* Theorem boxes, the claim-strength ledger, outlook material, and theorem-lane entries require the same status vocabulary: proved in finite algebraic model, proved degreewise stable, proved in weighted Tate coefficient model, computed to stated order, conditional on named QME/radial-parts/unweighted-regulator theorem, conjectural, open, or not asserted.

(xii) *Global and cross-volume transfer.* Any compact Calabi–Yau, global BCOV, Vol III chiral-homology, $\Phi_d$, Igusa/BKM/Borcherds, $K3 \times E$, or higher-dimensional derived-centre consequence requires a separate matched-conventions theorem containing the datum of Definition E.5. The local Hamiltonian BF/Moyal calculation supplies no such transfer by itself, and may be cited by sibling volumes only as a local $d = 2$ convention check unless they supply their own proof. The templates in Remark E.6 are obligations, not theorems; citing the template or the no-transfer lemma does not create a comparison morphism.

(xiii) *Rendered-page, table, and figure verification.* The rendered title and abstract pages, claim-strength ledger, local dictionary, principles, theorem lanes, open obligations, and first- and third-order figures require visual verification. The figure labels and captions must state the computed order and hypothesis they support. A successful TeX run by itself does not verify these pages.

(xiv) *Notation and macro hygiene.* Rendered and support notation files should contain no stale macros, duplicated definitions, informal text, or dead notation. A symbol should appear in the support files only when it is used with the same meaning in the manuscript or retained for a declared local convention.

(xv) *Primary-source anchoring.* For each use of Witten centrality, Costello graph/QME formalism, BCOV, Loday–Quillen–Tsygan, Procesi–Razmyslov, Matlis duality, and radial parts, record the source theorem, the local convention translation, what is used, and what is not used. A name citation alone does not transfer a theorem into the local model.

## 5 Outlook and convention firewalls

### 5.1 Protected twisted-M motivation and local reduction

Perturbative twisted M-theory in the sense of Costello [25] is used here as physical motivation for a protected sector near a brane point. It is not used as a global realization theorem. The mathematical assertion in this subsection is only the following reduction: if an ambient twisted-M background supplies the protected Hamiltonian data specified below, then the formal local theorem of §2.14 applies to that restricted sector.

*Definition* 5.1 *(Protected Hamiltonian sector at a brane point).* Let $(X, \omega)$ be a connected holomorphic-symplectic surface, let $p \in X$, and let $\mathcal{T}_{\mathrm{tw}M}(\mathbb{R} \times X; \mathbb{R} \times p)$ denote perturbative twisted-M-theory data on $\mathbb{R} \times X$ in the presence of a stack of $N$ one-dimensional topological branes supported on $\mathbb{R} \times p$. A *protected Hamiltonian sector at* $\widehat{p}$ is a restricted datum

$$\mathcal{T}_{\mathrm{tw}M}(\mathbb{R} \times X; \mathbb{R} \times p)\big|_{\mathrm{Ham},\widehat{p}} = \left(\mathcal{E}_{\mathrm{open,prot}}, \mathcal{E}_{\mathrm{cl,prot}}, \partial_{\mathrm{bb}}^{\mathrm{prot}}\right)$$

consisting of the protected open BV theory on the brane line, the protected closed BV theory coming from formal Hamiltonian operators at the symplectic disk $\widehat{X}_p$, and the protected bulk-to-boundary evaluation on the brane algebra.

This is a name for a restricted object. It does not assert that every ambient twisted-M-theory background admits such an object.

*Definition* 5.2 *(Admissible protected Hamiltonian restriction).* A protected Hamiltonian sector is *admissible for the local theorem* if it comes with the following data.

(i) *Formal Darboux Hamiltonian reduction.* The completed closed Hamiltonian algebra at $p$ is identified, up to formal symplectomorphism, with $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$, and the point brane Ext algebra is identified with $\mathbf{Ext}^*_X(\mathcal{O}_p, \mathcal{O}_p) \cong \Lambda^* T_p X$, with cyclic trace induced by $\omega_p$.

(ii) *Brane-defect locality.* The protected operator restriction to $\mathbb{R} \times p$ factors through the weighted Tate field complex and is compatible with the interval-wise factorization map $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$, the chain-level primitive projection, and the weighted Hadamard parametrix used in the local theorem.

(iii) *Scalar anomaly compatibility.* The centre-of-mass scalar anomaly of the protected brane stack maps to the class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{\mathcal{A}}, \mathbb{C})$ used in the local scalar-anomaly comparison.

*Remark* 5.3 *(Protected sector, not ambient theory).* The admissibility condition is an external input. It is not proved here from the full field content of twisted M-theory on $\mathbb{R} \times X$. Dependence on non-protected directions, holonomy along the topological direction, or gauge data not detected by the formal Hamiltonian jet at $p$ lies outside the local theorem.

PROPOSITION 5.4 *(Admissible protected-sector reduction).* Let $(X, \omega)$ be a connected holomorphic-symplectic surface and let $p \in X$. Suppose that a perturbative twisted-M background with branes on $\mathbb{R} \times p$ supplies an admissible protected Hamiltonian sector at $\widehat{p}$. Then, after the Darboux coordinate and scalar-anomaly normalizations included in the admissibility data, that restricted sector is identified with the formal local open–closed model of §2.14.

The open side is

$$\Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1],$$

the closed side is

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}), \qquad \mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[1], \qquad \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1,$$

and the bulk-to-boundary evaluation is

$$f \longmapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}.$$

Hence the proved Hamiltonian, factorization, and scalar-anomaly statements of the local theorem apply to the protected sector. The all-order one-antifield quantum extension is still conditional on vanishing of the mixed brane-defect QME obstruction of Problem 3.68 for that sector.

*Proof.* The formal Darboux part of admissibility identifies the completed Hamiltonian algebra at the brane point with $\mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$, after choosing formal Darboux coordinates. The point-brane Ext algebra and its cyclic trace give the open coefficient data of the Dirac probe model.

Brane-defect locality supplies the factorization-algebraic restriction to the brane line and matches it with the local factorization map, primitive projection, and weighted Hadamard reduction already used in the local theorem. These hypotheses put the protected sector in the same algebraic category as the local model. They do not prove QME-flatness for the one-antifield quantum extension.

Scalar anomaly compatibility identifies the brane centre-of-mass anomaly with the class $[\bar{c}]$ appearing in the local anomaly comparison. The three admissibility inputs therefore supply exactly the data on which the local theorem acts. Applying that theorem gives the asserted protected-sector reduction. □

LEMMA 5.5 *(Admissibility is not globalization).* An admissible protected Hamiltonian restriction is neither a globalization functor nor a comparison map from twisted-M backgrounds, compact Calabi–Yau threefolds, Vol III CY-to-chiral data, or Igusa/BKM data to the local theorem. It selects only the formal Hamiltonian disk, the Dirac brane Ext algebra, the brane-line factorization restriction, and the scalar anomaly class $[\bar{c}]$.

*Proof.* The three admissibility clauses record Darboux Hamiltonians at $\widehat{X}_p$, the point-brane Ext algebra, brane-defect locality on $\mathbb{R} \times p$, and scalar anomaly compatibility. They contain no compact Calabi–Yau threefold, no negative-cyclic Calabi–Yau class, no Vol III specialization datum $\Phi_d = \mathrm{Sp}_{\Sigma_{d-1},C} \circ \Phi_d^{\mathrm{FA}}$, no compact worldsheet or $S^3$ framing, no BCOV propagator or holomorphic anomaly equation, no Borcherds input Jacobi form, no Weyl chamber, and no compact Hall orientation. Two ambient theories can have the same admissible formal restriction and different choices of all those global data. There is therefore no functorial operation from the local restriction to any of those external structures. The admissible restriction cannot determine any compact $\mathrm{CY}_3$, global BCOV, Vol III, Igusa, Borcherds, or BKM theorem. □

*Remark* 5.6 *(BCOV comparison).* The original BCOV theory [1] is a theory on Calabi–Yau threefolds. The model in this paper is a formal holomorphic-symplectic surface model at $d = 2$, with closed fields $\bar{A} \oplus \bar{A}^\vee[1]$ and branes on the line $\mathbb{R}$. No compact $\mathrm{CY}_3$ BCOV theorem, anomaly cancellation theorem, or global Kodaira-Spencer theorem is imported by the protected-sector reduction above.

*Remark* 5.7 *(Uniformity after admissibility).* Once admissibility has been supplied, the reduction is uniform over connected holomorphic-symplectic surfaces and brane points. This is only the formal Darboux uniformity of the completed symplectic disk. It is not a statement about arbitrary ambient twisted-M backgrounds.

*Remark* 5.8 *(Dirac language inside the protected sector).* For an admissible protected Hamiltonian sector, the Dirac-thesis language of Remark 2.29 is a reading of the already identified formal local model. The brane stack probes the formal symplectic normal disk; Dirac reduction gives the open trace algebra; the closed Hamiltonian BF theory is the shifted cotangent theory of canonical transformations of that disk; and the closed observables identify with derived Poisson-central operations on the stable open Hamiltonian brane algebra only in the reduced local sense proved by Theorem 2.27. The scalar anomaly is the same class $[\bar{c}]$ on the closed and open sides. This is not an ambient twisted-M or BCOV centrality theorem.

*Remark* 5.9 *(Cross-volume firewall).* The protected-sector reduction is local at the formal symplectic disk. It carries no compact Calabi–Yau datum, no $\Phi_d$ functor, no $\kappa_{\mathrm{ch}}$ or $\kappa_{\mathrm{BKM}}$ consequence, no $K3 \times E$ specialization, no Igusa cusp form, no Borcherds input form, no BKM denominator identity, no global BCOV amplitude, no compact-$\mathrm{CY}_3$ chiral-homology computation, and no Stage-1 or Stage-2 $\Phi_d$ specialization. Any global CY-to-chiral, compact BCOV, or Igusa/BKM comparison requires a separate matched-conventions theorem internal to the sibling volume making that claim. This subsection can enter such a theorem only as a $d = 2$ formal-disk normalization after the external global data have already been supplied there.

### 5.2 *Cross-volume firewall*

The theorem proved in this paper is local on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}}$, with brane line $\mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{0\}$. It is a theorem about Hamiltonian BF theory on the formal symplectic disk and its Dirac brane probe. It is not a compact Calabi-Yau theorem, not a Borcherds or BKM theorem, and not a global twisted-M-theory theorem.

The formal Weyl/Moyal calculation fixes the Capelli coefficients on the reduced degree-zero Hamiltonian sector. The conditional Costello first/third graph-normalization test checks the same coefficients inside a putative brane-defect graph model. The all-order one-antifield quantum extension remains conditional on the mixed brane-defect QME class of Problem 3.68.

Appendix E records only a convention firewall. Any transfer to Vol III, Vol II, Vol I, the Igusa cusp-form program, a compact Calabi-Yau threefold, a $K3 \times E$ specialisation, or a BKM denominator identity requires a separate matched-conventions theorem. No such transfer is asserted here.

## Acknowledgements

This work belongs to the lineage of Bershadsky–Cecotti–Ooguri–Vafa 1993, Witten 1988, Costello 2011, and Beilinson–Drinfeld 2004; the local Hamiltonian BF analysis is a finite algebraic chain-level probe of the formal-disk corner of that program. Errors are the author's.

## A Matlis Principal Parts and the Continuous Cotangent Module

This appendix centralizes the reduced Matlis calculation used in Theorem 2.22. Its stable core is the following: the $m$-adic continuous dual of the reduced Hamiltonian Lie algebra is the reduced scalar-annihilator inside the top local-cohomology module; the residue pairing gives the coadjoint action; and no ordinary polynomial descendant module realizes this coadjoint representation. The Fourier–Rees comparison constructed below is only a Weyl-module and associated-graded label comparison. Over $\mathbb{C}[[\hbar]]$ its target is the Fourier delta Rees lattice inside local cohomology, not the full standard Cech lattice; the full principal-part lattice appears only after tensoring with $\mathbb{C}((\hbar))$. It does not identify $\Psi_{a,b}$ with $\rho_{a,b}$ as $\mathfrak{h}$-modules. The source conventions are Matlis duality [19], Grothendieck–Hartshorne local cohomology [20], and the power-series residue model of Kunz–Cox–Dickenstein, Chapter 5 [21].

*Definition* A.1 *(Topology and reduced Hamiltonians).* Put

$$R = \mathbb{C}[[z_1, z_2]], \qquad \mathfrak{m} = (z_1, z_2), \qquad \mathfrak{h} = R/\mathbb{C} \cdot 1.$$

The topology on $R$ is the $\mathfrak{m}$-adic topology. The topology on $\mathfrak{h}$ is the quotient topology, equivalently the topology transported through the zero-constant section

$$\mathfrak{h} \simeq \mathfrak{m}, \qquad \overline{f} \longmapsto f - f(0).$$

A fundamental system of neighbourhoods of zero is

$$F^N\mathfrak{h} = \mathfrak{m}^N, \qquad N \geq 1,$$

under this section. The Lie bracket is the Poisson bracket

$$\{f, g\} = \partial_{z_1} f \, \partial_{z_2} g - \partial_{z_2} f \, \partial_{z_1} g$$

followed by projection modulo constants.

Lemma A.2 *(Continuous dual).* The continuous dual of $\mathfrak{h}$, with $\mathbb{C}$ discrete, is

$$\mathfrak{h}^\vee_{\mathrm{cont}} = \bigcup_{N \geq 1} (\mathfrak{h}/F^N\mathfrak{h})^\vee = \bigoplus_{\substack{a,b \geq 0 \\ a+b>0}} \mathbb{C}\, \delta_{a,b},$$

where

$$\delta_{a,b}(z_1^m z_2^n) = \delta_{a,m}\delta_{b,n}.$$

*Proof.* A linear functional on the $\mathfrak{m}$-adic vector space $\mathfrak{h} \simeq \mathfrak{m}$ is continuous exactly when its kernel contains $F^N\mathfrak{h} = \mathfrak{m}^N$ for some $N$. Such a functional factors through the finite-dimensional quotient $\mathfrak{m}/\mathfrak{m}^N$, hence is a finite linear combination of Taylor coefficient functionals. Conversely every finite linear combination of the displayed $\delta_{a,b}$ kills all sufficiently high powers of $\mathfrak{m}$, hence is continuous. □

*Definition* A.3 *(Reduced Matlis principal parts).* The top local cohomology module of $R$ is written in Cech Laurent form as

$$H^2_{\mathfrak{m}}(R)\, dz_1 dz_2 = \bigoplus_{a,b \geq 0} \mathbb{C}\, z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2.$$

Equivalently this is the Cech quotient

$$R[z_1^{-1}, z_2^{-1}]/(R[z_1^{-1}] + R[z_2^{-1}])$$

after multiplication by $dz_1 dz_2$; the displayed direct sum is the standard representative basis in which both exponents are strictly negative. The reduced principal-part module is the $\mathbb{C}$-linear residue annihilator of constants,

$$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1) = \bigoplus_{\substack{a,b \geq 0 \\ a+b>0}} \mathbb{C}\, \rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2.$$

The residue pairing is

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \mathrm{Res}_{z_1=z_2=0}\, z_1^{m-a-1} z_2^{n-b-1}\, dz_1 dz_2 = \delta_{a,m}\delta_{b,n}.$$

Proposition A.4 *(Matlis realization of the continuous dual).* The residue pairing induces a continuous vector-space isomorphism

$$\mathcal{P} \xrightarrow{\sim} \mathfrak{h}^\vee_{\mathrm{cont}}, \qquad \rho_{a,b} \longmapsto \delta_{a,b}.$$

It is the scalar-reduced restriction of the Matlis evaluation pairing for the complete regular local ring $R$: after the trivialization $\omega_R = R\, dz_1 dz_2$, $H^2_{\mathfrak{m}}(R)\, dz_1 dz_2 \cong E_R(\mathbb{C})$. The subspace $\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1)$ is not asserted to be an $R$-submodule, and $R/\mathbb{C} \cdot 1$ is not an $R$-module quotient; it is the $\mathbb{C}$-linear scalar-annihilator dual to $\mathfrak{h} = R/\mathbb{C} \cdot 1$ and is stable for the Hamiltonian coadjoint action below.

*Proof.* The displayed residue formula identifies each basis vector $\rho_{a,b}$, $a + b > 0$, with the continuous Taylor coefficient functional $\delta_{a,b}$. Lemma A.2 shows that these coefficient functionals form the whole continuous dual. The exclusion of $\rho_{0,0}$ is exactly the condition that the functional vanish on the constant Hamiltonian 1, which has been removed from $\mathfrak{h}$. □

Proposition A.5 *(Residue coadjoint action).* Under the isomorphism of Proposition A.4, the coadjoint action of $f \in \mathfrak{h}$ is the principal-part residue action

$$f \cdot \rho = \pi_{\mathrm{pp}}\{f, \rho\},$$

where $\pi_{\mathrm{pp}}$ projects Laurent two-forms to the span of the principal parts. On monomials,

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1},$$

with the convention that any term with a negative index is zero.

*Proof.* Hamiltonian vector fields preserve the volume form $dz_1 dz_2$. Hence the residue pairing satisfies integration by parts,

$$\langle \{f, \rho\}, g \rangle = -\langle \rho, \{f, g\} \rangle,$$

which is exactly the coadjoint sign convention $(\mathrm{ad}_f^\vee \lambda)(g) = -\lambda(\{f, g\})$.

For $f = z_1^p z_2^q$ and $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2$,

$$\begin{aligned}\{f, z_1^{-a-1} z_2^{-b-1}\} &= p z_1^{p-1} z_2^q\, (-(b+1)) z_1^{-a-1} z_2^{-b-2} \\ &\quad - q z_1^p z_2^{q-1}\, (-(a+1)) z_1^{-a-2} z_2^{-b-1} \\ &= -(pb - qa + p - q) z_1^{p-a-2} z_2^{q-b-2}.\end{aligned}$$

Multiplication by $dz_1 dz_2$ gives $\rho_{a-p+1,b-q+1}$ precisely when the two new indices are nonnegative; otherwise the principal-part projection gives zero. □

THEOREM A.6 *(Residue pairing rigidity).* Up to multiplication by one scalar in $\mathbb{C}^\times$, the residue pairing is the unique nondegenerate continuous pairing

$$\mathcal{P} \times \mathfrak{h} \longrightarrow \mathbb{C}$$

of total degree zero which is equivariant for the coadjoint action above.

*Proof.* Let $\beta$ be another such pairing and write $\beta_{a,b;m,n} = \beta(\rho_{a,b}, z_1^m z_2^n)$. Total-degree zero gives $\beta_{a,b;m,n} = 0$ unless $a + b = m + n$. Equivariance for $z_1 z_2$ gives the second constraint $a - b = m - n$. Together these force $(a, b) = (m, n)$, so

$$\beta_{a,b;m,n} = c_{a,b}\delta_{a,m}\delta_{b,n}.$$

Equivariance for $z_1$ gives $c_{a,b+1} = c_{a,b}$, because $z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}$ and $\{z_1, z_1^m z_2^n\} = n z_1^m z_2^{n-1}$. Equivariance for $z_2$ gives $c_{a+1,b} = c_{a,b}$. All coefficients with $a + b > 0$ are therefore a single scalar $c$, and nondegeneracy forces $c \neq 0$. □

THEOREM A.7 *(No polynomial descendant realization).* Let $\mathcal{M}$ be a polynomial boundary descendant module on which the linear Hamiltonians $z_1, z_2$ act locally nilpotently. Then $\mathcal{P}$ is not isomorphic to any nonzero submodule of $\mathcal{M}$ as an $\mathfrak{h}$-module. In particular the polynomial classes $\Psi_{a,b}$ cannot be identified with the principal parts $\rho_{a,b}$ by an ordinary $\mathfrak{h}$-module isomorphism.

*Proof.* In any ordinary polynomial boundary model, the linear Hamiltonians act as constant coefficient Hamiltonian vector fields; iterating either $z_1$ or $z_2$ eventually exhausts the polynomial degree. Thus these actions are locally nilpotent.

On $\mathcal{P}$, Proposition A.5 gives

$$z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

Hence

$$z_1^N \cdot \rho_{a,b} = (-1)^N (b+1)(b+2)\cdots(b+N)\rho_{a,b+N} \neq 0$$

for every $N$, and similarly for $z_2$. Local nilpotence is preserved by module isomorphism and by restriction to submodules. Therefore no polynomial descendant module with locally nilpotent linear Hamiltonians realizes $\mathcal{P}$. □

THEOREM A.8 *(Fourier–Rees bridge, and the obstruction it does not remove).* Let

$$\mathcal{D}_\hbar = \mathbb{C}[[\hbar]]\langle z_1, z_2, \partial_1, \partial_2\rangle \big/ \big([\partial_i, z_j] = \hbar\delta_{ij}, [z_i, z_j] = [\partial_i, \partial_j] = 0\big)$$

be the Rees Weyl algebra. Let

$$\mathcal{O}_\hbar = \mathbb{C}[z_1, z_2][[\hbar]] \cong \mathcal{D}_\hbar/(\partial_1, \partial_2)$$

be the polynomial Rees module, and let

$$\Delta_\hbar^{\mathsf{F}} = \mathcal{D}_\hbar/(z_1, z_2)$$

be the Fourier delta Rees module, generated by a vector $e_0$ killed by $z_1, z_2$. The Cech local-cohomology lattice

$$\Delta_\hbar^{\mathrm{std}} = H^2_{\mathfrak{m}}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2[[\hbar]]$$

carries the standard action in which $z_i$ multiplies and $\partial_i = \hbar\partial_{z_i}$. Sending $e_0 \mapsto \rho_{0,0} = z_1^{-1}z_2^{-1}dz_1dz_2$ realizes $\Delta_\hbar^{\mathsf{F}}$ as the Rees sublattice

$$\iota_{\mathsf{F}}(\Delta_\hbar^{\mathsf{F}}) = \bigoplus_{a,b\geq 0} \mathbb{C}[[\hbar]]\,(-1)^{a+b}\hbar^{a+b}a!b!\,\rho_{a,b} \subset \Delta_\hbar^{\mathrm{std}}.$$

This inclusion becomes an isomorphism after tensoring with $\mathbb{C}((\hbar))$, but it is not an isomorphism of the two $\mathbb{C}[[\hbar]]$-lattices. The Rees Fourier automorphism

$$\mathsf{F}(z_i) = \partial_i, \qquad \mathsf{F}(\partial_i) = -z_i$$

identifies the Fourier twist $\mathsf{F}(\mathcal{O}_\hbar)$ with $\Delta_\hbar^{\mathsf{F}}$. On basis labels in the Cech realization,

$$z_1^a z_2^b \longmapsto \partial_1^a\partial_2^b e_0 \overset{\iota_{\mathsf{F}}}{\longmapsto} \partial_1^a\partial_2^b\rho_{0,0} = (-1)^{a+b}\hbar^{a+b}a!b!\,\rho_{a,b}.$$

After inverting $\hbar$, this gives a filtered Fourier–Rees label bridge between nonconstant polynomial labels and the reduced principal parts $\rho_{a,b}$, $a+b>0$. Over the Rees lattice before inverting $\hbar$, the bridge records the degree-$a+b$ shift by the explicit factor $\hbar^{a+b}$. The special fibre of the abstract Rees module keeps the labels $\partial_1^a\partial_2^b e_0$; the Cech inclusion into $\Delta_\hbar^{\mathrm{std}}$ sends the positive-degree labels into $\hbar\Delta_\hbar^{\mathrm{std}}$. Thus the comparison is an associated-graded label match, not an isomorphism of ordinary local-cohomology lattices.

This bridge is not an $\mathfrak{h}$-module bridge. The Hamiltonian action on polynomial descendants is by vector fields $X_f = \{f,-\}$ on $\mathcal{O}_\hbar((\hbar))$. In the Rees Weyl algebra the normalized linear operators are $\hbar X_{z_1} = \partial_2$ and $\hbar X_{z_2} = -\partial_1$. Their Fourier transforms are $\mathsf{F}(\hbar X_f)$, whereas the Matlis cotangent action on $\mathcal{P}((\hbar))$ is still $\hbar X_f$, acting on Laurent principal parts and projected to the reduced principal-part span. Already for the linear Hamiltonians,

$$\hbar X_{z_1} = \partial_2, \qquad \hbar X_{z_2} = -\partial_1,$$

but

$$\mathsf{F}(\hbar X_{z_1}) = -z_2, \qquad \mathsf{F}(\hbar X_{z_2}) = z_1.$$

The transformed linear operators are locally nilpotent on $\Delta_\hbar^{\mathsf{F}}$; the Rees-normalized coadjoint operators

$$\hbar X_{z_1}\rho_{a,b} = -(b+1)\hbar\rho_{a,b+1}, \qquad \hbar X_{z_2}\rho_{a,b} = (a+1)\hbar\rho_{a+1,b}$$

are not locally nilpotent on any nonzero reduced principal part after inverting $\hbar$. Dividing by $\hbar$ gives the same conclusion for the unnormalized Hamiltonian vector fields. Hence the Fourier–Rees label bridge cannot identify the polynomial descendant module with $\mathcal{P}$ as a module over the same reduced Hamiltonian Lie algebra.

*Proof.* The displayed formulas for $\mathsf{F}$ preserve the Rees Weyl relations, since

$$[\mathsf{F}(\partial_i), \mathsf{F}(z_j)] = [-z_i, \partial_j] = \hbar\delta_{ij}.$$

The polynomial module $\mathcal{O}_\hbar$ is cyclic with generator 1 annihilated by $\partial_1, \partial_2$. After Fourier twist the cyclic generator is annihilated by $z_1, z_2$, so the twisted module is $\mathcal{D}_\hbar/(z_1,z_2) = \Delta_\hbar^{\mathsf{F}}$. In the Cech model, $z_i\rho_{0,0} = 0$, and applying $\partial_i = \hbar\partial_{z_i}$ to $\rho_{0,0}$ generates the Rees lattice displayed in the theorem. It generates the full standard local-cohomology lattice only after $\hbar$ is inverted. Therefore $\mathsf{F}(\mathcal{O}_\hbar) \cong \Delta_\hbar^{\mathsf{F}}$, with the Cech realization $\iota_{\mathsf{F}}$.

Since $\partial_i$ acts as $\hbar\partial_{z_i}$ on Laurent representatives,

$$\partial_1^a\partial_2^b\rho_{0,0} = \hbar^{a+b}\partial_{z_1}^a\partial_{z_2}^b(z_1^{-1}z_2^{-1}dz_1dz_2) = (-1)^{a+b}\hbar^{a+b}a!b!\rho_{a,b}.$$

Removing the scalar label $a = b = 0$ gives the reduced label bridge after the stated Rees normalization. Since the factors $a!b!$ and $(-1)^{a+b}$ are units over $\mathbb{C}$, the only genuine Rees-lattice shift is the power $\hbar^{a+b}$.

For the obstruction, compare the two linear actions. On polynomial descendants the Rees-normalized operators $\hbar X_{z_1} = \partial_2$ and $\hbar X_{z_2} = -\partial_1$ are locally nilpotent; after inverting $\hbar$, the same is true for $X_{z_1}$ and $X_{z_2}$. Fourier sends the Rees-normalized operators to multiplication by $-z_2$ and $z_1$ on $\Delta_\hbar^{\mathsf{F}}$; multiplication by $z_i$ lowers pole order and is again locally nilpotent. The coadjoint Matlis action is not the Fourier-transformed action. It keeps the same vector fields, now acting on Laurent principal parts:

$$\hbar X_{z_1}\rho_{a,b} = \hbar\partial_{z_2}\rho_{a,b} = -(b+1)\hbar\rho_{a,b+1}, \qquad \hbar X_{z_2}\rho_{a,b} = -\hbar\partial_{z_1}\rho_{a,b} = (a+1)\hbar\rho_{a+1,b}.$$

After tensoring with $\mathbb{C}((\hbar))$, no nonzero vector in $\mathcal{P}((\hbar))$ is killed by a high power of either linear Hamiltonian: for a finite nonzero linear combination, every power of $X_{z_1}$ shifts each nonzero basis summand to a distinct $\rho_{a,b+N}$, with nonzero rising-factorial coefficient, and the same argument applies to $X_{z_2}$. Local nilpotence is invariant under module isomorphism. Thus any Fourier–Rees comparison that preserves the original Hamiltonian action is impossible; the actual Fourier comparison changes the polarization and supplies only the label-level bridge stated above. □

COROLLARY A.9 *(No flat same-action interpolation).* Let $M_\hbar$ be a $\mathbb{C}[[\hbar]]$-flat family whose generic fibre carries the polynomial descendant action of the linear Hamiltonians $z_1, z_2$, so that $X_{z_1}$ and $X_{z_2}$ act locally nilpotently after tensoring with $\mathbb{C}((\hbar))$. Then $M_\hbar((\hbar))$ is not isomorphic, as a module for the same Hamiltonian Lie algebra, to $\mathcal{P}((\hbar))$ with the Matlis coadjoint action. The same statement holds for the Rees-normalized operators $\hbar X_{z_i}$, since $\hbar$ is invertible on the generic fibre. Any honest bridge must therefore either change the Hamiltonian action by a Fourier twist, enlarge the boundary target to a distributional principal-part sector, or state a different deformation problem.

*Proof.* This is the local-nilpotence obstruction of Theorem A.7 after extension of scalars to $\mathbb{C}((\hbar))$. The two linear Hamiltonians remain locally nilpotent on the polynomial descendant generic fibre and remain non-locally-nilpotent on every nonzero element of $\mathcal{P}((\hbar))$. Multiplication by the invertible scalar $\hbar$ cannot change local nilpotence. □

*Remark* A.10 *(What the bridge proves).* The bridge above justifies the shared labels $\Psi_{a,b} \leftrightarrow \rho_{a,b}$ as Fourier–Rees associated-graded bookkeeping, with the factorial and $\hbar^{a+b}$ normalization displayed explicitly. It also records the lattice convention: the Fourier twist gives $\mathcal{D}_\hbar/(z_1, z_2)$, whose Cech realization is the Rees sublattice $\bigoplus \hbar^{a+b}\mathbb{C}[[\hbar]]\rho_{a,b}$, not the full $\bigoplus \mathbb{C}[[\hbar]]\rho_{a,b}$ before $\hbar$ is inverted. It does not weaken the no-polynomial theorem. The theorem says precisely where the apparent bridge breaks: Fourier sends the Hamiltonian vector fields to Fourier-transformed operators, while the Matlis cotangent module uses the original coadjoint vector fields on principal parts.

## B FACTORIZATION CURRENT CONVENTIONS ON THE BRANE LINE

This appendix fixes the current conventions used in Theorem 2.23. Its stable core is the degree-zero Hamiltonian current source, the continuous current dual, the CE coordinate which maps to $B_f(a)$, and the density sign under orientation reversal. It does not construct the unreduced BV cotangent factorization centre; that remains the obligation recorded in Remark 3.160.

*Definition* B.1 *(Hamiltonian current source).* For an open interval $I \subset \mathbb{R}$, the reduced Hamiltonian current Lie algebra is

$$\mathfrak{h}_I = \Omega_c^\bullet(I)\widehat{\otimes}\mathfrak{h}.$$

In the polynomial degree-zero formulas one may replace $\mathfrak{h}$ by its dense scalar-reduced polynomial subalgebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$. If $\alpha, \beta \in \Omega_c^\bullet(I)$ and $f, g \in \mathfrak{h}$, the bracket is

$$[\alpha \otimes f, \beta \otimes g] = \alpha \wedge \beta \otimes \{f, g\}.$$

For degree-zero test functions $a, b \in \Omega_c^0(I)$ this is

$$[a \otimes f, b \otimes g] = ab \otimes \{f, g\}.$$

The measurement notation writes the oriented density associated to $a$ as $a(t)\,dt$; this density notation does not change the de Rham degree of the source test function $a$.

*Definition* B.2 *(Current dual and shifted cotangent source).* Let

$$\mathcal{P} \simeq \mathfrak{h}_{\mathrm{cont}}^\vee$$

be the reduced Matlis principal-part module of Section A. The current dual used on $I$ is

$$\mathcal{P}_I = \mathcal{D}_c'(I)\widehat{\otimes}\mathcal{P},$$

with pairing

$$\langle B \otimes \rho,\ a \otimes f\rangle = B(a)\,\mathrm{Res}(\rho f).$$

If $B = b(t)\,dt$ is a regular compactly supported density, then $B(a) = \int_I a(t)b(t)\,dt$.

The interval shifted-cotangent Lie algebra is

$$\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1].$$

In degree zero the shifted cotangent generator $B \otimes \rho[1]$ is odd. The CE coordinate dual to such a shifted cotangent generator has degree zero. We denote the coordinate dual to the cotangent functional labelled by $a(t)\,dt \otimes f$ by

$$u_{a(t)dt\otimes f}, \qquad |u_{a(t)dt\otimes f}| = 0.$$

The Hamiltonian CE coordinate $c_{a\otimes f}$, dual to the unshifted Hamiltonian generator $a \otimes f$, has degree one.

Proposition B.3 *(Source of the smeared boundary Hamiltonian).* The length-one shifted-cotangent CE coordinate, not a shifted dual of the unshifted Hamiltonian current, is the source of the smeared Hamiltonian operation:

$$u_{a(t)dt\otimes f} \longmapsto B_f(a) = \int_I a(t)\,\overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\,dt.$$

With this convention the degree-zero current bracket is

$$\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab).$$

*Proof.* The CE algebra of $\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1]$ is the completed symmetric algebra on the continuous dual shifted by $[-1]$. A generator of $\mathcal{P}_I[1]$ has degree one in the degree-zero current sector; its CE coordinate therefore has degree zero and is written $u$. Under the CE/PV dictionary this coordinate maps to multiplication by the corresponding Hamiltonian boundary observable. Applying boundary evaluation $f \mapsto \overline{\operatorname{Tr} f(\phi_1, \phi_2)}$ and smearing against the compactly supported density $a(t)\,dt$ gives the displayed $B_f(a)$. The bracket identity is then the equal-time Poisson bracket smeared against $a(t)b(s)\delta(t-s)$, which gives the pointwise product $ab$. □

PROPOSITION B.4 *(Principal-part current bracket).* In the reduced $P_0$ current model after de Rham-current contraction,

$$A_\partial^{\mathrm{pp}}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{P}_I[1]),$$

the degree-zero brackets are

$$\begin{aligned}\{B_f(a), B_g(b)\} &= B_{\{f,g\}}(ab),\\ \{B_f(a), \Theta_\rho(B)\} &= \Theta_{f\cdot\rho}(aB),\\ \{\Theta_\rho(B), \Theta_\sigma(C)\} &= 0.\end{aligned}$$

Here $aB$ is multiplication of a compactly supported current by a smooth test function. On monomial labels and a regular current $B = b(t)\,dt$,

$$\{B_{z_1^p z_2^q}(a), \Theta_{\rho_{i,j}}(B)\} = -(pj - qi + p - q)\Theta_{\rho_{i-p+1,j-q+1}}(aB),$$

with negative-index targets equal to zero.

*Proof.* The first bracket is the Hamiltonian current bracket from Proposition B.3. The second bracket is the negative transpose of the first under the product of the de Rham-current pairing and the residue pairing:

$$\langle B \otimes \rho,\ a \otimes f\rangle = B(a)\operatorname{Res}(\rho f).$$

The residue factor gives the coadjoint action $f \cdot \rho$, computed in Proposition A.5; the line factor gives $aB$. The $\Theta$-sector is declared abelian in the reduced semidirect product, giving the third bracket. □

*Remark* B.5 *(Orientation reversal).* The orientation convention is the ordinary one for one-dimensional densities. Under an orientation-reversing coordinate change, $dt$ contributes its usual sign. No additional sign is inserted from the CE coordinate $u_{a(t)dt\otimes f}$, from the principal-part label $\rho$, or from the shifted cotangent notation. Thus the displayed operations are orientation-covariant after transforming the densities and currents in the usual way; if one reverses only the chosen orientation while holding the written density fixed, the sign is exactly the sign of $dt$.

*Remark* B.6 *(Residual convention boundary).* These conventions prove the reduced $P_0$ current bookkeeping. They do not supply the missing unreduced BV current complex, the primitive cumulant projection before reduction, or the homotopies making $\Theta_\rho$ central against arbitrary unreduced open observables.

## C UNREDUCED BV COTANGENT AND SCALAR QME OBSTRUCTIONS

This appendix records the obstruction which remains after the reduced principal-part current theorem and the weighted Hadamard reduction. It is a no-go statement for the current polynomial unreduced boundary category, not a construction of the missing analytic brane-defect graph realization.

*Definition* C.1 *(BV degrees and signs).* The open polynomial BV/Koszul model uses

$$|\phi_1| = |\phi_2| = 0, \qquad |\psi = A^*| = -1, \qquad Q\psi = [\phi_1, \phi_2], \qquad |Q| = 1.$$

A Hamiltonian generator $H_f$ has degree 0. A shifted cotangent generator $\eta_\rho = \rho[1]$ is odd of BV degree $-1$. The CE coordinates satisfy

$$|c_f| = 1, \qquad |u_\rho| = 0.$$

The reduced $P_0$ bracket has degree 0. The QME obstruction differential $Q + \{I_0, -\}$ has degree 1. The Poisson sign convention is $\{z_1, z_2\} = 1$, and the principal-part coadjoint action is

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1}.$$

Theorem C.2 *(Polynomial unreduced centrality no-go).* Let $M_{\mathrm{pol}}$ be any $Q$-stable subcomplex of the unreduced polynomial open BV observables in which every element has finite polynomial degree in $\phi_1, \phi_2, \psi$. There is no assignment

$$\rho_{a,b}[1] \longmapsto \Theta_{a,b} \in H^\bullet(M_{\mathrm{pol}}, Q)$$

and no system of centrality homotopies in $M_{\mathrm{pol}}$ whose induced $Q$-cohomology action realizes the principal-part module $\mathcal{P}$. Equivalently, the reduced map $\rho_{a,b}[1] \mapsto \Theta_{a,b}$ cannot be lifted to unreduced polynomial boundary representatives with homotopies satisfying

$$\{O_f, \Theta_\rho\} \simeq \Theta_{f\cdot\rho}, \qquad \{\Theta_\rho, \Theta_\sigma\} \simeq 0$$

for all polynomial Hamiltonians $f$.

*Proof.* The linear Hamiltonians act on polynomial open observables by the canonical scalar-translation derivations

$$O_{1,0} \mapsto \partial_{\phi_2}, \qquad O_{0,1} \mapsto -\partial_{\phi_1}.$$

These derivations commute with $Q$, since translating $\phi_1$ or $\phi_2$ by a scalar matrix leaves the moment-map equation $Q\psi = [\phi_1, \phi_2]$ unchanged. On every finite polynomial observable, repeated application of either derivation eventually vanishes. Hence their induced actions on $H^\bullet(M_{\mathrm{pol}}, Q)$ are locally nilpotent.

On the principal-part module the same two Hamiltonians act by

$$z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

Iterating gives

$$z_1^r \cdot \rho_{a,b} = (-1)^r (b+1)\cdots(b+r)\rho_{a,b+r} \neq 0,$$

and similarly for $z_2$, for every $r \geq 1$. Thus no nonzero principal part is locally nilpotent for the linear Hamiltonian action.

If the stated homotopies existed in $M_{\mathrm{pol}}$, then after passing to $Q$-cohomology the action of $O_{1,0}$ and $O_{0,1}$ on the classes $[\Theta_{a,b}]$ would be isomorphic to the displayed principal-part action. Local nilpotence is invariant under module isomorphism and under passage to a submodule. This contradicts the previous paragraph. Therefore the unreduced lift cannot live in the polynomial open BV/Koszul category. A corrected target must add genuinely distributional principal-part boundary fields and then prove new centrality homotopies there. □

PROPOSITION C.3 *(Scalar contact QME class).* Let

$$\omega(f,g) = [\{f,g\}]_0$$

be the constant Taylor coefficient cocycle of Lemma 2.2. For compactly supported $a, b \in \Omega_c^0(\mathbb{R})$, the unreduced scalar boundary contact is

$$\{I_\partial(a,f), I_\partial(b,g)\}_{\text{open}} = I_\partial(ab, \{f,g\}_{\bar{A}}) + N\,\omega(f,g)\int_{\mathbb{R}} a(t)b(t)\,dt.$$

In the quantum Weyl normalization it is equivalently represented by

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\ \mathrm{Tr}(A) \mod \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N).$$

After insertion of the central boundary ghost $\gamma_{\mathbf{1}}$, $|\gamma_{\mathbf{1}}| = 1$, the degree-one obstruction representative is

$$\mathrm{Ob}_{\mathrm{sc}}(\gamma_{\mathbf{1}}; a, f; b, g) = \hbar N\,\omega(f,g)\int_{\mathbb{R}} a(t)b(t)\gamma_{\mathbf{1}}(t)\,dt.$$

Its associated graded CE class is $\hbar N[\bar{c}]$. This class is not exact in the scalar-reduced Hamiltonian source.

*Proof.* The first displayed formula is Corollary 3.162. The second is the quantum version of the same central term: the Weyl convention gives $[X^i{}_j, Y^k{}_l] = \hbar\delta^i_l\delta^k_j$, and the normal ordered moment relation is $Y_\hbar X_\hbar - X_\hbar Y_\hbar + \hbar N I \equiv 0$. Hence $XY - YX = \hbar N I$ in the quantum Hamiltonian reduction, and tracing against $A$ gives the stated sign.

The scalar contact itself has degree 0. Multiplication by the central ghost $\gamma_{\mathbf{1}}$ makes the QME obstruction representative degree 1, the degree of the obstruction complex. The CE associated graded of this representative is the scalar extension cocycle $\hbar N\omega$, hence the class $\hbar N[\bar{c}]$.

If it were exact in the scalar-reduced Hamiltonian source, the leading scalar term would give a linear primitive $\bar{\eta} : \bar{A} \to \mathbb{C}$ with $\delta\bar{\eta} = \omega$. Pulling $\bar{\eta}$ back to $\mathbb{C}[z_1, z_2]$ and setting the value on 1 to zero would contradict the computation

$$\omega(z_1, z_2) = 1, \qquad \{z_1, z_2\} = 1.$$

Equivalently, the only primitive $\eta(f) = -[f]_0$ lives on the unreduced polynomial Hamiltonian algebra and does not descend to $\bar{A}$. Thus the scalar-reduced source has no counterterm whose associated graded kills the class. □

*Remark* C.4 *(Allowed escape routes).* A closed-exchange cancellation would have to produce a local graph contribution $W_{\text{closed exch}}$ and a counterterm $C$ satisfying

$$\mathrm{Ob}_{\mathrm{sc}} + W_{\text{closed exch}} + (Q + \{I_0, -\})C = 0,$$

with associated graded $[W_{\text{closed exch}}] = -\hbar N[\bar{c}]$. The current reduced closed source has no constant-Hamiltonian line from which this scalar primitive can be built. The other honest exits change the input data: use a supertrace theory with $\mathrm{Str}(1) = 0$, impose a central character $\chi$ with $\chi(I) = 0$, or reintroduce the unreduced scalar Hamiltonian and its primitive $\eta(f) = -[f]_0$. None is a proof of QME vanishing for the ordinary $\mathfrak{gl}_N$, scalar-reduced Hamiltonian model.

## D Radial Parts and the Moyal Normalization

This appendix records the proof spine for Theorem 2.25. The stable core is the scalar-reduced Weyl/Moyal normalization, the Capelli change of basis, and the stable connected trace bracket. The finite-$N$ all-$N$ radial-parts step is conditional on the Harish-Chandra–Wallach–Levasseur–Stafford radial-parts input in the normalization stated below. The Costello brane-defect graph realization is not part of this appendix.

*Definition* D.1 *(Formal Weyl/Moyal convention).* Let

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}.$$

The formal Weyl product on $\mathbb{C}[[z_1, z_2]][[\hbar]]$ is

$$f * g = m \circ \exp\left(\frac{\hbar}{2} P\right)(f \otimes g),$$

and

$$[f, g]_{\mathrm{raw}} = f * g - g * f, \qquad \{f, g\}_\hbar = \hbar^{-1}[f, g]_{\mathrm{raw}}.$$

The reduced quantum Hamiltonian Lie algebra is the scalar quotient

$$\bar{A}_\hbar = \mathbb{C}[[z_1, z_2]][[\hbar]]/\mathbb{C}[[\hbar]]$$

as a Lie algebra. It is not used as an associative quotient.

Proposition D.2 *(Moyal coefficients).* For $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$,

$$P^r(f, g) = \sum_{a=0}^{r} (-1)^a \binom{r}{a} (k)_{r-a} (l)_a (m)_a (n)_{r-a} z_1^{k+m-r} z_2^{l+n-r}.$$

Consequently

$$[f, g]_{\mathrm{raw}} = \sum_{\substack{r \geq 1 \\ r \text{ odd}}} \frac{2}{2^r r!} \hbar^r P^r(f, g),$$

so the first and third raw commutator coefficients are

$$\hbar P(f, g), \qquad \frac{\hbar^3}{24} P^3(f, g),$$

and all even raw commutator coefficients vanish.

*Proof.* The two summands in $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$ commute as differential operators on the tensor product. The binomial theorem gives

$$P^r = \sum_{a=0}^{r} (-1)^a \binom{r}{a} \partial_{z_1}^{r-a} \partial_{z_2}^{a} \otimes \partial_{z_1}^{a} \partial_{z_2}^{r-a}.$$

Applying this to the two monomials gives the falling-factorial formula. Interchanging $f$ and $g$ changes $P^r(f, g)$ by $(-1)^r$, so the commutator retains exactly the odd powers and contributes the coefficient $2/(2^r r!)$ in order $r$. □

*Definition* D.3 *(Matrix Weyl algebra and Weyl traces).* Let $\mathcal{W}_N$ be the completed Weyl algebra generated by entries of two $N \times N$ matrices $X, Y$ with

$$[X^i{}_j, Y^k{}_l] = \hbar\, \delta^i_l \delta^k_j, \qquad [X^i{}_j, X^k{}_l] = [Y^i{}_j, Y^k{}_l] = 0.$$

The normal-ordered quantum comoment for conjugation is

$$\widehat{\mu}_\epsilon = -\operatorname{Tr}([\epsilon, X]Y), \qquad \hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a.$$

For $f \in \mathbb{C}[z_1, z_2][[\hbar]]$, define the Weyl-ordered trace

$$J_N(f) = \operatorname{Tr} \operatorname{Op}_{\mathrm{W}}(f)(X, Y).$$

The scalar line is removed after passing to the degree-zero quantum Hamiltonian reduction

$$\mathrm{N}_{\mathcal{W}_N}(\mathcal{W}_N \widehat{\mu}(\mathfrak{gl}_N))/\mathcal{W}_N \widehat{\mu}(\mathfrak{gl}_N).$$

LEMMA D.4 *(Capelli–Weyl triangular normalization).* Let $T_{a,b} = \operatorname{Tr}(X^a Y^b)$ be the normal-ordered trace. In the quantum Hamiltonian reduction, the locally proved Weyl-normal-ordering formula is

$$J_N(z_1^a z_2^b) \equiv \sum_{r=0}^{\min(a,b)} \left(-\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} T_{a-r,b-r}.$$

The inverse triangular system is

$$T_{a,b} \equiv \sum_{r=0}^{\min(a,b)} \left(\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} J_N(z_1^{a-r} z_2^{b-r}).$$

*Proof.* Total Weyl symmetrisation is obtained from normal ordering by commuting $Y$'s past $X$'s. Each contraction contributes the Capelli contact term coming from

$$YX - XY + \hbar N I \equiv 0 \pmod{\mathcal{W}_N \widehat{\mu}(\mathfrak{gl}_N)}.$$

Choosing $r$ contracted $X$-slots and $r$ contracted $Y$-slots gives $\binom{a}{r}\binom{b}{r}$, pairing them gives $r!$, and Weyl ordering contributes $2^{-r}$. The sign is negative because the displayed moment relation rewrites a normal-ordering swap as the lower triangular term $-\hbar N$. The diagonal term is $r = 0$, hence the system is unipotent over $\mathbb{C}[[\hbar N]]$; inversion changes $-\hbar N/2$ to $+\hbar N/2$.

The cited Capelli literature supplies the standard source for the Capelli contact phenomenon. No primary source is used here for this exact bivariate trace formula; the displayed contraction count is the proof of the normalization in this manuscript. □

*Statement* D.5 *(External radial-parts input).* The finite-$N$ radial-parts step uses the following external Harish-Chandra radial-parts theorem for invariant differential operators, in the form refined by Wallach and by Levasseur–Stafford [8–11]. After identifying $\mathcal{W}_N \simeq \mathcal{D}_\hbar(\mathfrak{gl}_N)$, the Harish-Chandra map on invariants is assumed to have the quantum moment-map ideal as kernel, hence to inject the quantum Hamiltonian reduction by conjugation into Weyl-invariant differential operators on the regular Cartan. With $\lambda_1, \ldots, \lambda_N$ the Cartan coordinates and $\Delta = \prod_{i<j}(\lambda_i - \lambda_j)$, the radial-part map is assumed to be conjugated by $\Delta$. This appendix treats this precise Vandermonde-conjugated compatibility statement as source input; it is not re-proved here.

The input includes the exact image normalization for the Weyl trace generators used in this paper:

$$\mathrm{Rad}_0(J_N(f)) = \Delta^{-1}\left(\sum_{i=1}^{N} \mathrm{Op}_W(f)(\lambda_i, -\hbar\partial_{\lambda_i})\right)\Delta.$$

Lemma D.6 proves that a smaller source package would suffice: the pure $z_1$- and $z_2$-power images together with compatibility of the radial map with the Weyl adjoint shears. The Harish-Chandra–Wallach–Levasseur–Stafford anchors presently verify the homomorphism, regular-Cartan landing, and kernel/surjectivity context; they do not by themselves verify this exact Weyl-trace image convention. That convention remains the named conditional input.

Lemma D.6 *(Pure powers and shears determine the image normalization).* Fix $N$. Suppose a Vandermonde-conjugated radial map is defined on the reduced invariant quotient and is compatible with the following data:

$$\mathrm{Rad}_0(J_N(z_1^a)) = \Delta^{-1}\Big(\sum_i \lambda_i^a\Big)\Delta, \qquad \mathrm{Rad}_0(J_N(z_2^b)) = \Delta^{-1}\Big(\sum_i (-\hbar\partial_{\lambda_i})^b\Big)\Delta,$$

for all $a, b \geq 0$, and for every polynomial $p$ it intertwines the Weyl adjoint shear

$$X \mapsto X, \qquad Y \mapsto Y + p(X)$$

with the Cartan shear

$$\lambda_i \mapsto \lambda_i, \qquad -\hbar\partial_{\lambda_i} \mapsto -\hbar\partial_{\lambda_i} + p(\lambda_i).$$

With the sign convention $Y = -\hbar\partial_X^t$, this is the plus shear generated by $\mathrm{Tr}(X^{a+1})/(a+1)$. Then the full image formula

$$\mathrm{Rad}_0(J_N(f)) = \Delta^{-1}\left(\sum_i \mathrm{Op}_W(f)(\lambda_i, -\hbar\partial_{\lambda_i})\right)\Delta$$

holds for every polynomial $f \in \mathbb{C}[z_1, z_2]$.

*Proof.* It is enough to treat monomials. The case $z_1^a$ is one of the assumed pure-power images. For $b \geq 0$ and $a \geq 0$, apply the shear with $p_t(X) = tX^a$ to the pure generator $J_N(z_2^{b+1})$. Weyl quantization is linear in the commutative symbol and the shear sends the symbol $z_2$ to $z_2 + tz_1^a$, hence the coefficient of $t$ in the matrix Weyl trace is

$$(b+1)J_N(z_1^a z_2^b).$$

On the Cartan side the same shear sends $-\hbar\partial_{\lambda_i}$ to $-\hbar\partial_{\lambda_i} + t\lambda_i^a$, so the coefficient of $t$ in

$$\sum_i \mathrm{Op}_W\big((z_2 + tz_1^a)^{b+1}\big)(\lambda_i, -\hbar\partial_{\lambda_i})$$

is

$$(b+1)\sum_i \mathrm{Op}_W(z_1^a z_2^b)(\lambda_i, -\hbar\partial_{\lambda_i}).$$

Intertwining of the shears and the pure $z_2^{b+1}$-image therefore gives the required image for $z_1^a z_2^b$, after division by $b+1$. Linearity gives the result for all polynomials. □

THEOREM D.7 *(Finite-$N$ radial-parts identity, conditional on the input).* Under the radial-parts input of Statement D.5, put

$$S_N(f) = \sum_{i=1}^{N} \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, -\hbar\partial_{\lambda_i}).$$

Then

$$\mathrm{Rad}_0(J_N(f)) = \Delta^{-1} S_N(f)\Delta,$$

and for all $f, g$

$$D_N(f, g) = \hbar^{-1}[J_N(f), J_N(g)] - J_N(\{f, g\}_\hbar)$$

lies in the quantum moment-map ideal. Thus $D_N(f, g)$ vanishes in degree-zero quantum Hamiltonian reduction.

*Proof.* The first displayed identity is the exact image normalization included in Statement D.5. The assumed radial-parts input also makes the conjugated radial map injective on the reduced invariant quotient, so the commutator defect can be tested on the Cartan.

Lemma D.6 explains the minimal normalization package that would imply the full displayed formula: pure $z_1$-powers, pure $z_2$-powers, and compatibility with the plus shear $Y \mapsto Y + p(X)$ in the convention $Y = -\hbar\partial_X^t$, equivalently $z_2 = -\hbar\partial_\lambda$ on the Cartan. The present sources do not verify that package; it remains part of the external input.

Finally the $i$-th summand of $S_N$ is an ordinary one-particle Weyl algebra and distinct summands commute, hence

$$\hbar^{-1}[S_N(f), S_N(g)] = S_N(\{f, g\}_\hbar).$$

Conjugation by $\Delta$ preserves commutators, so $\mathrm{Rad}_0(D_N(f, g)) = 0$. Injectivity of the radial map on the reduced quotient puts $D_N(f, g)$ in the quantum moment-map ideal. □

COROLLARY D.8 *(Conditional stable scalar-reduced connected bracket).* Under the same radial-parts input as Theorem D.7, for each fixed polynomial degree the classes

$$\overline{J_N(f)}$$

have a stable connected large-$N$ limit after removing the empty trace and disconnected products. In that limit,

$$\hbar^{-1}[\partial_{\mathrm{bb},\hbar}(f), \partial_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \partial_{\mathrm{bb},\hbar}(\{f, g\}_\hbar).$$

*Proof.* The Capelli system of Lemma D.4 is triangular with leading term $T_{a,b}$. Its lower terms have smaller total degree and include the scalar empty trace only when both indices have been contracted away. The connected projection removes the empty trace and disconnected products, while preserving the single-trace $J_N$-basis. The finite-$N$ identity of Theorem D.7 is compatible with adjoining extra Cartan variables, because $S_{N+k}(f)$ is obtained from $S_N(f)$ by adding commuting one-particle summands. Therefore the bracket identity stabilizes degreewise in $N$. □

*Remark* D.9 *(Scalar quotient and the $U(1)$ anomaly).* Constants are central in the Moyal Lie algebra and are killed in $\bar{A}_\hbar$. Before this scalar reduction the linear commutator carries the central term $[z_1, z_2]_* = \hbar$, and on the brane side the corresponding matrix trace is the Capelli shift $\hbar N$. The conditional reduced theorem is the theorem after quotienting by this scalar line. The unreduced line is the $U(1)$ anomaly class of Theorem 2.3.

*Remark* D.10 *(Relation to the finite exact verification).* The finite exact verification is independent evidence for the local Moyal and Capelli normalizations, not a substitute for the external radial-parts input. It uses rational arithmetic to check:

Moyal monomial pairs with exponents $0, \dots, 10$ through order 11,

the Capelli triangular round trip for $a, b \leq 10$,

the direct $N = 2$ and $N = 3$ radial restrictions on 80 operator/test-polynomial pairs in each rank,

$[S_2(f), S_2(g)] = S_2([f, g]_*)$ on the primary and higher-order radial test pairs,

the leading connected-cumulant bracket in the primary cases,

and the open-line first/third graph weights.

The finite exact verification therefore checks the coefficients 1 and $1/24$, the vanishing of even commutator orders in the tested range, the Capelli signs, the rank-two Cartan shadow of the radial-parts identity, and the direct rank-two and rank-three restriction of $J_N(f)$ on the stated 80 operator/test-polynomial pairs in each of ranks $N = 2$ and $N = 3$. It does not prove injectivity of the all-$N$ Harish-Chandra radial-part map or source the exact image normalization beyond these low-rank tests.

*Remark* D.11 *(Residual graph problem).* The formal Weyl/Moyal product fixes the algebraic target. It does not prove that a Costello closed-open brane-defect model realizes this target to all orders. That requires the sector specialization, propagator, counterterm, anomaly, and QME data named in Problem 3.196.

## E Cross-volume convention firewall

The theorem proved in this manuscript is local on

$$\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}} = \mathbb{R}_{\mathrm{brane}} \times \mathbb{R}_{\mathrm{top,normal}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}},$$

with brane line $\mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{0\}$. Its closed sector is Hamiltonian BF theory on the formal holomorphic-symplectic disk together with the topological directions, and its open sector is the Dirac brane probe on the brane line. The statement is at $d = \dim_{\mathbb{C}} Y = 2$ in the holomorphic factor and uses the brane line $\mathbb{R}_{\mathrm{brane}}$ as its one-dimensional factorization direction.

This appendix fixes the boundary of that statement. It does not define a cross-volume dictionary. It does not record forward or backward transfer profiles. It only states what the local theorem does not assert.

*Remark* E.1 *(Non-assertions).* No compact Calabi–Yau threefold theorem is asserted here. No theorem of Vol III is asserted here. No Igusa cusp-form, Borcherds product, or BKM denominator identity is asserted here. No global twisted-M-theory realization is asserted here. No global CY-to-chiral statement, compact-$\mathrm{CY}_3$ chiral-homology statement, or global BCOV amplitude statement follows from the local Hamiltonian BF/Moyal calculation.

*Remark* E.2 *(Allowed use).* The local theorem may be used as a convention check: it fixes the $d = 2$ formal-disk normalization, the brane-line factorization convention, the scalar anomaly class $[\bar{c}]$, and the reduced degree-zero Moyal/Capelli coefficients. This is not a compact $\mathrm{CY}_3$ trace pairing, not an $S^3$ framing, not a specialisation to a chiral curve, and not an automorphic or Borcherds construction.

*Remark* E.3 *(Matched-conventions requirement).* Any global transfer to Vol III, Vol II, Vol I, the Igusa cusp-form programme, a compact Calabi–Yau threefold, a $K3 \times E$ specialisation, a compact-$\mathrm{CY}_3$ chiral-homology theorem, a global BCOV amplitude theorem, or a BKM denominator identity requires a

separate matched-conventions theorem. Such a theorem would have to state its holomorphic dimension, worldsheet or specialisation curve, trace pairing, framing datum, coefficient topology, anomaly normalization, deformation parameter, QME hypothesis, exact comparison morphism, and claim status. No such theorem is proved or assumed in this manuscript, and no sibling-volume theorem may use the local result or this appendix as a substitute for these data.

LEMMA E.4 *(No formal-disk transfer).* The local theorem at $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$, with brane line $\mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{0\}$, does not determine a compact Calabi–Yau threefold theorem, a global BCOV theorem, a Vol III CY-to-chiral or chiral-homology theorem, an Igusa cusp-form theorem, a Borcherds product theorem, a BKM denominator theorem, or a $K3 \times E$ specialization.

*Proof.* The local input consists of the formal Hamiltonian algebra $\mathbb{C}[[z_1, z_2]]/\mathbb{C}$, its continuous cotangent extension, the point brane Ext algebra, the brane-line factorization-current convention, the scalar quotient, and the Capelli/Weyl normalization. These data are invariant under the formal Darboux germ and omit compact topology, a negative-cyclic Calabi–Yau class, a worldsheet or specialization curve, an $S^3$ framing, a global BV propagator, the holomorphic anomaly equation, Borcherds input data, a Weyl chamber, a compact Hall orientation, and a compact chiral-homology object. Hence the local theorem can serve only as a $d = 2$ convention check until a separate matched-conventions theorem supplies the missing global data. □

*Definition* E.5 *(Matched-conventions theorem datum).* A cross-volume transfer theorem citing the local result must state at least the following datum: the holomorphic dimension, the ambient geometric object, the worldsheet or factorization-homology specialization curve, the brane or boundary condition, the trace pairing, the framing, the coefficient topology, the anomaly normalization, the deformation parameter, the QME or obstruction hypothesis, the exact comparison morphism, the claim status, and the precise role of the local theorem as convention check, source theorem, or unused comparison datum.

*Remark* E.6 *(Templates, not theorems).* The following are templates for future separate theorems. They are not assertions in this manuscript.

(i) *Vol III CY-to-chiral template.* State $d$, the compact or non-compact CY object, the CY class in negative cyclic homology, the stage-one object $\Phi_d^{\mathrm{FA}}$, the stage-two specialization $(\Sigma_{d-1}, C)$, the trace pairing, the framing, and the QME/anomaly witness. Then prove that the $d = 2$ formal-disk normalization used here is the stated local convention check for that theorem.

(ii) *Compact BCOV template.* State the compact $\mathrm{CY}_3$, holomorphic volume form, BCOV field complex, propagator, gauge-fixing, boundary or brane condition, holomorphic anomaly equation, zero-mode treatment, and counterterm scheme. Then prove the global amplitude statement inside that BCOV theory. The local Moyal coefficient can serve only as a boundary normalization after these choices are fixed.

(iii) *Igusa/BKM template.* State the Jacobi or modular input, Borcherds lift normalization, lattice and Weyl chamber, character, compact $K3 \times E$ or CHL host if one is claimed, Hall or chiral source, orientation line, primitive recognition theorem, and scalar DT/PT/OP normalization. Then prove the denominator or compact realization theorem in that setting. The local theorem supplies no Borcherds input form, no Weyl denominator, and no compact Hall orientation.

These templates are failure tests. If the listed datum is absent, the proposed cross-volume sentence is not a theorem and must be read as an open obligation.

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